

Feasibility study for a European asset registry in the context of the fight against money laundering and tax evasion

Final Report

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Abstract

This study assesses the feasibility of setting up a single access point to relevant asset registries in the EU Member States for competent authorities involved in the fight against money laundering and tax evasion. It first defines the relevant assets for money laundering and tax evasion, maps existing registries and identifies gaps. It then assesses three potential implementation scenarios' legal, operational and IT feasibility and provides a cost-benefit analysis. These scenarios are (1) the creation of additional asset registries at the national level and their interconnection at the EU level; (2) the interconnection of existing national asset registries at the EU level; and (3) the establishment of a centralised EU asset registry. The study further looks at the creation of new registries for assets where currently there are none, examining the possibilities from an operational, legal and cost-benefit perspective. It highlights existing data on these assets that do not stem from registries per se but from alternative sources of information. Further, the three scenarios for creating an EU asset registry are also compared from an operational, IT, legal, and cost-benefit perspective, highlighting the advantages and disadvantages of each. Finally, the study proposes technical options for an EU asset registry, including the user interface and needed architecture.

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List of abbreviations

AML	Anti-money laundering
AMLCAs	Anti-money laundering competent authorities
AMLD	Anti-Money Laundering Directive
APIs	Application programming interfaces
ARIS	Asset registers interconnection system
BDTI	Big Data Test Infrastructure
BO	Beneficial ownership
BORIS	Beneficial Ownership Registers Interconnection System
BRIS	Business Registers Interconnection System
CARIN	Camden Asset Recovery Inter-Agency Network
CCN/CSI	Common Communications Network and Common Systems Interface
CFT	Combating the financing of terrorism
CFR	Charter of Fundamental Rights
CRS	Common Reporting Standards
CSDs	Central securities depositories
CSRD	Central Security Depository Regulation
DAC	Directive on Administrative Cooperation
EAP	European access point
EBA	European Banking Authority
EBOCS	European Business Ownership and Control Structures
EBSI	European Blockchain Services Infrastructure
EC	European Commission
ECB	European Central Bank
ECHR	European Convention for the Protection of Human Rights and Fundamental Freedoms
ECJ	European Court of Justice
e-CODEX	e-Justice Communication via Online Data Exchange system
ECP	European central platform
ECRIS	European Criminal Records Information System
EDPS	European Data Protection Supervisor
eIDAS	Electronic identification, authentication and trust services
ELRA	European Land Registry Association
EP	European Parliament

ESMA	European Securities and Markets Authority
EU	European Union
EUCARIS	European Car and Driving Licence Information System
EUID	European Unique Identifier
EUR	Euro
FACTI	Financial Accountability, Transparency and Integrity
FATF	Financial Action Task Force
FIUs	Financial intelligence units
GDPR	General Data Protection Regulation
HTTPS	Hypertext Transfer Protocol Secure
IBAN	International Bank Account Number
IRI	Insolvency registers interconnection
JSON-LD	JavaScript Object Notation for Linked Data
LEAs	Law enforcement authorities
LED	Law Enforcement Directive
LEI	Legal Entity Identifier
LLCs	Limited liability companies
LRI	Land Registers Interconnection
ML	Money laundering
MS	Member State
NFT	Non-fungible token
RDF	Resource Description Format
REST	Representational State Transfer
SFTs	Securities financing transactions
SNRA	Supranational Risk Assessment
SOAP	Simple Object Access Protocol
STRs	Suspicious transaction reports
TESTA	Trans European Services for Telematics between Administrations
TEU	Treaty on European Union
TF	Terrorist financing
TFEU	Treaty on the Functioning of the European Union
TIN	Tax identification number
TR	Trade repositories

UTF	Unicode Transformation Format
W3C	World Wide Web Consortium

Glossary

Term or acronym	Meaning or definition
Anti-Money Laundering (AML) Supervisors of Financial Institutions	List of authorities designated by the European Banking Authority as responsible for the supervision of financial institutions' compliance with the EU Anti-Money Laundering Directive 2015/849.
Anti-Money Laundering Directive (AMLD)	EU legal framework that aims at preventing the use of the EU financial system for the purposes of money laundering and terrorist financing. There are currently five EU AML directives and a sixth proposed directive on AML and the countering of terrorist financing, on which a position was agreed by the Council in December 2022, and which will form the new AML framework once adopted.
Application programming interfaces (APIs)	A software interface, including protocols and definitions, that serves to facilitate interaction between two applications.
Asset recovery directive proposal	Proposal for a Directive of the European Parliament and of the Council on asset recovery and confiscation (COM (2022) 245 final).
Asset registry	Centralised and comprehensive record or database containing information about various types of assets owned by individuals or entities. An asset registry may include ownership information, asset types, description and specifications, value of assets, acquisition and disposal information, liabilities and encumbrances, etc. The information contained in an asset registry can vary depending on the intended purpose and types of assets being tracked. Asset registries are commonly used by governments, financial institutions, and regulatory authorities to monitor and regulate asset use and ownership. The purpose of an asset registry is to maintain transparency, accountability, and accuracy in tracking and managing assets.
Beneficial owner	Any natural person(s) who ultimately owns or controls a legal entity through direct or indirect ownership of a sufficient percentage of shares or voting rights or ownership interest in that legal entity, as defined in Article 3(6) of AMLD4.
BORIS	The Beneficial Ownership Registers Interconnection System, established under Directive (EU) 2015/849, is a decentralised system interconnecting the central, national registries on beneficial ownership and the European e-Justice Portal through the European Central Platform.
BRIS	The Business Registers Interconnection System, established under Directive 2012/17/EU and the implementing Regulation (EU) 2015/884 of 8 June 2015, is an information system that interconnects business registries in all EU Member States, Iceland, Liechtenstein and Norway. It allows users to search for and access business information from the national registries.
Building blocks	Existing functional components that can be used in the development of a registry.
Central nodes	Components of an IT system that have a higher degree of connectivity to other nodes in the system. Examples of central nodes are national and EU registries or interconnectors.
Central repository module	The part of the proposed system that ensures the storage of data and metadata. The main role of the module is to provide the functions for creation, maintenance, and operation of the EU centralised registries.
Competent authorities	There is no harmonised EU definition of competent authorities. For the purpose of this study, and according to Article 30(6) of the AMLD, competent authorities are those public authorities with designated

	responsibilities for combating money laundering or terrorist financing, as well as tax authorities, supervisors of obliged entities and authorities that have the function of investigating or prosecuting money laundering, associated predicate offences and terrorist financing, and for tracing, seizing or freezing and confiscating criminal assets.
Data acquisition module	The part of the proposed system that ensures the ingestion of data packets received from independent registries that are not capable of direct interconnection.
Data architecture	The ensemble of models, rules, standards, and policies that govern how the data used by an IT system are acquired, stored, integrated, and exploited.
Directive on Administrative Cooperation (DAC)	Consolidated version of Council Directive 2011/16/EU of 15 February 2011 on administrative cooperation in the field of taxation and repealing Directive 77/799/EEC, as successively amended by Council Directive 2014/107/EU (DAC2), Council Directive (EU) 2015/2376 (DAC3), Council Directive (EU) 2016/881 (DAC4), Council Directive (EU) 2016/2258 (DAC5), Council Directive (EU) 2018/822 (DAC6), and Council Directive (EU) 2021/514 (DAC7).
Domain agnostic architecture	An information architecture where all participating systems provide the same type of data (e.g. owner, location, date or record, type of record/transaction, entities connected and their relation and value), irrespective of the domain from which the data come.
EU portal	The component of a registry that is accessible and interactable for users of the registry, which are competent authorities with the requisite access rights granted. This takes the form of a set of web pages, functioning on the basis of World Wide Web methods and standards.
EUCARIS	The European car and driving licence information system is a system for the secure, EU-wide exchange of vehicle and driving licence registry data between authorised public authorities.
EU-LISA	European Union Agency for the Operational Management of Large-Scale IT Systems in the Area of Freedom, Security and Justice.
EU registry	Comprehensive asset registry containing information on assets and owners of the assets with an EU-wide scope.
Exchange protocol	A method and system that facilitates the exchange of information or data between stakeholders in a process by defining a set of common rules for communication that are agreed and applied by all participants.
Financial intelligence units (FIUs)	National authorities in Member States tasked with investigating, analysing and disseminating relevant money laundering information, whose functions and competences differ across Member States.
Gateway infrastructure	A system that facilitates the exchange of information between nodes, conducting a mapping between formats and/or functions.
Gateway module	The part of the proposed system that acts as a hub that securely interconnects and processes data across heterogeneous networks to and from functional domains, facilitating the exchange of information. It enables the direct exchange of data between registries at different levels and the central repository module.
GDPR	Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation) (Text with EEA relevance).

Intermediary nodes	Intermediary nodes are components or devices in a network, that have, among others, the role of facilitating the connection between other components of that network.
Internet-based approach	Method of building the communication mechanism of a system on open standards that form the foundation of the internet.
IT infrastructure	A set of information technology components, including but not limited to software applications, hardware components and network equipment that form the foundation of an IT service.
JSON-LD	A lightweight Linked Data format, extensively used in the exchange of data between systems.
Law enforcement authorities (LEAs)	Competent authorities in Member States tasked with enforcement of the law. Several types of LEAs exist, which can be operated by national and local agencies, responsible for ensuring that money laundering, terrorist financing and predicate offences are accurately investigated.
Metadata	A form of structured information that describes and explains a given set of data, providing context with details such as the source, type, owner and relations to other data sets.
Markets in Crypto-Assets Regulation (MiCA)	Regulation (EU) 2023/1114 of the European Parliament and of the Council of 31 May 2023 on Markets in Crypto-assets and amending Regulations (EU) No 1093/2010 and (EU) No 1095/2010 and Directives 2013/36/EU and (EU) 2019/1937. It establishes harmonised rules for crypto-assets at the EU level, thereby providing legal certainty for crypto-assets not covered by existing EU legislation. By enhancing the protection of consumers and investors as well as financial stability, the regulation promotes innovation and the use of crypto assets.
Markets in Financial Instruments Regulation (MiFIR)	Regulation (EU) No 600/2014 of the European Parliament and of the Council of 15 May 2014 on markets in financial instruments and amending Regulation (EU) No 648/2012.
Monitoring and analytics module	The part of the proposed system that facilitates the evaluation of operational status, by means of parameters, reports and statistics to support management and administrative decisions in relation to the information managed and the own system functions.
National workspace module	The part of the proposed system that allows independent creation, operation and access to Member State-specific registries.
Predicate offence	Criminal act whose proceeds can be used in a more serious crime (e.g. money laundering). The Financial Action Task Force lists designated categories of offences, which includes among others tax crimes.
RDF	Resource Description Format is the W3C standard for web data and a building block of the semantic approach to organising data.
Scenario 1	New registers/databases at national level in addition to the existing ones (Beneficial ownership, bank account) for sources of wealth for which registers do not exist at national level, and their interconnection at European level to form an EU asset registry.
Scenario 2 (interconnected EU registry)	Interconnection of only the existing national registers required under EU law (Beneficial ownership, bank account) and real estate registers with an EU asset registry.
Scenario 3 (centralised EU registry)	Central EU asset registry connecting the existing registers at national level and, for sources of wealth not covered by them, featuring a central EU register/database with national access portals.

Tax evasion	Criminal activity relating to direct and indirect taxes which constitutes a predicate offence to money laundering. The criminal act entails concealing income, which could be earned legally or illegally, from the detection and collection by tax authorities. There is no harmonised EU definition of tax evasion. The definition differs by EU Member State.
Tax Recovery Directive	Directive (EU) 2010/24/EU of 16 March 2010 concerning mutual assistance for the recovery of claims relating to taxes, duties, and other measures.
Transfer of Funds Regulation	Regulation (EU) 2023/1113 of the European Parliament and of the Council of 31 May 2023 on information accompanying transfers of funds and certain crypto-assets and amending Directive (EU) 2015/849.
User interface module	The part of the proposed system that supports the interaction of the users with the data and the functions provided.
VAT Directive	Directive (EU) 2006/112/EC of 28 November 2006 on the common system of value added tax.
VAT Implementing Regulation	Regulation (EU) 282/2011 of 15 March 2011 laying down implementing measures for Directive 2006/112/EC on the common system of value added tax.
Wizard-type interface	A type of interface that leads the user through a sequence of small steps, to configure a function or application, to a level of detail that would otherwise be difficult to achieve for non-technical users.
XML	The Extensible Markup Language is one of the most commonly used methods of describing information structures.

List of country codes

AT	Austria	**IE**	Ireland
BE	Belgium	**IT**	Italy
BG	Bulgaria	**LT**	Lithuania
CY	Cyprus	**LU**	Luxembourg
CZ	Czechia	**LV**	Latvia
DE	Germany	**MT**	Malta
DK	Denmark	**NL**	the Netherlands
EE	Estonia	**PL**	Poland
EL	Greece	**PT**	Portugal
ES	Spain	**RO**	Romania
FI	Finland	**SE**	Sweden
FR	France	**SI**	Slovenia
HR	Croatia	**SK**	Slovakia
HU	Hungary		

Executive Summary

This study aims to assess the feasibility of providing a unique access point for competent authorities involved in the fight against money laundering and tax evasion to relevant asset registers in the EU Member States. The study first defines a list of assets relevant for money laundering and tax evasion, then maps existing asset registries, identifies gaps and provides a feasibility assessment of three potential scenarios from an operational, legal, and IT perspective. In addition, a cost and benefit analysis is performed for each scenario.

Inventory of different assets and sources of wealth (Task 1)

In Task 1, an assessment was undertaken to identify the main relevant assets in terms of money laundering and tax evasion, and in terms of importance for tax recovery. The study used the following three relevance criteria:

- The relevance of the asset in terms of the size of the asset base/ risk of money laundering/tax evasion;
- The extent to which there are mitigating measures in place to combat money laundering/tax evasion with regard to an asset;
- How attractive the asset is for recovery of the proceeds from money laundering/tax evasion.

Based on these criteria, the following assets are considered relevant for the scope of the study:

Cash in hand, cash at bank, listed securities, unlisted securities, crypto coins/NFTs, properties/land, limited liability companies, in-game assets, high-value goods (works of art/jewellery/antiques/collector's items/firearms), commodities (precious raw materials), motor vehicles, ships, aircraft, safe deposits boxes at banks and safe deposit boxes at non-banks.

Mapping of existing registries (Task 2)

In total, 215 registries and alternative sources of information covering the relevant assets were identified across all EU Member States through desk research. Most of these registries (104) are managed by public authorities while the rest (111) are under private entities management, mostly containing some sort of recordings on transactions.

Further, a survey was conducted with the competent authorities (Financial Intelligence Units, law enforcement authorities, tax authorities and anti-money laundering supervisors of financial institutions) as regards the relevant assets identified in Task 1 and the access of competent authorities to ownership information. The answers were grouped to the following categories of assets: cash in hand, unlisted securities, in-game assets, vehicles, high-value goods, commodities (raw precious metals) and safe deposit boxes. Access was defined as occurring through a register or an alternative source, and access to be granted for investigation and/or litigation. While most Member State competent authorities did have access, there was still a significant minority of Member State competent authorities that did not have access which also varied considerably between asset categories. For example, in the case of the tax authorities, while 94% of tax authorities had legal right to access to information for vehicles for investigations purposes, this decreased to 40 % in the case of in-game assets.

Survey results showed that most of the mapped registries use digitally structured databases, which are in general continuously updated. In addition, around half of the respondents of the survey indicated using the standard Unicode encoding for the data, as well as recording the identity and type of legal owner of the asset together with a variety of other indicators. There was an absence of beneficial ownership data in the asset registers surveyed. However, a limited but increasing number of Member States have interconnected their national beneficial

ownership registers with their national asset registers which allows information to more easily available. The study also finds the lack of interconnection of the surveyed asset registries with other registries (at the national and EU level). For instance, only a minority of surveyed registries indicated being connected to other national registries, including the registries of beneficial owners. Most of the surveyed competent authorities report difficulties when searching or accessing the information on the asset and its owner, both in their country as well as across borders.

Mapping of potential registries – gap analysis (Task 3)

Task 3 identified the following characteristics as being essential for the operability of the asset registers: a digitally structured database, a uniform unique identifier for the asset, a uniform unique identifier for owner information, data documentation, adequate data quality, robust security measures, and inter-connectedness of registers.

The operational gaps regarding the additional harmonisation needed to ensure EU-wide sharing of asset data varies across Member States as well as across the assets considered for the gap analysis. For instance, in the case of aircraft and ship registers some countries like France show considerable gaps in terms of digitalisation. In contrast, other countries such as Czechia are much closer to achieving the necessary prerequisites for a possible interconnection of registries. In the case of vehicles, all interviewed authorities tasked with managing registries related to different subgroups of motor vehicles indicated great satisfaction with the already existing EUCARIS system for the cross-border exchange of data.

Operational and IT feasibility assessment (Task 4)

Task 4 assessed the operational and IT feasibility of the **following 3 scenarios:**

- Scenario 1: each Member State would establish and maintain a national asset register which would combine their existing national asset registers. Any new asset register created by a Member State would be included in this national asset register. Interconnection at EU level for competent authorities would be provided through a web-portal.
- Scenario 2: as with Scenario 1 each Member State would establish and maintain a national asset register. The scope of assets to be included in this national asset register would be more restricted than under Scenario 1. It would include the national real estate register(s) with the national beneficial ownership register and the national bank account register. Inter-connection at EU level for competent authorities would be done through a web-portal as under Scenario 1.
- Scenario 3: the EU would establish and maintain a centralised EU asset registry which, similar to the asset scope under Scenario 1, which would contain data from the existing national asset registers and data from any new national asset register set up by Member States. Access to this EU asset registry, with national access portals, for Member State competent authorities would be provided by the EU.

In addition, an alternative scenario drawing from Scenarios 2 and 3 is developed. This alternative scenario would interconnect the existing networks to a central EU portal. For the relevant assets for which no registries exist, MS would have the discretion to either set up registries at national level or directly create them on the EU portal.

Operational feasibility

The operational feasibility assessment highlighted the difficulties in establishing registries in particular for relevant assets were none currently exist (cash in hand, listed securities, unlisted

securities, in-game assets, works of art/jewellery/antiques/collector's items, commodities (precious raw materials), safe deposits at non-banks and crypto coins/NFTs).

The creation of fully-fledged registries denoting ownership and value of assets, which are comprehensive and complete, was mostly found to be operationally not feasible for this group of assets. However, the assessment also highlighted the possibility of leveraging the content of existing registers, as well as the possibility to use alternative sources of data mostly stemming from transaction information, for the fight against money laundering and tax evasion.

IT feasibility assessment of the three scenarios

The IT feasibility assessment shows that each of the three scenarios is technically feasible, but the drawbacks are specifically linked with elements that are more related with the flexibility, data coverage, the complexity of data integration and the security aspects. An advantage with Scenarios 1 and 2 is that they adopt IT technology already implemented at EU level, for example through the Beneficial Ownership Interconnection System (BORIS) and the Business Registers Interconnection System (BRIS). Scenario 3 presents significantly more challenges than Scenarios 1 and 2. The EU has never implemented an equivalent asset registry as required under Scenario 3 and has until now relied on inter-connecting national asset registers. Further, if new asset registers were to be established under Scenario 3, then the EU would need to ensure national compliance with data submissions by national intermediaries. Quality issues identified under money laundering and tax frameworks indicate that such monitoring, including sanctions, is a substantive task that would need to be undertaken at national level rather than at EU level. Scenario 4 could ease the integration process from a technical and administrative point of view.

Cost-benefit analysis

For each of the scenarios, the following costs have been assessed:

- Costs for the EU budget for establishing the registers as well as for Member States. The cost estimates were based on the interconnection of the existing asset registers;
- One-off costs (initial set-up costs), as well as recurring costs (annual costs).

The most important potential benefit of establishing the registers under the three scenarios would be as a means to allow the competent authorities to have real-time, or near to real-time, access to asset information. Tax recovery collection efforts, which are estimated to encompass EUR 1.3 trillion in total, of which EUR 250 billion are considered collectable, could especially be enhanced by asset register information.

Scenario 1 (building of national AML/anti-tax evasion registries, then interconnecting them) is seen as the most expensive option, as it involves the construction of additional national centralised registries for each Member State. The analysis finds that a centralised registry (Scenario 3) is the most beneficial solution purely from considering costs. Economies of scale (building less additional registries) drives down the cost of this option compared to Scenario 1. This is however somewhat offset by the more difficult monitoring, which would need to be handled on EU level if the new asset registers were to be established at a later point.

Scenario 2 is also found as the most cost-effective option amongst the surveyed and interviewed stakeholders. It yields cost advantages and is referred to as the quickest and least risky to implement, as it builds on existing registries. Conversely, it covers by far the least number of assets among the three options. However, this comparative disadvantage is mitigated somewhat through the inclusion of real estate and therefore a strong value coverage,

as the value of real estate in the EU is significantly higher than the asset values of the other assets surveyed (cross-border real estate in the EU is valued at EUR 1.4 trillion). Further, stakeholder consultations suggest that the interconnection of already established registries under Scenario 2 is the most preferred option amongst the three scenarios.

Legal feasibility assessment (Tasks 5 and 6)

EU legal framework and main legal constraints (Task 5)

Task 5 outlines the current EU framework to combat money laundering and tax evasion as well as the framework for tax recovery. Task 5 further analyses the main legal constraints that apply to all 3 scenarios. It shows that for the implementation of all the three scenarios, it is important to take into consideration the provisions of the GDPR, the Charter of Fundamental rights of the European Union and the general principles of EU law, including proportionality, necessity and subsidiarity. GDPR provides for the possibility for Union or Member States' law to restrict the scope of data subjects' rights ([1]) by way of a legislative measure to which the data controller or processor is subject. Such restrictions include an important economic or financial interest of the Union or of a Member State, including monetary, budgetary and taxation matters. As such, measures to combat money laundering and tax evasion would fall within this scope.

Legal feasibility of the 3 scenarios (Task 6)

Task 6 presents a detailed analysis of the three scenarios and briefly discusses an alternative scenario. When considering implementing any of the proposed scenarios, the interference with fundamental rights to privacy and data protection would require special attention.

From this perspective, particularly concerning is Scenario 3 (centralised registry), as the aggregation of data and high level of asset coverage increases the seriousness of the infringement upon fundamental rights to a greater extent than Scenarios 1 and 2. Another important point this Task highlights is the need to develop rules for access for the different pieces of information held in the proposed registries.

The Task further focuses on the GDPR obligations created under the different scenarios (data controller obligations/data processor obligations). For the EU, the creation of an EU asset registry under Scenario 3 would likely incur more accountability from a GDPR perspective in terms of processing, data security and data storage than would be the case if only a web-portal was set up to access the 27 national asset registers as is the case for Scenarios 1 and 2.

In terms of necessity, the effectiveness of the current mitigating measures in the EU frameworks that tackle money laundering and tax evasion like the Directive in Administrative Cooperation in Direct Taxation (the 'DAC') and the AML Directive, would need to be assessed. Under those frameworks, Member States already exchange information on a cross-border basis.

([1]) These are embedded in articles 12 to 22 of the GDPR.

Options for the design and functionalities of an EU asset registry (Task 7)

In terms of user interface, the study proposes a mechanism that makes use of the semantic relations between records to build queries. This mechanism would use the vague semantic relations between records to enable the construction of queries based on incomplete values (e.g. list legal entities that own a ship or an airplane located in a different Member State than the place of registration). The mechanism would be doubled by a written query-based system that allows advanced users to incrementally develop queries. This would enable users to include multiple sources in their search, and to work with sources that include properties that are outside the range of the central system (e.g., the main query at EU level can include subqueries run on properties that are specific to a domain that only one country uses).

Note de Synthèse

Cette étude a pour objectif d'évaluer la possibilité d'offrir un point d'accès unique aux autorités compétentes des États membres de l'Union Européenne impliqués dans la lutte contre le blanchiment d'argent et l'évasion fiscale.

L'étude commence par la définition des principaux actifs liés au blanchiment d'argent et à l'évasion fiscale. La liste est suivie par une cartographie des actifs et de l'identification des lacunes présentes dans les registres, ensuite complétée par une évaluation de faisabilité de trois différents scénarios. Les trois scénarios sont analysés d'un point de vue opérationnel, légal, informatique, et de rentabilité.

Inventaire des actifs et des sources de richesse (Tâche 1)

La Tâche 1 identifie les principaux actifs liés aux activités de blanchiment d'argent et d'évasion fiscale, une analyse basée sur le niveau de contribution de des actifs au recouvrement d'impôts. L'étude se base sur les trois critères suivants :

- La pertinence de l'actif lié au nombre d'actifs/ du risque de blanchiment d'argent/ risque d'évasion fiscale;
- Étendue des mesures existantes pour combattre le blanchiment d'argent/ l'évasion fiscale liés aux actifs;
- Valeur de l'actif pour un recouvrement de revenus liés au blanchiment d'argent/à l'évasion fiscale.

Sur base des critères mentionnées, les actifs suivants sont considérés pertinents :

Espèces en caisse, espèce en banque, titres cotés en bourse, titres non cotés, crypto-monnaies/NFT, propriétés privés (immobiliers et terrains), société à responsabilité limitée, bien virtuels (in-game assets), biens de grande valeur (œuvres d'art, bijoux/antiquités/pièces de collection/armes à feu), commodités (matières premières précieuses), véhicules motorisés, navires, aéronefs, dépôts sécurisés en banque, dépôts sécurisés non-bancaires.

Cartographie des registres (Tâche 2)

Un total de 215 registres d'actifs et sources d'information secondaires ont été identifiés à travers les États membres de l'Union Européenne. Une grande partie des registres (104) sont gérés par une autorité publique, et le reste (111), est géré par des acteurs privés et contiennent principalement des données de transaction.

Une enquête a également été faite auprès des autorités compétentes (cellules de renseignement financier, autorités de mise en œuvre, autorités fiscales et superviseurs de blanchiment d'argent d'institutions financières) au sujet des actifs, précédemment identifiés, auxquels ont accès les autorités sur la propriété des actifs. Les réponses des autorités ont catégorisé les actifs de la manière suivante : espèces en caisse, titres non cotés, biens virtuels (in-game assets), véhicules, biens de grande valeur, commodités (métaux précieux), et coffres sécurisés. Un accès est défini comme ayant recours aux informations du registre dans le cadre d'une enquête ou d'un litige, ce qui est de facto le cas dans la majorité de États membres. Il existe néanmoins, dans quelques pays, des autorités nationales sans accès aux données. Une variation entre les différentes catégories d'actifs a également pu être observé. Les autorités fiscales ont dans 94 % des cas le droit d'accéder aux informations liées aux véhicules

dans le cadre d'une investigation, un taux d'accès qui baisse à 40 % pour les biens virtuels (in-game assets).

Les résultats de l'enquête indique qu'une grande partie des registres sont constitués de données sur base digitale, ce qui impliquerait une mise à jour régulière des bases de données. Prêt de la moitié des participants à l'étude ont indiqué utiliser un Unicode standardisé pour enregistrer les identités et le statut des propriétaires légaux, en les combinant avec d'autres indicateurs. Limité, mais à la hausse, est le nombre d'États membres avec une interconnexion entre le registre de propriétaire bénéficiaire et le registre national d'actifs, un lien qui simplifie l'accès aux données. L'étude démontre également que le lien entre les différents registres est limité (au niveau national et de l'UE). Par exemple, un nombre limité des registres participant à notre enquête ont indiqué être connectés à d'autres registres nationaux, et cela même quand le registre des propriétaire bénéficiaires est considéré. La majorité des autorités indiquent aussi faire face à des difficultés de recherche et/ou d'accès aux informations des propriétaires d'actifs, que ce soit un accès national ou transfrontalier.

Cartographie de potentiels registres – analyse des lacunes (Tâche 3)

La Tâche 3 a identifié les caractéristiques suivantes comme essentielles pour le bon fonctionnement d'un registre d'actifs : une base digitale structurée, un identifiant uniforme et unique des actifs, un identifiant uniforme et unique par propriétaire, la documentation des données, un niveau de qualité des données, de fortes mesures de sécurité, et une interconnexion des bases de données.

Des lacunes opérationnelles sont comptabilisées dans l'analyse, cela en termes des besoins nécessaires pour pouvoir harmoniser et échanger des données d'actifs entre États membres. Cette disparité peut être illustrée en comparant la situation des bases de données des aéronefs et des navires en France et en Tchéquie. En France, les registres sont rarement numérisés, ce qui est le cas pour la quasi-totalité des donnés Tchèques. La Tchéquie serait presque en capacité d'interconnecter des registres.

Toutes les autorités participantes à l'étude ont indiqué une forte satisfaction avec le système EUCARIS, développé pour faciliter l'échange transfrontalier de données d'entre États membres.

Analyse de la faisabilité opérationnel et informatique (Tâche 4)

La Tâche 4 analyse la faisabilité opérationnel et informatique des trois scénarios suivants :

- Scénario 1 : Chaque État membre établirait et maintiendrait un registre d'actifs national, liant les différents registres existants. Tout nouveau registre établit au niveau européen serait inclus dans ce registre national. L'interconnexion des différents registres européens se ferait à travers un portail web au niveau de l'UE.

- Scénario 2 : Comme pour le Scénario 1, chaque État membre serait demandé d'établir et de maintenir un registre national d'actifs. Les actifs inclus dans ce registre national serait plus restreint que dans Scénario 1 et serait composé du registre national immobilier, du registre national des bénéficiaires et du registre national des comptes bancaires. Tout comme pour le Scénario 1, l'interconnexion entre autorités compétentes de l'UE serait rendue possible en utilisant un portail web au niveau de l'UE.

- Scénario 3 : L'Union Européenne établirait et maintiendrait un registre d'actifs centralisé, incluant les mêmes actifs que ceux du Scénario 1, qui combinerait les données des registres

nationaux existants aux nouveaux registres d'actifs nationaux établi par les États membres. L'accès à ce registre d'actifs établi au niveau de l'UE se ferait en utilisant des portails d'accès nationaux, fournis aux autorités compétentes des États membres, par l'UE.

Un quatrième Scénario est également analysé. Il se base sur le Scénario 2 et 3 et développerait une interconnexion entre les réseaux existants et un portail central de l'UE. Les États membres seraient demandés à développer, soit un registre des actifs non-répertoriés au niveau national, ou de créer un registre directement dans le portail.

Faisabilité opérationnelle

L'évaluation de la faisabilité opérationnelle met en avant les difficultés à développer un registre d'actifs de biens quand aucune base de données existe au préalable (espèces en caisse, titres cotés, titres non cotés, biens virtuels, œuvres d'art : bijoux/antiquités/articles de collection, matières premières (matières premières précieuses), dépôts sécurisés non bancaires et crypto-monnaies/NFT).

La création d'un registre complet indiquant la propriété et la valeur des actifs, est considéré comme impossible pour ce type de biens. L'étude a néanmoins démontré que les contenus des registres existantes et la possibilité d'utiliser des sources alternatives de donnés, principalement des données de transactions, pourrait contribuer à la lutte contre le blanchiment d'argent et l'évasion fiscale.

Faisabilité informatique des trois Scénarios

L'évaluation de faisabilité informatique a démontré que les trois scénarios sont possibles, mais avec des inconvénients liés à la flexibilité, l'étendue des données, la complexité à intégrer les données et les aspects de sécurité. L'un des avantages des Scénarios 1 et 2 est lié à leur utilisation de technologies informatique déjà implémentés au niveau européen, comme le Système d'interconnexion des registres des bénéficiaires effectifs (BORIS) et le Système d'interconnexion des registres du commerce ou des sociétés (BRIS). Le Scénario 3 représente plus de défis que les Scénarios 1 et 2. Un registre d'actifs, comme celui suggéré sous le Scénario 3, n'a jamais été développé. Ce système situé au niveau de l'UE, ne pourrait pas se baser sur un système d'interconnexion des registres nationaux. Un nouveau registre, développé sous le Scénario 3, obligerait l'UE de garantir la conformité nationale des données partagés par les intermédiaires nationaux. Ce Scénario ferait face à des problèmes de qualité liés à la surveillance du blanchiment d'argent et de l'évasion fiscale car elle nécessiterait de larges ressources financières. Ces mêmes ressources seraient utilisées de manière plus efficace au niveau national qu'au niveau européen. C'est dans ce cadre, que le Scénario 4 pourrait faciliter un processus d'intégration d'un aspect administratif et technique.

Analyse des avantages et des coûts

Pour chacun des scénarios, les coûts suivants ont été considérés :

- Coûts pour le budget de l'UE et des États membres liés à la mise en place des registres. Les estimations se basent sur l'interconnexion des registres d'actifs existants.
- Frais ponctuels (pour la mise en place d'un registre), et les coûts récurrents (coûts annuels).

Le bénéfice principal lié à la mise en place d'un registre, sur base des trois scénarios, est la création d'un accès aux donnés des actifs en temps réel (ou quasi-temps réel). Les efforts de

recouvrement d'impôts, estimé à EUR 1.3 mille milliards, dont EUR 250 milliards récupérables, pourrait être accru grâce aux informations des registres d'actifs.

Le Scénario 1 (création d'un registre national de blanchiment d'argent/évasion fiscale progressivement interconnecté) est considéré comme l'option la plus couteuse puisqu'elle nécessiterait la construction d'un nouveau registre dans chaque État membre. L'analyse conclu qu'il serait, d'un aspect purement économique, moins couteux de créer un registre central (Scénario 3) européen. Une économie d'échelle (limitant le nombre de nouveaux registres) ferait baisser les couts de ce scenario, en comparaison avec le Scénario 1. Cette option a cependant comme inconvénient la difficulté des contrôles, car cela demanderait à l'UE d'agir en tant que superviseurs quand un nouveau registre d'actifs est développé.

Les résultats de l'enquête faite parmi les différents acteurs, conclu que le Scénario 2 est considéré comme le moins couteux. Il est économiquement avantageux et peut être considéré comme le plus rapide et le moins risquée à implémenter, car il se base sur des registres existants, même si l'étendue des données des actifs est plus limitée que dans les autres scénarios. Ce désavantage est partiellement réduit grâce à l'inclusion des biens immobilier, d'une valeur largement supérieure à celle des autres actifs passés en revue (la valeur des biens immobiliers transfrontalier de l'UE est estimée à EUR 1.4 mille milliards). Les consultations avec différents acteurs indiquent que l'utilisation des interconnexions existantes sous le Scénario 2 serait la meilleure option des trois scénarios.

Évaluation de la faisabilité juridique (Tâches 5 et 6)

Cadre juridique de l'UE et principales contraintes juridiques (Tâche 5)

La Tâche 5 schématise le cadre de l'UE sous lequel s'organise le combat contre le blanchiment d'argent et l'évasion fiscale tout comme celui du recouvrement d'impôts. Elle analyse également les limites légales des trois scénarios. Elle démontre que pour une implémentation des trois scénarios, il faut considérer le RGPD, la Charte des droits fondamentaux de l'Union Européenne et les principes généraux du droit européen, ce qui inclus la proportionnalité, la nécessité d'intervention et la subsidiarité. Le RGPD offre la possibilité aux législations de l'UE et des États membres, de limiter l'étendue des droits des sujets soumis à une vérification dans le cadre de cette législation, en utilisant des mesures législatives, qui s'appliquent aux contrôleurs ou au processus de contrôle. Ces restrictions incluent d'importants intérêts économiques et financiers de l'Union et des États membres, ce qui englobe les aspects monétaires, budgétaires et fiscaux, vu que le combat contre le blanchiment d'argent et l'évasion fiscale se trouverait dans ce cadre limité.

Faisabilité juridique des trois Scénarios (Tâche 6)

La Tâche 6 analyse de manière détaillée les trois scénarios et discute brièvement d'un scénario alternatif. Quand on considère l'implémentation des différents scénarios, il faut impérativement vérifier leurs liens aux droits fondamentaux de la vie privé et la protection des données.

Le Scénario 3 (mise en place d'un registre centralisé), comparé aux Scénarios 1 et 2, devient particulièrement préoccupant car le regroupement de données et l'amplification de la quantité de données risque de fortement augmenter les infractions liées aux droits fondamentaux. Un autre aspect important établie par l'étude est le besoin de développer des règles d'accès aux données du registre.

La Tâche 6 se concentre également sur les obligations définies sous le RGPD liés aux différents scénarios (obligation de contrôlés des données/obligations liées aux processeurs des données). Pour l'UE, la création d'un registre d'actifs sous le Scénario 3 risque de créer des responsabilités de processus, de sécurité et de stockage de données vis-à-vis du RGPD. Ces risques sont liés à la mise en place d'un portail en ligne pour accéder aux données des 27 États membres, stockés dans des registres nationaux, comme sous les Scénarios 1 et 2.

En termes de besoins, il serait nécessaire de revoir le cadre des différentes mesures de l'UE qui essayent de combattre le blanchiment d'argent, comme la directive relative à la coopération administrative ('DAC') et la directive sur le blanchiment d'argent du domaine fiscal ('AML'). La révision aurait pour but de combattre et mitiger le blanchiment d'argent et l'évasion fiscale. C'est sous ces domaines que le États membres ont déjà développés un échange d'informations transfrontalier.

Options pour la conception et les fonctionnalités d'un registre des actifs de l'UE (Tâche 7)

En ce qui concerne l'interface utilisateur, l'étude propose un mécanisme qui utilise la relation sémantique entre les bases de données à faire des requêtes. Le mécanisme utiliserait cette relation pour permettre la construction de requêtes basé sur des valeurs incomplètes (liste des entités juridiques propriétaire de navire, d'aéronefs, situé dans un autre États membre que celui où il est enregistré). Le mécanisme serait revu par un système de requêtes manuels, ce qui permettrais aux utilisateurs avertis de faire des requêtes. L'utilisateurs pourrait grâce à ce système avoir accès à plusieurs sources d'information dans son travail, ce qui inclus des sources de données externes du système centralisé (par exemple la demande principale au niveau UE pourrait inclure de sous-requêtes basés sur des propriétés spécifiques du domaine unique à un État membre).

1. Introduction

Study and background objectives

The objective of this study is to assess the feasibility of providing a unique access point for competent authorities involved in the fight against money laundering and tax evasion to relevant asset registries in the EU Member States.

Competent authorities play a key role in combating money laundering and tax evasion. Despite the fact that there is no EU-harmonised definition of competent authority ([2]), it generally refers to public authorities designated by each Member State with responsibilities, powers, and obligations for combating money laundering and/or terrorist financing. The competent authorities considered in this study include financial intelligence units (FIUs), law enforcement authorities (LEAs), tax authorities and anti-money laundering (AML) supervisors of financial institutions ([3]).

These authorities perform different tasks in relation to the fight against money laundering and tax evasion. For instance, LEAs are in charge of enforcement activities, such as initiating criminal investigations and applying sanctions. FIUs are responsible for analysing and reporting suspicious financial transactions to other competent authorities. Authorities with competences for AML supervision of financial institutions are mostly related to the supervision of the financial market and its participants. Tax authorities are responsible for the administration, analysis, and collection of taxes, but also for enforcement of any law related to taxation.

Governments, financial institutions and regulatory authorities use asset registries to monitor the ownership and use of assets. An asset registry is a centralised and comprehensive record or database that contains information about various types of assets owned by individuals or entities. The purpose of an asset registry is to maintain transparency, accountability and accuracy in tracking and managing assets. Registries play a significant role in preventing fraud and ensuring compliance with legal and regulatory requirements. Governments may establish national or regional asset registries to track and manage assets, and some jurisdictions may require individuals and businesses to disclose their assets for tax or regulatory purposes.

- The information contained in an asset registry can vary depending on the type of asset being tracked and the intended purpose of the registry. In general, the key features of an asset registry include:
- ownership information (details about individuals or entities that own the asset, such as name(s) and address(es), etc.);
- asset types;
- specifications and descriptions of the asset (e.g. the model, size and other identifying characteristics);
- value of the asset, which can be estimated or actual;
- acquisition and disposal information, referring to the date when the asset was acquired or transferred to another party;

([2]) European Commission (2021b), https://data.europa.eu/doi/10.2874/068191

([3]) The supervisory bodies empowered by the Anti-Money Laundering Directive (AMLD) cover the financial and non-financial sector. For the purpose of this study, only those authorities tasked with the AML supervision of financial entities are considered.

- liabilities and encumbrances associated with the asset (e.g. mortgages or loans secured by certain assets).

In line with its objective, the study first defines the assets deemed relevant for money laundering and tax evasion in the context of this study. It then maps existing registries and alternative sources of information and identifies gaps. The study moves on to provide a feasibility assessment of three potential scenarios from a legal, operational and IT perspective, along with a cost-benefit analysis:

- Scenario 1 – creating new registers/databases at national level in addition to the existing ones (Beneficial ownership, bank account) for sources of wealth for which registers do not exist at national level, and their interconnection at European level to form an EU asset registry;
- Scenario 2 – interconnecting only the existing national registers required under EU law (Beneficial ownership, bank account) and real estate registers with an EU asset registry;
- Scenario 3 – establishing a central EU asset registry connecting the existing registers at national level and, for sources of wealth not covered by them, featuring a central EU register/database with national access portals.

The remainder of this report presents the methodology, elaborates the findings of the study by task in dedicated sections and puts forward conclusions and recommendations. The report is complemented by annexes, which provide additional information on the methodology and findings of the study.

2. Research methodology

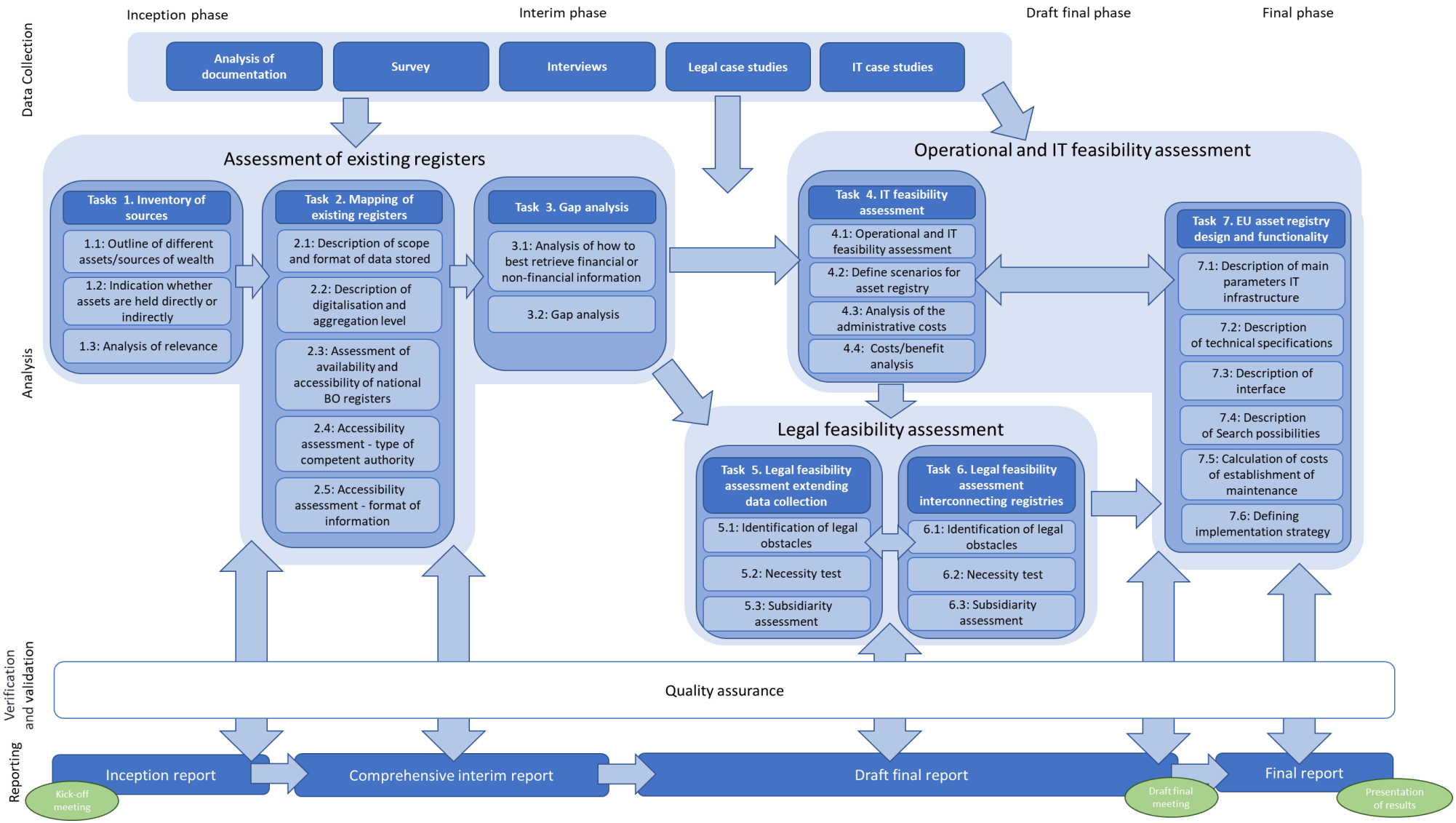
The study has been conducted based on the methodology presented in Table 1. The methodology was refined through discussion with the Commission services during the inception phase.

Table 1: Proposed tasks

Tasks	Title
Task 0	Data collection
Task 1	Inventory of different assets/sources of wealth
Task 2	Mapping of existing registries and alternative sources of information
Task 3	Mapping of other potential asset registries and alternative sources of information
Task 4	Operational/IT feasibility assessment and cost-benefit analysis
Task 5	Legal feasibility assessment: EU law framework and legal constraints
Task 6	Legal feasibility assessment: Analysis of the three scenarios
Task 7	Technical options for the design and functionalities of an EU asset registry

The approach has aimed at optimising data and information collection to provide a good understanding of the needs, existing registries and alternative sources of information and gaps. This has underpinned assessment of the legal, operational and IT feasibility as well as potential design and functions of an EU asset registry (see Figure 1).

Figure 1: Methodological framework for the study



Source: Authors' compilation.

2.1 Country coverage

The following section sets out the rationale for the selection of Member States for which structured interviews and case studies were conducted. The objective was to assess the feasibility of providing a single point of access for competent authorities – FIUs, LEAs, tax authorities and AML supervisors of financial institutions – to various sources of asset ownership in the EU. Data were collected on indicators covering several relevant characteristics to ensure that the selection of countries was representative, enabling the results to be extrapolated across the EU.

The project approached the country selection broadly with a proposal of **threefold rationale** by focusing on countries which:

- are at risk of money laundering, terrorist financing and tax evasion (source countries);
- are at risk of facilitating money laundering, terrorist financing and tax evasion (international financial centres), and
- presented **clear gaps** in the analysis conducted under Tasks 1 and 2.

The initial selection of countries was based on the first two criteria above. For interviews in Member States, a wide sample of 15 countries was drawn by using a country matrix that mapped the different selection criteria. We considered the following additional factors to be the most relevant:

- **Member State characteristics.** The sample of Member States was to be as representative as possible and cover a wide variety of countries. We considered the population size, the geographical region (north, east, south or west) and how long the country had been in the EU (if it had joined before 2004 or after).
- **Quantitative indicators.** Indices such as the Basel AML Index, Financial Secrecy Index (FSI), Corruption Perceptions Index (CPI) and KnowYourCountry (KYC) point at countries which could most benefit from setting up a centralised asset registry at the EU level.
- **Type of legal system.** To further ensure a representative sample of the Member States (especially with respect to the legal feasibility assessment), countries were classified by legal system into civil law, common law and mixed jurisdictions.

Our approach to the country selection prioritised countries at risk of money laundering and terrorist financing (source countries) or at risk of facilitating these activities (international financial centres). The rationale for the selection was to encompass the cross-border aspects of financial crime and to provide better networks and understanding on the international characteristics of it. The initial country selection was based on mapping where access to and linking of asset registries could be most beneficial.

For Tasks 3, 4 and 7, an additional quantitative indicator was used (the percentage of enterprises that use enterprise resource-planning software as a proxy for the level of digital development in a country ([4])). For the selection criteria for Tasks 5 and 6, recent high-level court cases of money laundering or predicate offences in the jurisdictions were used. Then, the type of legal system was considered specifically for the gap analysis under Task 3. As a

([4]) https://ec.europa.eu/eurostat/statistics-explained/index.php?title=E-business_integration

final step to validate the country selection, the study team followed the survey response rate and quality from questionnaires circulated across all EU Member States. The selection criteria for case studies under each task included additional variables and indicators as described in dedicated sections and Annex A.

Based on all these steps, interviews and case studies were conducted in 15 Member States. The final sample included Croatia, Cyprus, Czechia, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Luxembourg, Malta, the Netherlands, Romania and Spain. The countries selected represent new and old Member States, small and large countries, and the EU's north, east, west and south. Among the 15 are legal case studies in a sub-group of 8 Member States (Czechia, Cyprus, Hungary, Malta, the Netherlands, Luxembourg, Latvia and Romania) and IT case studies in a sub-group of 6 (Cyprus, Ireland, Italy, Malta, the Netherlands and Romania) (see the country selection matrix in Annex A).

2.1.1 Quantitative indicators: AML Index, FSI, CPI and KYC

We used a number of sources that provide cross-cutting indices to estimate which Member States should be subject to a feasibility assessment for a single point of access to all asset registers for the relevant authorities. The selected indicators have a strong focus on money laundering and terrorist financing risks in the respective jurisdictions. The following sources were used:

- Basel AML Index ([5])
- FSI ([6])
- CPI ([7])
- KYC rating ([8]).

These indices provide data with clear values that can be used to estimate the risks of money laundering and terrorist financing, and the state of play related to the predicate offences usually associated with money laundering/tax evasion risks. The description of the indicators is further explained in Table 2, which shows their relevance in assessing the Member States that could benefit most from the establishment of a centralised asset registry at the EU level.

Table 2: Description of sources for the country selection criteria

Selection source	Description

([5]) https://baselgovernance.org/sites/default/files/2020-07/basel_aml_index_2020_web.pdf
([6]) https://fsi.taxjustice.net/en/introduction/fsi-results
([7]) https://www.transparency.org/en/cpi/2019/results
([8]) https://www.knowyourcountry.com/country-ratings-table

Basel AML index	The Basel AML Index is an annual ranking by Basel Institute's International Centre for Asset Recovery, which assesses the risk of money laundering and terrorist financing. It considers five key areas that provide an overall risk score: quality of framework for anti-money laundering and the countering of terrorist financing, bribery and corruption, financial transparency and standards, public transparency and accountability, and legal and political risk. These categories are based on multiple sources, including recommendations by the Financial Action Task Force (FATF) for compliance by country (35 %) but also the FSI (10 %) and CPI (5 %), among others. The index uses a 0-10 scale, with 10 being the value for countries at highest risk.
Secrecy Financial Index (FSI)	The FSI was created and is managed by the Tax Justice Network. The FSI is the only selected indicator that is not based on other indices. Instead, it draws the secrecy rating from 20 secrecy indicators under the following categories: ownership registration, legal entity transparency, integrity of tax and financial regulation, and international standards and cooperation. These indicators include, for example, the availability of beneficial ownership information, public company information and the removal of harmful or secretive financial structures. Some of the indicators also overlap. The full scale of the FSI ranges from 0 (as most transparent) to 100 (fully opaque) ([9]).
Corruption Perceptions Index (CPI)	The CPI is a data aggregate index managed by Transparency International that focuses on providing an overview of corruption perception by experts. The index is built on 13 data sources: (1) African Development Bank Country Policy and Institutional Assessment 2018; (2) Bertelsmann Stiftung Sustainable Governance Indicators 2018; (3) Bertelsmann Stiftung Transformation Index 2020; (4) Economist Intelligence Unit Country Risk Service 2019; (5) Freedom House Nations in Transit 2018; (6) Global Insight Country Risk Ratings 2018; (7) IMD World Competitiveness Center World Competitiveness Yearbook Executive Opinion Survey 2019; (8) Political and Economic Risk Consultancy Asian Intelligence 2019; (9) the PRS Group International Country Risk Guide 2019; (10) World Bank Country Policy and Institutional Assessment 2018; (11) World Economic Forum Executive Opinion Survey 2019; (12) World Justice Project Rule of Law Index Expert Survey 2019; (13) Varieties of Democracy (V-Dem) 2019. The CPI gives a rating from 0 to 100, where 0 is the most corrupt perception and 100 is the least.
KnowYourCountry (KYC) rating	KnowYourCountry is an independent organisation that produces its own Country Rating Index. It prioritises the weight of money laundering and terrorist financing risks (56 %) in the scoring system while also taking into account risk indicators from international sanctions, corruption risks, governance indicators of individual crimes and listings of opaque jurisdictions. The rating uses the following indicators ([10]): FATF Uncooperative/AML Deficient, FATF Compliance with the Mutual Evaluation Report Recommendation Assessment, US State Money Laundering Assessment, the US Secretary of State list of sponsors of terrorism, international sanctions, corruption risks, World Governance Indicators, narcotics major list, human trafficking, EU tax blacklist and offshore finance centres. The rating ranges from 0 to 100, with 0 being at most risk and 100 being at least risk.

Source: Authors' compilation.

The quantitative indicators do not use the same scale; hence they are standardised to ensure comparability:

- the Basel AML Index is multiplied by 10 and reversed in the scale of 0-100, so that 100 is the country at least risk;
- the FSI rating is reversed in the scale of 0-100, so that 100 is the country at least risk;

([9]) https://fsi.taxjustice.net/PDF/FSI-Methodology.pdf

([10]) https://www.knowyourcountry.com/methodologies-sources

- CPI and Know Your Country ratings remain unchanged.

The detailed scores of the EU Member States are available in Annex A, including the calculated average scores per country. These scores offer a basis for the qualitative evaluation of the countries, leading to the final selection. The qualitative analysis for countries at risk of money laundering and terrorist financing and countries at risk of facilitating these activities is described further in the report.

2.2 Data collection

Due to the complex nature of the topics studied, the assignment required a combination of data collection techniques, as detailed below:

- analysis of documentation (Task 2)
- desk research and literature review (all tasks, in particular Tasks 1, 2 and 3)
- a survey (questionnaire) (Tasks 2 and 4)
- semi-structured interviews with relevant stakeholders (Tasks 1, 3 and 7)
- structured interviews with relevant stakeholders (Tasks 3, 4, 5, 6 and 7)
- case studies (Tasks 3, 4, 5, 6 and 7).

2.2.1 Desk research and literature review

Conducting a targeted review of the available relevant literature at the regional and national levels was necessary for a well-grounded, evidence-based analysis. In addition, a good understanding of the existing knowledge base helped to avoid duplication of research work later in the study. The desk research was not limited to academic and grey literature, but additionally considered reports from relevant stakeholders (e.g. NGOs, FIUs, competent national authorities and European institutions) and legal documents.

2.2.2 Surveys and interviews

As part of Task 1, a questionnaire was sent to the authorities involved in the fight against money laundering and tax evasion, to determine which assets are most relevant for them in such a context, as well as the authorities' current options for accessing information related to them. Comprehensive and targeted questionnaires were sent to each type of competent authority. This included FIUs, LEAs, tax authorities and AML supervisors of financial institutions. A questionnaire was also sent to authorities involved in the management of registries, or users of registries, to collect information concerning the registries – such as access, their level of digitalisation, the types of data stored, identification of the beneficial owners of assets and the way information is retrieved. Additionally, the questionnaire included relevant questions on the feasibility of interconnection among registries.

The survey was drafted in English as the stakeholders are used to reporting in English. However, the country experts, covering all EU languages, supported the contact points in completing the survey correctly at the request of stakeholders. The survey was converted into an Excel format after being confirmed by DG FISMA/DG TAXUD. In comparison with Word, Excel offers key features, such as pre-completed cells, checks and cross-referencing, which provided greater exactitude and manipulability of the answers. Excel surveys also provide broader flexibility to the respondents compared with online surveys, as it is easier to circulate the survey within a company (especially as several people within an organisation provide input) while saving the progress already made on the answers.

The questionnaires were sent for the first time in mid-October 2022; two reminders were sent to the stakeholders in November and December ([11]). In total, the number of questionnaires sent was 242, of which the majority were sent to asset registries (105), followed by AML supervisors of financial institutions (56) and the rest of the stakeholder groups (27 questionnaires were sent to each). The total number of stakeholders contacted was slightly lower (237), as certain bodies belong to two different categories of authorities at the same time (e.g. tax authorities and asset registries). This implies that some stakeholders received more than one survey for response.

In this regard, the observed response rate is under a third of the surveyed stakeholders (29 %). In particular, LEAs answered the fewest questionnaires ([12]) (6) but still represented around a fifth (22 %) of this group of stakeholders. Asset registries had the lowest response rate (12 %), with only a few surveys completed (13) ([13]). In contrast, more than half of the FIUs responded (18) ([14]), representing the highest response rate among all stakeholders (67 %). The same number of tax authorities ([15]) and AML supervisors of financial institutions ([16]) completed the questionnaire (17), with the second-highest response rate for the former (63 %) and around a third for the latter (30 %).

After completing the second round of collection of survey responses, follow-up interviews were structured (to gain detailed – quantitative and qualitative – information and understand the different points of view of stakeholders on a given topic) or semi-structured (in order to explore new areas of interest during the course of the interview). The interviews were conducted over video conferencing (Microsoft Teams, Zoom, WebEx, Skype). The interviews completed the collection of information from all relevant stakeholders.

It is important to note that the response-gathering process entailed some difficulties, because of a lack of either a response by stakeholders or knowledge by the respondents. Consequently, the results obtained are not fully representative of the complete sample of the competent authorities.

(11) CEPS asked the ACDT expert group for assistance in contacting the tax authorities that had not responded by the beginning of December 2022.

(12) The LEAs responding to the survey represented the following countries: DK, EE, HR, MT, NL and RO.

(13) Surveys were sent solely to asset registry holders that are managed under public authorities (excluding registries for cash at bank and safe deposit boxes at bank). These are vehicles and firearms. Asset registry holders responding to the survey represent the following countries: BG, CY, CZ, DK, HR, IE, LT, LU, LV, RO, and SE.

(14) FIUs responding to the survey represented the following countries: CY, CZ, BE, EE, ES, DE, DK, FI, HU, IT, LU, MT, NL, PL, SE, SI and SK. There are more questionnaires than countries because the Spanish authorities responded to two different surveys.

(15) Tax authorities responding to the survey represented the following countries: AT, BE, BG, CY, CZ, DK, EE, EL, FI, HR, LT, LV, PL, PT, SE, SI and SK.

(16) AML supervisors of financial institutions responding to the survey represented the following countries: AT, BG, EE, FI, HR, LT, LU, LV, PT, RO, SE, SI and SK.

3. Task 1: Inventory of different assets and sources of wealth

3.1 Methodology

Methodologically, the Task 1 objectives were first approached by identifying the **sources of wealth and assets** usually recognised as relevant to fighting money laundering, terrorist financing and tax evasion. This led to identification of the **types of registries and alternative sources** that hold information on these assets and sources of wealth, providing an overview of the overall asset registry landscape.

This identification of types of registries supported **the mapping of the national registries** in Task 2. The work also directly contributed to Task 3, of identifying potential new registries, by **pinpointing which asset registries** were not found during a comparison of general types of registries and alternative sources of information with existing ones in EU countries. An overview of this process is presented in Figure 2 below.

Figure 2: Process of identifying the relevant registries and alternative sources of information



Source: Authors' compilation.

The Task 1 mapping exercise used desk research and interviews to build an inventory of registries and alternative sources of information. The desk research also led to a draft inventory of assets that can be found in Table 3 below. The asset definitions can be found in Annex B.

Table 3: List of assets identified from desk research

Assets for which interconnection is already envisaged in the Anti-Money Laundering Directive and for which there are interconnection projects/studies (hence excluded from Tasks 2-3, but relevant for Tasks 4-7)	
Cash at bank and in hand	Cash at bank
Intangible assets	Crypto assets
Financial assets	Limited liability companies
Tangible assets	Real estate
Other assets	
Cash at bank and in hand	Cash in hand
Financial assets	Listed securities
	Unlisted securities
	Crowdfunding-platform intermediated assets
	Life insurance policies/Pension products
	Asset insurance policies
	Gambling currency (land-based and online casinos)

	Prepaid cards and vouchers
Intangible assets	Intellectual property
	Concessions
	In-game assets
Tangible assets	Vehicles
	Other high-value goods
	Animals
Commodities	Commodities (precious raw materials)
Other assets	Safe deposit boxes

Source: Authors' compilation.

Analysis of the relevance of assets for AML and preventing tax evasion was conducted based on literature review as well as empirical analysis. The preliminary findings were first gathered from the literature review and were then validated through the empirical findings collected via surveys across all relevant authorities such as FIUs, tax authorities, LEAs and AML supervisors of financial institutions. Task 1 led to a list of relevant assets for AML and preventing tax evasion and for their potential inclusion in a single EU asset registry.

3.2 Literature review

The literature review considered reports from authorities as well as academic literature and news articles. The assets in this section are defined as analogous to the literature. In many cases, the literature follows rather broad definitions (such as for securities) or sectors (such as the insurance sector as a whole). The list of assets considered for the single EU asset registry follows this broad approach, which later on, under the preliminary findings of Task 2, is connected to the asset definitions (see Annex B) as used by the existing registries or alternative sources of information, to facilitate the potential operationalisation of the single EU asset registry.

The assets and sources of wealth identified in literature have been classified according to their relevance in the fight against money laundering and tax evasion. The asset relevance is defined based on the following criteria:

The relevance of the asset in terms of money laundering/tax evasion. This assessment includes the size of the asset bases and cross-border impacts – that is, when the owner is resident in a Member State different to the Member State where the asset is located. Further, the assessment also includes the inherent vulnerability of the asset for money laundering/tax evasion. These vulnerabilities are asset-specific, for example, in the case of listed securities, one vulnerability is that many intermediaries are involved in the ownership chain. The Supranational Risk Assessment (SNRA) on money laundering and terrorist financing of 2021 is the main source document used to assess inherent vulnerabilities that could potentially facilitate predicate offences like tax evasion. A high rating means that the asset is materially significant and has high inherent vulnerabilities.

The extent to which there are mitigating measures to combat money laundering/tax evasion with regard to an asset. The money laundering and terrorist financing SNRA of 2021 is the main source document used to assess mitigating measures. An example of a mitigating measure in money laundering is that intermediaries who are involved in the transactions of the

asset are obliged entities under the AMLD framework, which means that they are required to apply AML rules to any transaction undertaken by the entity on behalf of a customer including ownership. In the case of tax evasion, this would involve national or international reporting obligations, for example through the DAC, to exchange information with the competent authorities of Member States and between the competent authorities of Member States. A high rating means there are robust mitigating measures in force while a low rating means that mitigating measures are lacking.

How attractive the asset is for recovery of the proceeds from money laundering/tax evasion. Traceability of the transaction and the owner, the mobility of the asset and value on liquidation of the asset are the relevant issues for this criterion. A high rating for this criterion means that the real or beneficial owner can easily be traced, the location of the asset is more difficult to change, i.e. land and property, and the asset represents a higher value which can be realised in liquidation.

If there was no literature connecting the asset group to either tax evasion or money laundering, the asset is classified as having no information. This does not necessarily indicate an overall low level of relevance. Literature may be forthcoming or abuse with these assets has not been discovered yet.

The assessment of asset relevance in the context of enforcement and recovery has been carried out in a similar manner.

The following sections discuss the assets that might have some relevance for money laundering and tax evasion and which could potentially be covered by a single EU asset registry.

3.2.1 Cash at bank and in hand

3.2.1.1 Cash in hand

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Cash in hand	High	High	High

Cash, commonly defined as currency and bearer-negotiable instruments, can be used to purchase goods and services in a rapid manner without trace (EC, 2019). The difficulty of detecting the parties involved in cash transactions make it particularly attractive for money laundering and tax evasion.

Criminal activities can generate great quantities of cash (FATF, 2015). Moreover, especially high-value banknotes can be used as a store for the proceeds from criminal activities (OECD, 2019; EC, 2019c). As to the size of this asset, the total value of banknotes in circulation at end-

2021 was about EUR 1.5 billion ([17]) (ECB, 2022). In addition, it is estimated that 25 % of all euro banknotes issued circulate in countries that are outside the euro area (ECB, 2014). Importantly, cash in hand for the purpose of this study relates to cash controls under the EU Cash Controls Regulation ([18]). In this regard, statistical data on cash controls ([19]) show a total value of almost EUR 61 billion being transported through the EU borders between June 2021 and June 2022. Moreover, competent authorities in charge of implementing the Cash Control Regulation were involved in more than 78 000 cases where cash was identified at EU borders, entering or leaving the EU.

Criminals also use the cash obtained from criminal activities to launder money through payments (EC, 2019c), for example, by using cash to purchase high-value goods, such as real estate, luxury vehicles, jewellery, etc. (see also Section 3.2.4.3 on high-value goods) (OECD, 2009). In most of the Member States, the use of cash for high-value transactions is restricted at least for some transactions. The thresholds applied range between EUR 1 000 in France and EUR 15 000 in Poland (Ecorys and CEPS, 2018). Cash from criminal activities and legitimate origins is further carried across borders to jurisdictions with lax supervision and stronger regulations on banking secrecy (EC, 2019c; FATF, 2015).

In addition, holding amounts of cash constitutes one of the preferred methods to avoid declaring income and hide wealth. This practice is still commonly applied in many countries. For example, cross-border cash transactions are widely used in Lithuania. However, to avoid controls on the physical transportation of cash, people in Lithuania buy other assets, such as crypto assets, to transfer abroad and sell. Similarly, transferring cash across borders to hide wealth offshore is also frequent in countries such as Portugal, Romania, Poland and Latvia. There is evidence that the estimated value of hidden wealth in cash is higher than the reported wealth in cash for countries such as Spain and France. Holding cash is also the chosen mechanism for less wealthy people, in contrast with wealthier individuals who prefer investing in other financial assets or using bank accounts (Ecorys, 2021).

The holders of cash are not recorded in a dedicated registry, unlike high-value cash transactions and cross-border trafficking. More specifically, the EU Cash Control Regulation requires that any traveller entering or leaving the EU carrying EUR 10 000 or more in cash must declare it to the authorities. Competent authorities can also register lower detected amounts when they have a suspicion that the cash is related to criminal activities. In addition, entities accepting cash are required to conduct customer due diligence checks when accepting cash payments of EUR 10 000 or more under AMLD4 ([20]).

([17]) It is not indicated whether the banknotes are circulating physically within the euro area or outside the euro area (ECB, 2022).

([18]) Regulation (EU) 2018/1672 of the European Parliament and of the Council of 23 October 2018 on controls on cash entering or leaving the Union and repealing Regulation (EC) No 1889/2005.

([19]) Cash controls statistical data for 3 June 2021 to 2 June 2022 (inclusive) according to Article 18 of Regulation (EU) 2018/1672 on controls on cash entering or leaving the Union, https://taxation-customs.ec.europa.eu/system/files/2022-11/20221125%20-%20Cash%20Controls%20Statistical%20Data%203%20June%202021%20-2%20June%202022.pdf

([20]) Directive (EU) 2015/849 of the European Parliament and of the Council of 20 May 2015 on the prevention of the use of the financial system for the purposes of money laundering or terrorist financing, amending Regulation (EU) No 648/2012 of the European Parliament and of the Council, and repealing Directive 2005/60/EC of the European Parliament and of the Council and Commission Directive 2006/70/EC.

The use of cash is one of the preferred payment methods for criminals to launder money and evade taxes. The main characteristics of cash transactions are anonymity, the ease of carrying large amounts and speed of disbursal, which make cash a hard asset to seize, as criminals may easily hide it or convert it into other goods. Generally, criminals seek to move the generated cash to other locations (e.g. countries where placing the illegal proceeds is easier). Considering this threat, the EU Cash Controls Regulation provides a control system to prevent and detect money laundering from any person entering or leaving the EU with a cash value of EUR 10 000 or more, which contributes to making cash an easily traceable asset for its recovery. Also, it is the most relevant asset to seize in terms of its liquidity, as it can be very easily used to satisfy tax burdens for example.

Indeed, a potential cash registry could help to identify the holders of cash and recover the proceeds from criminal activities.

3.2.1.2 Cash at bank

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Cash at bank	High	High	High

Bank accounts are one of the main channels through which money originating from illicit activities enters the financial system, through cash deposits and withdrawals (EC, 2019c). Bank accounts are generally defined as available accounts commonly used to make payments, deposits or withdraw cash, as well as to conduct exceptional operations. The total deposits held by monetary financial institutions ([21]) in the euro area from non-financial corporations and households amounted to about EUR 15 trillion at end-2021 (ECB, 2021). Once the funds are in a bank account, the bank account can be used as a vehicle to transfer funds across countries (EC, 2019c) and carry out tactics to shield the funds from their source. For example, the funds in the bank accounts can be used to make purchases or invest in high-value assets, or they can be moved to different accounts ([22]).

Bank accounts are not an asset in themselves but facilitate the holding of cash in electronic form. The identification of the owner of a bank account is important for detecting irregularities related to the ownership structure of the asset (EC, 2019c) as well as for the future reclaim and confiscation of assets (EC, 2020). For example, the ultimate beneficiaries behind bank accounts can hide behind 'money mules', which are often used to move funds across jurisdictions (EC, 2019c).

([21]) This excludes the Eurosystem.

([22]) See FATF, https://www.fatf-gafi.org/faq/moneylaundering/

The use of bank accounts to hide wealth is widespread in many EU countries. For instance, in Italy, the money generated through criminal activities is often transferred across countries and deposited in bank accounts (Ecorys, 2021). In fact, there is evidence that cash accounted for 5 % of the total voluntarily disclosed assets in Italy in 2009-2010 (Pellegrini et al., 2016). Likewise, the use of foreign bank accounts is one of the preferred practices to hide wealth in France, the Netherlands, Slovenia and Slovakia (Ecorys, 2021).

There are already bank account registries which have been established under AMLD5 ([23]) to give FIUs and LEAs access to account information to combat money laundering and financial crime. In addition, DAC2 ([24]) requires financial institutions in the EU to collect and send to their own tax authorities financial account information, including account balances, with regard to customers resident in other Member States. This information is then automatically exchanged with the tax authority in the EU where the customer is resident. Equivalent information is automatically exchanged at the international level under the OECD Common Reporting Standard ([25]). Although the inherent risk of money laundering connected to cash at bank in the EU is largely mitigated by credit institutions, there are still limitations when dealing with offshore structures and high-risk country customers (EC, 2019c). Nevertheless, recent interviews carried out in some Member States revealed that the use of bank accounts to hide wealth offshore has lost attractiveness in recent years due to stricter AML legislation and other national measures (Ecorys, 2021).

Bank accounts are an important asset to be seized due to the funds that are stored in them. As there are already bank account registries, identifying the holder of the bank account is currently possible. Furthermore, the confiscated funds are maintained in interest-bearing accounts (Brun et al., 2021).

'Cash at bank' has already been investigated by other studies commissioned by the European Commission and therefore are not considered further in the context of Tasks 2-3 of this study but will be considered in Tasks 4-7.

3.2.2 Financial assets

3.2.2.1 Securities

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Listed securities	High	Moderate	High
Unlisted securities	High	Moderate	High

([23]) Directive (EU) 2018/843 of the European Parliament and of the Council of 30 May 2018 amending Directive (EU) 2015/849 on the prevention of the use of the financial system for the purposes of money laundering or terrorist financing, and amending Directives 2009/138/EC and 2013/36/EU.

([24]) Council Directive 2014/107/EU on the automatic exchange of financial account information.

([25]) https://www.oecd.org/ctp/exchange-of-tax-information/standard-for-automatic-exchange-of-financial-account-information-for-tax-matters-9789264216525-en.htm

The existing literature on the relevance of securities for money laundering and tax evasion considers securities without differentiation of the types. More specifically, securities cover assets such as financial instruments settled by central securities depositories (CSDs), derivatives, undertakings for collective investment in transferable securities (UCITS) and alternative investment funds, bearer securities and other financial instruments (FATF, 2018). The European Securities and Markets Authority (ESMA) reported an overall trading volume of equity, bonds and bond-like instruments across the EEA30 countries of about EUR 26 trillion in 2020 (ESMA, 2021). One important aspect is the cross-border holding of these securities. In 2018, the equity markets of 23 Member States showed a foreign ownership share for listed and unlisted securities of more than 30 % (Oxera, 2020) ([26]). This includes the large market economies of Spain, France and Italy. International financial centres such as Luxembourg and Cyprus boast a share of foreign holdings in excess of 90 % and 80 % respectively.

The variety in products and variability of legislation across Member States leads to great complexity in the international markets for securities. This allows for the exploitation of securities for money laundering and tax evasion (FATF, 2018; EC, 2019). The highest risk of exploitation lies within the relationships between investors and intermediaries. Intermediaries may help to conceal the beneficial ownership of a security, who may be a politically exposed person or part of a larger money-laundering operation. This is especially concerning when securities are traded via private banks, as reporting and oversight in those types of banks seems to be much weaker compared with corporate and investment banks (EC, 2019c).

While securities in general are part of the EU AML framework with reporting requirements for all involved entities, the huge complexity and variety of products and actors may still allow for money laundering and tax evasion (EC, 2019c).

Moreover, securities in CSDs are subject to specific legislation, with information about security holders and transfers being kept. However, this does not necessarily provide information about the beneficial ownership of the securities, as many CSDs only list large omnibus clients as asset owners, in turn comprising a large number of smaller individual owners (ECSDA, 2015). One mitigating factor at least for companies listing their shares on regulated exchanges is the rights and obligations laid down in the Shareholder Rights Directive II ([27])). This directive allows companies listing their shares to demand the full ownership chain of their shares, up to the beneficial owner, thus increasing transparency in terms of ownership chains involving financial intermediaries. While this information is not centralised or held by the companies in question, nor is it mandatory that it be collected for each shareholder, it still can be beneficial in identifying beneficial owners, thus easing enforcement and mitigating the risk factors.

Still, the lack of universally centralised and comprehensive beneficial ownership information could be exploited to obscure the owners of securities, as well as the potentially illicit funds used to buy them.

([26]) European Commission (2020), *Primary and secondary equity markets in the EU*: *Final report*, Directorate-General for Financial Stability, Financial Services and Capital Markets Union, Publications Office, Luxembourg, https://data.europa.eu/doi/10.2874/496492

([27]) Directive (EU) 2017/828.

Similarly, trades of derivatives are recorded in trade repositories. While the change of ownership is recorded, any substantial information about the current direct or beneficial owner is not, which may introduce a risk of abuse.

In terms of mitigating measures to combat tax evasion, under DAC2 information on asset values, gross proceeds and passive financial income like dividends and interest income from debt and equity securities held by a custodian is exchanged cross-border between the tax authority of the Member State where the financial account linked to the security is located and the tax authority of the Member State where the investor is resident for tax purposes. Such information is not currently automatically exchanged under DAC2 for non-custodial securities within the EU.

Securities that are not traded on conventional exchanges, but rather on over-the-counter (OTC) markets present an additional risk. Money laundering could be conducted by settling options for cash, or by agreeing upon an option price that is vastly different from conventional markets (FATF, 2009). Furthermore, OTC markets in general face much lower regulatory requirements in many jurisdictions in terms of disclosing trades, membership or listings (FATF, 2009).

Overall, listed securities face much higher regulatory requirements for transparency and record-keeping compared with non-listed securities. This added transparency diminishes their usefulness as tools for money laundering and tax evasion. While universally centralised information on ownership and beneficial ownership is still missing, the current regulatory environment allows for the identification of ownership and thus increases the ease of enforcement and recovery.

The same is less true for unlisted securities, where direct and beneficial ownership can still be obscured due to the lower number of regulatory transparency and disclosure requirements when compared with listed securities. They present a better opportunity for money laundering and tax evasion without being detected and are a greater risk in terms of seizure as ownership can often be unclear. However, even closely held shares (i.e. a company with very few shareholders or even a sole one) can pose an additional risk of financial crime, as illegal activities can more easily be concealed with a limited number of owners.

Listed securities are not further assessed in Task 1, particularly as part of the survey analysis, given their high relevance for inclusion in a potential EU asset registry. In addition, as the existing registries are not held by national authorities but rather by private companies, listed securities are not assessed in Task 2. All the same, interviews with responsible authorities have been carried out for Task 3, in order to assess the current degree and further possibilities of standardisation of the stored data.

3.2.2.2 Crowdfunding-platform intermediated assets

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Crowdfunding-platform intermediated assets	Moderate	Low	Moderate

Crowdfunding-platform intermediated assets concerns those assets involved in financial transactions that take place through crowdfunding platforms. A crowdfunding platform is

commonly defined as an open call to the public for raising money to fund a particular project (EC, 2019c). The interaction between fundraisers and investors is done through the crowdfunding platform. Crowdfunding intermediated assets are linked to different crowdfunding categories, such as equity or investments, lending, invoice trading, reward- and donation-based crowdfunding (EC, 2019c). Data from the CCAF ([28]) annual survey revealed the existence of around 632 crowdfunding platforms in 2018 in Europe, with a financing volume of EUR 6.5 billion. The disaggregated total differs by type of crowdfunding, with lending-based crowdfunding accounting for the greatest share, EUR 4.7 billion, in 2018 (PwC, 2021).

Equity-crowdfunding intermediated assets are associated with equity crowdfunding contracts, which constitute one of the most relevant investments for money laundering. For example, investors can launder money through equity crowdfunding contracts by agreeing with the fundraiser to invest in the securities of a sham enterprise (Robock, 2014). Charitable crowdfunding contracts can also be used as a means to launder illegal money. In this case, criminals can create donation campaigns and use the money raised to finance illegitimate activities. The feasibility of carrying out this type of activity lies in the difficulty LEAs have in identifying the last recipient and determining the use of the funds raised (EC, 2019c).

Crowdfunding platforms can be authorised or unauthorised depending on whether they are in the scope of the crowdfunding legislation. The possibility to carry out anonymous transactions on certain platforms increases the inherent money laundering risk. The latter is also an issue for regulated platforms, where traceability of the sources of funds and the scope of the project are not necessarily completely clear. In addition, supervision of the crowdfunding platforms is particularly challenging as many of them are located in countries other than those where they actually operate (EC, 2019c).

Even though certain aspects of crowdfunding are at risk of criminal activity, there is limited evidence that crowdfunding platforms are used to launder money. This may be explained by the expertise needed to successfully raise money through the platforms, as well as the costs involved in the process. According to the LEAs, the crowdfunding sector is not the focus of money laundering, but rather a way to commit fraud or conduct a scam (EC, 2019c).

The EU obligations on anti-money laundering/combating the financing of terrorism (AML/CFT) do not apply to crowdfunding platforms as such, but only to certain crowdfunding services, such as investment-based and lending-based crowdfunding, depending on whether the platform is supervised as a financial services provider (EC, 2022). Furthermore, some Member States included crowdfunding platforms in their national transpositions of MiFID II and PSD II. Yet, this is not applied uniformly across Member States (EC, 2019c). As not all types of crowdfunding services are in the scope of AML/CFT, the risk of money laundering is higher. Crowdfunding registries that list the parties involved in the crowdfunding contracts for every type of crowdfunding could help to mitigate the risk.

It is difficult to identify the legitimacy of the parties involved in crowdfunding financial contracts. Also, the capability to assess the veracity of the projects involved in the contracts is limited. This highlights the necessity of establishing a registry for crowdfunding-platform intermediated assets. Such a registry would enable the identification of the persons involved and action to

([28]) Cambridge Centre for Alternative Finance.

freeze and confiscate the crowdfunding intermediated assets. Even so, the asset recovery process associated with crowdfunding contracts may be challenging. This is because crowdfunding platforms are often located in jurisdictions different from those where they operate, meaning that the crowdfunding-platform intermediated asset could be difficult to locate and thus seize.

3.2.2.3 Life insurance policies and pension products

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Life insurance policies and pension products	Moderate	High	High

Life insurance policies are defined as contracts with the central aim of financially protecting the policy holders and their families or other beneficiaries against future events such as critical illness or death (FATF, 2018). The premiums for life insurance amounted EUR 630 billion in the EU in 2020 ([29]). Policies with pay-out conditions not involving critical life events are especially vulnerable to money laundering (EC, 2019c). For example, illicitly obtained funds can be used to pay the life insurance premiums. After time has passed, the policy may then be cancelled, or its pay-out criteria could be met. This which would generate a lump sum of clean money (FATF, 2018; EC, 2019c).

The potential for money laundering is further increased when the relationships between the parties involved in the contract are unclear. Compromised insurance brokers with an unknown relationship to the insured party pose further money laundering risks. Similarly, the involvement of trusts, funds or other entities helping to obscure the beneficial owners of a policy can also facilitate money laundering (EC, 2019c). Also, the obscured policy beneficiary may be foreign entity, thus enabling cross-border money laundering (FATF, 2018).

Life insurance policies are already part of the EU AML framework, ensuring awareness of the issue and enabling supervising authorities to conduct checks on insurers. This concerns the reporting of suspicious transactions as well as ensuring that sufficient customer due diligence checks are conducted to identify all potentially problematic policy holders (EC, 2019c).

Turning to relevance for tax evasion, generally only small amounts of wealth are hidden abroad using life insurance policies. Normally, classic life insurance products, where a pay-out is dependent on a critical life event, are included under automatic information exchange under DAC1. But an exchange can only take place if information is available in the first place in the Member State where the insurance contract is located. In contrast, information on cash-value insurance products (i.e. non-life insurance policies) is exchanged cross-border under DAC2, as these insurance products are comparable to other financial products and produce a return for investors. While this provides some coverage, differing national definitions of ‘life insurance’

([29]) https://www.eiopa.europa.eu/tools-and-data/insurance-statistics_en#premiums-claims-and-expenses

and the lack of coverage by some life-insurance policies of non-residents still leave gaps where tax evasion may take place (Ecorys, 2021).

Additionally, in nearly all cases, bank accounts are necessary to facilitate a life insurance policy (EC, 2019c). Bank accounts are already regulated in AMLD5, which mandates bank account registries for all Member States. This leads to easier identification of policy holders and beneficial policy holders.

These circumstances are also viable for pension products, which are not life-insurance based. These products similarly aim to provide income upon retirement by capital which was accumulated over a longer term by the retiree. In the case of pension products involving contributions from an employer, several documented cases of tax evasion have been reported, where income and pension contributions have been purposefully altered in order to lower the tax burden (Mineva & Stefanov, 2018). An additional complication arises in the case of many Eastern European countries, where record keeping has been lacking in the past and previous transgressions can no longer be reliably verified. While pension income is also exchanged cross-border, these gaps in record keeping can reduce detection and redress of tax evasion.

The same avenues for life insurance money laundering can also be used for pension products. Like other financial assets, life insurance policies and pensions are relevant assets to seize. In this regard, as these contracts are in general linked to a bank account, it is easier to identify the owner of the asset as well as to freeze the asset.

3.2.2.4 Asset insurance policies

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Asset insurance policies	Low	Low	Low

The purpose of asset insurance policies is to insure valuable assets against damage or loss. In these cases, the insured party receives a payment from the insurance company, which depends on the conditions agreed upon by both parties and the extent of the damage/loss. According to Insurance Europe, the premiums for these kinds of insurance policies amounted to EUR 280 billion in the EU in 2020.

Asset insurance policies are not widely used for money laundering and tax evasion. The amount of effort and complexity involved in order to launder money or evade taxes directly makes asset insurance policies an unattractive vehicle. This is further evidenced by LEAs reporting no such cases involving this type of insurance in the European Commission's Supranational Risk Assessment (EC, 2019c).

Nor was evidence found in the literature of asset insurance policies being used directly for tax evasion. The only use case would be the same as for money laundering, namely the indirect detection of insured assets. This form of insurance is not a part of the EU AML/CFT framework, but is instead subject to national legislation (EC, 2019c).

While seldom used directly, there are reported cases of asset insurance policies for high-value assets leading to the detection of money laundering. One example involves a car salesman insuring his cars. After failing to prove that the vehicles were obtained in a legal fashion during

the due diligence process conducted by the insurance company, the case was reported as potentially involving money laundering (MONEYVAL, 2010). This case may be exemplary for a wider array of illegally obtained assets, and similar to the existence of these assets stored in free ports for example.

For the same reasons, asset insurance policies can give valuable information for asset seizures. If the insurance policy and the owner can be identified, the insured asset can also be identified more easily. This is due to the information contained in the policy, such as characteristics of the insured asset and the owner of the asset.

3.2.2.5 Gambling currency

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Land-based and online casinos	High	High	Low
Lotteries and betting transactions	Low	Low	Low

Gambling currency is used in services that involve betting with monetary value in games of chance (EC, 2019c). Gambling currency is used, for instance, in casinos (physical and online), lotteries, and betting transactions. The revenue generated from regulated physical and online gambling services in the EU amounted to EUR 81 billion in 2020, of which more than EUR 30 billion is from online gambling ([30]) (EGBA, 2022). The money laundering risk in the gambling sector is significant but varies across the different types of gambling (EC, 2019c). In physical casinos, players can purchase gambling currency (chips) with, for instance, cash or prepaid cards, in order to play. As this purchase is anonymous, the money used to buy the chips can come from illicit activities. The player can exchange these chips for legitimate money, laundering money. In order to reduce the vulnerability to money laundering, many land-based casinos have systems where the customer due diligence is automatically done within the identification process when players buy chips above a certain amount (EC, 2019c).

Online casinos also provide a variety of ways to launder money, since identifying the sources of the funds that are placed in the gambling accounts and checking the identity of the user is quite difficult ([31]). For example, gamblers can use cryptocurrency or other anonymous payment methods to buy gambling currency to play. Once they receive a payout, the money is converted

([30]) Here, the EU includes the EU27 + the UK.
https://www.egba.eu/uploads/2021/12/European-Online-Gambling-Key-Figures-2021-Edition.pdf
([31]) https://www.amlc.eu/online-gambling-as-a-money-laundering-method/

into legitimate money. Other practices in online gambling consist of people agreeing among themselves to bet money, and if one of them wins, then the other will be favoured and vice versa, thus manipulating the game (AMLC, 2021). It is noteworthy that the AML legislation applies to all providers of gambling services. Gambling service providers must carry out customer due diligence measures for transactions of EUR 2 000 or more. This can concern a single operation or many operations that seem to be connected. Although there is no sector-specific legislation on gambling, Member States should supervise the gambling providers and give the competent authorities the faculty to monitor them accordingly (EC, 2019c). Despite the legislative frameworks in place, not all gambling services are subject to the same AML requirements within the EU, because Member States have the possibility to grant exemptions to certain services (except for casinos) based on an AML and terrorist financing risk assessment ([32]). For example, bingo providers are exempted in Czechia, Denmark, Germany and Ireland ([33]).

Gambling is not included under the DACs. Member States are responsible for gambling legislation, which results in different betting and gambling taxes across Member States. For example, the Netherlands applies a tax on online betting and gambling as well as taxes on gains ([34]). Others, such as Denmark, Germany, Spain and Sweden, have gambling taxes depending on the gambling service. For example, in Germany sports bets are subject to taxes if either the betting takes place, or the bettor resides in the country. Lottery tickets bought in foreign countries are also subject to tax once they enter the country ([35]). This implies that individuals gambling outside the country are evading taxes if not declared.

For online gambling, the chances of seizing the asset are higher compared with land-based gambling services. Indeed, in order to gamble online it is often required to link a bank account to the gambling account. The gambler could be identified through the identification of the bank account, thus enabling the asset recovery process. However, if the gambler uses another payment method to buy gambling currency (e.g. cryptocurrency), then the possibilities for identifying the gambler, as well as seizing the asset, are reduced.

As the level of risk related to money laundering varies depending on the type of gambling service, the study focuses on gambling at casinos (both land-based and online).

([32]) https://www.european-lotteries.org/parliamentary-questions-0

([33]) National exemptions for gambling service providers are from the National Provisions Transposing Directive (EU) 2015/849 (AMLD).

([34])https://www.europarl.europa.eu/RegData/bibliotheque/briefing/2013/130509/LDM_BRI%282013%29130509_REV1_EN.pdf

([35])https://ec.europa.eu/taxation_customs/tedb/legacy/taxDetail.html?id=134/1424159159&taxType=Other+indirect+tax

3.2.2.6 Prepaid cards and vouchers

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Prepaid cards and vouchers	Moderate	Moderate	Low

Prepaid cards are generally defined as a payment method where monetary value can be stored to make purchases. Prepaid cards constitute an alternative to cash, debit/credit cards and other means of payment (FATF, 2013; EC, 2016). These cards can be classified among those that can be used to make purchases at one particular merchant or a restricted number of retailers ('closed-loop' as is the majority of the gift cards and vouchers) and those used for general purposes ('open-loop'). The risk of money laundering is mostly associated with the latter, as the former are characterised by being non-reloadable and not allowing for withdrawals (FATF, 2013). In addition, both types of prepaid cards can be used for payments online and in stores (EC, 2019c; Euro Banking Association, 2020). Based on ECB statistics, the market for prepaid instruments was estimated at around EUR 19.3 billion in 2014 ([36]) (EC, 2016; EC, 2019). The European e-Money Association estimated that about 106 million cards and vouchers were issued in the EEA in 2015 (EC, 2016).

Prepaid cards can be used by criminals to place illicit funds into the financial system as well as to make cross-border transactions. For example, some open-loop prepaid cards are globally accepted and allow for the transfer of money from person to person, making prepaid cards attractive for laundering money (FATF, 2013). Furthermore, the anonymity of a customer purchasing, using or reloading a prepaid card increases the possibilities for financing criminal activities and money laundering (FATF, 2013; EC 2016). Some prepaid cards can be funded using cash, where detecting the identity of the customer is quite difficult (FATF, 2013). As a vehicle for tax evasion, prepaid cards and vouchers have not been cited for this purpose in the existing literature.

Certain prepaid instruments show a higher risk of money laundering than others. In particular, the AML legislation only covers the general-purpose prepaid cards and not the closed-loop ones, as the latter encompass a lower AML risk (EC, 2019c). For general-purpose prepaid cards, AMLD5 reduces the threshold for which obliged entities are allowed to forego customer due diligence checks (EC, 2019c), but only if certain conditions are met. This applies to prepaid cards used exclusively to purchase goods or services that are not reloadable, or which have a maximum monthly payment transaction limit of EUR 150 to be used in one Member State ([37]). Open-loop prepaid cards are also under the scope of the E-money Regulation ([38]) (EC, 2019c). The definition of cash in the EU Cash Control Regulation likewise contemplates prepaid cards,

([36]) This estimate does not include the limited network market (e.g. the gift card market). The EU market includes the EU27 + the UK.

([37]) Directive (EU) 2018/843 of the European Parliament and of the Council of 30 May 2018 amending Directive (EU) 2015/849 on the prevention of the use of the financial system for the purposes of money laundering or terrorist financing, and amending Directives 2009/138/EC and 2013/36/EU.

([38]) Directive 2009/110/EC of the European Parliament and of the Council of 16 September 2009 on the taking up, pursuit and prudential supervision of the business of electronic money institutions amending Directives 2005/60/EC and 2006/48/EC and repealing Directive 2000/46/EC.

in particular anonymous prepaid cards that are not linked to any bank account (EC, 2019c). Despite the fact that some entities require the registration of the customer before using the prepaid card, this does not happen uniformly.

In addition, the possibilities for seizing a prepaid card or voucher depends on the previous information provided when buying or using it. If the owner of the card or voucher has not been asked for any type of personal information, then the chances of seizing the funds are nearly nil.

3.2.2.7 Limited liability companies

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Limited liability companies	High	High	High

Limited liability companies (LLCs) are entities with share capital and legal personality. They often own separate assets. The exact definition and classification vary depending on legislation in each Member State. According to the European Commission, currently almost 26 million of these enterprises are active across the EU. LLCs have an overall high potential to be abused for money laundering and tax evasion, by creating opaque chains of ownership and obscuring the overall ownership structure. This enables the establishment of illicit business relationships, where illegally obtain funds are transferred from one company to another as part of a fake purchase and thus laundered (EC, 2019c). Similar set-ups and fake invoicing can easily be abused to evade taxes, where funds are funnelled through different jurisdictions in order to conceal the original source (Ecorys, 2021).

While LLCs are covered under the EU AML framework and have to disclose their beneficial ownership in a centralised and interconnected register, the involvement of offshore jurisdictions in the chain of ownership makes enforcing this requirement difficult (EC, 2019c). This leaves a gap, through which illicit activities using LLCs can still persist.

One particular issue is the ownership of minority stakes in a business being used for illicit activities. In some Member States with more relaxed disclosure requirements, more complex ownership structures involving different levels of majority and minority ownership may successfully be used to obscure the true beneficial ownership (EC, 2019c). Several different set-ups are imaginable, for example using strawman ownership. The actual beneficial owner may only have a minority stake in the entity used for illicit activities, while having a strawman listed as a majority and beneficial owner. This tactic is also employed to tie financial service professionals, such as accountants and lawyers, into ownership structures, acting as strawmen for the true beneficial owner as well (Ricardi & Savona, 2013).

The issues involved in ownership tracing also increase the difficulty of seizing either the ownership stake or the company as a whole if an asset owner commits a crime or uses stakes to launder money. In cases where ownership is accurately displayed and not obscured, it is very worthwhile to seize or freeze ownership of these companies, as the stakes invested in the company may carry substantial value as well as stop any illicit business activity.

LLCs have already been investigated by other studies commissioned by European Commission services and are therefore not considered further in the context of Tasks 2-3 of this study but will be considered within the context of Tasks 4-7.

3.2.3 Intangible assets

3.2.3.1 Intellectual property rights

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Intellectual property rights	Low	Low	Low

Intangible assets are generally defined as assets that do not physically exist but are capable of generating revenue. Intellectual property rights such as patents, trademarks and copyrights are included in this category. The European Patent Office (EPO) received more than 188 000 applications for patents in 2021 ([39]), while trademark applications received by the European Union Intellectual Property Office (EUIPO) were around 101 000 for the first 6 months of 2021 (EUIPO, 2021). Money laundering connected to intangible assets concerns 'services-based money laundering' (FATF, 2020).

The intangible nature of these assets and the existence of different valuation methods of intellectual property rights ([40]) leads to potential overvaluations of the asset price. This overestimation in price increases their vulnerability to money laundering and tax evasion. Criminals may use intangible assets to move funds across borders, to give an appearance of legitimacy to the revenue generated by the assets, and to evade taxes (Eurasian Group, 2012). For example, patents can be used to launder money by selling the patent to front businesses outside the country of origin, while the economic benefits generated return to the local economy in a legal manner (paying taxes for them). Later, the legal money can be sent abroad via other illegal networks ([41]).

Intellectual property rights are not commonly cited in the existing literature as a way to evade taxes. Instead, some literature highlights the role of the strategic location of intellectual property as a channel through which multinational corporations can avoid and reduce the tax burden of the entire organisation. For example, enterprises can carry out research and development

([39]) https://www.epo.org/en/news-events/press-centre/press-release/2022/452146#:~:text=Munich%2C%205%20April%202022%20%E2%80%93%20The,in%202020%20(%2D0.6%25).

([40]) There are different methods for estimating the value of intangible assets, which can lead to different results.

([41]) https://economictimes.indiatimes.com/news/economy/policy/money-laundering-trade-patents-a-new-tool-for-bringing-back-black-money/articleshow/51220405.cms?from=mdr

activities in one country and transfer the patent ownership to a low-tax jurisdiction (IMF, 2018; Dudar & Voget, 2016).

Given the nature of the sector, detecting suspicious activities represents a challenge for competent authorities due to the difficulty of evaluating the behaviour of the parties involved in the arrangements (FATF, 2020). In addition, intangible assets are not contemplated in the AML legal framework, meaning that there is no legal obligation for the parties involved in an intellectual property contract to comply with AML rules. Nevertheless, EU and international anti-tax avoidance rules will contribute to counteracting tax evasion through intangibles.

While tax avoidance is not the focus of this study, the anti-tax avoidance measures such as stricter reporting may also help in identifying cases of tax evasion. Directive 2016/1164 ([42]) on rules against tax avoidance practices that directly affect the functioning of the internal market aims to ensure that taxes are paid where the profits and value are generated. It states that a Member State of a taxpayer must include in the tax base the non-distributed income of the controlled foreign company, or the income of the permanent establishment that is derived from the royalties or any other income generated from intellectual property. Furthermore, OECD BEPS actions establish minimum standards with requirements for intellectual property regimes with the objective of countering tax avoidance. Member States of the OECD are required to use the 'nexus approach' for intellectual property regimes. This only allows a taxpayer to receive the benefit of an intellectual property regime if the taxpayer has incurred the R&D expenditures which originated the intellectual property income (EP, 2015; OECD, 2019).

Intellectual property rights of patents, trademarks and copyrights can be seized. However, determining the real value of the intangible asset in order to confiscate it is quite difficult. What is more, copyright owners are not always registered, which would increase the difficulty of identifying the asset owner in order to proceed with the asset recovery process.

3.2.3.2 Concessions

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Concessions	No information	No information	No information

No evidence in the literature could be identified on the use of concessions, such as franchising departments, for money laundering and tax evasion.

([42]) https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32016L1164&from=EN

3.2.3.3 Crypto assets

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Crypto assets ([43])	High	Low	Low

Crypto assets such as crypto coins and non-fungible tokens (NFTs) are commonly referred to as either virtual currency or virtual assets respectively. In terms of EU law, there is a difference as NFTs are not necessarily considered crypto assets if they are unique and non-fungible. However, both still share many characteristics that make them broadly comparable, notwithstanding differences in treatment in EU legislation. The general characteristics of crypto assets usually entail a digital representation of value, not including digital fiat currency, which can be traded and is generally accepted by natural persons or by legal persons as payment (EC, 2019c; FATF, 2021). The ECB estimates the market capitalisation of the largest cryptocurrencies totalled around EUR 2.5 trillion in 2021, which represented a sevenfold increase compared with the previous year (ECB, 2022).

These assets lend themselves very well to tax evasion, money laundering and terrorist financing due to the high degree of anonymity involved and the lack of tracing possibilities (EC, 2019c; FATF, 2021).

As transactions can be made anonymously and cross-border via the internet, there is ample opportunity to transfer illicitly obtained money across jurisdictions with lax regulatory standards, where it can be exchanged into fiat currency (FATF, 2021). A similar pathway can be used for terrorist financing, by sending the support anonymously via virtual currencies (EC, 2019c). It is currently not possible for competent authorities to know the content of e-wallets. Therefore, authorities may not get a gauge on all the exact transactions conducted with either assets stored in the wallet itself or in an intermediary wallet (FATF, 2021).

AMLD5 obliges e-wallet providers to register and conduct customer due diligence, but this is not enough to prevent criminal activity conducted with crypto assets (EC, 2019c). Placing more regulations on providers is currently the subject of the proposed DAC8 ([44]). This directive would in effect require each registered crypto provider to report all transactions undertaken by customers residing in the EU, including both national and cross-border transactions. While intended to combat tax-related issues, information could also be vital in combating money laundering, as chains of transactions could then easily be traced. Conversely, obscured ownership and chains, which may only partly involve wallet providers, could nonetheless easily allow for illegal money to be transferred and placed in the regular economy or to finance illicit activities. The speed of innovation of virtual currencies and the lack of comprehensive centralised information is further worsening this issue. As ownership of crypto assets cannot

([43]) The ratings for crypto assets with regard to their relevance for money laundering and tax evasion are based on information from the European Commission's Supranational Risk Assessment (SNRA) 2021, and therefore they do not take into account recent reforms in the AML framework and DAC8.

([44]) The DAC8 proposal was published by the European Commission in December 2022. It expands the reporting and information exchanges between national tax authorities within the EU, in order to cover transactions made through the use of crypto assets; see https://ec.europa.eu/transparency/documents-register/detail?ref=COM(2022)707&lang=en

reliably be traced, reclaiming or freezing these assets is fairly difficult. Neither direct nor beneficial ownership are clear in many cases, and easily obscured in others. Therefore, assigning crypto assets to a criminal in order to seize them is complicated. At the same time – depending on the price – a lot of value may be stored in the crypto assets, making them a potentially interesting target for asset recovery.

Cryptocurrencies have been investigated by a separate study commissioned by European Commission services. In particular, that study suggested that with the introduction of the new MiCA Regulation, in addition to the Transfer of Funds Regulation, a more unified framework and increased transparency of transactions and wallet holders within the EU Member States will be established, generally providing a homogeneous and balanced environment for the crypto assets market ([45]). Still, as these regulations have not yet been fully implemented, the overall ratings for this asset are based on the European Commission's Supranational Risk Assessment of 2021.

Given the existence of the study on these assets (as discussed above), crypto assets are not considered further in the context of Tasks 2-3 of this study but will be further explored in Tasks 4-7.

3.2.3.4 In-game assets

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
In-game assets	High	Low	Moderate

In-game assets are generally considered a special form of virtual currency/virtual assets, which can theoretically solely be spent or used in the context of one specific videogame, as they share all the characteristics with the aforementioned group, except that they entail no mechanism to directly convert them into fiat currency (FATF, 2014). Usually, these assets are found in video games and could encompass a number of different types, such as in-game currency, character customisation items or complete accounts. According to the Interactive Software Federation of Europe, in 2021 the European video game market generated EUR 23.3 billion in revenues, of which 40 % were generated from the sale of in-game assets.

In-game assets can be used for money laundering via a secondary market and associated exchanges (Moiseienko & Izenman, 2019). On these marketplaces, items can be traded for real money. These exchanges are often very loosely regulated, thus enabling the exchange of illicitly acquired funds with in-game assets and vice versa. This may happen very easily across borders, as the transactions occur online (FATF, 2014).

There are initiatives to further regulate these kinds of assets by some EU Member States. They oblige the companies developing the game assets to introduce restrictions on sale and re-sale, so misuse of assets could be reduced. For the most part, this task has been taken up by

([45]) https://op.europa.eu/en/publication-detail/-/publication/dbaa8b83-5bc3-11ec-91ac-01aa75ed71a1

marketplaces like eBay and video game developers like Blizzard themselves in trying to reduce the illicit trading of in-game assets (FATF, 2014; Moiseienko & Izenman, 2019).

In-game assets could technically be used for asset recovery or freezing, but there is no indication that this is currently taking place. Although substantial funds may be held in these assets, the difficulty of identifying beneficial owners severely complicates efforts to accurately trace asset owners and claim any assets of monetary value.

3.2.4 Tangible assets

3.2.4.1 Real estate

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Real estate	High	Moderate	High

Real estate (including properties and land) is an often-used asset to launder money and evade taxes, due to the overall lack of transparency and oversight involved in the sector (FATF, 2022).

There are several characteristics of the real estate market contributing to this. For example, real estate transactions may still be conducted using cash in some of the Member States (EC, 2019c). This obscures the origin of the funds used to acquire the asset, therefore making it comparatively easy to use illicit funds to purchase real estate. This is further exacerbated by comparatively low reporting standards and requirements in some jurisdictions (FATF, 2022). Similarly, while real estate is part of the AML framework, oversight of transactions is often insufficient and varies across Member States in order to effectively curb money laundering and tax evasion (EC, 2022). Furthermore, the parties involved in a sale may agree on a sales price which is significantly lower than the market value, with the buying party paying the seller additional funds in cash or otherwise undetectable sources, thus obscuring the origin of the money (EC, 2022). These aspects result in large amounts of real estate held in a cross-border setting, with intermediaries helping to veil beneficial ownership. Specifically, across the EU more than EUR 1.4 trillion was held in cross-border ownership in 2018 (Ecorys, 2021).

Obscuring of the beneficial ownership of real estate is a general issue regardless of the payment method, as supervisory authorities often lack oversight of the specific ownership of assets (EC, 2019c; FATF, 2022). This can also easily be exploited in terms of cross-border money laundering given that beneficial owners may reside in a different country. Similarly, this complicates asset recovery as well (FATF, 2022). In addition, unclear ownership may facilitate tax evasion on earnings through rent or sales (Ecorys, 2021).

While real estate ownership will be made public to the relevant authorities, as laid out in AMLD5, the lack of a centralised registry and different reporting rules and conventions across Member States make identifying potentially troubling transactions difficult (FATF, 2022). In many countries (e.g. France, Germany and Italy), existing registries lack harmonisation on even a national level (EC, 2022). In the case of France, the registry is not even fully digitised. What is more, in countries with existing registries, beneficial ownership information is often lacking (FATF, 2022). This can easily be exploited in the commercial real estate sector, as renters can be legal entities, with potentially opaque relationships with landlords (EC, 2019c). Coupled with the low requirements in terms of the skills needed to participate in the real estate

sector, this makes it overall fairly likely to be abused for money laundering. Again, the lack of beneficial ownership information or incorrect direct ownership information, often makes seizure of real estate difficult. As ownership cannot reliably be traced back to a potential criminal, it is not certain whether there is a basis to seize an asset or not.

Real estate has already been investigated by a separate study commissioned by European Commission services ([46]). The issued report highlights the main money laundering risks associated with real estate transactions, for instance, not knowing the real origin of the funds used to purchase it, concealing the owner or using complex loan and credit schemes. Based on the analysis, the interconnection of real estate registries could be a useful option to fight money laundering and tax evasion in this sector. However, the objective would not be completely achieved if the data stored in the registries were not harmonised in all Member States. Even though such a project entails complexities, there has been a pilot project called 'Land Register Interconnection Member State Connection' (LRI Member State Connection) funded by the EU Justice Programme, where two Member States, Austria and Estonia, have technically developed and successfully implemented the connection of their national land registries to a Central Land Register Interconnection Service at the EU Justice Programme. This LRI project has been expanded to 'LRI 2', which will incorporate other Member States such as Latvia, Hungary, Portugal and Spain.

Given the other study, real estate is not considered further in the context of Tasks 2-3 of this study but will be explored in Tasks 4-7.

3.2.4.2 Vehicles

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating	Recovery
Vehicles	High	High	High

Vehicles are means of transport used to carry people or goods. Vehicles include cars, ships and aircraft.

Criminals launder money through buying/selling vehicles, in particular high-value ones. Similar to other high-value goods, the only reporting obligation for vehicle sales involves transactions of more than EUR 10 000. This requirement is difficult to enforce across all vehicle dealerships and easily skirted (EC, 2019c). As cash is still widely used when trading vehicles, it is extremely easy for a criminal to purchase them using proceeds from criminal activities, and due diligence checks may not be conducted. Alternatively, a network of illicit dealerships may be set up across Europe in order to conduct car sales among themselves for both money laundering and VAT tax evasion, as uncovered in a recent coordinated investigation across nine Member States ([47]).

([46]) https://op.europa.eu/en/publication-detail/-/publication/77c10664-d3cc-11eb-895a-01aa75ed71a1/language-en/format-PDF/source-227909909

([47]) See https://www.eurojust.europa.eu/news/eurojust-supports-major-crackdown-vat-fraud-car-sales

Overall, the lack of enforcement, oversight and regulation targeted towards that sector result in ample opportunities to launder money or evade taxes using vehicles.

High-value vehicles are especially relevant for asset seizure, because they are expensive. However, many high-value vehicles are purchased through cash transactions which diminishes the possibilities to trace the identity of the owner. Also, high-value vehicles such as yachts can be difficult to seize since they can be moved quickly from one jurisdiction to another. There have been recent cases where criminals have moved their yachts into international waters to hide them and avoid their confiscation. Moreover, some criminals change the ownership structure of the vehicles when moving them outside the EU in order to avoid seizure. In turn, the asset recovery process for vehicles presents some challenges, particularly the costs involved between the seizure and confiscation stages. The value of vehicles can rapidly depreciate, meaning that their market value may be questionable (Brun et al., 2021).

3.2.4.3 High-value goods

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
High-value goods	Moderate	Moderate	Moderate

Despite the fact that there is no binding definition of high-value goods, they can be defined as luxury products that have specific characteristics in terms of their quality and aesthetics, and whose existence is scarce (Vigneron & Johnson, 2004). For the purpose of this study, high-value goods include works of art, collectors' items, antiques, jewellery and worked specimens, as these assets are often mentioned in the context of money laundering and tax evasion. Firearms are also considered part of this group of high-value items.

High-value goods are identified as goods that are sold for a value of EUR 10 000 or more in a single transaction or a series of linked transactions ([48]). The aggregated sales of the global art and antiques market has been estimated at USD 65.1 billion in 2021, of which the EU market share was about 14 % (Art Basel & UBS Report, 2022). The illegal trade of antiquities and works of art has one of the largest financial values (EC, 2019c). Numerous groups of criminals laundering money are involved in illegal trafficking (e.g. firearms ([49])).

Most of the literature investigating the link between money laundering and high-value goods focuses on the transactions of criminals laundering dirty money by buying and selling high-value goods. These transactions can be carried across countries legitimately or illegally. For example, the illegal traffic of antiques is one of the leading criminal trade activities (EC, 2019c). This is also closely connected to tax evasion, as perpetrators can buy or sell these assets in cash without being identified or registered. Indeed, it is uncommon to register transactions in high-value goods (EC, 2019c). The small size of some of the high-value goods such as

([48]) The threshold of EUR 10 000 is based on the value of the transaction that determines persons trading or acting as intermediaries in the trade of works of art as obliged entities under Directive (EU) 2018/843 of the European Parliament and of the Council. Persons trading or acting as intermediaries include art galleries and auction houses as well as free ports.

([49]) Estimations indicate that EU civilians owned around 35 million illicit firearms in 2017 (EC, 2020).

jewellery make them attractive for money laundering, as pieces can easily be stored and transported across borders (FATF, 2013). Often, there is no publicly quoted price for the goods available, especially when the transactions are conducted privately rather than through auction houses (Hufnagel & Colin King, 2020; EC, 2019c). Furthermore, most of the high-value goods are purchased using cash, which complicates the tracing of the transactions. Indeed, the underreporting by entities and dependence on cash payments increase the risks of money laundering and tax evasion (EC, 2019c).

In most Member States, the use of high-value goods to hide wealth offshore is not among the most common practices. This is understandable as there is no direct return derived from these assets. Yet, France and the Netherlands seem to be exceptions, as rich individuals often invest in fine and modern art to hide their wealth abroad (Ecorys, 2021). Many of the high-value goods are stored in freeports in countries like Luxembourg or Switzerland. Yet, some evidence suggests that, for example, hiding wealth by storing works of art at Luxembourg Freeport is unlikely, as AML and tax officials are present all the time (Ecorys, 2021).

To alleviate the risks, AMLD5 includes traders in works of art such as art galleries, auction houses and freeports as obliged entities for transactions of EUR 10 000 or more. Obliged entities have to report suspicious transactions to FIUs and implement customer due diligence to identify customers and mitigate any potential risk. Even though measures are in place, monitoring all the transactions is almost impossible as many of the purchases are made through informal channels (EC, 2019c). Information can also be exchanged cross-border under the DAC. This is available upon request by a foreign tax authority, if that authority is aware that a high-value good is owned by a taxpayer. Given that no systematic collection of ownership information for high-value goods takes place, establishing ownership of these assets remains an issue which can be exploited.

The features of high-value goods (expensive, easily portable and storable) make them a good target for asset recovery. But where there is no named ownership and given that the transportation of most high-value goods is quite easy, they can simply be obscured. Given that they are often portable (artworks, expensive watches, etc.) they can be moved and hidden without leaving traces or possibilities of detection. Locating these assets and assigning definite ownership to potential criminals makes seizure fairly complicated. Additionally, expertise is needed to verify and value the assets in order to seize and confiscate them (Brun et al., 2021).

3.2.4.4 Animal assets

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Animal assets	Low	Low	Low

Animal assets include livestock ([50]), equines, racing pigeons and exotic animals ([51]). Generally, these animals are held for the income that can be derived from them, for example, as food and so on (FAO, 2011). Livestock numbers were estimated at around 130.2 million units in the EU ([52]) in 2013, with France, Germany and Spain having the highest shares (Eurostat, 2016). In addition, the mean total population of equines in the EU was estimated at 7 million in 2015 (World Horse Welfare & Eurogroup for Animals, 2015).

There is evidence that criminals tend to use legitimate channels, such as import and export of animals, to move and hide illicit funds. Such illicit funds may be derived from the illegal trade of animals (FATF, 2020). The illegal trade of animals constitutes a major international crime, where criminals generally launder the proceeds from the illegal traffic of animals in both the source and destination countries (FATF, 2020). The illegal trade of wild animals generates profits that may well involve more than USD 20 billion per year (FATF, 2020). Tax evasion as well as complex fraud and corruption are also commonly committed by criminal groups (FATF, 2020). In 2019, a particular case in Bulgaria evidenced practices of money laundering through a fake declaration of ownership of lands and ownership of livestock (EP, 2021).

Animals are not included in the AML legislation. However, regulatory mechanisms to register animals are already in place within the EU. The EU regulation on transmissible animal disease (Animal Health Law) ([53]) states that Member States shall implement a database to identify and register animals, as well as gather information about the establishments where the animals are kept.

Animals that have a high value (e.g. racehorses) may also be subject to asset seizure. But in practice, seizing animals is not easy. This is explained by the necessary costs that are involved in maintaining certain types of animals such as livestock (e.g. veterinary costs and feeding) (Brun et al., 2021). A lack of information exchange between agencies from different countries also limits the possibility to trace and seize the animals (FATF, 2020).

3.2.5 Commodities

3.2.5.1 Commodities (precious raw materials)

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Commodities (precious raw materials)	High	High	Moderate

([50]) This group of animals covers farm animals such as cattle, pigs, sheep, goats, rabbits, poultry and hens, and bees from a beekeeping farm.

([51]) Exotic animals are defined as species that are not native to the area where they are (Eurogroup for Animals, 2013); this group of animals includes invertebrates, fish, amphibians, reptiles, exotic birds, etc.

([52]) The EU27 + the UK.

([53]) Regulation (EU) 2016/429 of the European Parliament and of the Council of 9 March 2016 on transmissible animal diseases and amending and repealing certain acts in the area of animal health ('Animal Health Law').

Commodities are generally defined as precious raw materials such as gold bars, uncut diamonds and other fungible high-value assets. As assets, the main physical characteristics of commodities make them easy to store and transport, which in turn facilitates money laundering (FATF, 2008). For many years, commodities have been used to launder illicit funds as well as to evade taxes (Gilmour, 2017). For example, the tangible features of diamonds (small and low weight), as well as the fact that they are not captured by metal detectors, enables the transport of large amounts of value (FATF, 2008). Criminals choose commodities to launder dirty money as they do not involve high storage costs and are easy to convert into cash or other commodities (EC, 2019c).

The possibility of purchasing commodities in cash increases the money laundering and tax evasion risks. Commodities such as gold and diamonds can also be used as means of payment. However, gold is the only commodity that is considered under the EU Cash Control Regulation to be a cash instrument (EC, 2019c). Based on this Regulation, any person entering or leaving the EU with a cash amount (including gold) of EUR 10 000 or more must declare it (EC, 2019c). In contrast, diamonds do not need to be declared by people carrying them across borders, which make them hard to detect (EC, 2019c). Moreover, the level of compliance of the sector with FIUs is insufficient and dealers of diamonds do not seem to be fully aware of the AML risks (FATF, 2013). Some private entities have indicated that the use of cash in diamond transactions has decreased with the introduction of limits on high-value transactions at the national level. For gold, there is no information available in this regard (EC, 2019c). Commodities are not included in the DAC provisions on automatic exchange of information.

Commodities that are comparatively easy to transport and hide, such as gold bars or uncut diamonds, are prime targets for asset recovery. On the one hand, they have a very high value and can ably be physically seized and stored. On the other hand, these assets can be readily transported cross-border and change owners without notice. Verifying ownership and thus seizing them is very difficult. There are currently no registries identifying ownership of the commodities, which further complicates identification and seizure.

3.2.6 Other assets

3.2.6.1 Safe deposit boxes

Relevance	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery
Safe deposit boxes at banks	Moderate	High	High
Safe deposit boxes at non-banks	Moderate	Moderate	High

Safe deposit boxes are normally defined as objects that are used to store currency, monetary instruments or other high-value physical items like precious metals or documents of title. Money laundering and tax evasion risks appear when the contents of the safe deposit boxes are hidden with the aim of taking them out of the financial system. As the contents of safe deposit boxes are often only known to renters and sometimes their relatives, most providers of safe deposit boxes have no information regarding what is kept within them. In addition, the service can be provided by banks or by private companies (safe custody service providers). If

the latter is the case, then the money laundering threat increases. Moreover, the level of expertise needed to launder money through safe deposit boxes is low, which makes them an interesting tool for criminals (EC, 2019c).

Storing assets in a safe deposit box does not imply incorporating them into the legal economy (Teichmann & Falker, 2020). In fact, evidence suggests that using safe deposit boxes to hide wealth (e.g. storing cash) is a common practice in Greece (Ecorys, 2021). Furthermore, the evidence suggests that cash held offshore in safe deposit boxes in Switzerland, or the Cayman Islands makes up around 5 % of the total amount of offshore wealth (Ecorys, 2021).

AMLD5 requires Member States to introduce central registries to identify natural or legal persons holding safe deposit boxes. The directive also states that Member States shall prohibit credit and financial institutions from keeping anonymous safe deposit boxes. Moreover, under the AML framework financial institutions have the responsibility to carry out customer due diligence checks and comply with monitoring obligations. Nevertheless, assessing the origin of the funds and supervising their movement is quite challenging as the institutions are not aware of the contents of the deposit boxes.

Non-bank undertakings providing safe deposit boxes or similar storage services currently do not fall under the scope of the AML legal framework. Criminals may store assets involved in predicate offences, money laundering or tax evasion in a safe deposit box. Therefore, they represent an attractive target for recovery, as the assets stored within them could be seized.

'Safe deposit boxes at bank' have already been investigated by another study ([54]) commissioned by the European Commission services and therefore not considered further in the context of Tasks 2-3 of this study but will be considered in Tasks 4-7. Information on safe deposit boxes at non-banks is not currently being centralised or investigated, and will therefore be included in Tasks 2-3, in addition to Tasks 4-7.

3.2.7 Summary of the literature review

Overall, 21 assets (or groups of assets) have been identified that could be relevant for inclusion in a single EU asset registry. Table 4 lists and summarises these assets (or groups of assets) and whether they are considered for inclusion in the register in the rest of this study.

The remaining part of this task assesses assets that are highly relevant based on at least one of the criteria of the literature review or moderately relevant for all three criteria. There are 10 assets (or groups of assets) with moderate or high relevance for a single EU asset register: cash in hand; listed securities; unlisted securities; crowdfunding-platform intermediated assets; life insurance policies/pension products; in-game assets; tangible assets like cars, ships and aircraft; high-value goods; commodities (precious raw materials); and safe deposit boxes at non-banks. A further 5 assets are also relevant for inclusion in a single EU register (cash at bank, crypto assets, limited liability companies, real estate and safe deposit boxes at banks) but have not been considered in Tasks 2-3. They have only been considered in Tasks 4-7 of this study. This is because the interconnection of some of these assets, (e.g. cash at bank) is already envisaged in the latest AMLD proposal, while others (e.g. crypto assets and real

([54]) https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52019DC0372&from=EN

estate), have already been assessed for their feasibility of interconnection in other projects/studies.

In addition, any assets (or group of assets) that have low relevance for one or more criteria or no information available are disregarded in the further analysis. More specifically, this covers the following 6 assets: asset insurance policies, gambling currency (land-based and online casinos), prepaid cards and vouchers, intellectual property, concessions and animals.

Table 4: List of assets and their relevance

Category	Asset group	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery	Relevance: Assets considered for inclusion in an EU registry
Assets for which interconnection is already envisaged in the AMLD and for which there are interconnection projects/studies (hence excluded from Tasks 2-3, but relevant for Tasks 4-7)					
Cash at bank and in hand	Cash at bank	High	High	High	Yes
Financial assets	Crypto assets	High	Low	Low	Yes
	Limited liability companies	High	High	High	Yes
Tangible assets	Real estate	High	Moderate	High	Yes
Other assets	Safe deposit boxes at banks	Moderate	High	High	Yes
Other assets					
Cash at bank and in hand	Cash in hand	High	High	High	Yes
Financial assets	Listed securities	High	Moderate	High	Yes
	Unlisted securities	High	Moderate	High	Yes
	Crowdfunding-platform intermediated assets	Moderate	Low	Moderate	Yes
	Life insurance policies/pension products	Moderate	High	High	Yes
	Asset insurance policies	Low	Low	Low	No
	Gambling currency (land-based and online casinos)	High	High	Low	No
	Prepaid cards and vouchers	Moderate	Moderate	Low	No
Intangible assets	Intellectual property	Low	Low	Low	No
	Concessions	No information	No information	No information	No
	In-game assets	High	Low	Moderate	Yes
Tangible assets	Vehicles	High	High	High	Yes
	High-value goods	Moderate	Moderate	Moderate	Yes
	Animals	Low	Low	Low	No
Commodities	Commodities (precious raw materials)	High	High	Moderate	Yes

Category	Asset group	Importance and inherent vulnerability to money laundering/tax evasion	Mitigating measures	Recovery	Relevance: Assets considered for inclusion in an EU registry
Other assets	Safe deposit boxes at non-banks	Moderate	Moderate	High	Yes

Source: Authors' compilation.

3.3 Findings from the survey

To validate the relevance of the assets identified in the literature review, competent authorities (FIUs, LEAs, tax authorities and other AML supervisors of financial institutions) were surveyed.

The relevance of the assets in the context of the fight against money laundering and tax evasion is defined based on the following criteria:

- the importance (based on, for instance, the size of the asset base) of each asset in the investigation and enforcement activities of the surveyed competent authorities' organisations;
- the risk or inherent vulnerability (on a scale of 0 to 4) of each asset used for money laundering and tax evasion;
- indication of which assets the surveyed competent authorities would add or remove from the list of relevant assets identified above (see Table 4).

The survey responses were collected in two rounds. Overall, FIUs had the highest response rate with more than half of the authorities responding to the questionnaire (18 or 67 % of the FIUs), followed by tax authorities (17 or 63 % of the tax authorities) and LEAs (6 or 22 % of the LEAs) ([55]). In addition, 56 AML supervisors of financial institutions ([56]) were identified, of whom approximately a third responded to the questionnaire (17 or 30 % of the AML supervisors of financial institutions).

3.3.1 Importance of the assets in competent authorities' investigation and enforcement activities

Competent authorities perform different tasks in relation to the fight against money laundering and tax evasion. For instance, while LEAs are in charge of enforcement activities, such as

([55]) The rate of response for FIUs, tax authorities and LEAs is calculated based on the total number of questionnaires sent per group of authorities, which is 27.

([56]) The AML supervisors of financial institutions identified are based on the 'List of authorities designated as competent for the AML/CFT supervision of financial institutions in the EU' by the European Banking Authority.

initiating criminal investigations and applying sanctions, FIUs are responsible for analysing and reporting suspicious financial transactions to other competent authorities. The tasks of AML supervisors of financial institutions are mostly related to supervision of the financial market and its participants, while tax authorities are responsible for the administration, analysis and collection of taxes, but also for the enforcement of any law related to taxation.

To understand the importance of the assets when conducting the respective activities, competent authorities were asked to indicate this for each group of assets in the investigation and enforcement activities of their organisations ([57]). Responses range from 'not at all important' to 'very important', also including a category of assets that are not under the scope of the competent authority ('not applicable').

3.3.1.1 FIUs

More than half of the FIUs responding to this question consider cash in hand, limited liability companies, crypto assets, real estate, vehicles and safe deposit boxes important or very important in investigative and enforcement actions (see Table 5). In contrast, life insurance and pension products, as well as commodities, are considered slightly or not important by around half of FIUs. Securities and in-game assets are identified by several respondents as assets of moderate importance within the investigation and enforcement activities of FIUs.

Table 5: Importance level of assets for FIUs

Assets	FIUs						Number of observations
	Not at all	Slightly	Moderately	Important	Very important	Not applicable	
Cash in hand	0 %	0 %	8 %	17 %	75 %	0 %	12
Unlisted securities	8 %	0 %	77 %	15 %	0 %	0 %	13
Crowdfunding-platform intermediated assets	8 %	33 %	33 %	8 %	8 %	8 %	12
Life insurance policies/pension products	8 %	42 %	33 %	17 %	0 %	0 %	12
Limited liability companies	0 %	9 %	9 %	45 %	36 %	0 %	11
Crypto assets	0 %	0 %	8 %	42 %	42 %	8 %	12
In-game assets	10 %	20 %	40 %	10 %	0 %	20 %	10
Real estate	0 %	0 %	0 %	38 %	62 %	0 %	13
Vehicles	8 %	0 %	23 %	46 %	23 %	0 %	13
Other high-value goods	8 %	25 %	17 %	50 %	0 %	0 %	12

([57]) For this question, assets were presented based on asset groups.

Commodities (precious raw materials)	17 %	25 %	33 %	8 %	8 %	8 %	12
Safe deposit boxes	0 %	8 %	38 %	38 %	15 %	0 %	13

Notes: The total number of FIU respondents differ by type of asset and are shown in the last column of the table ('number of observations'). The total percentage per asset might not add up to 100 % due to rounding. 'Not applicable' means outside the scope of the authority concerned.

Source: Authors' compilation.

3.3.1.2 Tax authorities

Most of the tax authorities surveyed consider that all assets are at least slightly important for enforcement and investigation purposes on money laundering and tax evasion (see Table 6). The exception is life insurance policies and pension products, where a minority of the tax authorities responding to this question consider these assets of no importance (8 %).

Tax authorities show greater consensus compared with other stakeholders surveyed. A majority of the tax authorities view real estate as very important in investigation and enforcement activities (79 %), followed by crypto assets (70 %), vehicles (67 %) and cash in hand (64 %). In addition, a significant number of tax authorities indicated that unlisted securities and limited liability companies are important. This is in line with the literature review, which suggests that these assets are highly relevant for money laundering and tax evasion. Furthermore, crowdfunding-platform intermediated assets are also mentioned as important assets for most tax authorities, which is in line with the findings in the literature review for certain types of crowdfunding.

Table 6: Importance level of assets for tax authorities

Assets	Tax authorities						Number of observations
	Not at all	Slightly	Moderately	Important	Very important	Not applicable	
Cash in hand	0 %	0 %	7 %	21 %	64 %	7 %	14
Unlisted securities	0 %	8 %	8 %	62 %	23 %	0 %	13
Crowdfunding-platform intermediated assets	0 %	11 %	11 %	56 %	11 %	11 %	9
Life insurance policies/pension products	8 %	8 %	23 %	38 %	23 %	0 %	13
Limited liability companies	0 %	0 %	7 %	60 %	33 %	0 %	15
Crypto assets	0 %	0 %	0 %	30 %	70 %	0 %	10
In-game assets	0 %	22 %	33 %	44 %	0 %	0 %	9
Real estate	0 %	7 %	0 %	14 %	79 %	0 %	14
Vehicles	0 %	7 %	7 %	20 %	67 %	0 %	15
Other high-value goods	0 %	20 %	20 %	40 %	20 %	0 %	10

Commodities (precious raw materials)	0 %	9 %	36 %	27 %	27 %	0 %	11
Safe deposit boxes	0 %	0 %	38 %	31 %	23 %	8 %	13

Notes: The total number of tax authority respondents differ per type of asset and are shown in the last column of the table ('number of observations'). The total percentage per asset does not add up to 100 % due to rounding. 'Not applicable' means outside the scope of the authority concerned.

Source: Authors' compilation.

3.3.1.3 LEAs

The level of importance of the assets expressed by LEAs is shown below (see Table 7). It should be noted that there were significantly fewer LEA responses compared with other stakeholder categories, which makes comparison more difficult, particularly in percentage terms.

At least a third of the LEAs responding to this question regard crowdfunding-platform intermediated assets (33 %) and in-game assets (50 %) as not important at all for their enforcement and investigation purposes on money laundering and tax evasion. In contrast, at least half of LEAs see cash in hand, limited liability companies, crypto assets, real estate and vehicles as important or very important when conducting their activities. This is in accordance with the responses previously collected from the tax authorities and FIUs. In addition, there is one type of asset that a significant number of LEAs mentioned as important, in line with the responses of other competent authorities: other high-value goods.

Table 7: Importance level of assets for LEAs

Assets	LEAs					Number of observations
	Not at all	Slightly	Moderately	Important	Very important	
Cash in hand	0 %	0 %	50 %	0 %	50 %	4
Unlisted securities	0 %	33 %	33 %	33 %	0 %	3
Crowdfunding-platform intermediated assets	33 %	33 %	33 %	0 %	0 %	3
Life insurance policies/pension products	0 %	33 %	33 %	33 %	0 %	3
Limited liability companies	0 %	0 %	25 %	25 %	50 %	4
Crypto assets	0 %	0 %	0 %	40 %	60 %	5
In-game assets	50 %	0 %	0 %	50 %	0 %	2
Real estate	0 %	0 %	20 %	0 %	80 %	5
Vehicles	0 %	0 %	20 %	40 %	40 %	5
Other high-value goods	0 %	0 %	25 %	75 %	0 %	4
Commodities (precious raw materials)	0 %	67 %	0 %	0 %	33 %	3
Safe deposit boxes	0 %	50 %	0 %	25 %	25 %	4

Notes: The total number of LEA respondents differ by type of asset and are shown in the last column of the table ('number of observations'). The total percentage per asset does not add up to 100 % due to rounding.

Source: Authors' compilation.

3.3.1.4 AML supervisors of financial institutions

AML supervisors of financial institutions reveal more diversity in responses regarding the importance of particular assets that are part of the enforcement and investigation activities on money laundering and tax evasion (see Table 8). One of the main observations is the significant share of AML supervisors of financial institutions affirming that most of the assets are not under the scope of these authorities, which is shown under the 'not applicable' category. Among those assets that are under the scope of these authorities' activities, most are deemed slightly important. For instance, in contrast with previous results from surveyed competent authorities, only a third of AML supervisors of financial institutions view vehicles as slightly important in enforcement and investigation activities for money laundering and tax evasion. Cash in hand is also mentioned as slightly important for some AML supervisors of financial institutions (20 %), in contrast with previous results. Among the assets considered to have more importance for these authorities are crypto assets (27 %) and life insurance policies/pension products (21 %). However, in both cases, the same share of competent authorities indicated such assets as only slightly important, creating ambiguity in the result.

Table 8: Importance level of assets for AML supervisors of financial institutions

Assets	AML supervisors of financial institutions						Number of observations
	Not at all	Slightly	Moderately	Important	Very important	Not applicable	
Cash in hand	0 %	20 %	10 %	10 %	10 %	50 %	10
Unlisted securities	0 %	15 %	15 %	31 %	8 %	31 %	13
Crowdfunding-platform intermediated assets	0 %	18 %	18 %	9 %	9 %	45 %	11
Life insurance policies/pension products	7 %	21 %	7 %	7 %	21 %	36 %	14
Limited liability companies	0 %	17 %	8 %	17 %	17 %	42 %	12
Crypto assets	0 %	27 %	0 %	9 %	27 %	36 %	11
In-game assets	17 %	0 %	0 %	17 %	0 %	67 %	6
Real estate	0 %	20 %	20 %	0 %	10 %	50 %	10
Vehicles	0 %	33 %	0 %	0 %	11 %	56 %	9
Other high-value goods	25 %	0 %	0 %	0 %	13 %	63 %	8
Commodities (precious raw materials)	17 %	0 %	0 %	17 %	0 %	67 %	6
Safe deposit boxes	9 %	27 %	9 %	0 %	9 %	45 %	11

Notes: The total number of AML supervisors of financial institution respondents differ by type of asset and are shown in the last column of the table ('number of observations'). The total percentage per asset does not add up to 100 % due to rounding. 'Not applicable' means outside the scope of the authority concerned.

Source: Authors' compilation.

3.3.1.5 Summary

Overall, the authorities surveyed indicated that most of the assets are at least moderately important, with a low share deeming assets as not at all or slightly important. Surveyed FIUs, tax authorities and LEAs consider vehicles, cash in hand, limited liability companies, real estate and crypto assets to be important or very important to their investigation and enforcement activities in relation to money laundering and tax evasion. This is largely in line with the literature review findings, whereby assets such as cash in hand and vehicles were classified as highly relevant for money laundering and tax evasion. Notably, around a third of FIUs, LEAs and AML supervisors of financial institutions agree that life insurance policies and pension products are not important or just slightly important in this context. This differs somewhat for tax authorities, among whom most rank them as at least moderately important, which is in line with the literature review findings above. Finally, AML supervisors of financial institutions reflect a certain ambiguity due to a significant number of authorities indicating that the assets were not under their scope of competence.

3.3.2 Risk of asset use in money laundering and tax evasion

Identifying, evaluating and understanding the risk of assets being used for money laundering and tax evasion is key for competent authorities to develop and implement the corresponding measures ([58]). Where competent authorities identify that a certain asset presents a higher risk, it is necessary to ensure that the AML regime addresses those risks. Assets that are easy to conceal or immerse in a non-regulated market are examples of risky assets. Based on this, competent authorities were asked to indicate the level of risk of each of the assets being used for money laundering and tax evasion, according to their understanding within their respective organisations. Responses range from ‘not at all risky’ to ‘high risk’, including a category for assets that are not under the scope of the competent authority (‘not applicable’).

3.3.2.1 FIUs

The majority of the FIUs surveyed hold that the most inherently high-risk assets for tax evasion are cash in hand (83 %) and crypto assets (83 %) (see Table 9). That is, these assets are easier to conceal or are part of non-regulated markets. The ease of concealment may stem from the main characteristics of the transactions where these assets are involved, in particular, speedy and anonymous transactions. In addition, at least half of the FIUs regard limited liability companies (58 %), real estate (54 %) and other high-value goods (50 %) as assets at high risk of money laundering and tax evasion. Securities and crowdfunding-platform intermediated assets are perceived as moderately important by at least half of the FIUs responding to this question, while insurance policies and pension products are deemed by around two thirds of the FIUs to be assets with low or no risk at all (66 %). Only around a quarter of the FIUs classifies commodities, crowdfunding-platform intermediated assets, unlisted securities and in-game assets as low or no risk at all for money laundering and tax evasion. In contrast, these assets are considered moderately risky for most FIUs.

([58]) FATF (2013a).

Overall, assets that are riskier according to FIUs have also been classified previously as the most relevant and/or important assets.

Table 9: Risk of asset use for money laundering and tax evasion purposes (FIUs)

Assets	FIUs						Number of observations
	Not at all	Low	Moderate	High	Not applicable	No opinion	
Cash in hand	0 %	0 %	17 %	83 %	0 %	0 %	12
Unlisted securities	0 %	23 %	54 %	23 %	0 %	0 %	13
Crowdfunding-platform intermediated assets	0 %	25 %	50 %	17 %	8 %	0 %	12
Life insurance policies/pension products	8 %	58 %	25 %	8 %	0 %	0 %	12
Limited liability companies	0 %	17 %	25 %	58 %	0 %	0 %	12
Crypto assets	0 %	0 %	8 %	83 %	8 %	0 %	12
In-game assets	11 %	11 %	44 %	22 %	11 %	0 %	9
Real estate	0 %	0 %	38 %	54 %	0 %	8 %	13
Vehicles	8 %	0 %	46 %	38 %	0 %	8 %	13
Other high-value goods	0 %	17 %	33 %	50 %	0 %	0 %	12
Commodities (precious raw materials)	8 %	17 %	42 %	25 %	8 %	0 %	12
Safe deposit boxes	0 %	15 %	38 %	38 %	0 %	8 %	13

Notes: The total number of FIU respondents differ by type of asset and are shown in the last column of the table ('number of observations'). The total percentage per asset does not add up to 100 % due to rounding. 'Not applicable' means outside the scope of the authority concerned.

Source: Authors' compilation.

3.3.2.2 Tax authorities

As in the case of FIUs, more than two thirds of the tax authorities surveyed agree that crypto assets and cash in hand present a high risk in the context of money laundering and tax evasion (see Table 10). Compared with FIUs, the share of tax authorities considering crypto assets and cash in hand to be high risk is smaller. Yet, these assets are also among those that are important for tax authorities' enforcement and investigation activities. A significant number of tax authorities consider that real estate has a high risk of money laundering and tax evasion (69 %). This is also the case for vehicles, other high-value goods and safe deposit boxes (50 % in each case).

There are certain assets that are perceived as moderately important by the majority of the tax authorities. These assets are life insurance policies and pension products (55 %), limited liability companies (54 %), crowdfunding-platform intermediated assets (56 %) and commodities (56 %). In line with FIUs, commodities are considered to be of moderate to high risk by the majority of the tax authorities.

Table 10: Risk of asset use for money laundering and tax evasion purposes (tax authorities)

Assets	Tax authority	

	Not at all	Low	Moderate	High	Not applicable	No opinion	Number of observations
Cash in hand	0 %	8 %	8 %	69 %	8 %	8 %	13
Unlisted securities	0 %	17 %	33 %	42 %	0 %	8 %	12
Crowdfunding-platform intermediated assets	0 %	11 %	56 %	11 %	11 %	11 %	9
Life insurance policies/pension products	0 %	36 %	55 %	0 %	0 %	9 %	11
Limited liability companies	0 %	8 %	54 %	31 %	0 %	8 %	13
Crypto assets	0 %	0 %	20 %	70 %	0 %	10 %	10
In-game assets	0 %	22 %	44 %	22 %	0 %	11 %	9
Real estate	0 %	15 %	8 %	69 %	0 %	8 %	13
Vehicles	0 %	7 %	36 %	50 %	0 %	7 %	14
Other high-value goods	0 %	10 %	30 %	50 %	0 %	10 %	10
Commodities (precious raw materials)	0 %	0 %	56 %	33 %	0 %	11 %	9
Safe deposit boxes	0 %	10 %	20 %	50 %	10 %	10 %	10

Notes: The total number of tax authority' respondents differ by type of asset and are shown in the last column of the table ('number of observations'). The total percentage per asset does not add up to 100 % due to rounding. 'Not applicable' means outside the scope of the authority concerned.

Source: Authors' compilation.

3.3.2.3 LEAs

In line with previous results, the majority of the LEAs surveyed state that cash in hand (75 %), crypto assets (100 %) and real estate (60 %) are high risk for money laundering and tax evasion (see Table 11). Most of the LEAs responding to this question indicated that assets such as unlisted securities (67 %), crowdfunding-platform intermediated assets (100 %) and vehicles (60 %) are at moderate risk. Similar to FIUs, the vast majority of the LEAs responded that life insurance and pension products are low-risk assets (67 %).

In contrast with FIUs and tax authorities, most LEAs responding to this question hold that commodities present a low risk for money laundering and tax evasion (67 %).

Table 11: Risk of asset use for money laundering and tax evasion purposes (LEAs)

Assets	LEAs				Number of observations
	Not at all	Low	Moderate	High	
Cash in hand	0 %	0 %	25 %	75 %	4

Unlisted securities	0 %	0 %	67 %	33 %	3
Crowdfunding-platform intermediated assets	0 %	0 %	100 %	0 %	3
Life insurance policies/pension products	0 %	67 %	33 %	0 %	3
Limited liability companies	0 %	25 %	25 %	50 %	4
Crypto assets	0 %	0 %	0 %	100 %	5
In-game assets	0 %	0 %	50 %	50 %	2
Real estate	0 %	20 %	20 %	60 %	5
Vehicles	0 %	20 %	60 %	20 %	5
Other high-value goods	25 %	0 %	25 %	50 %	4
Commodities (precious raw materials)	0 %	67 %	0 %	33 %	3
Safe deposit boxes	0 %	50 %	25 %	25 %	4

Notes: The total number of LEA respondents differ by type of asset and are shown in the last column of the table ('number of observations'). The total percentage per asset does not add up to 100 % due to rounding.

Source: Authors' compilation.

3.3.2.4 AML supervisors of financial institutions

Similar to the results on the importance of assets, AML supervisors of financial institutions provided varied responses, with a high number of assets being outside the scope of these authorities. This is explained by the type of activity that these competent authorities conduct, mostly related to the supervision of credit and financial institutions and oversight of AML/CFT compliance. In line with previous responses, cash in hand and crypto assets are considered high risk by the majority of the respondents (40 % and 55 %, respectively) (see Table 12). Among the assets that are under the scope of AML supervisors of financial institutions, only one category was regarded as having no risk at all for money laundering and tax evasion: life insurance policies and pension products (7 %). In relation to this asset, more than half of these authorities also indicated it as low risk, which is in line with the opinion of other authorities.

In addition, certain assets are perceived as moderately risky for a significant number of respondents. These are unlisted securities (46 %) and safe deposit boxes (45 %). These results slightly contrast with the literature findings, where such assets are considered high risk for money laundering and tax evasion.

Table 12: Risk of asset use for money laundering and tax evasion purposes (AML supervisors of financial institutions)

Assets	AML supervisors of financial institutions	

	Not at all	Low	Moderate	High	Not applicable	No opinion	Number of observations
Cash in hand	0 %	20 %	0 %	40 %	30 %	10 %	10
Unlisted securities	0 %	23 %	46 %	8 %	23 %	0 %	13
Crowdfunding-platform intermediated assets	0 %	9 %	27 %	27 %	36 %	0 %	11
Life insurance policies/pension products	7 %	57 %	7 %	0 %	29 %	0 %	14
Limited liability companies	0 %	33 %	25 %	0 %	33 %	8 %	12
Crypto assets	0 %	0 %	18 %	55 %	27 %	0 %	11
In-game assets	0 %	17 %	33 %	0 %	50 %	0 %	6
Real estate	0 %	30 %	10 %	10 %	40 %	10 %	10
Vehicles	0 %	22 %	33 %	0 %	44 %	0 %	9
Other high-value goods	0 %	25 %	13 %	13 %	50 %	0 %	8
Commodities (precious raw materials)	0 %	33 %	0 %	17 %	50 %	0 %	6
Safe deposit boxes	0 %	9 %	45 %	9 %	27 %	9 %	11

Notes: The total number of AML supervisors of financial institution respondents differ by type of asset and are shown in the last column of the table ('number of observations'). The total percentage per asset does not add up to 100 % due to rounding. 'Not applicable' means outside the scope of the authority concerned.

Source: Authors' compilation.

3.3.2.4.1 Summary

The authorities surveyed indicated that most of the assets are at least moderately risky, with a low percentage (20 % or less) deeming assets not at all risky. At least half of the surveyed competent authorities reported that cash in hand and crypto assets are among the riskiest assets. Moreover, FIUs, tax authorities and LEAs hold that real estate and limited liability companies also present a high risk in relation to money laundering and tax evasion, while AML supervisors of financial institutions qualify these assets as low to moderately risky.

In contrast, life insurance policies and pension products are viewed as assets with low or no risk at all by most competent authorities. Still, tax authorities alone assign it a higher degree of risk, perceiving it as having a moderate risk in relation to money laundering and tax evasion.

3.3.3 Assets to be added or removed from the study

3.3.3.1 Assets to be added to the relevant list of assets

A majority of FIUs agree that the list of assets is exhaustive and that there are no other relevant assets that should be included in the list. Only one FIU suggested adding a relevant asset in the context of money laundering and tax evasion not included in the current asset list: gambling currency.

However, a limited number of tax authorities suggested adding other relevant assets in the context of money laundering and tax evasion not included in the current asset list: intellectual property rights such as trademarks (e.g. registry in EE) as indicated by two tax authorities, and farm animals (e.g. racehorses), as indicated by three tax authorities. In relation to the latter, two LEAs surveyed also suggested that farm animals (e.g. racehorses) should be added to the list of relevant assets.

In line with previous responses from other competent authorities, animals (in particular, racehorses) are mentioned by one AML supervisor of financial institutions as an asset that should be included in the list of assets due to its relevance in relation to money laundering and tax evasion.

All assets suggested by competent authorities for inclusion in the list of relevant assets were previously analysed in the literature review (see Section 3.2). In all cases, such assets were found to be less relevant in the context of money laundering and tax evasion, and hence outside a potential EU asset registry.

3.3.3.2 Assets to be removed from the relevant list of assets

Competent authorities were asked to indicate which assets should in their view should be removed from the list of relevant assets (see Annex C). Each column of the table in Annex C shows the total share of competent authorities supporting the removal of each asset, with the exception of the last column, which indicates the total share for all competent authorities. It is important to note that, in the case of LEAs, the number of respondents is relatively lower than the other competent authority categories. This means that shares could be bigger for this group, as the denominator is smaller.

A relatively large share of authorities from each group stated that life insurance policies and pension products can be removed from the relevant assets list. The highest percentages are exhibited by AML supervisors of financial institutions for life insurance policies (18 %) and pension products (24 %), followed by tax authorities (12 % and 6 %, respectively) and FIUs (6 % and 11 %, respectively). This is in line with responses on the low riskiness of life insurance and pension products for tax evasion and money laundering detailed above. In particular, some tax authorities state that there is a lack of previous occurrences of life insurance policies being used for money laundering and tax evasion.

- It is possible to see that FIUs presented varied opinions as to which assets should be removed. The greatest consensus is for some high-value goods and commodities. In this sense, around a fifth of FIUs supported the removal of worked specimens (see Section 3.2.4.3 on high-value goods in the literature review) and commodities (precious raw materials). Apart from pension products, around a tenth of FIUs surveyed agreed on removing collectors' items and safe deposit boxes at non-banks. Some competent authorities sustain that certain of the above assets must be removed, particularly worked specimens, collectors' items, commodities and safe deposit boxes at non-banks. In the case of worked specimens, even though they are part of high-value goods which were previously qualified as having moderate to high importance and risk, among all high-value goods they are the only ones whose removal is supported by a significant number of other competent authorities. For this reason, this asset is a candidate for removal.
- On the other hand, commodities were previously identified as being of moderate importance and risk by the majority of the competent authorities, and highly relevant among the literature, suggesting it is a relevant asset in relation to the fight against money laundering and tax evasion. Likewise, safe deposit boxes were identified by

most competent authorities as having moderate to high importance and risk, which supports the relevance of this asset in the context of money laundering and tax evasion. For these reasons, these two assets will not be removed from the relevant list of assets.

- Aside from life insurance policies, a quarter of tax authorities stated that cash in hand should also be removed. This is due to the fact that, among other reasons, building registries for cash in hand is difficult because it is hard to detect parties in a cash-in-hand transaction. This opinion is supported by more than a tenth of the AML supervisors of financial institutions (12 %) and a minority of FIUs (6 %). However, given the high importance and risk that this asset entails for competent authorities based on survey responses and literature findings, it will be still considered a relevant asset in relation to money laundering and tax evasion.
- Around a third of LEAs flagged in-game assets for removal, while less than a fifth suggested it for firearms (17 %). Also, certain FIUs, tax authorities and AML supervisors of financial institutions also agreed on removing in-game assets (6 % in each case). But given the degree of importance and risk attributed by most competent authorities (moderate to high), as well as the literature findings, they will be considered among the relevant assets.
- Finally, some AML supervisors of financial institutions supported the removal of life insurance policies and pension products, as well as cash in hand, worked specimens and safe deposit boxes at non-banks. As previously noted, worked specimens have already been removed from the list of relevant assets. In addition, crowdfunding-platform intermediated assets were specified as an asset to remove by AML supervisors of financial institutions, as well as by FIUs and tax authorities. As most competent authorities identified this asset as having low to moderate importance and risk, it too will be excluded from the relevant list of assets.

Overall, based on the survey responses, crowdfunding-platform intermediated assets, life insurance policies, pension products and worked specimens are deemed eligible for removal, and are excluded from the study.

3.3.4 List of relevant assets

Generally, the survey results verified the literature review findings (see Table 13), suggesting that cash in hand, vehicles, real estate and limited liability companies are relevant to money laundering and tax evasion. Other assets – notably life insurance policies, pension products and crowdfunding-platform intermediated assets – seem to be of low to moderate relevance for most competent authorities, in line with the results from the literature review (which finds them of moderate relevance for money laundering and tax evasion). Based on the literature review and survey findings, leading to the removal of several assets from the study as described in the previous section, the list of relevant assets is presented below.

Table 13: List of relevant assets retained for the purpose of this study

Category	Asset group	Asset
Cash at bank and in hand	Cash in hand	Cash in hand
	Cash at bank	Cash at bank
Financial assets	Securities	Listed securities
		Unlisted securities

	Limited liability companies	Limited liability companies
Intangible assets	Crypto assets	Crypto coins
		NFTs
	In-game assets	In-game assets
Tangible assets	Real estate	Properties
		Land
	Vehicles	Motor vehicles
		Ships
		Aircraft
	High-value goods	Firearms
		Works of art
		Collectors' items
		Antiques
		Jewellery
Commodities	Commodities (precious raw materials)	Commodities (precious raw materials)
Other assets	Safe deposit boxes	Safe deposit boxes at banks
		Safe deposit boxes at non-banks

Note: As 'cash at bank', 'safe deposit boxes at banks', 'limited liability companies', 'crypto assets' and 'real estate' registries are either being interconnected, or there are previous or ongoing studies to interconnect such registries, they are therefore not considered further in the context of Tasks 2-3 of this study but will be considered in Tasks 4-7.

Source: Authors' compilation.

4. Task 2: Mapping of existing registries and alternative sources of information

4.1 Methodology

Task 2 includes an assessment of existing registries and other sources of information on asset ownership in terms of the data stored, digitalisation level and access of competent authorities (FIUs, LEAs, tax authorities and AML supervisors of financial institutions) for carrying out their respective tasks. A common approach has been used to develop the database with comprehensive information on the existing registries and alternative sources. Desk research first identified and mapped the existing registries for the relevant assets determined under Task 1. Subsequently, a survey analysis was conducted based on the responses of both the administering entities of the registries and the competent authorities, complemented with desk research where necessary. The total number of asset registries reviewed as part of the analysis is based on the number of responses received on the part of existing asset registries,

excluding the five asset registries that either are being interconnected (e.g. the BORIS project) or are the subject of studies on the interconnection of such registries ([59]).

To proceed with the mapping of existing registries and alternative sources of information it was essential to identify them. In total, 215 registries and other sources of information on asset ownership were identified that cover the listed assets determined through Task 1. It is important to note that these asset registries and alternative sources of information on asset ownership are registries/databases managed by public authorities (e.g. national ministries) or by private entities (e.g. CSDs, trade repositories, high-value goods shops). Overall, the majority of the registries and alternative information sources (160 registries/alternative sources, or 74 %) cover one asset. Almost a fifth (32 registries/alternative sources, or 15 %) cover two assets. The remainder, approximately a tenth of them (23 registries/alternative sources, or 11 %) cover three to five assets. For example, some high-value goods databases recording transaction data often cover several types of assets (e.g. works of art, collectors' items and antiques).

Turning to assets, four types can be distinguished in terms of coverage by registries and alternative sources (see Table 14 and Table 15):

- First, there are three assets ([60]) for which information is available for all 27 EU Member States in registries that **cover these assets in each Member State**. This is the case for motor vehicles, aircraft and firearms.
- Second, there are another three assets that all Member States cover, but in at least some countries **more than one registry/alternative source covers the asset**. The assets concerned include ships, collectors' items and commodities.
- Third, there are four assets that **only some Member States cover**. This group of assets can be divided into two sub-groups: those that are almost fully covered and those that are only partially covered. Listed securities registries were found in nearly all of the Member States (24 countries), yet with full coverage at EU level. The following three assets were found to be covered in only a few Member States by alternative sources of information recording transaction data: works of art, antiques and jewellery.
- Fourth, for three assets there are **currently no registries** or other sources of information **available** on asset ownership. These are cash in hand, unlisted securities and in-game assets.

([59]) Registries for cash at bank, safe deposit boxes at banks, crypto assets and real estate.

([60]) Providers of safe deposit boxes at non-banks encountered in each Member State are at least one per country.

Table 14: Number of existing asset registries and alternative sources of information, by Member State

Category	Asset group	Asset	AT	BE	BG	CY	CZ	DE	DK	EE	EL	ES	FI	FR	HR	HU	IE	IT	LT	LU	LV	MT	NL	PL	PT	RO	SE	SI	SK	EU	Total number of registries/other sources	Total number countries with registries/other sources	Share
Cash at bank and in hand	Cash in hand	*Cash in hand*	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0 %
Financial assets	Securities	*Listed securities*	1	1	1	0	1	1	1	0	1	1	1	1	1	1	2	1	0	3	1	1	2	2	1	1	1	1	2	0	30	24	89 %
		Unlisted securities	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0 %
Intangible assets	In-game assets	*In-game assets*	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0 %
Tangible assets	Vehicles	*Motor vehicles*	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	27	27	100 %
	High-value goods	*Ships*	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	1	1	1	1	1	1	0	28	27	100 %
		Aircraft	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	27	27	100 %
		Firearms	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	27	27	100 %
		Works of art	0	3	0	0	0	0	1	0	2	0	1	1	0	0	0	3	0	1	1	2	3	0	2	0	2	0	0	0	22	12	44 %
		Collectors' items	1	2	2	1	1	1	2	2	2	1	1	2	2	1	1	3	1	2	2	2	4	1	4	1	2	1	1	0	46	27	100 %
		Antiques	0	2	0	0	0	0	1	0	2	0	0	1	0	0	0	3	0	1	0	2	2	0	3	0	2	0	0	0	19	10	37 %
		Jewellery	1	2	0	0	0	0	1	1	2	0	2	1	0	0	0	3	0	1	1	0	3	0	3	0	2	0	0	0	23	13	48 %
Commodities	Commodities (precious raw materials)	*Commodities (precious raw materials)*	1	1	2	1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	1	1	1	1	1	1	1	1	1	0	29	27	100 %
Other assets	Safe deposit boxes	*Safe deposit boxes at non-banks*	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	27	27	100 %

Notes: Two ship registries are found in the Netherlands, one belonging to the Kadaster and the other to the Netherlands Vehicle Authority (RDW), which is in charge of registering different types of vehicles, such as vessels. At least one non-bank safe deposit box provider was encountered in each EU Member State.

Source: Authors' compilation.

Table 15: Coverage by existing asset registries and alternative sources of information, by Member State

Category	Asset group	Asset	AT	BE	BG	CY	CZ	DE	DK	EE	EL	ES	FI	FR	HR	HU	IE	IT	LT	LU	LV	MT	NL	PL	PT	RO	SE	SI	SK	Total (count)	Total share %
Cash at bank and in hand	Cash in hand	*Cash in hand*																												0	0 %
Financial assets	Securities	*Listed securities*	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	27	100 %
		Unlisted securities																												0	0 %
Intangible assets	In-game assets	*In-game assets*																												0	0 %
Tangible assets	Vehicles	*Motor vehicles*	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	27	100 %
		Ships	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	27	100 %
		Aircraft	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	27	100 %
	High-value goods	*Firearms*	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	27	100 %
		Works of art		X					X		X		X	X				X		X	X	X	X		X		X			12	44 %
		Collectors' items	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	27	100 %
		Antiques		X					X		X			X				X		X		X	X		X		X			10	37 %
		Jewellery	X	X					X	X	X		X	X				X		X	X		X		X		X			13	48 %
Commodities	Commodities (precious raw materials)	*Commodities (precious raw materials)*	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	27	100 %
Other assets	Safe deposit boxes	*Safe deposit boxes at non-banks*	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	27	100 %

Source: Authors' compilation.

The administration in charge of the different asset registries and alternative sources of information varies depending on the type of asset, as well as on the country (see Table 16). Out of the 215 asset registries and other sources of information, 104 were identified as managed by a public authority, whereas 111 were identified as under private entity management. There are certain asset registries that, in all Member States, are managed mostly by public administration entities. These assets are vehicles, ships, aircraft and firearms. Alternative sources of information on asset ownership are administered entirely by private entities. This is the case for listed securities, and some high-value goods such as collectors' items, works of art, antiques, jewellery and commodities. For instance, in the case of listed securities, CSDs act as the primary holders of these securities databases.

The sources of information identified for most high-value goods are not registries per se but rather private companies holding private transaction data with regard to the respective asset. This difference may have an impact when analysing the possible interconnection of different asset registries as

some of them already exist, while others do not. In addition, the asset registries managed by public authorities are generally based on national legislation establishing the requirement to register the assets for the administration of duties. For instance, for vehicles, there is a legal obligation to register them in the country of residence. In general, vehicle registries are managed under the national ministries (i.e. ministries of interior or transport) of each Member State.

Table 16: Administering entity of existing asset registries and alternative sources of information, by Member State (based on country coverage)

Category	Asset group	Asset	AT	BE	BG	CY	CZ	DE	DK	EE	EL	ES	FI	FR	HR	HU	IE	IT	LT	LU	LV	MT	NL	PL	PT	RO	SE	SI	SK	Total %		
																														PU	PR	No registry
Cash at bank and in hand	Cash in hand	*Cash in hand*																												0 %	0 %	100 %
Financial assets	Securities	*Listed securities*	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	0 %	100%	0 %
		Unlisted securities																												0 %	0 %	100 %
Intangible assets	In-game assets	*In-game assets*																												0 %	0 %	100 %
Tangible assets		*Motor vehicles*	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	100 %	0 %	0 %
	Vehicles	*Ships*	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	100 %	0 %	0 %
		Aircraft	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	100 %	0 %	0 %
	High-value goods	*Firearms*	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	PU	100 %	0 %	0 %
		Works of art		PR					PR		PR		PR	PR				PR		PR	PR	PR	PR		PR		PR			0 %	44 %	56 %
		Collectors' items	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	0 %	100%	0 %
		Antiques		PR					PR		PR			PR				PR		PR		PR	PR		PR		PR			0 %	37 %	63 %
		Jewellery	PR	PR					PR	PR	PR		PR	PR				PR		PR	PR		PR		PR		PR			0 %	48 %	52 %

Commodities	Commodities (precious raw materials)	*Commodities (precious raw materials)*	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	0 %	100%	0 %
Other assets	Safe deposit boxes	*Safe deposit boxes at non-banks*	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	PR	0 %	100%	0 %

Note: 'PU' means public administering entity of the asset registries; 'PR' means private administering entity of the asset registries and other sources of information on asset ownership.

Source: Authors' compilation.

The scope of the asset registries and other sources of information on asset ownership differs depending on the type of asset and the country coverage (see Table 17). The scope of the registries and alternative sources could be at the national, regional or company level.

National level. There are many asset registries and other sources of information at this level. This is the case for motor vehicles, ships, aircraft and firearms (100 % in each case).

Regional level. No asset registries or other sources of information were found at the regional level. Nevertheless, real estate registries (outside the scope of this Task 2) have regional scope in certain countries ([61]).

Company level. All asset registries and other sources of information on asset ownership identified for commodities (precious raw materials), collectors' items, antiques, works of art, jewellery and safe deposit boxes at non-banks are characterised by being at company level. This implies that there are no existing registries managed by any public competent authority for these assets, but rather private entities holding private transaction data. Registries of listed securities, particularly those administered by CSDs or trade repositories, are also held by companies but some contain information with EU-level coverage (e.g. trade repositories in IE, LU, NL and PL).

Task 2 maps the existing asset registries that correspond to the relevant list of assets, which are managed under national public authorities. A summary of the registries is in Table 18. For the mapping exercise, the authorities managing the asset registries listed below were approached to complete a survey.

([61]) European Commission (2021b), https://data.europa.eu/doi/10.2874/068191

Table 17: Scope of the existing asset registries and alternative sources of information, by Member State (based on country coverage)

Category	Asset group	Asset	AT	BE	BG	CY	CZ	DE	DK	EE	EL	ES	FI	FR	HR	HU	IE	IT	LT	LU	LV	MT	NL	PL	PT	RO	SE	SI	SK	Total %		
																														N	C	No registry
Cash at bank and in hand	Cash in hand	*Cash in hand*																												0 %	0 %	100 %
Financial assets	Securities	*Listed securities*	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	%	100%	0 %
		Unlisted securities																												0 %	0 %	100 %
Intangible assets	In-game assets	*In-game assets*																												0 %	0 %	100 %
Tangible assets	Vehicles	*Motor vehicles*	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	100%	0 %	0 %
		Ships	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	100%	0 %	0 %
		Aircraft	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	100%	0 %	0 %
	High-value goods	*Firearms*	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	N	100%	0 %	0 %
		Works of art		C					C		C		C	C				C		C	C	C	C		C		C			0 %	44%	56 %
		Collectors' items	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	0 %	100%	0 %
		Antiques		C					C		C			C				C		C		C	C		C		C			0 %	37%	63 %
		Jewellery	C	C					C	C	C		C	C				C		C	C		C		C		C			0 %	48%	52 %
Commodities	Commodities (precious raw materials)	*Commodities (precious raw materials)*	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	0 %	100%	0 %
Other assets	Safe deposit boxes	*Safe deposit boxes at non-banks*	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	C	0 %	100%	100 %

Notes: 'C' means scope of asset registry at the company level; 'N' means scope of asset registry at the national level.

Source: Authors' compilation.

Table 18: Identified registries of relevant assets under public authorities

Category	Asset group	Asset	Number of registries *
Cash at bank and in hand	Cash in hand	Cash in hand	NA
Financial assets	Securities	Listed securities	NA
		Unlisted securities	NA
Intangible assets	In-game assets	In-game assets	NA
		Motor vehicles	27
	Vehicles	Ships	28**
		Aircraft	27
Tangible assets	High-value goods	Firearms	27
		Works of art	NA
		Collectors' items	NA
		Antiques	NA
		Jewellery	NA
Commodities (raw precious materials)	Commodities (raw precious materials)	Commodities (raw precious materials)	NA
Other assets	Safe deposit boxes	Safe deposit boxes at non-banks	NA

Notes: As 'cash at bank', 'safe deposit boxes at banks', 'limited liability companies', 'crypto assets' and 'real estate' registries are either being interconnected, or there are previous or ongoing studies to interconnect such registries, they are not therefore considered further in the context of Tasks 2-3 of this study but will be considered in Tasks 4-7.

* This column shows the number of existing registries in the EU for each of the assets under public authorities' management in Member States (e.g. if there is more than one registry of the same asset in the same country, all registries will be included in the total number of asset registries). However, it does not consider the coverage of the asset registry.

** Two ship registries were found in the Netherlands, one belonging to the Kadaster and the other to the Netherlands Vehicle Authority (RDW) which is in charge of registering different types of vehicles, including vessels.

Source: Authors' compilation.

This task is divided into five main sub-tasks, as presented and discussed below:

- description of the scope and format of the data stored
- description of digitalisation and aggregation level
- assessment of national beneficial ownership registries
- accessibility assessment – type of competent authority
- accessibility assessment – format of information.

4.2 Scope, format of the data stored and data protection requirements

To describe the scope and format of data stored in registries, a survey was conducted across the public authorities responsible for administering registries for the identified relevant assets. In total, 105 questionnaires were sent for the asset registries that are covered in this task, resulting in a total of 13 responses. It should be noted that questionnaires were only sent to asset registries that are managed by public authorities, which includes vehicles and firearms.

Below are the findings from the survey responses. The number of responses received per country as well as the assets covered are shown in Table 19.

Table 19: Number of survey responses, by Member State

Country	Number of responses (per asset registry)	Asset(s) covered
Lithuania	2	Motor vehicles
		Ships
Sweden	2	Aircraft
		Ships
Bulgaria	1	Aircraft
Croatia	1	Aircraft
Cyprus	1	Firearms
Czechia	1	Aircraft
Denmark	1	Motor vehicles
Ireland	1	Aircraft
Latvia	1	Ships
Luxembourg	1	Ships
Romania	1	Ships

Source: Authors' compilation.

The legal bases for data collection are EU law, national laws, acts and regulations.

The collection, storage and exchange of data under the AMLD involve conducting customer due diligence, monitoring, identifying suspicious transactions and beneficial owners of legal entities or legal arrangements, among other tasks ([62]).

Access to and the processing of the data must comply with EU data protection law and the General Data Protection Regulation (GDPR) ([63]). Data collection and its subsequent processing are necessarily limited to the objectives of such Directive, which implies that its use for any other purpose is prohibited. In particular, FIU data protection regimes depend on the nature of the FIU. Some of them follow the GDPR, while others the Law Enforcement Directive ([64]).

Consequently, the collection of data in national asset registries is delimited by the objectives of the national acts, laws and regulations that establish them. For instance, aircraft registries indicated collecting data based on civil aviation acts, ordinances of affiliation, and registering the aircraft in the national registry of civil aircraft. Firearms registries gather data based on firearms regulations such as the EU Firearms Directive ([65]), as well as other non-firearms

([62]) https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32015L0849&from=EN

([63]) https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32016R0679

([64]) European Parliament (2022).

([65]) Directive (EU) 2021/555 of the European Parliament and of the Council of 24 March 2021 on control of the acquisition and possession of weapons

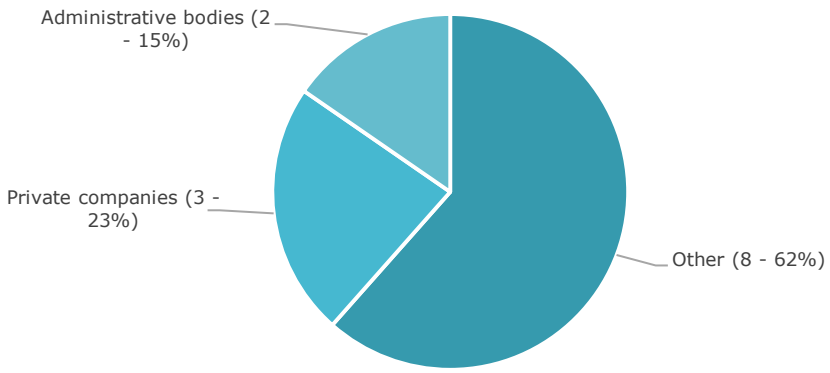
regulations that are part of administrative regulatory acts, while cars and other motor vehicle registries do it based on national laws on vehicle registration and safe car traffic on roads. This implies that national vehicle registries, for instance, were created to facilitate the administration of traffic, not as a mechanism to fight against money laundering.

There are certain main obligations imposed on the registries to ensure data are processed properly. For instance, in the case of a particular aircraft registry, documents should be stamped and signed in order to register or re-register the aircraft, which must be done by the responsible authority of the organisation. The registry should reflect the factual and legal status of the aircraft, in accordance with the respective ordinance. Another important point is linked to the security of the data, which is usually ensured by data protection laws for most of the asset registries covered by the survey.

The providers of data to the asset registries differ among the respondents (see Figure 3). An important share of the registry entities responding to the survey indicated that sources other than administrative bodies or private companies supply data for the registries (8 or 62 % of the registries). Other sources of information include natural and legal persons, tax offices and other relevant offices, and beneficial ownership registries, as well as both private and public entities. Yet, some revealed that they receive data solely from private companies (3 or 23 % of the registries) or administrative bodies (2 or 15 % of the registries).

In general, vehicle registries obtain data directly from asset owners through applications and documents provided by private persons. At the same time, these registries also obtain data from other entities such as inspection companies, vehicle importers, insurance companies and the national police forces, among others.

Figure 3: Types of organisations supplying data to registries (number of observations)

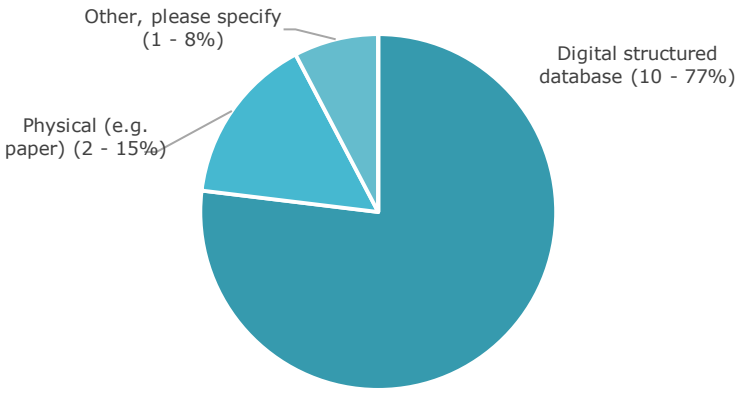


Source: Authors' compilation.

The way the data are stored in asset registries varies (see Figure 4). The large majority of respondents indicated that the data in the registry are stored digitally, such as in XML and Microsoft SQL formats, generally using digital structured databases (DSDs) (10 or 77 % of the registries). However, there is still a minor but relevant proportion using physical data storage, such as on paper (2 or 15 % of the registries). This suggests a lack of accessibility by machines, such that data cannot be retrieved by their identifiers using a standardised communication protocol. Only one respondent reported using a different structure (1 or 8 % of the registries). This refers to scanned documents. In these latter cases, potential difficulties may arise when analysing the interconnection of asset registries due to the divergences in

digitalisation levels. This aspect is later analysed in the operational feasibility assessment (Task 4).

Figure 4: Format of the data stored in the registry (number of observations)



Source: Authors' compilation.

Respondents were asked to specify the categories of information available in the registry, with the possibility of indicating one or more categories in their response (see Table 20). Most of the respondents selected at least four of the categories shown below. The majority indicated transactional data and historic data as included in the information available in the registry (85 % each). Furthermore, more than three quarters of the respondents also indicated the availability of master data and logs data (77 % each), followed by metadata (69 %) and reference data (62 %). Finally, fewer than half of the respondents to this question indicated the use of temporary data.

In addition, a large majority of respondents indicated that data entry points are unique, with only one having a process in place to avoid double entries.

Table 20: Share of asset registries collecting each type of data

Categories of information available in the registry	Share
Transactional data	85 %
Historical data	85 %
Master data	77 %
Logs data	77 %
Metadata	69 %
Reference data	62 %
Temporary data	46 %

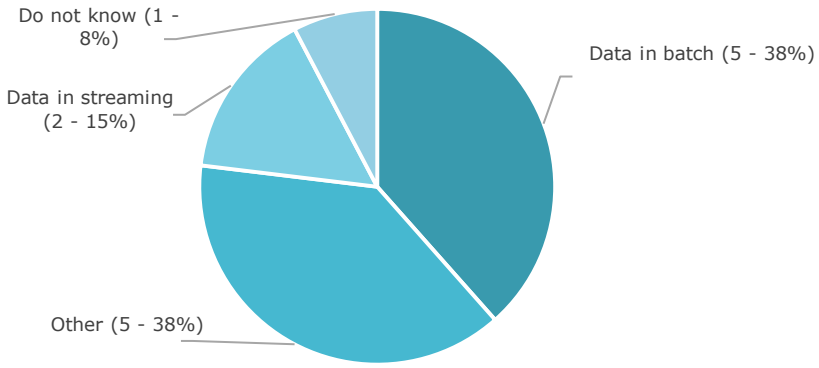
Note: The total number of responses for asset registries is 13.

Source: Authors' compilation.

Administering authorities were requested to indicate the format of data available in the registry (see Figure 5). Most registries reported having either data in batch or other formats of data (5 or 38 % of the registries, respectively). Among the respondents who indicated other formats of data, one uses a paper-based system, who indicated that batch download is only possible

upon request. Fewer than a fifth of the respondents have data in streaming (2 or 15 % of the registries), while one was not aware of the format of data available in the registry (1 or 8 % of the registries).

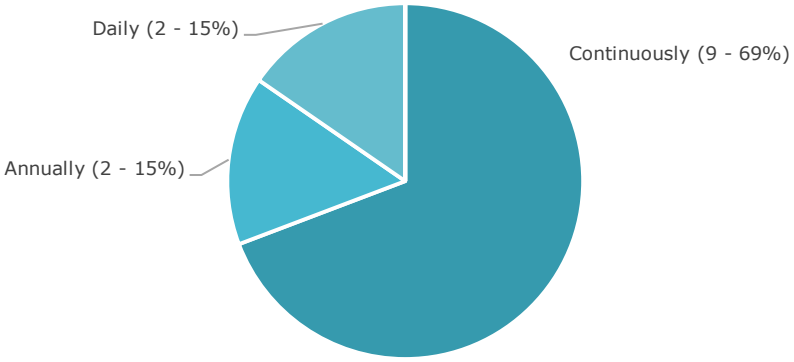
Figure 5: Type of data available in the registry (number of observations)



Source: Authors' compilation.

Most of the respondents indicated that asset registries are updated continuously in real time (see Figure 6) (9 or 69 %). In contrast, fewer than a fifth of the respondents have annual or daily updates of the information available in the registries (2 or 15 % of the registries, respectively). The frequency of registry updates is an important factor to consider when assessing the possibility of interconnecting registries or building an EU centralised asset registry. The higher the frequency of these individual registry updates, more accurate and up to date will be the information available for competent authorities to carry out their tasks.

Figure 6: Frequency of registry updates (number of observations)



Note: The total percentage per asset does not add up to 100 % due to rounding.

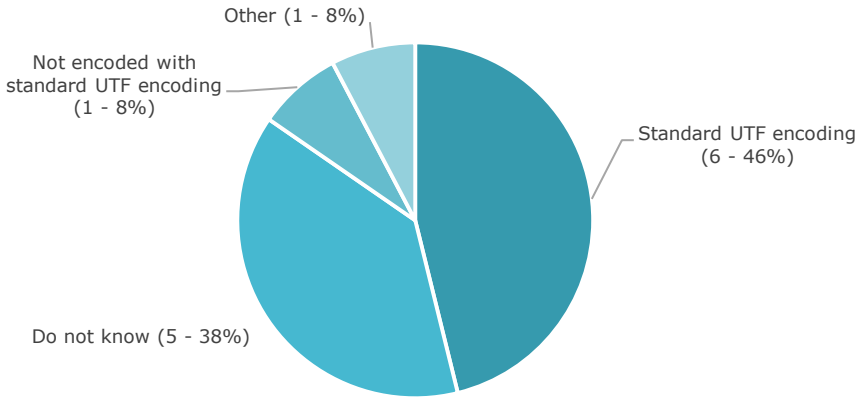
Source: Authors' compilation.

Registries were requested to indicate whether the standard UTF encoding is used as the encoding method for the registry (see Figure 7). The responses are mixed and show potential obstacles in standardising registries. The results show that the standard UTF encoding (UTF-8) is employed in around half of the asset registries (6 or 46 % of the registries). Still, more

than a third of the respondents were not aware of whether such an encoding method is used (5 or 38 % of the respondents).

Furthermore, one respondent stated that the registry does not use the standard UTF encoding (1 or 8 % of the registries), while another respondent said the respective registry has different type of encoding. In both cases, the responses are explained by the fact that such registries are paper-based.

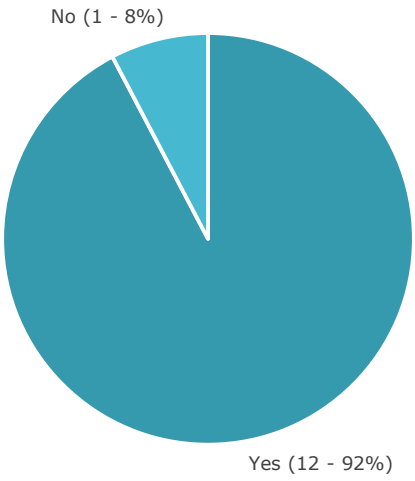
Figure 7: Registry encoding methods (number of observations)



Source: Authors' compilation.

In most of the cases, data entries are unique within the registry (see Figure 8) (92 % of the registries). This means, for instance, there is no duplication of entries. Only one respondent reported that the asset registry data are not unique within the registry (8 % of the registries). This is explained by the fact that the registry is implementing and migrating historical data into other software.

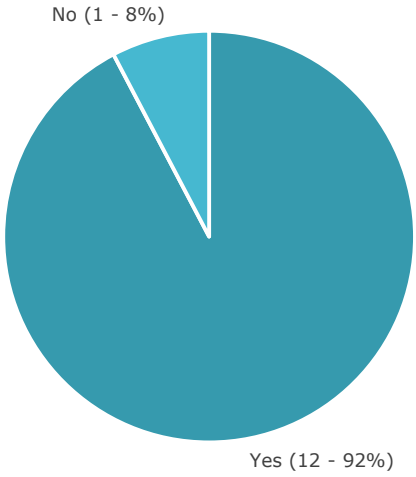
Figure 8: Unique data within the registry (i.e. no double entries) (% of asset registries)



Source: Authors' compilation.

Administering authorities were also requested to specify whether the databases are structured in a way that is easily used by both humans and machines (see Figure 9). Following the above trend, the majority affirmed that databases are structured to be used by both (92 % of the registries). Only one ship registry indicated a different situation, in which its database does not seem to be structured for easy use by both humans and machines, without providing further explanation (8 % of the registries).

Figure 9: Asset registry database is structured in a way that is easy to use by both humans and machines (% of asset registries)



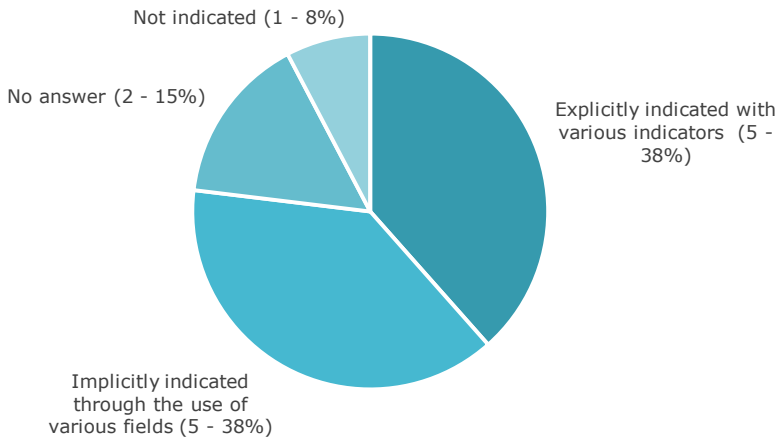
Source: Authors' compilation.

Holders of asset registries were asked about whether the registries include the identity and type of legal owner of the asset. In this regard, the results are varied (see Figure 10). More than a third of the respondents revealed that the identity and type of legal owner of an asset is explicitly indicated in their registry with assorted indicators related to the legal owner (5 or 38 % of the registries). The types of owners include natural persons, legal persons such as limited liability companies and foundations, and legal arrangements such as trusts. In particular, explicit reference is made by motor vehicle registries on this point.

The same number of respondents also expressed that the type of legal owner of an asset is implicitly indicated through the use of various fields (5 or 38 % of the registries in each case). This includes first and last name, as well as company name. Most of the respondents stated they recorded all historic information on previous legal owners of assets, whereas others include it only from a specific date, or for a particular period. Among those recording all historic information, the majority also expressed that the indicators recorded are the same as those for present owners.

Finally, only one respondent for an asset registry noted that the identity and type of legal owner of the asset is not shown in the registry, and two respondents did not provide an answer (2 or 15 % of the registries).

Figure 10: Indication of the identity and type of legal owner of the asset in the registry (number of observations)



Note: The total percentage per asset does not add up to 100 % due to rounding.

Source: Authors' compilation.

Asset registries include different indicators for the stored data. In fact, respondents mentioned a variety of indicators used in the databases. The existence of several indicators among the national registries is important because they help to accurately identify the assets.

All registries supplying this information belong to the vehicles category – in particular, aircraft, cars and other motor vehicles and ships. With regard to the former, it is possible to verify certain indicators present in all aircraft registries, notably serial number, manufacturer, year of manufacture, model, registration mark, date of registration and registered owner of the aircraft. The majority of these asset registries hold other data, such as those related to the aircraft engine, registered operator of the aircraft, insurance information and existing mortgages. One of the aircraft registries reportedly has certain data regarding natural owners, which are not disclosed in the registry due to the GDPR.

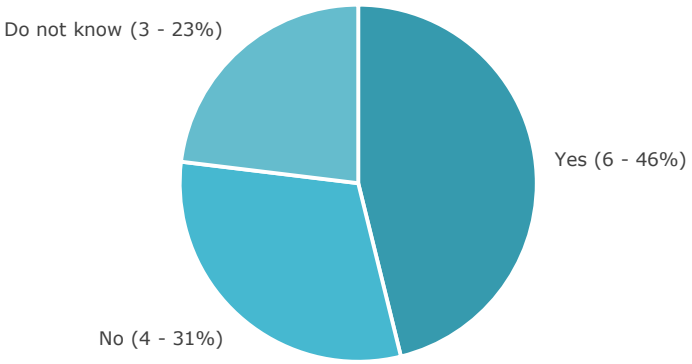
Compared with aircraft, the ship registries reportedly include more indicators regarding both technical aspects of the asset and ownership information. Similar to aircraft, all ship registries cover the category, type, date of registration, year of manufacture and registration number of the asset. Ship registries also include the port of registration, ship dimensions and machinery. One ship registry highlighted that indicators differ depending on the search type. That is, ships can be searched by person, ship or engine, which implies the presence of different indicators for each search. Finally, ship registries also include ownership indicators, such as the registered owner, the principal owner, type and company name if it is the owner, address, email and phone number among others.

Lastly, only one motor vehicle registry provided a list with indicators covered in the registry. However, it is not detailed and only grouped in wider categories such as information regarding vehicle registration, technicalities, insurance, periodic tests of road worthiness, and calculated tax and excises related to vehicle registration.

In order to make data more searchable, registries can offer data catalogues, which help users by describing the data the user can collect (see Figure 11). Almost half of the respondents use catalogues to describe the data (6 or 46 % of the registries), while around a third do not use any data catalogue (4 or 31 % of the registries). Fewer than a quarter of respondents indicated not knowing whether a data catalogue is used or not (3 or 23 % of the registries). Among the

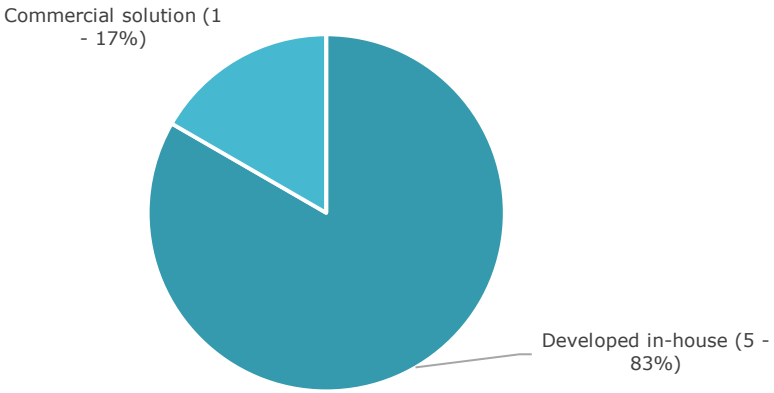
registries that use a data catalogue (see Figure 12), for the vast majority such catalogues were developed in-house (5 or 83 % of the registries). The remaining one uses a commercial solution to catalogue data (1 or 17 % of the registries).

Figure 11: Use of a data catalogue to describe data in an organised way (number of observations)



Source: Authors' compilation.

Figure 12: If a data catalogue is used, the type of data cataloguing (number of observations)



Source: Authors' compilation.

4.2.1 Summary

Overall, the data in the registries stem from a variety of sources – such as tax offices and other relevant offices, non-profit legal entities and natural or legal persons – depending on the type of asset. From an operational perspective, this means that asset registries require set-ups or arrangements that allow for the transfer of such data from the data supplier to the registry. The format of the data stored in the registry is mostly through the use of a digitally structured database, with a minority of the asset registries storing data in physical (paper) or scanned form.

The data are for the large part updated regularly, with some updated annually or daily. In terms of the encoding method, around half of the respondents indicated using the standard UTF, while a significant number were not aware of the method used.

The identity and type of legal owner of the asset is primarily identifiable either explicitly with various indicators or implicitly through the use of different fields. Most of the registries also record all historic information of previous legal owners, which includes the same indicators that are applicable to present owners.

Finally, registries include a variety of indicators, from the most general ones such as name, registration date, model, serial number and category, to some more specific. Almost half of the respondents use a data catalogue to describe the data in the registry. Among them, the majority use one developed in-house.

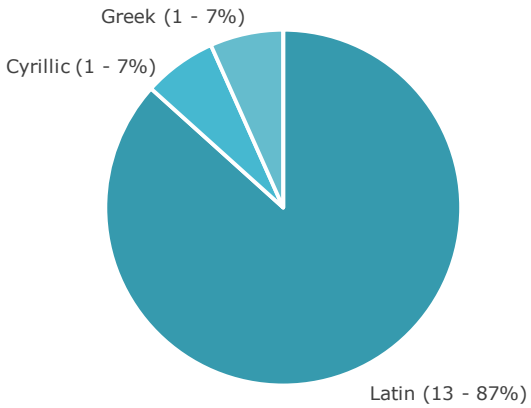
4.3 Digitalisation and aggregation level

To describe the level of digitalisation and aggregation of the data stored in registries, a survey was carried out across the authorities responsible for administering registries. Below are the findings collected through the survey responses.

One important part of digitalisation is the searchability of a registry. The searchability can further be enriched by allowing for smart search capabilities within the search functions, accounting for approximate matches and for slight differences in spelling. Another part of digitalisation is the degree to which the spelling of text-based information in the registry is harmonised. Having an agreed-upon standard of spelling that takes into account special characters found in some languages or different alphabets, such as Cyrillic or the Greek alphabet, is very important in terms of conducting searches for entries. In the absence of a harmonised standard of spelling, searching for entries may lead to mismatches simply because the user is doing so with information that is spelled using a convention different from that used in the registry.

Holders of asset registries were requested to indicate the alphabet(s) in which the entries are displayed in the registry (see Figure 13). All responded that such entries are shown in the Latin alphabet (13 registries). Among them, two indicated that they also display the entries in different alphabets, in Cyrillic and in Greek (1 or 7 % of the registries in each case).

Figure 13: The alphabets used in the registry (number of observations)

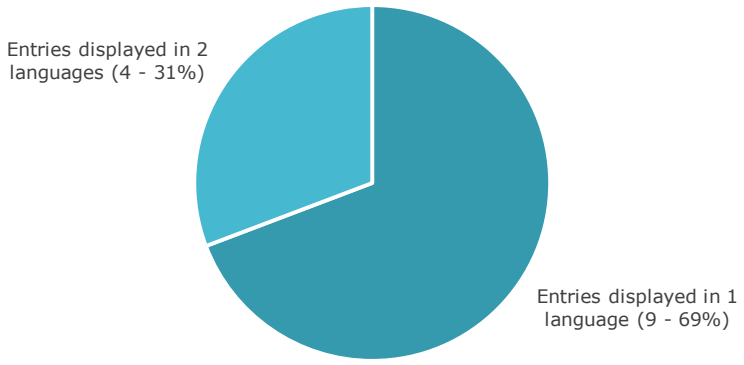


Notes: The total percentage per asset does not add up to 100 % due to rounding; question with multiple options.

Source: Authors' compilation.

In general, entries in the registries are displayed in only one language (see Figure 14) (9 or 69 % of the registries). This language is always the official language of the country where the registry is established. Around a third of the respondents display the entries in two languages (4 or 31 % of the registries). That is, entries are displayed in the local language but also in English in all cases. For instance, this can be seen in certain registries from countries such as Bulgaria, Czechia, Cyprus and Lithuania.

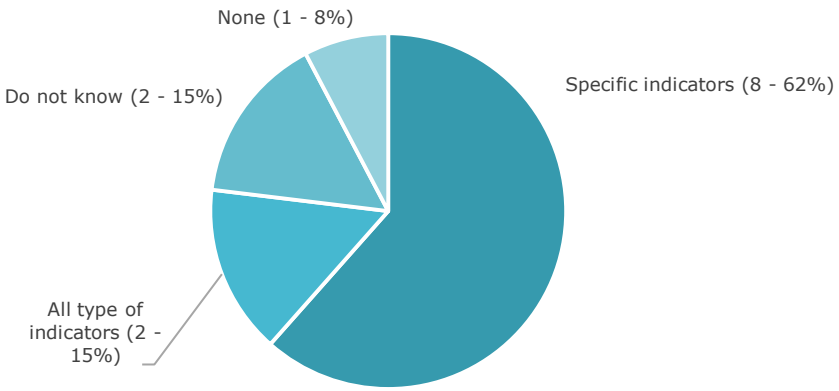
Figure 14: The number of languages in which the entries are displayed in the registry (number of observations)



Source: Authors' compilation.

Various indicators can be used to search within the registries (see Figure 15). The majority of the respondents note that not all but rather specific indicators may be used (8 or 62 % of the registries). For instance, aircraft and ship registries can be searched by the name of the asset, registration mark, registration number, aircraft model, category, registration date and registered owner, among others. For cars and other motor vehicles, it is possible to search with the vehicle identification number, registration number and/or legal or natural person code. Finally, some indicators that can be used to search in firearms registries are ID number, name and surname, registration number of the asset, model, serial number, among others.

Figure 15: Indicators that can be used to search in the registry (number of observations)

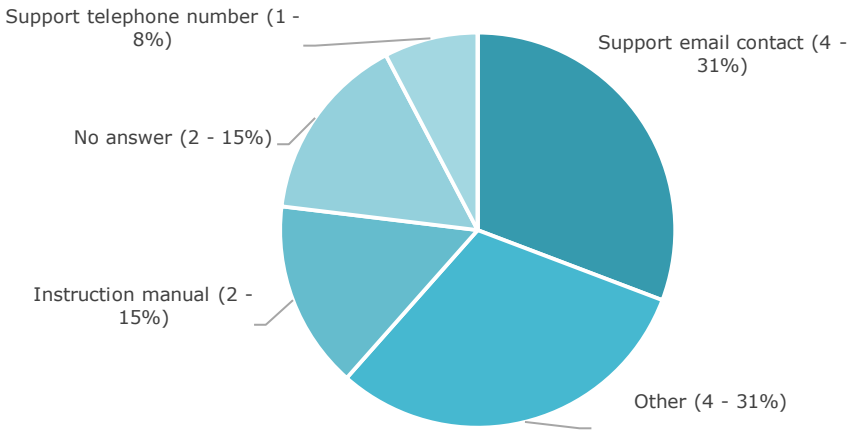


Source: Authors' compilation.

Holders of asset registries were also asked to convey whether any supporting documentation or help is available on the registry website(s) (see Figure 16). It is apparent that asset registries offer different support services for accessing documentation. The most common types offered are guidance through a support email contact and 'other types of support' (noted by 4 or 31 % of the registries, in each case). The latter includes support through online texts, general help centres and direct support for the extraction of information from the database. Less than a quarter of the respondents provide an instruction manual, while the same number of respondents did not provide any answer (2 or 15 % of the registries in each case).

Finally, fewer than a tenth of the respondents offer a telephone number to support the use of the registry (1 or 8 % of the registries).

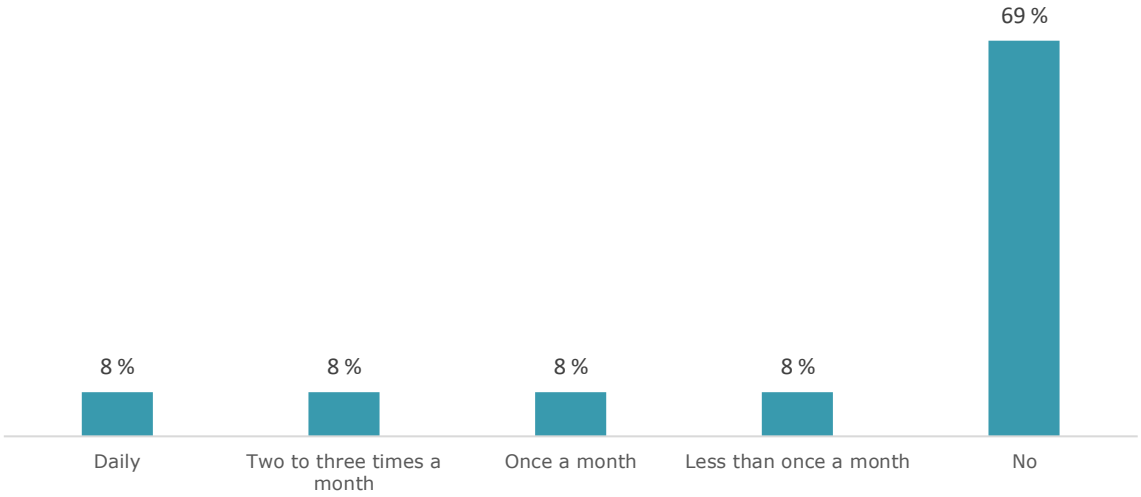
Figure 16: Provision of any documentation or help to support the use of the registry (number of observations)



Source: Authors' compilation.

Due to the lack of existing interconnection among registries, some administering authorities may have to manually search other registries to obtain information (see Figure 17). Overall, this is not a common practice among the respondents (69 % of the registries). However, around a third of the registries responding to the survey reported manually searching other asset registries. Among this group, around a tenth manually search other registries on a daily basis (8 % of the registries), while the rest do it either two to three times a month, once a month or less than once a month (8 % of the registries, respectively). In particular, these registries record aircraft and ships. For instance, holders of certain ship registries manually search population registries, legal persons registries, foreigners' registries and beneficial owners' registries.

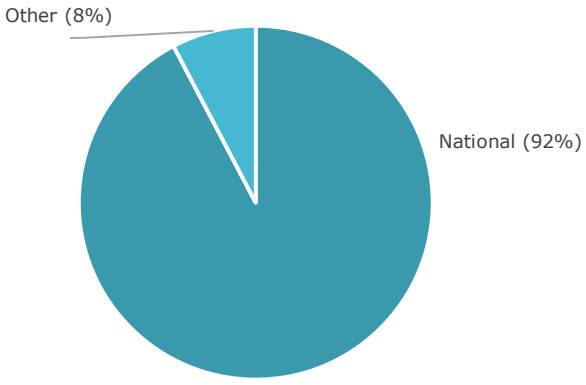
Figure 17: Asset registry authorities manually search other registries and frequency (% of asset registries)



Source: Authors' compilation.

Administering authorities were requested to indicate the scope or level of aggregation of the registry (see Figure 18). Results suggest that most of the registries have a national scope (92 % of the registries) while a minority indicated having a different one, without mentioning the level of aggregation of the registry (8 % of the registries). National registries have the advantage of obtaining information for an entry in an easier way, given that the search has only to be conducted once, instead of searches in, for instance, each regional registry. In addition, the existence of national registries means that the regional interconnection of registries can be skipped, since all entries are already centralised.

Figure 18: Scope of the registry (% of asset registries)



Source: Authors' compilation.

4.4 Assessment of the availability and accessibility of national beneficial ownership registries

The AMLD defines 'beneficial owner' as 'any natural person(s) who ultimately owns or controls the customer and/or the natural person(s) on whose behalf a transaction or activity is being conducted'. (Article 3(6) of (EU) 2015/849). For corporate entities, this means that any natural person who owns directly or indirectly a shareholding or ownership interest of more than 25 %,

will be identified as a beneficial owner (Article 3(6) of (EU) 2015/849). The directive also requires all Member States to establish central beneficial ownership registries with accurate and current information on beneficial owners of corporate or legal entities (Article 30(1-4) of (EU) 2015/849).

In this regard, the European Commission has set up a Beneficial Ownership Registers Interconnection System ('BORIS'), a decentralised system to interconnect the beneficial ownership registries. BORIS is being implemented and will serve as a central search service to make available information on beneficial owners in line with the mentioned AMLD, where Member States are required to interconnect the national beneficial ownership registries through the European Central Platform. The BORIS platform would enable searches of companies and other legal entities, trusts or similar arrangements through their national registration numbers or company registration numbers (if different). For competent authorities, this has the advantage of accessing up-to-date and accurate beneficial ownership information, which is of utmost relevance for detecting suspicious activities related to money laundering and tax evasion.

It is important to note that the AMLD requires corporate and legal entities to record adequate and accurate information on their beneficial ownership. This means that if the legal owner ([66]) of a certain asset is a legal entity or legal arrangement and it is recorded in a particular asset registry, competent authorities may be able to identify the beneficial owner by checking the beneficial ownership registry. However, this concept does not apply if the legal owner of an asset is a natural person.

As stated in the AMLD ([67]), FIUs and other competent authorities should be able to access in a timely and unrestricted manner the beneficial ownership information included in the central registry (Article 30 of (EU) 2015/849). FIUs and other competent authorities are able to access such information without previous notice to the concerned entity, in order to conduct their investigations, monitor and report suspicious transactions.

4.4.1 Findings based on survey results

Administering entities of asset registries were asked different questions regarding beneficial ownership information. They were first requested to indicate whether the asset registries are connected to other registries, both international and national, with the latter including beneficial ownership registries. The connection between asset registries (e.g. business registries) and beneficial ownership registries can be extremely valuable for potential AML purposes. In the case of the interconnection of beneficial ownership registries with business registries, this may enable, for instance, searches for an entity with immediately available financial information and its beneficial owners. Furthermore, this connection may also allow for searches of beneficial owners and a list of its beneficially owned entities.

The survey results regarding the interconnection of beneficial owner registries and asset registries show that most are not connected to other registries, to either international or national ones, including those on beneficial ownership (see Figure 19). This means that they are

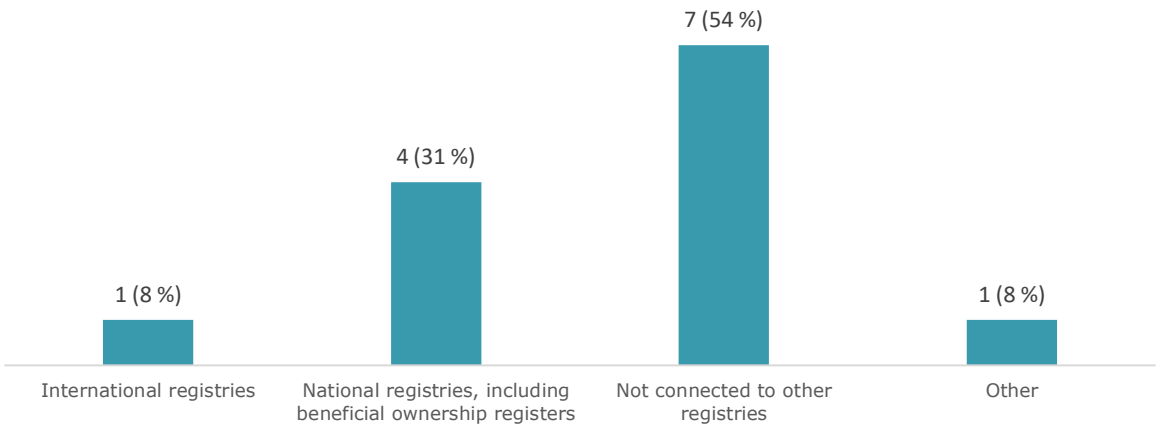
([66]) The legal owner of a company may be different from the beneficial owner who ultimately controls the company.

([67]) https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32015L0849&from=EN

standalone registries and are grouped in the 'not connected to other registries' category (7 or 54 % of the registries). About a third of the respondents indicated that the registries are connected to national registries (including beneficial ownership registries) (4 or 31 % of the registries). These asset registries cover vehicles (aircraft (1), ships (1), other motor vehicles (1)) and firearms (1). These registries indicated to be connected to other national registries such as population registries (motor vehicles and ships), registry of taxpayers (motor vehicles), legal entities registries (motor vehicles, ships and aircraft) and property seizure acts registries (motor vehicles and ships), among others. One aircraft registry holder indicated being connected to other national registries through a platform called 'Regix', used to interconnect several national registries (62 registries) in a central infrastructure in Bulgaria ([68]).

Notably, among of the 4 asset registries connected to other national registries, 3 were explicitly mentioned as being connected to legal entity registries of their countries. They receive from such registries information about the direct owner and beneficial owner, as the beneficial ownership registry represents in practice a component of the legal entity registries in these countries (BG, LT).

Figure 19: Asset registry connections to other registries (number of survey responses)



Note: All relevant assets considered, except for 'cash at bank', 'crypto assets', 'limited liability companies', 'real estate' and 'safe deposit boxes at banks'.

Source: Authors' compilation.

4.5 Accessibility assessment – Type of competent authority

Access conditions and the right to access an asset registry differ depending on legal competences conferred to an authority based on both its type and the Member State where it

([68]) https://e-gov.bg/wps/portal/agency-en/digital-government-infrastructure/information-systems-applications/regix

is situated. Importantly, it is up to each Member State to designate the competent authority as well as its responsibilities in combating money laundering and/or tax evasion.

For FIUs, the AMLD states that FIUs should have access to all administrative, financial and law enforcement information they need in order to conduct their tasks, such as carrying out suspicious transaction reports and cross-border investigations. Nevertheless, not all FIUs across Member States have access to the same sources of information ([69]), since FIUs are not fully harmonised in their functions and competences across the EU. The literature shows that the majority of FIUs in the EU follow an administrative model ([70]) (21 Member States), while the rest have adopted a hybrid ([71]) (5 Member States) or a judicial ([72]) model (1 Member State) ([73]). This means that FIUs function differently across the EU, and not all FIUs need to have access to the same sources of information to conduct their tasks.

At the EU level, the AMLD states that Member States need to ensure access to the central beneficial ownership registry of the competent authorities with designated responsibilities for combating money laundering or terrorist financing, tax authorities, supervisors of obliged entities and authorities with the functions of investigating and prosecuting money laundering, associated predicate offences and terrorist financing ([74]). Moreover, the AMLD establishes the direct and immediate access to centralised bank account registries for FIUs and competent authorities, although there is no explicit indication of who these competent authorities are. Directive 2019/1153/EU ([75]) on laying down rules facilitating the use of financial and other information for the prevention, detection, investigation or prosecution of certain criminal offences requires Member States to designate and empower competent authorities – such as tax authorities competent for prevention, detection, investigation or prosecution of criminal offences, LEAs and asset recovery offices – to access the information held in centralised bank account registries.

Similar to FIUs, the extent to which LEAs, tax authorities and AML supervisors of financial institutions are granted access to information depends on the powers and competences given to them by each Member State, meaning that access rights may differ across the competent authorities.

A survey analysis has been conducted across each of the authorities relevant in the context of money laundering and tax evasion (FIUs, LEAs, tax authorities, AML supervisors of financial institutions and asset registries) to assess the nature and conditions of access to the different registry data. The survey questionnaire circulated among the competent authorities allowed the collection of inputs with regard to the legal right of the competent authorities (FIUs, LEAs,

([69]) https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:52019DC0371&from=EN

([70]) Egmont Group defines the FIU administrative model as a 'centralised, independent, administrative authority, that received and processes financial sector information and transmits disclosures to judicial or law enforcement authorities for prosecution'. See https://egmontgroup.org/about/financial-intelligence-units/

([71]) Egmont Group defines a hybrid FIU as a model that 'serves as a disclosure intermediary and a link to both judicial and law enforcement authorities, combining at least two FIU models'. See https://egmontgroup.org/about/financial-intelligence-units/

([72]) Egmont Group defines FIU judicial model as one where an FIU 'is established within the judicial branch of government'. See https://egmontgroup.org/about/financial-intelligence-units/

([73]) EU FIUs Platform (2016), 'Mapping exercise of Gap Analysis on FIU's powers and obstacles for obtaining and exchanging information', 5-7, cited in Mouzakiti (2020), p. 354.

([74]) Article 30(6) of AMLD4.

([75]) https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX %3A32019L1153

tax authorities and AML supervisors of financial institutions) to access information on the assets and their owners in order to carry out their tasks (for instance, conducting investigations, litigation). In addition, the current access to data by the competent authorities is analysed. The findings of the survey analysis for each relevant stakeholder are presented in two parts. First, the analysis is focused on the competent authorities' responses. Then, the responses from the holders of asset registries are analysed.

4.5.1 Survey responses from competent authorities

Competent authorities (FIUs, LEAs, tax authorities and AML supervisors of financial institutions) were requested to indicate whether they have the legal right to access information on certain assets and their owners for the purpose of their tasks, as well as the way in which they access that information (through registries, alternative sources, etc.). Survey questions enquired about two types of access. First, the competent authorities were asked whether they are legally entitled to access the information on the asset and its owner. Second, the survey asked whether the competent authorities currently have means to access such information. Both aspects are assessed below for each group of competent authorities.

For the following questions in Task 2, assets were grouped into categories, i.e. vehicles include motor vehicles, ships and aircraft. Other high-value goods include works of art, jewellery, antiques, collectors' items and firearms. Safe deposit boxes include safe deposit boxes at banks and non-banks.

For FIUs, survey results show that more than half of the FIUs have the legal right to access information for investigation and, to a lesser extent, for both investigation and litigation ([76]) (see Table 21). As previously stated, the type of activities FIUs conduct differ among them, depending on their nature or status. In general, FIUs conduct financial investigations to detect suspicious transactions related to money laundering. For example, some FIUs do not carry out investigations or litigation but rather intelligence analyses. However, they still have the legal right to request information for this purpose. As a consequence, the exact powers of the FIUs depend on the statute granting these under national legislation, as does their access to asset information.

Furthermore, the assets for which most surveyed FIUs indicated having the legal right to access data include safe deposit boxes, cash in hand and high-value goods. For instance, safe deposit boxes at banks are accessed from centralised contact points, whereas for non-banks there are differences across countries on whether they are obliged entities under the AMLD. Some Member States can retrieve information on non-banks from obliged entities active in cash-in-transit and valuables safekeeping, whereas others cannot. At the other end, a significant number of the respondents indicated that they do not have the legal right to access

([76]) This falls under procedural law; litigation means filing a lawsuit. The initiation of legal proceedings before a court qualifies as 'litigation'.

in-game asset information, while for some FIUs some of the assets are outside their scope (e.g. cash in hand and other high-value goods).

In terms of legal rights, Member States have to ensure that FIUs have access either directly or indirectly to all financial, administrative and law enforcement information they need to perform their tasks, as stated in Article 32(4) of the AMLD ([77]). This means that FIUs should be able to obtain all necessary information they need to conduct their tasks related to suspicious activities from the obliged entities under this directive. The possibility to access certain data then depends on the specific functions and competences assigned to FIUs in different Member States.

Table 21: Legal right to access information on assets and their owners (FIUs)

Assets	FIUs				Number of observations
	Yes, for investigation and litigation	Yes, but only for investigation	No access	Not applicable	
Cash in hand	18 %	64 %	0 %	18 %	11
Unlisted securities	25 %	50 %	17 %	8 %	12
In-game assets	0 %	50 %	40 %	10 %	10
Vehicles	31 %	54 %	8 %	8 %	13
Other high-value goods	25 %	58 %	0 %	17 %	12
Commodities (precious raw materials)	9 %	55 %	18 %	18 %	11
Safe deposit boxes	25 %	67 %	0 %	8 %	12

Notes: The total number of responses for FIUs is 18. The total percentage per asset does not add up to 100 % due rounding. 'Not applicable' means outside the scope of the authority concerned.

Source: Authors' compilation.

The responses regarding the type of access to asset information and their owners provided by FIUs is shown below (see Table 22). In this case, more than two thirds of the FIUs responding to this question indicated having access to information on vehicles (motor vehicles, ships and aircraft) and their owners through a registry. As indicated in Table 18, in all Member States these types of assets are recorded in national or regional registries, managed by public authorities. Still, two FIUs reported they did not have any type of access to vehicle information, without giving further details.

There is certain asset information for which some FIUs indicated they did not currently have access through a registry or through an alternative source. Such information is related to in-

([77]) https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32015L0849&from=EN

game assets, unlisted securities and commodities. In this regard, there are no existing registries of these assets through which the authorities can access the information. Nevertheless, for unlisted securities, some FIUs noted they access the asset information through alternative sources, such as CSDs, the analytical database on financial instruments and the national financial services authority. Moreover, half of the FIUs who responded to this question indicated access to information about commodities (precious raw materials) through sources other than registries, such as from the obliged entities (e.g. traders in gold and diamonds), a register of economic activities and via OSINT, for instance, through shipping documents.

In addition, more than half of the FIUs indicated having some type of access to information on cash in hand, through registries or through an alternative source. In the first case, this mainly refers to information on financial transactions that involve the use of cash, and which are carried out through a financial institution. Access to information on cash in hand can also be obtained from the registry of declarations on the cross-border transportation of cash, as disclosed by certain respondents. Lastly, one FIU provided explanations about the alternative source used to access cash-in-hand information, which relates to obliged entities providing the information, while another indicated using an internal database which includes cash transactions for amounts of EUR 10,000 or more on a monthly basis, transmitted by financial entities.

Finally, some FIUs can access information on high-value goods through registries. This is, for instance, the case for firearms, whose registries are held by national authorities in charge of weapons registration (e.g. ministry of interior). In addition, two FIUs indicated that information on jewellery, works of art and antiques could be accessed upon request to the obliged entities (traders in diamonds, art galleries and auction houses).

Overall, most of the respondents expressed use of alternative sources different from registries to access to information on the assets and the owners, which is in most cases explained by the fact that no registries exist for certain assets.

Table 22: Type of access to information on assets and their owners (FIUs)

Assets	FIUs				Number of observations
	Through a register	Through an alternative source	No access	Not applicable	
Cash in hand	25 %	33 %	25 %	17 %	12
Unlisted securities	0 %	64 %	36 %	0 %	14
In-game assets	0 %	36 %	55 %	9 %	11
Vehicles	71 %	14 %	14 %	0 %	14
Other high-value goods	46 %	31 %	15 %	8 %	13
Commodities (precious raw materials)	0 %	50 %	33 %	17 %	12
Safe deposit boxes	57 %	36 %	7 %	0 %	14

Notes: The total percentage per asset does not add up to 100 % due to rounding. 'Not applicable' means outside the scope of the authority concerned.

Source: Authors' compilation.

Among tax authorities, the majority of respondents have the legal right to access data on vehicles (motor vehicles, ships and aircraft), upon request to the respective authorities, for both investigation and litigation ([78]) activities (see Table 23). A significant number of the tax authorities also indicated having the legal right to access information on unlisted securities for investigation and litigation. For instance, one of the authorities mentioned that obliged entities related to the trade of these assets (e.g. unlisted derivatives, financial instruments not settled by CSDs, and other financial instruments such as non-custodial and unlisted securities) have reporting obligations derived from the Tax Reporting Act. This means that if a reporting obligation exists, then the tax authority may have access to the report. Moreover, if there is a current oversight case in progress, tax authorities have greater scope to request information.

Regarding high-value goods, most tax authorities indicated having the legal right to access information for investigation and litigation, as well as to safe deposit boxes information. In the latter case, information is accessed upon request. It is important to bear in mind that safe deposit boxes at banks are subject to different legal requirements from safe deposit boxes at non-banks. In particular, under the 5th AMLD, the establishment of central registries identifying the holders of safe deposit boxes at banks applies to credit and financial institutions. However, there is no requirement for non-banks to create such registries. Another tax authority indicated having the legal right to access information on safe deposit boxes at banks for both investigation and litigation through the bank account registry in the EU. One tax authority conveyed having a legal right to access information on safe deposit boxes at banks solely for investigations, based on the reporting obligations that credit, payment and electronic money institutions have under the national Tax Administration Law.

Similar to other authorities, half of the tax authorities surveyed have no legal right to access information on in-game assets. Yet, a significant number of tax authorities also indicated having the legal right to access such information for both investigation and litigation purposes upon request.

Lastly, one tax authority pointed out that the legal right to access information on all assets and their owners when no asset registry exists is linked to tax recovery purposes. However, the authority still has the legal right to access all information that is reported from the debtors.

Table 23: Legal right to access information on assets and their owners (tax authorities)

Assets	Tax authorities				Number of observations
	Yes, for investigation and litigation	Yes, but only for investigation	No access	Not applicable	
Cash in hand	60 %	13 %	20 %	7 %	15
Unlisted securities	79 %	7 %	14 %	0 %	14

([78]) In general, tax authorities have the power to lodge a complaint which then leads to a judicial procedure for tax evasion or a tax fraud investigation. A tax authority can constitute itself as a 'civil party' to the case to substantiate its complaint during the judicial investigation.

In-game assets	40 %	0 %	50 %	10 %	10
Vehicles	81 %	13 %	6 %	0 %	16
Other high-value goods	75 %	0 %	25 %	0 %	12
Commodities (precious raw materials)	67 %	0 %	25 %	8 %	12
Safe deposit boxes	71 %	7 %	21 %	0 %	14

Notes: The total percentage per asset does not add up to 100 % due to rounding. 'Not applicable' means outside the scope of the authority concerned.

Source: Authors' compilation.

The type of access to information on the assets and the owner for tax authorities seems to be more limited compared with the other groups of competent authorities (see Table 24). The majority of the tax authorities responding reported not having any type of access to information on in-game assets, other high-value goods or commodities. For some assets such as firearms, access through alternative sources to firearms registries are upon request to the national registry held by national authorities. In the case of commodities, one tax authority mentioned that private companies dealing with these kinds of assets may be an alternative source to access information on precious raw materials.

In contrast, most tax authorities indicate having access to information on vehicles (motor vehicles, ships and aircraft) either through a registry or through an alternative source. Furthermore, almost half of the tax authorities have access to information on safe deposit boxes through alternative sources, such as through enquiries with banks and non-banks, as well as financial agencies.

Table 24: Type of access to information on assets and their owners (tax authorities)

Assets	Tax authorities				Number of responses
	Through a register	Through an alternative source	No access	Not applicable	
Cash in hand	20 %	33 %	40 %	7 %	15
Unlisted securities	21 %	29 %	50 %	0 %	14
In-game assets	0 %	10 %	90 %	0 %	10
Vehicles	56 %	31 %	13 %	0 %	16
Other high-value goods	9 %	18 %	73 %	0 %	11
Commodities (precious raw materials)	0 %	27 %	64 %	9 %	11
Safe deposit boxes	21 %	43 %	36 %	0 %	14

Notes: The total percentage per asset does not add up to 100 % due to rounding. 'Not applicable' means outside the scope of the authority concerned.

Source: Authors' compilation.

Turning to LEAs, half or more than half of the respondents indicated having the legal right for both investigation and litigation ([79]) to information on several assets (see Table 25). These assets include cash in hand, vehicles, other high-value goods and safe deposit boxes. For cash in hand, two LEAs expressed that data can be accessed through a request to the respective authorities. One LEA indicated having the legal right to access cash-in-hand information solely for investigations, and doing it through tax return declarations, which depend on individual initiative. Furthermore, all of the respondents reported having the legal right to access other high-value goods information, specifically firearms.

In contrast, none of the LEAs seem to have the legal right to access information on in-game assets (effectively on ownership of in-game items on the servers of video game companies offering these items) for investigation or litigation, while a significant number of respondents also revealed not having it for unlisted securities and commodities. In the case of in-game asset information, the possibility of identifying the asset owner as well as the monetary value that is linked to the asset is quite low given that their exchanges are rarely regulated (see Section 3.2.3.4).

Finally, one LEA disclosed having the legal right to access unlisted securities information through a request, while another LEA expressed having the legal right to requisition it if there are suspicions of criminal offences related to bearer securities.

Table 25: Legal right to access information on assets and their owners (LEAs)

Assets	LEAs			Number of responses
	Yes, for investigation and litigation	Yes, but only for investigation	No access	
Cash in hand	50 %	25 %	25 %	4
Unlisted securities	33 %	0 %	67 %	3
In-game assets	0 %	0 %	100 %	2
Vehicles	83 %	17 %	0 %	6
Other high-value goods	100 %	0 %	0 %	5
Commodities (precious raw materials)	33 %	0 %	67 %	3
Safe deposit boxes	67 %	33 %	0 %	6

Note: The total percentage per asset does not add up to 100 % due to rounding.

Source: Authors' compilation.

([79]) Law enforcement authorities can initiate litigation in relation to AML and tax evasion cases since judicial authorities are included as law enforcement agents (e.g. see Recital (11) of the Directive (EU) 2016/680 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data by competent authorities for the purposes of the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, and on the free movement of such data, and repealing Council Framework Decision 2008/977/JHA).

It is important to note that, as law enforcement mechanisms are determined by each Member State, the extent to which LEAs have access to information will depend on the national legislation of each country. Survey results suggest that most of the surveyed LEAs do not have any type of access to information on certain relevant assets and their owners (see Table 26). This applies to unlisted securities, in-game assets and commodities information, which is aligned with the previous results on the legal right to access the information on these assets.

However, the majority of LEAs have access to information on vehicles (motor vehicles, ships and aircraft) through the existing registries, while some have it through other sources. An example of alternative sources is the tax data managed by the revenue service authorities. Furthermore, more than half of the LEAs revealed they have access to information on cash in hand and safe deposit boxes through other sources different from registries. For cash in hand, this is through tax return declarations and making enquiries to the respective tax authorities. For safe deposit boxes at banks, this implies asking the financial institutions for information.

Table 26: Type of access to information on assets and their owners (LEAs)

Assets	LEAs			Number of observations
	Through a register	Through an alternative source	No access	
Cash in hand	0 %	60 %	40 %	5
Unlisted securities	0 %	33 %	67 %	3
In-game assets	0 %	0 %	100 %	2
Vehicles	67 %	33 %	0 %	6
Other high-value goods	60 %	20 %	20 %	5
Commodities (precious raw materials)	0 %	33 %	67 %	3
Safe deposit boxes	20 %	80 %	0 %	5

Notes: The total percentage per asset does not add up to 100 % due to rounding. 'Not applicable' means outside the scope of the authority concerned.

Source: Authors' compilation.

Responses from AML supervisors of financial institutions significantly differ from those of other groups of competent authorities. This is strictly related to the scope of activities and functions of these competent authorities (see Table 27). As supervisory authorities have different tasks such as carrying out financial inspections, most of these authorities conveyed that either they did not have access, or the question was not applicable to them.

Still, half of the AML supervisors of financial institutions indicated having the legal right to access unlisted securities information, if only for investigation activities, whereas about a tenth indicated having it for both investigation and litigation purposes. In addition, a third of these authorities have the legal right to access cash in hand information for investigations, which in many cases is achieved by requesting information from supervised entities.

Some of this group of competent authorities indicated having a legal right to access to information on vehicles for investigation and litigation, while around a third just for investigation purposes. This is also the case for safe deposit boxes at banks.

Table 27: Legal right to access information on assets and owners (AML supervisors of financial institutions)

Assets	AML supervisors of financial institutions				Number of observations
	Yes, for investigation and litigation	Yes, but only for investigation	No access	Not applicable	
Cash in hand	15 %	31 %	15 %	38 %	13
Unlisted securities	14 %	50 %	0 %	36 %	14
In-game assets	0 %	22 %	33 %	44 %	9
Vehicles	20 %	30 %	0 %	50 %	10
Other high-value goods	0 %	20 %	20 %	60 %	10
Commodities (precious raw materials)	0 %	25 %	13 %	63 %	8
Safe deposit boxes	23 %	31 %	8 %	38 %	13

Notes: The total percentage per asset does not add up to 100 % due to rounding. 'Not applicable' means outside the scope of the authority concerned.

Source: Authors' compilation.

Similar conclusions can be drawn with regard to the type of access to asset information (see Table 28). For most AML supervisors of financial institutions, having access to information on certain assets and their owners is not a requirement for fulfilling their tasks, which is to be expected due to the nature of their activities. Regarding their legal right to access the information, more than half of these authorities indicated having access to unlisted securities data through an alternative source. For instance, in countries like Slovenia, the CSD is mentioned as the alternative source to access information for unlisted securities and bearer securities. In other countries, like in Estonia, this information is extracted from intra-institutional registries. Elsewhere, other authorities indicated accessing information via requests to the supervised entities.

About a third of the AML supervisors of financial institutions expressed having current access to information on vehicles through registries while a tenth has it through an alternative source. These registries are in general managed by public authorities, such as the ministry of interior, maritime administration, transport agencies, etc.

Table 28: Type of access to information on assets and their owners (AML supervisors of financial institutions)

Assets	AML supervisors of financial institutions				Number of observations
	Through a register	Through an alternative source	No access	Not applicable	
Cash in hand	9 %	27 %	27 %	36 %	11
Unlisted securities	8 %	54 %	15 %	23 %	13
In-game assets	0 %	14 %	43 %	43 %	7
Vehicles	33 %	11 %	22 %	33 %	9
Other high-value goods	13 %	13 %	38 %	38 %	8

Commodities (precious raw materials)	0 %	17 %	33 %	50 %	6
Safe deposit boxes	18 %	45 %	18 %	18 %	11

Notes: The total percentage per asset does not add up to 100 % due to rounding. 'Not applicable' means outside the scope of the authority concerned.

Source: Authors' compilation.

4.5.2 Survey responses from registry holders

Asset registry holders were requested to indicate which national competent authorities (FIUs, LEAs or tax authorities) have access to the registries and to what extent the registries grant them access. In particular, some competent authorities may be granted full access, limited access or simply no access at all. Moreover, if some access is granted, it could be that it is given just to certain information and not to the complete information available in the registry.

Around half of the surveyed asset registries (6 registries) indicated that full access is granted to the national FIUs. A third of respondents (4 registries) indicated that the national FIU has limited access to asset registries. Limitations include conditionality of access, such as legitimate interest, or other requests that are linked to the performance of the FIU's direct functions. Nevertheless, the circumstances and conditions for there is legitimate interest are difficult to define ([80]). Also, as the status of FIUs differs across the EU Member States, so does their power in terms of accessing the asset registries, in accordance with the national laws. This means that a certain asset registry may be accessible for one FIU in a given Member State, while another FIU in a different Member State may not have the same access within its country. In the case of FIUs, it is essential that they receive quality data from the obliged entities in order to analyse and detect suspicious transactions related to money laundering. More specifically, access to relevant information, when appropriate, is of utmost importance for suspicious transaction reports and cross-border analysis ([81]).

Around half of the respondents indicated granting full access to national tax authorities (the same authorities as for granting FIUs access). More than a third of respondents (5 registries) disclosed that access is possible but with limitations. For instance, legitimate interest is considered a key requirement for granting access (3 registries) ([82]). One aircraft registry noted that as part of the registration process, every time an aircraft is registered, a notification is sent to the tax authority. The remainder indicated that access to the asset registries is not granted to national tax authorities, with one registry indicating that there is no right of access to data unless requested for a legal investigation, which applies to every competent authority.

Access given by asset registries to national LEAs is similar to the access granted to FIUs and tax authorities. A significant number of registry holders indicated that full access is given to LEAs (5 registries). But some expressed that LEAs only have limited access to their registries (4 registries). Two asset registry holders indicated that no access is granted to the national LEA, while another was not aware of whether the LEA has access to the asset registry. Finally,

([80]) https://curia.europa.eu/jcms/upload/docs/application/pdf/2022-11/cp220188en.pdf

([81]) European Commission (2019a).

([82]) See also https://curia.europa.eu/jcms/upload/docs/application/pdf/2022-11/cp220188en.pdf, and refer to Sections 7 and 8 on the legal feasibility assessment of the study.

one registry holder did not provide an answer, but the national LEA reported that it had access to the asset registry.

To access data, some national competent authorities can be required to fulfil certain requirements. These requirements vary across different types of registries.

Starting with tax authorities, in some cases they are required by asset registries to enter specific personal credentials in order to gain access to information (2 registries). Other registries provide information upon request (3 registries), while still others do it based on data supply agreements (2 registries). Yet, certain registries do not have any specific requirements in place for accessing the registry information, which coincides with public registries. However, limited data are available on the registry websites for public users. In some cases, e-IDs are required (1 registry). Finally, one asset registry body conveyed that in order to access vehicle data belonging to natural persons, the GDPR legal basis (Article 6) should be met, while there is no legal restriction on accessing data from legal entities.

As in the case of tax authorities, FIUs can be requested to enter information in order to access data in the registries. For certain respondents, this requirement refers to entering specific personal data. Other requirements can include the submission of a direct request or a signed agreement from the registry.

For AML supervisors of financial institutions, the requirements to access registries seem to be similar to those for tax authorities and FIUs. Still, as previously indicated by AML supervisors ([83]), in general, they do not have access to most of the relevant asset registries since it is not a prerequisite for performing their tasks.

4.5.3 Difficulties encountered by competent authorities when accessing asset registries while carrying out their respective tasks (inside the Member State and cross-border)

Access to information is essential for competent authorities in order to fulfil their functions in a sufficient and efficient manner. Previous sections have analysed the competent authorities' legal right to access the relevant asset information, through registries or alternative sources. The present section focuses on difficulties that competent authorities encounter while accessing asset registries while carrying out their respective tasks. The analysis is based on a survey of competent authorities, where the respondents were asked to indicate the difficulties that they currently face when they want to check different, relevant asset registries in their country as well as cross-border.

Notably, competent authorities' powers and legal capabilities differ when it comes to accessing information. Consequently, access is not envisaged for all types of competent authorities, but only for those required to have it under national law.

As illustrated by Table 29, most LEAs and tax authorities perceive accessing information as difficult due to the *existence of different registries* for different types of assets. On the cross-

([83]) See Section 3.2.5.1.

border aspects, for certain LEAs, accessing information from other Member States involves requests that should be made to the respective authorities, which usually take a considerable amount of time and entails inefficiencies in the collection of the necessary data for their investigations. In the absence of a centralised or interconnected asset registry at the EU level, the possibility for competent authorities to access other Member State registries depends on the legal rights conferred to them by national law, in line with the competences and powers of each authority.

In addition, certain FIUs expressed that they encounter obstacles when accessing the registries in their respective countries. This is mostly explained by the lack of single asset registries at the national level (or the existence of multiple registries), which generally applies to Member States that are constituted as federal states, like Germany. In this case, data are stored in a decentralised manner and under different regional authorities' management. For instance, in terms of asset data, real estate information is often stored in regional registries ([84]).

Communication issues with competent authorities, especially with FIUs from other Member States, are among the difficulties experienced by the respondents when accessing asset registries. Among the authorities, FIUs most often perceived this difficulty. On the one hand, for FIUs, communication with other Member State FIUs seems to be challenging due to the obstacles in obtaining all the required information they need to perform their tasks. This process is generally time-consuming and directly impacts on the effectiveness of their analysis, as well as on the dissemination of the results to other competent authorities (e.g. LEAs and tax authorities). AMLD4 recommends use of the FIU.net ([85]) system as the communication and exchange channel between FIUs to support cross-border cooperation in the fight against money laundering and tax evasion. On the other hand, the fact that FIU structures and powers differ across Member States makes it more difficult for some FIUs to share information with other FIUs, since not all of them have access to the same information sources ([86]). The survey results also found that the level of communication between AML supervisors of financial institutions and FIUs in other Member States is quite low or non-existent, which is to be expected due to the nature of the functions of these competent authorities.

Furthermore, a considerable number of authorities indicated encountering difficulties as a consequence of some *registries combining different data categories* (e.g. employment income and directors' fees). One FIU stated that this does not represent a significant issue at the national level, however, the opposite applies when considering cross-border availability and access to registries, which is also linked to the existence of multiple registries with different data categories.

Competent authorities also mentioned *technical obstacles to extracting data* from the current national registries. Tax authorities flagged issues such as *differences in the format used among the diverse asset registries*, which make it necessary to edit the data before using them for another purpose.

([84]) European Commission (2021b).

([85]) FIU.net is a sophisticated system that allows FIUs to exchange information between them.

([86]) Commission Staff Working Document on Improving Cooperation between EU Financial Intelligence Units, European Commission, Brussels, SWD(2017) 275 final.

Finally, even though many AML supervisors of financial institutions indicated encountering difficulties when accessing the information, the magnitude of the issue for these authorities is not as large as for FIUs, LEAs and tax authorities. That is because AML supervisors of financial institutions usually get information through direct requests to supervised entities, and they are not in charge of investigations with regard to private and legal persons.

Table 29: Difficulties encountered when checking asset registries (share of competent authorities)

Difficulties encountered	FIUs	Tax authorities	LEAs	AML supervisors of financial institutions
Difficulties in accessing information (e.g. separate registries for different types of assets)	36 %	48 %	33 %	67 %
Communication issues with FIUs from other Member States	44 %	17 %	31 %	17 %
Registries combining different categories of data	20 %	34 %	31 %	17 %

Notes: Question with multiple responses; the total number of responses per type of competent authority: FIUs (25), tax authorities (29), LEAs (12) and AML supervisors of financial institutions (12).

Source: Authors' compilation.

4.6 Accessibility assessment – Format of information

This sub-task assesses how information in the registries is presented, and whether the format of presentation varies depending on different levels of access. Using the list of relevant assets defined in Task 1, the assessment is conducted based on the information collected from the surveys run across the asset registries.

Survey responses indicated that some competent authorities may have access to only part of the information contained in a registry, and others may have access to, for instance, a complete searchable portal displaying all information. Access to a searchable portal displaying information is especially relevant for competent authorities (particularly FIUs), as it increases the flexibility with which entities can be investigated and identified. As previously analysed, the level of access to asset registries differs per competent authority. Moreover, it also differs when considering the access given to national and other Member State authorities. In this regard, three asset registries indicated that there should be a legal basis for requesting access to information for both national and non-national authorities, without specifying whether the conditions differ. Nevertheless, the majority of the respondents indicated that there are no differences (4 registries) or they were not aware of differences in the access conditions for national and non-national authorities (2 registries).

With regard to the format of information, some asset registry holders indicated that the data available in the registry is mostly in batches of data (5 registries), while a minority had data in streaming (2 registries). Some stated that data are entered manually (e.g. by an inspector) or uploaded via data files. For digital registries, their size depends on the type of data. For instance, some aircraft registries can reach around 5 000 MB, while others indicated the size by number of pages (about 10 000 pages). As to the number of records, the responses ranged from 3 000 for certain aircraft registries, to more than 8 million records (including historical data) for vehicles like cars and motorcycles.

Around half of the asset registry holders surveyed stated that the data entries are not protected through encryption of the storage support, while the others either protect the entries with encryption or are not aware of it. Other methods for data protection include regular backups, saving historical changes.

The way in which users can access information is primarily through human access via a user interface, with some registries providing an automated extract from the registry and others manually listing specific information. The format can be either physical or digital. A small number of registries also provide programmatic access. The asset registries' management of this access is mainly through a centralised identity and access management solutions, as well as through online requests.

Preventing, identifying, detecting and investigating money laundering and tax evasion threats requires gathering personal data. The AMLD states that Regulation 2016/679 (GDPR) ([87]) applies to the processing of personal data. Given the differences in the nature and powers of FIUs across the EU, the data protection laws applying to them differ. Some FIUs apply Regulation 2016/679 as the basis to process personal data, while others apply Directive 2016/680 ([88]) (Law Enforcement Directive). This divergence has an impact on the access that FIUs have to information, as well on how FIUs process personal information.

Public access to relevant asset registries may be important when competent authorities need to rely on it to obtain information. In this regard, Article 86 of the GDPR states that access to public information and its subsequent processing needs to be done in accordance with EU or Member State law to which the authority is subject to reconcile public access to official information with the right to protection of personal data. Under EU law, if the information is publicly accessible, it should be 'limited, clearly and exhaustively defined, and must be of a general nature, so as to minimise the potential prejudice' (case C-37/20) to asset owners. So, if competent authorities rely on public access (e.g. information that is available to the public at large) the data available should be very superficial, otherwise it is likely to amount to an unlawful infringement of the fundamental rights to data protection as enshrined in the GDPR (Article 5 in particular). In addition, where registries or alternative sources of information are private (e.g. administered by a private entity, as is the case for example for listed securities or commodities), registries are still bound by the GDPR requirements and the data they process will have to be limited to what is necessary for the relevant purpose of the private database.

([87]) https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32016R0679&from=EN

([88]) https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX %3A32016L0680

Public access to data depends on the type of asset registry, the Member State where the registry is established and whether the data are not in conflict with the data protection rules. For instance, some ship registries are open to the public in certain countries (e.g. LU), while not in others (e.g. LT). The same applies to some aircraft registries. For other motor vehicles (cars, motorcycles, etc.), some registries may allow the general public to access certain information, but for some specific data an e-ID is required.

5. Task 3: Mapping of other potential asset registries

5.1 Methodology

Task 3 naturally follows the two previous tasks, as it assesses other assets and sources of wealth whose asset ownership information cannot be retrieved through the existing registries or other sources of information. It also evaluates which new data collection obligations would be needed in order to potentially establish an asset registry and evaluate the current possibilities to access other sources of information on asset ownership and their suitability to fight financial crime. As part of this task, interviews were carried out with ESMA and the relevant authorities in five Member States – Czechia, Germany, France, Latvia and Romania. The authorities in these countries represent assets for which national authorities naturally have established a registry and administer it as well (see Table 18). This would include the broad groups of vehicles (motor vehicles, aircraft, boats, etc.) and potentially firearms. The countries were selected from the preliminary list of the sample based on geographical coverage and quantitative ratings (see Annex A). The degree of digitalisation of the databases was also taken into account for the final selection. As a result, the selected five countries represent both the small and large, as well as new and old EU Member States, with a low as well as high degree of digitalisation of the databases.

In addition, interviews were conducted with the relevant EU-level professional associations involved in AML or tax evasion. These associations represent members involved with assets for which no asset registries exist so far, or whose asset records are only kept at a single company level, such as the European Fund and Asset Management Association, the International Association of Dealers in Ancient Art and CINOA. An overview of the interviewed associations as well as the authorities managing the registries can be found in Annexes D and E.

This task has been divided into two main sub-tasks, as presented below:

- Sub-task 3.1 – analysis of how best to retrieve financial or non-financial information,
- Sub-task 3.2 – gap analysis for the interconnection.

5.2 Analysis of how best to retrieve financial or non-financial information

For each type of asset, the corresponding sources of data have been analysed and mapped in an analytical grid, both of which were identified in Task 1. The identified sources of information for these data may already be in existing registries at a national or regional level, or registries or other sources of information administered and held by private companies.

Yet, for a group of assets no registries may be identified. For asset classes for which no official registry is identified, or registries that are only maintained and administered by private companies, interviews with representatives of EU-level professional associations related to

these assets and sources of wealth were conducted, as noted above. In these cases, an assessment was made of whether there are possibilities to standardise or interconnect existing privately-held registries and furthermore if there are any alternative available data sources which could be used for the purposes of fighting money laundering and tax evasion.

Table 14 and Table 15 present the mapped registries according to the analytical grid.

5.3 Gap analysis

For each of the selected countries and selected registries, an analytical grid was established to illustrate the requirements which should be in place to facilitate either an interconnection or centralisation of the asset registry in question (see Table 30). The requirements do not represent a full or exhaustive list of requirements, but merely the core basics that need to be in place, on top of which further harmonisation and centralisation could be built upon solely for already established, national authority-administered registries. This analysis serves to highlight the overall situation in the country in question related to general digitalisation and interconnection, as well as potential measures for improvement with the eventual aim of achieving EU-wide interconnection, as well as aiding the work of authorities dealing with money laundering and tax evasion.

The most crucial requirements revolve around the registry already being in the format of a digitally structured database, as otherwise no simple and automated exchange of data can be facilitated. Likewise, the usefulness for relevant authorities would be greatly diminished if they could not use, for example, more sophisticated search functions that are found in database implementation, as opposed to simple Excel files. Furthermore, a database already implies a level of technical sophistication related to the entry and exchange of data.

The ability to uniquely identify the owned asset and the corresponding asset owner in a cross-border context is key. National unique alphanumerical identifiers such as tax IDs are of great use. Ideally, the assets are listed with either an extensive description or a standardised identifier such as the VIN number for vehicles.

Extensive data documentation and manuals are important preconditions for any potential data-standardising or harmonisation exercise. If registries are clearly documented and their contents and structure well defined, spotting differences – especially in terms of semantics and particularities of national definitions of datapoints across registries – becomes much easier. This plays a large part in potentially creating an EU-wide registry.

Table 30: List of requirements for interconnection

Requirement
Digital structured database
Uniform unique identifier asset
Uniform unique identifier of an asset owner/owner information
Data quality and security measures
Data documentation
Interconnectedness of registries

Source: Authors' elaboration.

Furthermore, there should already be best practices surrounding data security and backups in place, as well as procedures to ensure data quality. Data need to be high quality and stored securely in order to ensure their usefulness and public trust in any analysis conducted with the data in question.

Finally, an existing structure of interconnected databases in a country lessens the amount of effort required to connect the registry at an EU level, as the best practices employed and learned during the national interconnection process are in place. Additionally, this translates into less effort in general as it means that fewer registries need to be interconnected in total, as the national interconnection has already been achieved. This also stretches to issues surrounding the harmonisation of datapoints.

5.3.1 Findings based on survey responses and interviews with registry administering authorities

5.3.1.1 Romania – Ships

Requirement	Current set-up	Gap
Digital structured database	Digitally structured database	No gap
Uniform unique identifier asset	Vessel name, engine unique identification number exists, in both national and international standards	No gap
Uniform unique identifier asset owner/owner information	Indicators specifying the legal owner, including natural persons, legal persons, and legal arrangements; unique person and vessel identification number at a national and international level	No way of automatically retrieving the information from other national registries
Data documentation	In-house data catalogues are used	No gap
Data quality and security measures	Data encryption of the data entries' storage support is used, and data protection solutions were implemented; the origin of each entry is documented; Technical aspects of data are checked on a monthly basis; regular backups are made while access control, firewall, encryption, physical software, regular software updates and monitoring are employed as security measures	No gap
Interconnectedness of registries	Data are available upon request from relevant authorities; no national or international interconnection with other registries	No centralised data platform or data interconnection

Source: Authors' elaboration.

The level of interconnection of data presently held in registries in Romania is fairly low. Romanian authorities mention that they are currently conducting a feasibility study on connecting national registries, focusing on a connection to the business registry. This holds true for the naval registry as well.

In terms of the data stored, the registry provides the necessary information to link it to the business registry and to uniquely identify both assets and owners.

The Romanian authorities need to file separate information requests for each registry they would like to retrieve data from. Processing these requests takes time, often around a month. Authorities note that the added time is especially detrimental to their investigations, as the investigations in terms of money laundering in particular are very dependent on speed. This is simply due to the fact that potential criminals may take steps to flee the country or move the assets in question if they suspect any investigation of them or in general to simply cover their tracks. At least from an efficiency perspective, the interviewed Romanian authorities would appreciate an interconnection at both the national and international levels.

5.3.1.2 Latvia – Motor vehicles

Requirement	Current set-up	Gap
Digital structured database	Digitally structured database	No gap
Uniform unique identifier asset	Vehicle registration numbers that are unique are collected	No gap
Uniform unique identifier asset owner/owner information	Registration numbers on legal entities are collected as well as technical information on the vehicles	No gap
Data documentation	Data are well documented; the procedures connected to the use of national registries are written in government legislation	No gap
Data quality and security measures	Daily updates of data points with error checks and backups being conducted regularly; security measures are taken to ensure the data are protected	No gap
Interconnectedness of registries	Data exchange infrastructure is in place to allow for direct access by competent authorities to interconnected registries (particularly the tax authority and police); There is interconnection among registries such as the registry on legal entities and beneficial ownership registries where access is direct; Part of EUCARIS services on CBE and VAT information exchange	No gap

Source: Authors' elaboration.

In this case, all the data related to the aforementioned assets are kept in the same registry. Latvia exhibits a generally high degree of digitalisation within its registries. All relevant registries are already interconnected at a national level. This means that all relevant information stemming from company or beneficial ownership registries is already exchanged across all registries, such as the vehicle registry.

Hence, uniquely identifiable ownership information is readily available for all vehicles registered in Latvia.

Latvia furthermore participates in the EUCARIS CBE (cross-border exchange) services. This allows other Member States to request vehicle-holder/owner information easily and quickly. It is important to note that under the CBE service, this is only possible while investigating traffic

offences and not money laundering or tax evasion ([89]). Similarly, EUCARIS VAT services allow the transmission of such information, including vehicle holders across the lifetime of the vehicle. This technically allows the tracing and change of ownership information, with the possibility of identifying suspicious changes, although again only for VAT-related purposes as set out in the relevant legislation ([90]). Moreover, access to these kinds of data are possible for LEAs through specific means not using the EUCARIS system. This system just presents an added value in terms of speed of access to harmonised information. Monetary figures are not attached to the changes of ownership. In all, Latvia already exhibits the infrastructure and the means to further interconnect its vehicle (and possibly other registries) on an EU level.

5.3.1.3 Germany – Motor vehicles

Requirement	Current set-up	Gap
Digital structured database	**Digitally structured database**	**No gap**
Uniform unique identifier asset	Vehicle registration numbers that are unique are collected	No gap
Uniform unique identifier asset owner/owner information	**In Germany only the asset holder is recorded, not the owner**	**No asset ownership information**
Data documentation	Data are well documented	No gap
Data quality and security measures	**Continuous data updates whenever a change of the underlying data occurs with error checks and backups being conducted regularly; security measures are taken to ensure the data are protected**	**No gap**
Interconnectedness of registries	Vehicle registry is not connected to other registries; regional authorities do not have their own registries, but rather possess the ability to add and edit new observations to the national registries; Part of EUCARIS services on CBE and VAT information exchange, though no ownership information is exchanged, as it is not stored in the registries	No interconnection or exchange of ownership information with other national and international registries

Source: Authors' elaboration.

In terms of the vehicle database, which includes all assets mentioned above in a single database, Germany's core limitations are that according to national law, only information on the vehicle holder is stored. This does not cover the legal aspect of ownership or transfer of ownership; no information on the owner of the car is noted. Thus, ownership chains or transfer of ownership cannot be established in Germany and are not recorded. The authorities in particular note that they faced the same issue when trying to seize assets owned by sanctioned

([89]) Directive (EU) 2015/413.

([90]) Council Regulation (EU) 2018/1541.

entities in the wake of recent sanctions levied against Russia. As the two concepts are different, and data are collected only for the holder of the vehicle, no assessment based on ownership can be made. This relates to the person possessing the right to operate the vehicle, as well as bearing responsibility for any crime committed while operating the vehicle. This is legally distinct from the person or legal entity possessing ownership of the vehicle, which is not recorded anywhere in the national registry.

Information on holders alone is of limited use for AML or tax evasion purposes.

This limitation of the national laws documenting vehicle registration in Germany seems easily the biggest obstacle to the interconnection and exchange of information on asset ownership.

5.3.1.4 France – Aircraft

Requirement	Current set-up	Gap
Digital structured database	**Digital structured database**	**No full digitalisation**
Uniform unique identifier asset	Unique airplane number exists	No gap
Uniform unique identifier asset owner/owner information	**Sufficient personal information on owner is recorded**	**No gap**
Data documentation	No in-house data documentation	No documentation
Data quality and security measures	**The validity of the data is checked through verification of original documents provided; data are protected by security measures**	**No external validation of the data**
Interconnectedness of registries	Registry is not connected to other registries	No interconnection or exchange of information for validation or investigative purposes

Source: Authors' elaboration.

The airplane registry in France faces multiple obstacles to being interconnected on either a national or international level.

The core gap is that not all information is fully digitised, thus making a fully automated exchange of information impossible. Furthermore, its stand-alone nature prohibits automated data-validation measures such as automated checks and changes of personal information stored in the registry.

As a consequence, further interconnection of the French airplane registry will require considerable efforts in digitalisation, standardisation and additional validation.

5.3.1.5 Czechia – Motor vehicles, ships

Requirement	Current set-up	Gap
Digital structured database	Digitally structured databases	No gap

Uniform unique identifier asset	Unique vehicle registration as well as boat registration numbers are collected	No gap
Uniform unique identifier asset owner/owner information	Uniquely identifiable ownership indicators are collected for both assets in addition to historical ownership	No gap
Data documentation	Data are well documented according to national guidelines	No gap
Data quality and security measures	Daily updates of data points with error checks and backups being conducted regularly; security measures are taken to ensure the data are protected	No gap
Interconnectedness of registries	Registries are connected to the registry of legal entities, such that when an update occurs in that registry (e.g. change of address) it is automatically reflected in the vehicle/boat registries; Vehicle registry is part of EUCARIS services on CBE and VAT information exchange	No gap

Source: Authors' elaboration.

In this case, every mentioned asset apart from ships are kept in the same registry, with a separate registry for ships. Czechia in general already exhibits a high degree of interconnection with the link to the legal person registry, which both registries use to update personal information on previously registered owners.

Similarly, EUCARIS integration allows for an easy and efficient cross-border transfer of car ownership data, of which historical records are kept.

5.3.2 Conclusions

Overall, the gaps for interconnection of asset registries differ quite a lot across the countries as well as the assets considered in the case studies.

Concerning vehicles, all the interviewed authorities indicated great satisfaction with the present EUCARIS system for the cross-border exchange of data. All EU Member States already subscribe to the EUCARIS system. There remain a few caveats, which relate to a legal specificity in Germany where only car holders and not owners are considered. Thus, at least from a technical and operational point of view, an interconnection and exchange of data are already well established as functional. Other sections in this study will further elaborate on the legal aspects of this system and its use for the purposes of the fight against money laundering and tax evasion.

The state of aircraft and ship registries differs considerably across countries. While some countries like France show considerable gaps in terms of digitalisation, other countries such as Czechia are much closer to achieving the necessary prerequisites. The gaps in digitalisation can in some cases be explained by the comparatively small size of the registries in question, as boating and plane registries in some countries only store a small number of observations. Hence, it may not be deemed economical in these countries to consider a wider degree of interconnection or digitalisation.

Box 1: Summary of inventory of different assets and sources of wealth, mapping existing and potential registries (Tasks 1-3)

The objective of this study was to assess the feasibility of providing a unique access point for competent authorities involved in the fight against money laundering and tax evasion to relevant asset registers in the EU Member States. The study first defined the relevant assets for money laundering and tax evasion, then mapped existing registries, identified gaps, and provided a feasibility assessment of three potential scenarios from the perspectives of the legal, IT and cost and benefit analysis:

Scenario 1: Creation of new asset registries/databases at national level, in addition to the already existing ones, and their interconnection at EU level to form an EU asset registry.

Scenario 2: Interconnection of only the existing national asset registers required under EU law (Beneficial ownership, bank account) and real estate registers, entailing standardisation of queries at the EU level by obtaining information directly from national registries.

Scenario 3: Establishment of a centralised EU asset registry, implying the creation of a new database with national access portals, where data is stored across the EU.

The starting point for the analysis was to conduct a review of the literature, legal documents, and relevant stakeholders' reports, based on which more than twenty assets were identified for potential inclusion in a single EU asset registry. Such assets were analysed regarding their relevance, entailing legal enforcement and recovery in the context of money laundering and tax evasion, classifying them into high, moderate, and low relevance categories. Based on this classification, fifteen assets were considered for inclusion in the single EU asset registry. To further validate the relevance of these fifteen assets, surveys were distributed among Financial Intelligence Units (FIUs), Law Enforcement Authorities (LEAs), Tax Authorities, and Anti-Money Laundering supervisors of financial institutions. Based on the literature review and survey findings, the final list of relevant assets was compiled, including cash at bank and in hand, financial assets such as listed and unlisted securities, Limited liability companies (LLCs), crypto assets, in-game assets, real estate, vehicles, high-value goods, commodities, and safe deposit boxes at banks and at non-banks (Task 1).

Once the list of relevant assets was defined, the existing asset registries and other sources of information were mapped and assessed. Through a focused desk research, 215 registries and alternative sources of information were identified in the EU. The objective of this task was to assess the stored data, the accessibility of competent authorities to the registries, their digitalisation and the level of interconnection with other registries. To this end, surveys were sent to public authorities responsible for administering asset registries. Results suggested that most registries use digitally structured databases, which are generally regularly updated. In addition, around half of the respondents indicated using the standard UTF as the encoding method and recording the identity and type of legal owner of the asset together with a variety of other indicators. The study also found a lack of interconnection between the surveyed asset registries and other registries (both at the national and EU levels). Furthermore, results from the surveys sent to competent authorities (FIUs, LEAs, tax authorities and AML supervisors of financial institutions) evidenced difficulties when searching or accessing the information on the asset and its owner, both in their country and cross-border. This creates inefficiencies in accessing and collecting the necessary data, which is essential for authorities to perform their investigative functions.

As a next step, the study has identified organisational or technical gaps hindering an option of EU-wide centralisation or harmonisation of information held in the national registries. Within this task, interviews were carried out with ESMA (European Securities and Markets Authority) and the relevant authorities in five Member States representing both the small and large, new and old EU Member States, as well as the low and high degree of digitalisation of the databases. The analysis illustrated that existing gaps are quite heterogeneous across the countries as well as across the assets considered for the gap analysis. For instance, in the case of aircraft and ship registers, some countries, like France, showed considerable gaps in terms of digitalisation. In contrast, other countries such as Czechia are much closer to achieving the necessary prerequisites. In the case of vehicles, all interviewed authorities indicated great satisfaction with the already existing EUCARIS system for the cross-border exchange of data, which was found to facilitate data exchange across the EU member states for the fight against money laundering and tax evasion. For assets for which no asset registries exist so far or whose asset records are only kept at single company level, interviews were conducted with the relevant EU-level professional associations involved in AML or tax evasion, such as the European Fund and Asset Management Association, the International Association of Dealers in Ancient art and CINOA. Interviews suggested that for some classes of assets (e.g. listed securities), measures, systems, and harmonisation are already in place, and therefore, it is feasible to create interconnected or centralised registries. However, no such data exist for other assets, such as in-game or cash (Task 3).

6. Task 4: Operational/IT feasibility assessment and cost-benefit analysis

6.1 Introduction

This section begins with an operational assessment, the first part of which lays out and considers general operational issues relating to the current landscape of registries as well as possible approaches to overcome them. The second part of that operational assessment deals with each relevant asset in terms of the operational feasibility of either creating a new registry where one does not exist for a given source, or the requirements existing registries need to fulfil in order to be interconnected at the EU level. This analysis is valid for multiple scenarios and therefore is conducted in that separate section.

Following this, the goal of the IT assessment is to the derive the necessary architectural features any interconnected/centralised EU asset registry should possess to achieve its core goal (fighting against money laundering and tax evasion). This assessment begins with setting out the necessary technical functions any interconnected/centralised EU asset registry should have. In light of this, existing systems of interconnection, centralisation and networks of exchange of asset information are analysed to determine best practices and the technical necessities for an EU-wide asset registry.

Using the requirements derived from these two analyses (operational and IT), four scenarios describing concrete architectural set-ups for EU asset registries are developed. The advantages and disadvantages of each scenario are analysed in terms of how well they fulfil the previously laid out operational and IT requirements and overcome the challenges. Then, a cost-benefit analysis compares each scenario from the perspective of what is needed to establish each option.

6.2 Operational assessment

As explained, the operational assessment is split into two parts. The first part broadly identifies operational challenges that need to be overcome when considering any type of interconnection of registries across Member States. The needs and concerns expressed by the surveyed authorities in Task 2 are then categorised. Several approaches for addressing their expressed difficulties are analysed in terms of how well they can resolve the indicated issues.

The second part of the operational assessment deals with the operational feasibility of building new registries for assets where there are none. It also assesses existing registries in terms of their ability to contribute to the goals of an EU asset registry, in addition to laying out operational hurdles in interconnecting them at an EU level. This assessment informs the selection of assets considered for the scenarios throughout the remainder of the study.

6.2.1 Assessment of challenges that might affect implementation

6.2.1.1 Data quality

Having data that is of high enough quality is crucial for any registry. A realistic operational assessment must start from the data to be entered into the registries, before they are further harmonised or interconnected in a cross-border context. Therefore, issues stemming from inexistent, partial, inaccurate or outdated data are important to consider.

Information in the case of other significant assets (e.g. crypto assets, in-game assets or unlisted securities) in Tasks 1 and 3 has shown that significant limitations (the speed of transactions and technical developments, and the lack of regulations) do limit the actual capacity to capture or properly use the necessary information (e.g. the situation of real estate registries in France, Germany and Italy).

Additionally, it is important to note that data quality issues have surrounded the implementation of current EU-wide interconnection efforts. For the DAC in particular, the European Court of Auditors highlights the issues with data quality in a report on the functioning of the DAC ([91]). Crucially, it found a lack of auditing at the Member State level of reporting entities, leading to potentially low-quality data being exchanged in the system. In particular, incomplete information on unique identifiers (in this case the TIN) was noted, as was the issue of record duplication. Similar points, notably the lack of Member State quality assurance, were raised in the 2018 evaluation of the DAC implementation. Furthermore, snags with the timeliness of the data were highlighted. Both reports recommend an exchange of feedback among sending and receiving Member States, as well as a more harmonised approach to data quality assurance.

Such issues persist for beneficial ownership data as well, for which the registries are due to be interconnected via BORIS. A report by Transparency International France and the Anti-Corruption Data Collective analysed the quality of data in the French beneficial ownership registries and found significant data quality issues ([92]). The issues mostly boil down to a large

([91]) https://www.eca.europa.eu/lists/ecadocuments/sr21_03/sr_exchange_tax_inform_en.pdf

([92]) https://www.aefinfo.fr/assets/medias/documents/5/3/533043.pdf

degree of non-reporting by companies obliged to indicate their beneficial owner (31 % of legal entities in France). Of particular concern is that especially entities with unknown or undeclared beneficial owners seem to own a worryingly large share of real estate, thus presenting a serious obstacle in determining asset owners. The lack of verification and enforcement of existing beneficial ownership rules seem to present a real barrier in ensuring data quality at the Member State level, which would carry over if such data are shared via an EU-wide interconnected system.

6.2.1.2 Lack of data harmonisation

Another large operational challenge that needs to be overcome in order to interconnect registry data at the EU level arises from the diversity of existing asset registries across Member States and corresponding data structures that will have to be harmonised.

As illustrated by the gap analysis in Section 5.3 (and previous sections of the study) this diversity manifests itself in how the asset registries or other sources of information at the Member State level differ, in terms of:

- data structure and dynamics (due to market needs, domain specificity, regulatory requirements, traditions, etc.);
- differing working frameworks (national/local, state/private, regulated/unregulated); or
- existence or non-existence.

These differences are indicated in the following types of challenges:

- **Legislative.** Differences in the legal frameworks of the Member States relate to (i) laws on competences/access rights of competent authorities and (ii) laws on the obligation to declare asset information across Member States (see Section 4.5). Small differences in either of these elements can make the harmonisation, if not necessarily impossible, harder and costlier to achieve.
- **Technical.** Discrepancies between the data architecture of assets across the Member States relate to structural differences in how properties of the records associated with particular assets are stored by registries in different Member States.

6.2.1.3 Inconsistencies

Another group of issues arises from operational inconsistencies within and across registries:

- **Inconsistent data in the same registry.** Legislative and market developments that necessitate a change in the format of certain indicators in a registry (such as switching to a more precise representation of the time and date of an indicator) cause issues, as older entries (with an older format) live alongside the entries with a new format in the same registry. As a result, there are two different formats for the same indicator, potentially complicating data harmonisation.
- **Different update frequencies of registries for the same assets across Member States.** A timely update frequency (especially for assets with frequent ownership changes) is key for reliably identifying the current asset owner at any one time. Different registries for the same assets across Member States with varying update frequencies therefore lead to inconsistent data, which needs to be remedied in order to ensure harmonisation.

6.2.2 Operational feasibility of creating new registries or leveraging existing ones

To assess the operational obstacles that stand in the way of either creating new registries or interconnecting existing registries, several aspects of the current state of the data landscape surrounding a given asset need to be evaluated. For this purpose, first the evaluation criteria are explained. This is followed by an analysis of the operational feasibility of establishing new registries for assets where there are none as well as the feasibility of interconnecting existing registries. The study then examines how well they can serve the goals of speeding up access for competent authorities and facilitating cross-border cooperation.

To create new asset registries, it is necessary to consider several aspects, chiefly the presence or absence of reporting obligations as an important precondition for generating any data in the first place. Additionally, a reporting infrastructure needs to be created, allowing for reporting to occur from any reporting entities, as well as data to be stored, evaluated and analysed for quality and for ensuring complete and comprehensive information. Furthermore, consideration needs to be given to the strict enforcement of obligations to produce up-to-date and high-quality data.

Another aspect is mostly tied to the nature of the assets themselves and relates to the operational feasibility of collecting data which are complete, accurate and current. Complete data are necessary especially to fully establish ownership of assets, as well as any changes of ownership. Naturally, information on every asset should be recorded, meaning every parcel, every car in use, etc. Related to this is either the value of an asset, or alternatively, enough detailed information on an asset to potentially derive its value. Accurate data are necessary to reliably identify the correct owners of an asset, which might be obscured through a series of legal persons like private LLPs or legal arrangements like trusts. This highlights the need for current data, which are crucial for assets where ownership changes at a rapid pace. Having up-to-date data are key for capturing ownership at any given time, especially in cases where seizures of assets are required for tax recovery purposes.

In this analysis, it is deemed necessary for an asset registry to contain complete and comprehensive information on both the owners and the value of every asset under consideration. This necessarily implies that the registry contains data on the stock of ownership rather than the transfer or transaction of ownership. The operational assessment therefore focuses on whether building such a registry is indeed feasible. It further highlights cases where building a fully-fledged registry is not feasible, but a more centralised collection of transaction data might be, which could also be of use in the fight against money laundering and tax evasion. Such an alternative, however, should not be considered a registry within the context of this study and hence is not assessed based on its costs and benefits or IT feasibility. Rather it is identified as a potential mitigating measure (particularly in the legal feasibility assessment of this study) or a possibility for further research into how and to what extent these data can be of use in combating financial crime.

6.2.2.1 Operational feasibility of creating new registries

This section reviews the assets from the list of relevant assets (presented earlier in Table 13, in Section 3.3.4) for which registries do not exist. This includes high-value goods and commodities, listed and unlisted securities, cash in hand, in-game assets and safe deposit boxes at non-banks. The assessment entails five steps.

In a first step, a hypothetical asset registry and the data it would contain to be effective in combating money laundering and tax evasion are **described**. Step two dives into the **concrete operational requirements** for constructing such a registry. Step three outlines any **existing data on the asset** in question and information on ownership and value, which have partly been gathered in Task 3. Step four outlines the **gap** between the existing information and the state of data needed to construct a comprehensive and complete registry. Finally, step five **concludes** whether achieving this state is operationally feasible, and if not, points out the extent to which existing sources of information may be leveraged as an effective tool to combat money laundering and tax evasion.

6.2.2.1.1 Cash in hand

Description of required data for an asset registry

To construct a complete and comprehensive asset registry, a cash registry would need to track all physical cash ownership (or at the least ownership of large sums of cash) of all natural and legal persons in the EU at all times. This would also entail tracking all changes in ownership of large amounts of cash.

Operational requirements to build an asset registry

Effectively, all natural and legal persons in the EU would have to report the current amount of cash ownership after every change in the amount of cash owned, be that after purchases, withdrawals or other events changing money in hand. Therefore, a reporting mechanism would have to be built in which every cash owner would have to report the amount of physical money owned at any point in time, which would entail an unprecedented effort in terms of reporting in both frequency of change and scope. Ensuring that data and reporting are accurate, complete and timely is extremely unlikely due to the fact that cash can easily be transported and exchanged at high frequency and moved across borders.

Existing data on the asset

The only potential source from which information on cross-border cash movements could be gathered is customs declarations, as laid out in the Cash Controls Regulation (2018/1672/EU). When detected and declared, this could provide a source of data on ownership of cash. That said, the Schengen agreement and the EU's open internal borders make illicitly transporting cash very feasible with little danger of being caught and therefore such records are highly likely to be incomplete.

Gap between existing state of data and state of data required to build an asset registry

Data reported via the Cash Controls Regulation are not nearly comprehensive and complete enough to be built into an asset registry. As mentioned, these data only cover a partial aspect of cash movement and do not track ownership consistently. There is a striking lack of quality in current tracking according to the Supranational Risk Assessment, mostly due to cash being easy to hide as well as the EU's open borders. Even increased enforcement of this regulation would not address these issues, as post-declaration, cash movement would no longer be tracked. Thus, this gap cannot reasonably be closed.

Conclusion

Based on the nature of physical cash, the anonymity it provides, how easily it can be transported without detection and the unrealistic operational effort it would take to consistently track ownership and ensure comprehensive data, creating a cash registry is not operationally feasible. Data from the Cash Control Regulation do exist and could be the subject of study for further centralisation and data sharing across EU borders; however, the resulting information would not be comprehensive and complete.

6.2.2.1.2 Listed securities

Description of required data for an asset registry

For a comprehensive registry of listed securities ownership in the EU, data describing both the value and the ownership of a security are needed at any given moment. For simple shares this would mean that all shareholders of a given company are identified and catalogued at all times. Information on the ownership chain is also crucial, especially when dealing with non-custodial UCITS or similar funds, for which data on beneficial shareholding are often not clear and, in many cases, defined differently across Member States. Similarly, a value for a given security would have to be determined and recorded.

Operational requirements to build an asset registry

To build a registry on listed securities ownership, either the owners or the issuers of securities would be responsible for reporting. For a fully comprehensive registry, effectively every owner of listed securities would need to be identified, and their current ownership and changes of ownership would have to be reported. This could be done by the asset owners themselves by self-reporting to a hypothetical registry or by the entities issuing these securities. In theory, issuers could be mandated to always be aware and report a complete and exhaustive list of owners of issued securities, down to single shareholders for example. Due to the rapid nature of changes in ownership of listed securities, this would require an extensive reporting infrastructure and wide-reaching data collection efforts to effectively cover every single owner of a listed security.

Similarly, ensuring the data are up to date would also be difficult due to the large volume of transactions. Connected to the same issue is the question of how to ensure data quality, as conducting checks on the rapid changes in data would be extremely challenging. Finally, deriving a value for the recorded asset would be complicated as well. Prices for securities can often fluctuate wildly and are not necessarily dependent on the initial price paid or the issuance price. A method would have to be devised that always takes into account current market prices for all securities to derive their value, which would be crucial for a comprehensive registry.

Existing data on the asset

Several existing pieces of EU legislation (MiFID II ([93]), MiFIR ([94]) and EMIR ([95])) mandate the collection, standardisation and transmission of transaction data to national regulators and ESMA on listed securities or securities with a trading and clearing obligation. Specifically, transaction data stemming from MiFIR reporting that is transmitted and shared via the Transaction Reporting Exchange Mechanism (TREM) is another valuable source of standardised information.

The information collected and transmitted to regulators and ESMA differs depending on the security in question, mostly based on whether the asset is a derivative or whether it is another security.

For derivatives, trade repositories already transmit highly EU-standardised information on the changes of ownership of the derivatives mentioned in the relevant pieces of EU legislation. This includes detailed information on the parties involved in a trade, down to national identifiers such as social security numbers. ESMA further notes that these data do not just represent a current snapshot of ownership status, but rather represent the possibility to trace the flow of ownership as well. In addition, ESMA receives ownership data to a high degree of granularity with the possibility of deducing even the beneficial ownership of parties involved with a sale. Thus, the centrally collected data by ESMA would be a viable option as a source of information for ownership of derivatives traded on exchanges as it is already highly standardised using an EU framework and offers a high degree of granularity. It is important to note that the purpose of this kind of reporting is to tackle financial markets abuse.

For data reported for other securities, the situation differs. While standardised data are submitted to ESMA, by national competent authorities for example, the level of ownership information is appreciably lower than for derivatives. According to the Final Report on Cum/Ex and Cum/Cum tax abuse, ESMA concluded that the data currently exchanged for market abuse reporting are not suitable to counteract withholding tax abuse, and specific tax information would need to be reported as well ([96]). ESMA mentions that differences in national legislation defining the beneficiaries of large funds or omnibus accounts conducting transactions prevent a standardisation of ownership information. Thus, while more ownership information is available it is only at the national level, kept by national regulators and CSDs themselves. This is also a crucial obstacle to the usefulness of the data collected by ESMA for money laundering or tax evasion purposes, as the inability to identify beneficial owners hinders the accurate tracing of transactions as well as the potential seizure of assets to satisfy a tax burden. Current legislative proposals, such as the proposed revision of the Shareholder Rights Directive II ([97]) could in this case help increase the transparency of ownership data, especially for shareholdings.

([93]) https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex %3A32014L0065

([94]) https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex %3A32014R0600

([95]) https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex %3A32012R0648

([96]) esma70-155-10272_final_report_on_cum_ex_and_other_multiple_withholding_tax_reclaim_schemes.pdf (europa.eu)

([97]) https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX %3A32017L0828

Gap between existing state of data and state of data required to build an asset registry

While potentially a worthwhile source of information, the data at the disposal of ESMA and other national financial supervisory authorities are unlikely to contain enough key datapoints to act as a proxy for a fully-fledged, comprehensive and complete asset registry for listed securities. That these data are transaction-based is a key limiting factor, as are weaknesses in fully determining beneficial ownership – especially when funds or omnibus accounts act effectively as intermediaries in trades. Similarly, the value determination is not given, as the value of an asset would only be tracked based on the amount paid for the latest trade. Given the volatile price fluctuations for securities, the information therefore is at a risk of becoming outdated fairly quickly. In terms of the effectiveness of these data in uncovering financial crime, ESMA notes that data from CSDs in particular lack the necessary granularity in terms of ownership information to be of use. Still, it notes that better experiences are had with sharing the data collected through TREM (ESMA, 2020), thus constituting a potential avenue to leverage this information, albeit not in the form of a registry.

Conclusion

While the construction of a fully-fledged registry on ownership of listed securities does not seem to be operationally feasible due to the extremely heavy burden on reporting and validation placed either on companies issuing securities or on entities owning them, valuable information on this matter does exist. While not fully comprehensive and complete (mainly because the real owners of assets often cannot be reliably identified), harmonised transaction data on these assets are widely collected by exchanges, CSDs and trade repositories, and are transmitted to both national regulators and ESMA. Therefore, improvements in the identification of owners could be considered for these data, although a large harmonisation effort in terms of adapting national legislation would be necessary to achieve such a result.

6.2.2.1.3 Unlisted securities

Description of required data for an asset registry

For the purposes of this assessment two main sub-groups of unlisted securities are assessed:

(i) **Closely held unlisted securities**. This means that the securities are owned by one individual or by a small group of controlling shareholders. In the context of EU AML rules, 'controlling' in terms of legal persons like private limited liability companies can be defined as having 25 % or more direct or indirect control of the company.

(ii) **Widely held unlisted securitie**s. In contrast to sub-group (i), these securities are issued by large companies in which thousands of investors may own shares. Therefore, the controlling interest of any investor would be less than 25 %.

Notably, for sub-group (ii), widely held securities would have the same features as listed securities assessed above, as they would normally be held through a custodial chain by financial intermediaries.

Operational requirements to build an asset registry

For sub-group (i), each company would need to report the identity of its owners, and any subsequent update of owner information, to an asset registry.

For sub-group (ii), widely held unlisted securities, the operational requirements would also be comparable to listed securities, possibly with added complexity because the potential issuers of such securities would also have to be identified, whereas with listed securities this is known due to trades occurring on exchanges which collect information.

Existing data on the asset

Although there is currently no asset registry of unlisted securities in the EU, other national and EU registries can provide information on closely held securities under sub-group (i). National company databases, interconnected at the EU level through the Business Registers Interconnection System (BRIS), are required to identify the legal owner of a legal entity like a private limited liability company. Under the 4th and 5th EU AMLD, beneficial ownership information is required to be held by national registries and be accessible for competent authorities responsible for combating money laundering and predicate offences like tax evasion. Therefore, unlike widely held unlisted securities, national competent authorities can access such information already albeit more slowly than if legal and beneficial ownership of a security were held in an asset registry.

Gap between existing state of data and state of data required to build an asset registry

To produce complete and usable data, every securities issuer would have to report ownership data. As ownership changes are not catalogued on exchanges (as is the case for listed securities), the burden would fall solely on either issuers or financial intermediaries to provide such information. This would result in an extremely heavy burden – not just on the issuers themselves to produce accurate and up-to-date data, but would also require reporting mechanisms, infrastructure and oversight to ensure that the reporting obligations were followed. Given that ownership of unlisted securities can change fairly rapidly, capturing accurate information would be very problematic as well. Moreover, the operational difficulties for unlisted securities are much greater than for listed securities, simply based on the larger number of entities issuing them. Further, as indicated in Task 1, unlisted securities are not subject to the breadth of national and EU legislation as listed securities are, so any incremental reporting obligations would be greater for them compared with listed securities.

Conclusion

For the same reasons as for listed securities, building a comprehensive asset registry is not operationally feasible. Yet here too, data on shareholdings are available, particularly for closely held shareholdings, which could be accessible via use of company registries. Thus, while building a comprehensive registry may not be feasible, more efforts in ensuring accurate and up-to-date data in company registries for example could prove worthwhile for improving the information used to combat financial crime surrounding these assets in particular.

6.2.2.1.4 Works of art, antiques, collectors' items and jewellery

Description of required data for an asset registry

For an asset registry containing information on art, antiques, collectors' items or jewellery ownership, effectively every asset in these categories in existence would have to be catalogued and its ownership determined, be it by natural persons or other entities such as enterprises or museums. Changes of ownership would also need to be captured and a determination made of value. Valuations of art can be extremely subjective and vary significantly depending on the art market in general, or the particular markets and valuations for other assets in this category.

Operational requirements to build an asset registry

Effectively, all art ownership and ownership of other high-value assets would have to be reported to a hypothetical registry. This would entail unrealistic requirements for enforcement in order to achieve a comprehensive registry, as efforts would have to be invested in determining the ownership of each piece of art in existence. Changes in ownership would also have to be tracked across all owned art. Finally, a method to derive value would have to be developed, an extremely vague undertaking given the subjective value of art. This would pose an even more complex issue for art or other high-value assets which have never been sold and thus there is no sales price on which to base their value.

Existing data on the asset

The Fifth AMLD requires art dealers and auction houses to conduct extensive customer due diligence on high-value transactions (more than EUR 10 000). These measures do already entail the collection of extensive personal information on prospective buyers using official documentation, as well as the collection of beneficial ownership information where necessary.

There are several smaller databases and attempts to catalogue art and related assets (e.g. object ID ([98])), but none could claim to be comprehensive. Typically, when conducting sales, associations note that traders include a detailed description of the asset sold and potentially a photograph which serves to uniquely identify the good in question. While some of this information could be standardised (e.g. the name of the painter or printed serial numbers on watches), a vast, standardised and harmonised framework would need to be developed to uniquely identify these assets in a cross-border context.

For antiques, collectors' items and jewellery, there is no dedicated reporting obligation in the EU's AML framework apart from the general obligation businesses to becoming obliged entities when a sale of a good is made in excess of EUR 10 000 in cash. The European Commission's Supranational Risk Assessment (2022) mentions that this provision is worded much too vaguely, as is the term 'good'. Therefore, businesses very often are not even aware that they are required to conduct due diligence checks. Further, the requirement that payments have to be made in cash ignores all transactions being paid by transfers of money. As a result, reporting for these assets is basically non-existent at present.

Additional sources of information – such as import licensing documentation, VAT declarations and insurance information – could facilitate the tracing of assets.

Gap between existing state of data and state of data required to build an asset registry

As any current obligations apply to traders of the above-named assets, only data on legally conducted transactions using this intermediary would be captured in some form. This is not sufficient to constitute a fully comprehensive and complete registry of asset ownership, but rather just scattered information on changes of ownership transacted through legitimate traders. As previously mentioned, deriving a value for a piece of art is also extremely difficult. Even the harmonisation and reporting of such transaction information from intermediaries to

([98]) https://icom.museum/en/resources/standards-guidelines/objectid/

competent authorities would come with significant costs to reporters, which could total up to EUR 100 000, not to mention the significant efforts of authorities to verify and enforce the obligations.

Conclusion

Due to the large (in many cases either unknown or impossible to determine) number of pieces of art, antiques, collectors' items or jewellery, it would not be operationally feasible to construct a registry which contains ownership information of all assets in existence falling under these categories. In fact, even defining the assets in these categories, apart from tying them to their sales price, seems extremely difficult and efforts in legislation to do so are often profoundly vague. This naturally adds to the difficulty in terms of constructing a complete registry, as when the population of assets is not determinable, neither is the share of recorded ownership. While this may be the case, data could be collected from the traders involved in the transaction of these assets, as highlighted above. This would not constitute a fully-fledged registry, and gaps in coverage would arise from the problem of tracking ownership across transactions and issues in determining potential reporters, as well as ensuring the accuracy of reporting.

6.2.2.1.5 Commodities (precious raw materials)

Description of required data for an asset registry

Similar to the previous assumptions, a full database containing information on all owners of any commodities would be necessary, including valuations of the commodities in question.

Operational requirements to build an asset registry

Effectively, the totality of commodities ownership in the EU would need to be identified to build a fully comprehensive and complete registry. This would necessarily include any stock held by private individuals or other legal entities. Verifying especially the smaller, more valuable stocks would entail excessive operational efforts, as would capturing changes of ownership. A reasonable valuation could nonetheless be made, as commodities have fairly stable market prices, which can easily be tracked.

Existing data on the asset

The Supranational Risk Assessment (2022) states that at least for the largest group of diamonds traded in the EU, AML reporting obligations do exist. The Cash Controls Regulation now also applies to gold bars, nuggets and pure gold coins. Hence, when detected and enforced properly, at least some data on ownership of these assets should exist. Yet, the risk assessment also mentions the large divergence in legislation surrounding these assets across Member States and the need for harmonisation, adding additional operational burdens. To address this issue, the Council proposed including entities dealing with gold and diamonds in the group of obliged entities during the negotiations for AMLD.

In terms of reporting such transaction information, there is a precedent for this in Italy (see the case study in Annex F), where all transactions surrounding pure gold need to be catalogued and transmitted to the Bank of Italy specifically to combat money laundering.

Gap between existing state of data and state of data required to build an asset registry

All present reporting is only based on transaction data and therefore does not fully capture all existing ownership, which is necessary for a comprehensive and complete asset registry. No assumption can be made that any captured or reported transaction data are complete, as other avenues for trade may exist which could evade capture.

Conclusion

Due to the nature of the assets in question, especially considering that in many cases even small and portable amounts of which can hold high value, a full capture of all ownership of these assets is not operationally feasible. However, alternative data sources based on transaction information, which is recorded for example in Italy, could still provide a valuable source of information for combating financial crime. It is further important to consider the large reporting burden which would be placed on the traders reporting such information, as well as authorities verifying it. These data, while potentially valuable, are not considered complete and comprehensive enough to constitute a registry within the context of this study. This is mainly due to the lack of completeness of ownership data, as well as potential gaps arising from data that are not up to date due to changes of ownership taking place outside the entities obliged to report them.

6.2.2.1.6 Safe deposit boxes at non-banks

Description of required data for an asset registry

In practice, all safe deposit box providers would be required to keep and transmit full records of their customers to competent authorities in order to gain a full picture of the users of safe deposit boxes. This necessarily would need to be accompanied by data such as the duration of rental, among others. While the safe deposit box itself is not the asset in question with which money laundering or tax evasion is committed, it is nevertheless used as a vehicle to hide other assets and therefore just tracking the renters could provide a valuable benefit.

Operational requirements to build an asset registry

As mentioned, effectively all providers of storage options would be required to fully report all records of their customers renting their storage space at any point in time. This process would involve (or define) the population of potential reporters, as well as build up the infrastructure for enterprises to report such information. Verification efforts for such data would be extremely extensive, especially given the potential large number of storage providers, as well as the high volume of changes in customers, as many rentals may only be short term.

Existing data on the asset

Private providers of safe deposit boxes are not obliged entities under the EU AML framework. An additional reporting obligation for these service providers would be necessary to gather information on their clients that could then be used to build up a registry. While non-bank safe deposit providers are not part of EU money laundering legislation, the Netherlands consider them obliged entities. There, all safe deposit boxes are supervised by the national bank and both types have to fulfil the same obligations in terms of keeping customer records and relationships and carrying out due diligence.

Gap between existing state of data and state of data required to build an asset registry

As previously mentioned, there is currently no data collection on non-bank safe deposit box ownership whatsoever. Building up a registry would require an appropriate reporting infrastructure, providing possibilities for harmonised data transmission from the providers to the registry, as well as oversight of correct enforcement of the reporting obligation. Bearing in mind the extensive efforts required to ensure a complete registry, especially given the potentially large number of such storage providers as well as the high turnover of customers, this would create a huge operational burden.

Conclusion

Overall, the buildup of a registry of renters of safe storage boxes at non-banks is not deemed operationally feasible, as the systematic data collection and verification necessary to build up and maintain such a database would place unrealistic burdens on both the operators of these services and potential administrators. This situation starkly differs from banks, where centralised data collection already takes place, as naturally there are far fewer banks compared with general storage providers and for which conducting such a task is operationally not nearly as burdensome as for a non-bank provider.

6.2.2.1.7 Crypto coins and NFTs

Description of required data for an asset registry

Data on crypto ownership would necessarily include complete information on all wallets administered by crypto providers as well as any unhosted wallets to assure a full and comprehensive registry. Furthermore, it would have to provide an indication of the assets stored within the wallets as well as a valuation of the same.

Operational requirements to build an asset registry

To build such a full and comprehensive registry, theoretically every crypto provider administering a wallet to citizens of the EU would be obliged to fully report the ownership information of such a wallet to the registry, as well as its contents. Value determination in particular is extremely difficult for crypto assets, as there is a large number of them with rapidly changing valuations. In addition, every individual or legal entity would have to self-report any unhosted wallets along with their contents to gain a complete picture of asset ownership. Operationally, determining the population of self-hosted asset owners similarly would be extremely difficult.

Existing data on the asset

The recently adopted MiCA and Transfer of Funds Regulations, once implemented, could significantly improve and harmonise the regulatory environment and transparency related to crypto assets as well as generate some data on ownership of wallets and transaction reporting.

Gap between existing state of data and state of data required to build an asset registry

In 2021, the European Commission published a study investigating the possibility of setting up an EU registry containing user identities and addresses of wallets (EC, 2021a). The study concludes that while technically it is feasible to build such a registry, there are very significant operational obstacles which are unlikely to be addressed any time soon. Chief among them is

the lack harmonised rules on how exactly to register wallet holders and the lack of unified rules or systems in place on how to systematically report information on transfers of ownership to the relevant authorities. Given the large degree of complexity, diversity and quickly changing nature surrounding crypto assets, this presents itself as a major operational obstacle requiring a large effort in harmonisation. The MiCA and Transfer of Funds Regulations still do not provide for a systematic transfer and collection of information on wallet holders and particularly their transactions to the degree necessary to build a registry. Furthermore, ensuring that such information is accurate and current is also very challenging, given the high speed and volume of transfers of crypto assets.

Additionally, the study notes that the data held in an envisaged registry would still not be complete in terms of covering all types of wallets. Most crucially, a feasible technical solution has not yet been put forward that would account for transfers between unhosted wallets (which are wallets hosted by the users themselves, foregoing the need for a third party and related reporting obligations). This has also been noted as a major source of concern for the FIUs surveyed in the study, as it still leaves the door open for illicit, hard-to-detect transactions.

Conclusion

The aforementioned study commissioned by the European Commission already points out the high operational requirements for building a registry even just containing information on wallet ownership and addresses. It further points out and explicitly excludes an assessment on unhosted wallets due to the lack of legislative clarity on their treatment. Just these two aspects alone would be needed as a bedrock for a registry that would fulfil the requirements of being a registry on crypto ownership, not just wallet ownership. To track the contents of these wallets, which would be relevant for an asset registry on crypto ownership, much more would be necessary. Due to the properties of these assets (highly diverse, frequent changes of ownerships, etc.), along with the lack of harmonisation and extremely demanding technological requirements needed to centralise data on them, the creation of such a registry for the purpose of fighting financial crime, at least given the present state of progress is deemed not possible.

6.2.2.1.8 In-game assets

Description of required data for an asset registry

For a complete and exhaustive list of in-game asset ownership, all providers would have to fully report information on their clients as well as the extent of ownership of the assets in question. The value of such assets would also need to be determined.

Operational requirements to build an asset registry

As elaborated, in order to build a registry a reporting infrastructure would be necessary that would allow all companies offering such an asset to upload the data on both the ownership and valuation of all assets they offer. Furthermore, high-quality information on asset ownership information would need to be ensured, indicating the verified real name or other unique identifiers of asset owners.

Existing data on the asset

There is no obligation or regulation of any kind compelling video game companies to report information on the ownership or value of these assets.

Gap between existing state of data and state of data required to build an asset registry

One major gap in the buildup of a hypothetical asset registry is that often the value of these assets is determined by secondary market prices, where they are often traded, or by dynamically changing factors that are related to the game mechanics. Establishing such a registry would require every company selling such assets to report data on basically their whole userbase, which often can be in the millions. Even the data these companies hold may not always be accurate. As user accounts can be created with an anonymous email and a fake name, the real owner of these assets cannot always be accurately determined. There is often no verification system that reliably ties a natural person to a video game user. Introducing these obligations would result in a very high burden on reporting, coupled with the inherent difficulties of ensuring accurate data.

Similarly, payment methods in which the ultimate beneficiary can easily be obscured present an issue, as platforms may not be regulated, and payments could be made using less-regulated providers by spoofing the origin of the asset purchaser using software such as virtual private networks.

Conclusion

Due to the high frequency of change of ownership and the very proprietary and diverse nature of in-game assets, establishing a central registry denoting ownership is not operationally feasible. More stringent regulation on the providers is necessary to further store the transfer of funds, as well as potentially verify the accounts on the platforms so that asset owners can definitely be linked to their in-game funds.

6.2.2.2 Operational feasibility of leveraging existing registries

6.2.2.2.1 Cash at bank

Registries denoting the ownership of bank accounts are already envisaged to be interconnected at the EU level under the scope of the AML framework.

6.2.2.2.2 Limited liability companies

As the business registries in the EU are already interconnected through BRIS, containing relevant data for the fight against money laundering and tax evasion, no further operational assessment is necessary.

6.2.2.2.3 Properties, Land

The European Commission published a study assessing the potential to interconnect land/real estate registries (European Commission (2021b)). The study concluded that interconnection of these registries is indeed feasible, albeit with some caveats. Due to the difference in digitalisation of these registries, as well as different set-ups stemming from historical and political traditions in a given country, a larger operational effort is required to share harmonised data across all Member States.

Another key point of the study is that the harmonised data in an EU registry may not be exhaustive and cover all datapoints required for a competent authority to perform its duties. This again is based on national differences in terms of which registries store which information. Particularly key in this regard is a potentially missing value or purchase price of a piece of land or real estate, which is a core point of interest for money laundering or tax evasion-related

investigations. To ensure exhaustive and relevant data, in some cases the national real estate registries might also be combined with other registries containing the relevant information and only then can EU-wide interconnection be considered. This adds to the operational burden and associated costs of reorganising some national registries, in addition to the present operational challenge of harmonising and interconnecting data EU-wide.

The Land Registers Interconnection project can serve as a useful blueprint for the harmonisation of land registries in particular, providing best practices and potentially reducing the operational burden of interconnection to a small degree. But the number of participating countries in this project is comparatively low, and the inherent diversity of these registries across Member States still calls for significant effort for harmonisation and potential interconnection.

6.2.2.2.4 Motor vehicles

Data on motor vehicles is already shared EU-wide through the EUCARIS platform for which harmonisation and technical implementation already exist. As such, while not the full breadth of data contained in the national registries are connected, Member States have the possibility to connect different datapoints to the overall data sharing system. The data which is being shared already contains many relevant datapoints for efforts to investigate money laundering and tax evasion (see assessment in Section 5.3). Information is immediately updated whenever new vehicle registrations occur and given the regulations and requirements surrounding car registration ensure that the data is correct. Lacking from this database is the value of the vehicle. While many factors play into the value of a vehicle, it can usually be derived from information already in the register such as the type of vehicle, the model and year of production. Therefore at least an estimation of the price of a vehicle can be made with the already existing data. As such the operational burden itself of expanding the system would be manageable. However, it is important to point out that the legal purpose of this database is not to aid in combatting financial crimes, but rather to ensure the proper functioning of road traffic across the EU. The appropriateness to use this database the purposes laid out in the study will further be considered in the legal assessment.

6.2.2.2.5 Ships

As laid out in Section 5.3 and the use cases in Annex F, the level of digitalisation and approach to registering ships varies quite a bit for the analysed cases. Often, this is for historical and cultural reasons and because of the fact that some countries are landlocked and thus naturally might have fewer ships to register.

Considerable efforts in harmonisation would need to be undertaken to gather the correct and relevant information in each Member State. Regarding the completeness of the data, none of the analysed registries contains information on the value of the ships. Similar to cars, this could be addressed by deriving the value from the technical specifications of the ship in question. Therefore, while interconnection and harmonisation are in principle possible, work would be needed to reconcile and identify the correct information to share in each Member State.

6.2.2.2.6 Aircraft

Aircraft registries, similar to ship registries, vary quite a lot in their set-up and functions depending on the Member State. Similar operational obstacles would be encountered in ensuring that the relevant information is gathered, harmonised and interconnected.

6.2.2.2.7 Firearms

Firearms are covered by dedicated registries in each Member State. As firearms are highly regulated assets owing to their inherent danger, data quality by and large can be expected to be fairly high and comprehensive due to stringent enforcement.

6.2.2.2.8 Safe deposit boxes at banks

Information on clients renting safe deposit boxes at banks is already standardised and its interconnection at the EU level along with the interconnection of information related to bank accounts is envisaged. Thus, information on the renters of these boxes is known and validated through the bank account number and additional due diligence checks. While the renters of safe deposit boxes provide a very valuable datapoint in fighting financial crime, it should be emphasised that none of the reporting extends to the assets contained in such boxes.

6.2.2.2.9 Conclusion

The operational feasibility assessment of establishing new asset registries or interconnecting existing ones informs the rest of the study.

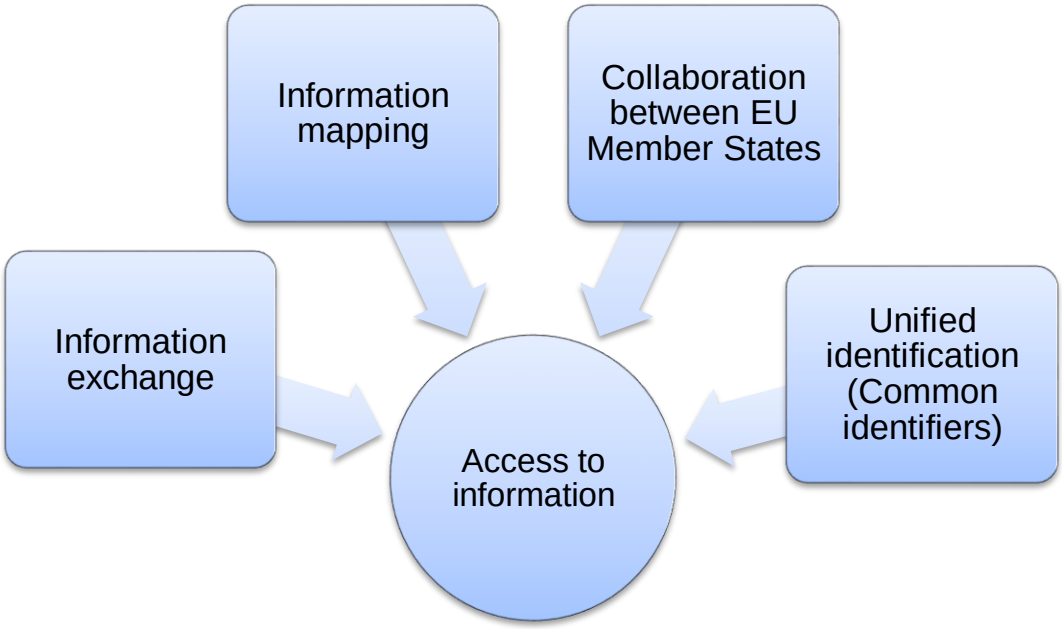
For registries that are either already interconnected or a platform for data exchange exists, no additional operational effort is necessary. For motor vehicles, EUCARIS serves to interconnect motor vehicle registries. Registries covering real estate, ships, aircraft, firearms and safe deposit boxes either require additional datapoints or a larger degree of harmonisation to make them fit for the requirements of an EU asset registry in addition to the operational steps involved in interconnection (save for safe deposit boxes, for which interconnection is already envisaged as part of the planned bank account registries interconnection).

6.3 IT feasibility assessment

The main aim of the IT feasibility assessment is to identify, outline and evaluate the architectural aspects that could influence the development of the infrastructure and services required for an EU asset registry, regardless of the scenario in which it is implemented. The registry would be used to aid investigations with the following ways:

- detection of cross-border money laundering schemes,
- identification of cross-border tax evasion situations,
- freezing and confiscation of assets,
- tax recovery.

To assist with these aspects, the registry would need to provide the functions illustrated below.



These functions build on the mapping exercise of the registries and reported difficulties encountered by the competent authorities discussed earlier in the study (see Table 29). Below are the findings from Task 2 (Section 4) on defining registry characteristics and limitations that can be considered critical for the development of the connecting architecture:

- Different asset registries offer different support services.
- The frequency of the registry updates differs (from daily to monthly, or longer).
- Not all relevant information is included in the present registries (e.g. identity and type of owner or value).
- The encoding of the data does not follow a consistent approach (UTF).
- Research indicators of the registries are sector-specific. (See the expressed need for data catalogues and the different domain indicators as detailed in Section 4.2.)
- Data catalogues are used by a small number of registries.
- The general accessibility of the registries for long-term machine access is insufficient.
- There are differences in the legal frameworks across Member States on access to data.

Based on the list of relevant assets held under public authorities (see Table 18) a selection has been made of representative case studies on intangible assets, tangible assets and commodities (precious raw materials) that, due to their specificities and the status of the registries, could be considered for inspiration in the design of an EU asset registry. The case studies cover crypto currencies (Cyprus), works of art and antiques (Malta), movable values (Romania), real estate (Netherlands), commodities (Italy), ships (Ireland) and aircraft (Malta) (see Annex F).

The wide range of approaches seen in the use cases and across the information collected in Tasks 1 and 2 depict a heterogenous situation. The variations range from simple web pages (Cyprus, for crypto) and non-digitised data coupled with structured databases (France, for aircraft and real estate) or PDF files (Malta, for aircraft) to highly modern, interconnectible semantic approaches (the Netherlands, for real estate). It would be unrealistic to expect to unify this level of diversity from the bottom up. The assessment thus focuses on the subject from a centralised position, looking for the elements and actions that could serve as foundation

blocks and identifying how the architecture could be designed to accommodate this degree of heterogeneity in the long term.

6.3.1 Use of existing interconnection systems as references for implementation

The idea of implementing exchange systems between various platforms containing information that are primarily managed by the Member States is not new. A number of platforms already integrate data at the domain level:

- European Criminal Records Information System (ECRIS)
- European Car and Driving Licence Information System (EUCARIS) ([99])
- EU-wide interconnection of insolvency registries (IRI 2.0)
- Business Registers Interconnection System (BRIS) ([100])
- Beneficial Ownership Registers Interconnection System (BORIS)
- Land Registers Interconnection (LRI) ([101])
- European Business Ownership and Control Structures (EBOCS) ([102])
- e-Justice Communication via Online Data Exchange system (e-CODEX).

These systems have already been implemented, independently or on a shared infrastructure (e.g. BRIS and BORIS), by a range of between 6 and 27 Member States and provide practical lessons for issues that might affect implementation ([103]). Interconnection with these existing systems would be essential in order to minimise the time required for implementation and costs.

Each of these systems is tailored for a particular field. Consequently, they tend to handle a relatively uniform set of information within their specific domain. Contrasting these specialised systems, which concentrate on managing data related to a specific type of domain or asset, with the registry under consideration, a notable distinction emerges. Unlike the other systems that focus on a single type of information, this registry must integrate information about different entities (organisations or individuals) and various assets that hold significance. In essence, it deals with a broader range of information, making its data management more complex.

([99]) EUCARIS the basics, 18 October 2022, Idske Dijkstra, Netherlands Vehicle Registration Authority (RDW), Senior Advisor, Head of EUCARIS Secretariat: Technical vehicle data exchanged between 20 countries (including IS, LI, NO); vehicle owner/holder data exchanged by all 27 EU Member States + CH, IS, LI, NO.

([100]) Commission Staff Working Document, Impact Assessment Report, Proposal for a Directive of the European Parliament and of the Council: BRIS interconnects all Member States' business registers.

([101]) https://lri-ms.eu/news_2; Member State Authorities connected as of LRI II: Austria, Estonia, Latvia, Spain, Hungary and Portugal

([102]) EBOCS Final report, Document. ref.: EBOCSIII.WP1.D1.6.V1.0: Countries involved in the EBOCS project – Estonia (EE), Ireland (IE), Italy (IT), Latvia (LV), Romania (RO) and Spain (ES).

([103]) Impact assessment report Accompanying the document Proposal for a DIRECTIVE OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL amending Directives 2009/102/EC and (EU) 2017/1132 as regards further expanding and upgrading the use of digital tools and processes in company law, SWD(2023).

6.3.2 Existing EU projects as a basis for new development

As with any IT architecture, a future registry would be part of an ecosystem. The registry could try to encapsulate just the information received from the members of that ecosystem or use different elements of the environment as building blocks.

In addition to looking at the existing interconnection systems presented in Section 6.3.1, the sections below list some additional building blocks that could serve the construction of the registry both in terms of both effort reduction and costs.

6.3.2.1 Technical implementation of services and other technical developments that could serve as components

A number of developed EU systems and services could be considered for integration in different ways:

- **Trans European Services for Telematics between Administrations** (TESTA) – a private network run by the European Commission completely separately from the public internet;
- **eTranslation** – an online, free, neural network-based, machine translation service;
- **eDelivery** – an architectural component to support the exchange of electronic data and documents;
- **eIDAS** – the system of electronic identification, authentication and trust services for electronic transactions;
- **eSignature** – a set of standards, tools and services that enable the creation and verification of signatures legally valid in all EU Member States;
- **eInvoicing** – a digital solution that enables public sector contractors and companies to receive and process electronic invoices;
- **Context Broker** – a digital platform component that enables the integration of data gathered from multiple systems;
- **ISA² Core Vocabularies** – a set of controlled vocabularies (business, location, person, etc.) that ensure interoperability for the exchange of metadata ([104]) between systems and users;
- **Publications Office and ESTAT reference data resources** – a range of multilingual controlled vocabularies, spreading from authority tables of countries, places and languages to taxonomies of services and product classification, which can help unify the exchange of information between systems and users;
- **Big Data Test Infrastructure** (BDTI) – a set of architectural components that offer applications, virtual machines and/or databases for use in the context of analytics;
- **European Blockchain Services Infrastructure** (EBSI) – a network of distributed nodes assembled to deliver cross-border public services by the application of blockchain technology;
- **Common Communications Network and Common Systems Interface** (CCN/CSI) – a network that ensures transmissions by electronic means between competent authorities in the area of customs and taxation.

([104]) A form of structured information that describes and explains a given set of data, providing context with details such as the source, type, owner and relationships with other data sets.

Each of these existing components can fulfil essential functions in a large infrastructure system providing a level of interoperability with a similar EU system, which would otherwise be hard to achieve, while also reducing the costs and overall time for implementation.

6.3.2.2 Network projects

Various initiatives have been developed across the EU at the domain level. Members of a respective ecosystem have an obvious tendency to communicate among themselves and to form networks. Such initiatives can develop into large architectures entities that in turn become actors in their own way and succeed in their effort to formalise their presence. Examples of relevant networking projects are listed below and described in Annex G:

- European Land Registry Association (ELRA)
- Camden Asset Recovery Inter-Agency Network (CARIN).

These networks act like unifying agents in their respective fields, providing support for the development of EU projects (ELRA is a main actor on the LRI 2 project) and facilitate access to information in a faster way (see the CARIN table in Annex G).

6.3.3 IT standards and models

The analysis conducted on implementation at the EU Member State level has shown a range of standards and models that are not necessarily much different in terms of the technologies used as they are in terms of generational differences.

Typically, the systems use a relational database model, and build on top of it a layer of business rules and user interface components. The components in charge of business rules and business intelligence develop as the solution becomes more complex in architectural terms (advances associated with the development of the particular market and regulations in place). The components of business intelligence relatively reduce as reporting in such systems grows less important in comparison with the actual data management functions.

Data exchanged between Member States under the DAC provides an example of how a reporting mechanism for automatic exchange of information is currently operating. DAC1 uses IP exchanges, meaning exchanges on income and the ownership of property/land between the Member States where the property and land is located and the Member State where the owner is resident. An XML format is used to exchange information based on a pre-determined set of criteria. In this way information can be standardised without the need to harmonise the data contained in the national asset register. The extent to which national asset registers need to be harmonised or existing data can be used and then be reported in a standard format would need to be further assessed. In terms of value, 13 of 16 Member States that exchanged information on property/land in 2022 under DAC1 included a value of the land/property in the exchanges.

Considering the total number of registries in use and the limited amount of information obtained on the exact architecture used, it is not possible to identify a clear advantage in terms of the number of specific implementations per technology. It is conclusive to say that the predominant use of relational databases does offer the possibility to consider a reasonable intersection in terms of data organisation but not in terms of data models. This common technical aspect can represent a facilitation element for integration in a pilot phase but in the long term can also induce a series of difficulties (due to the development of each independent platform).

6.3.4 Potential interoperability of existing registries

While the use of interconnectors is the preferred option to connect registries (as in the cases of BRIS, BORIS, EUCARIS, IRI and LRI), the observed lack of common standards at the level of some domains (e.g. real estate) indicates that direct exchanges of information between existing registries is not an option that can always be realistically considered. Yet bearing in mind the range of fields that are of interest for the fight against money laundering and tax evasion, interaction based on dedicated REST APIs (application programming interfaces that conform to the design principles of the REST, or Representational State Transfer Architectural Style) could be considered. The specific conditions of the exchange between the local registries to the top nodes make this scenario relatively easy to implement, if not already available for some of the existing registries.

More developed cases like the Netherlands land registry (Kadaster) show that migration from domain standards to web standard and linked data is already taking place. Finding common ground on this principle is easily done even for older architectures. Such an approach would ensure interoperability despite the technical differences between platforms.

In terms of interoperability, the project has to first consider the existing EU-wide platforms that connect Member State data on specific asset types (see Table 31).

Table 31: Existing EU platforms and asset

EU platform	Asset group covered
Business Registers Interconnection System (BRIS)	Limited liability companies
Beneficial Ownership Registers Interconnection System (BORIS)	Limited liability companies ([105])
European Car and Driving Licence Information System (EUCARIS)	Motor vehicles
EU-wide interconnection of insolvency registers (IRI 2.0)	Limited liability companies
Land Registers Interconnection (LRI)	Real estate (limited to land properties)
European Business Ownership and Control Structures (EBOCS)	Limited liability companies

Source: Authors' elaboration.

While not all of these platforms are prepared for integration in terms of offering the necessary services, the potential advantages of their integration are high.

It can also be observed that while most of the targeted asset groups are covered by these platforms, their scope is limited to specific asset types. The following assets fall outside these platforms' coverage:

- aircraft

([105]) All corporate and other legal entities incorporated within a Member State and all trusts/other types of legal arrangements established in a Member State are required to report their beneficial ownership information to the national beneficial ownership registry. Thus, their beneficial ownership information will also be available through the interconnection (BORIS). Yet in the context of the present study only limited liability companies are part of the scope.

- ships
- real estate, other than land (e.g. apartments, commercial buildings)
- securities (while ESMA compiles a number of databases these are not entirely comparable with the rest of the services).

With the exception of securities, the respective assets are also covered by dedicated registries at the level of Member States (see Table 14 and Table 15). Yet interfaces for searches and general interoperability between them are still desirable objectives, making them prime candidates for the construction of interconnectors (for transactions involving higher volumes like real estate) or EU-wide registries (for aircraft and ships).

6.3.5 Requirements

The ability to provide the necessary information to ensure the feasibility of the project is dependent on a set of technical and operational factors that should be approached from the beginning of the process.

6.3.5.1 Architectural requirements

Beyond the standard requirements of *security*, *scalability*, *reliability*, *traceability* and *availability* expected of any large IT infrastructure service, the purpose of the EU asset registry in aiding the fight against money laundering and tax evasion requires particular attention in approaching certain aspects with an impact on the IT architecture:

- **Facilitate an open-ended extensibility at the level of asset properties and relations.**

While the study has tried to identify all asset categories that are relevant for the subject, the research has shown that the probability of a perfectly interoperable, complete description of an asset class, applicable long term at the EU level, is fairly low. In this context, it is more realistic to consider an architecture where new assets and interconnections are continuously added as the need, or the necessary information appears in any of the individual nodes of the network. A feasible response to this need is the use of a semantic approach to storing properties and relations (defining the various data models as ontologies and extensively using controlled vocabularies).

- **Accommodate extensive factors of destabilisation.**

The experience of previous interconnection systems shows that the architecture has to ensure the necessary level of flexibility to accommodate a partial, temporary disconnect of various registries (at either the category or national level) while ensuring the persistence and integrity of the data already shared between nodes.

- **Use the minimum viable set of information (metadata and corresponding identifiers), to ensure the protection of identity.**

The amount of personal data involved and the massively distributed heterogenous network to be used (a system employing a large number of nodes that are not centrally located, are different in terms of architecture and data, and are exchanging data on an open network), present a critical risk in terms of security and protection of identity. A principle of self-contained data units at the most atomised level should be considered (a minimal set of information details about a record that are sufficient to perform an action) to isolate the identity of the owners and keep it at the source until the last moment (see also the legal limitations described in Sections 7 and 8, for Tasks 5 and 6).

6.3.5.2 Operational requirements

While connectivity and access to data are viewed in principle as technical aspects, there are also several operational requirements necessary to achieve these goals. This implies satisfying three major requirements:

- access to the necessary data,
- agreement on the types of properties associated with each type of record,
- establishment of the necessary governance to facilitate the appropriate access by the relevant authorities.

All these have to be fulfilled in accordance with the need to find and connect information that is usable in meaningful ways. The scope of this is to enable the exploitation of the information for both AML and prevention of tax evasion. That implies the need to:

- establish a network of relations between entities and assets;
- maintain the respective network in good shape in terms of data quality (data are accurate, complete and up to date);
- ensure the ability to navigate the respective network.

Subsequently, the network – comprising entities (legal and natural persons) and assets – has to provide sufficient information to aid investigations.

To ensure this, it is essential that each of the registries connected to the system (either by interconnectors or by providing direct dumps of information) provide data that are accurate, complete and up to date.

Connecting registries that do not provide a complete picture of the assets managed implies that a potential user of the system would see a partial or erroneous picture of a situation. In that respect, having variable coverage would make the system insufficiently credible for the task but at least knowing what part is missing would still be satisfactory. For that matter, it would be preferable to start with a stable and well-defined set of asset categories (like bank account information, road vehicles and limited liability companies) and extend this range to additional categories as they become available, stable and credible.

It is also important to ensure that the set of asset information of a given country covers a specific time range. This is needed to build a complete picture of the relations between entities and assets for that time interval. That means working either with real time data (which is unrealistic at the scale of the given system) or working with clearly defined batches of past metadata that are regularly transferred to central nodes.

From an operational perspective, this need for validation implies that transaction details covering a certain range, and covering a given set of asset categories, have to be available to the registry in scope. That of course could be in contradiction of the regulations on storage of personal data if the whole set is stored for long periods (see Tasks 5 and 6). To avoid such a situation, only metadata containing the relevant information should be centrally stored and the respective metadata should be stripped of any personal details. The nature of this information would have to vary according to the asset in question and would need to be determined on a case-by-case bases. Such cleansed data could offer the necessary control checks but also ensure conformity with personal data regulation. Accordingly, a reliable set of data – which means complete, with known limits and validated to the best criteria – could be kept available.

The reliability of the data is also important for the infrastructure because most of the data would remain stored in the source registries. Ensuring their operational availability at all times would be paramount.

Finally, project management-wise, the operationalisation of the system would be dependent on implementation of the techno-functional factors outlined below.

6.3.5.3 Techno-functional requirements

A set of functional objectives have to be considered for any type of successful data interconnection or integration into an EU asset registry:

- certify an information flow to comply with the specific requirements of the function performed;
- define and test a minimally viable set of data in accordance with necessary usage scenarios;
- organise the data (from a storage, model and serialisation perspective);
- agree among participants the integration components (e.g. network, exchange systems, protocols ([106]) necessary to ensure the connectivity between registries;
- describe and agree among participants the functional features of the user interface serving the search functions;
- define the technical specifications of the architecture.

Once the aspects around the definitions are formalised, considering the scale and complexity of the system and the discrepancies observed between Member States, this study concludes that the optimisation of the data to support the integration process is the most important element. This could be attained by choosing one of the following options:

(i) **Optimised information exchange.** Ensure compliance with a standardised service protocol that enables clear exchange at the national level and furthermore at the EU level, between a central interface and each separate registry.

(ii) **Centralisation at the EU level.** Create a structured process for regularly submitting information to a central system, using a mechanism that reduces the risks linked to difficulties in connecting to sub-nodes.

(iii) **A combination of the previous two options.** – with central asset registry at the national level and an interface at the EU level.

6.3.5.4 Architectural principles

An EU asset registry (regardless of the implementation scenario) that is supposed to speed up access for competent authorities to the cross-border data they require to perform their duties, and which mostly deals with sensitive information should fulfil the following criteria to ensure long-term adaptability to changes caused by new legislation and general practicality:

([106]) A method and system that facilitates the exchange of information or data between stakeholders of a process by defining a set of common rules of communication that is agreed and applied by all participants.

- *Flexibility*. The architecture should be designed in such a way that it can easily adapt to changes in reporting at either the Member State or individual registry level, in terms of technical connectivity, data availability and final needs over a longer time horizon.
- *Simplicity of the submission/exchange processes*. A complex process would make integration a less appealing prospect for a newly created registry, by creating possible difficulties in terms of the financial or human resources necessary for the interconnection.
- *Security of the data exchang*e. The set-up should protect the personal information managed by the system and ensure compliance with the GDPR.

These principles can be translated into the following requirements:

- **The process for submitting data to the system, by the individual registries at the Member State level, has to be simple, clear and easy to comply with.**

Any complexity added here in the form of data structures that are hard to obtain through a simple export from a relational database, risk the potential of nonparticipation by individual registries, and should be excluded. It is preferable to consider a form of enrichment of the structure at the level of the gateway or on the central server (if that is the case).

- **The protocol of communication has to be standardised in terms of serialisation but allow for flexibility in terms of the actual data model (possible solution: open-ended, linked data).**

In this regard it is important to avoid the trap of 'complete and structured data' and focus on the actual ability to transfer the metadata that are relevant for searches and encapsulate the rest.

- **Only relevant information should be included in the exchanges between nodes.**

It is of course safer to transfer just the relevant data for the search function as they are more likely to be generic information. Keeping the complete record in the original registry has the advantage of inheriting the security of that system. But this principle is also relevant from a much broader perspective. Keeping the complete data at the source has the added advantages of reducing the complexity of the data transferred, reducing the legal implications associated with the transfer and minimising the size of the data to be transferred (making them easier to encrypt).

- **The exchanges have to be transparent for all parties involved.**

The architecture of any large-scale system tends to become complex. Given the level of vertical integration to be developed, the present system is even more prone to such a tendency. To avoid that, the general architecture of the system has to be designed around the model of data.

6.3.6 Elements enabling easy or automatic data searches

A major issue affecting multinational projects aiming for interoperability is related to data homogeneity. While in principle homogeneity can be found among asset-category data models at the Member State level, the practical issue lies in the implementation of the models. The differences are not just in subjective factors (engineering preferences or technological platforms) but also in the preferred approach for searches and the organisation of data following different keys, induced by:

- differences in business rules,
- level of market development, or
- legal constraints.

Yet even under such conditions the data that define the entities involved, either legal or natural persons, are principally the same. Differences at this level are down to the use of different national identifiers by Member States. Examples of such differences can be seen in the use of one of the following identifiers:

- for natural persons, use of an identity card serial number (in Romania), vs personal identification number (in Luxembourg), or passport serial number (Denmark does not issue identity cards that are valid as travel documents);
- for legal persons, use of a VAT code, national registration code or EORI code.

The area where differences are most common is associated with the parameters that define the characteristics of the various assets involved (specifically in the case of non-monetary assets). This is also the area where most efforts are consumed in approaching interoperability. Still, given the needs of the system, it is assumed that in most cases these asset properties (e.g. car brand, model and production year) are important, especially for non-financial assets. That is because they can serve as the basis to estimate the value of an asset that would otherwise be missing and is highly relevant for combating money laundering and tax evasion and for asset recovery. It is therefore relevant to accommodate multiple search criteria. As a result, the possibility to conduct a much more granular search needs to be considered.

6.3.7 Scope and level of harmonisation required

As a result of the responses gathered in Task 2 and the subtasks of this section, the scope of the source registries could be at the national level, regional level, company level or a combination of these.

Considering the replicable functional needs that every record of a registry has to fulfil, at the sector level, a **de facto harmonisation that facilitates interconnection between all registries cannot be realistically expected**. Nevertheless, **development of a strategy for harmonisation** is of utmost importance to ensure positive results from the perspective of preventing money laundering and tax evasion. Such a strategy could be built on specific incentives like:

- technical and administrative facilities for the set-up of new or rebuilt registries;
- mechanisms to ensure data exchanges benefiting each individual registry (not just the central nodes);

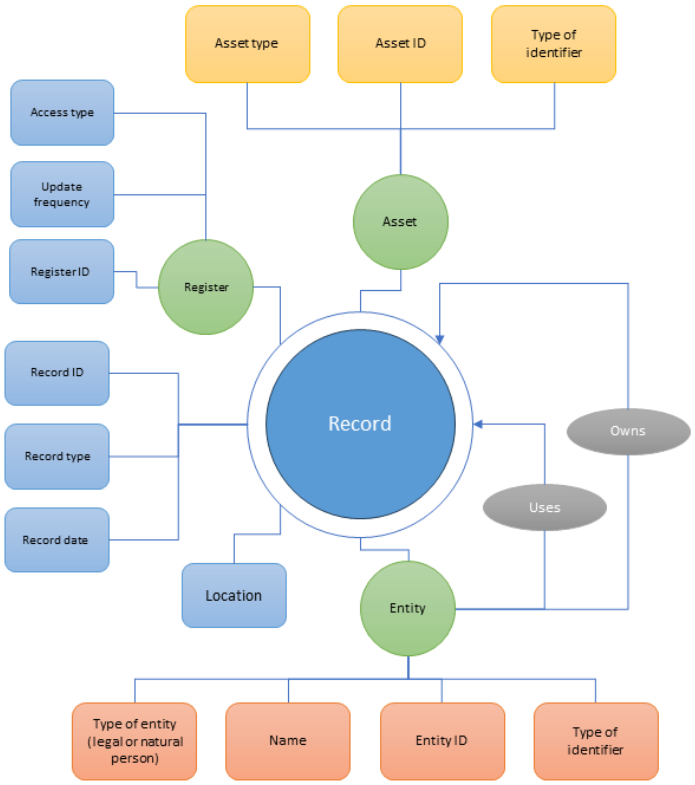
freedom of choice in setting up the individual policies (architecture, security, etc.) of each individual registry.

The scope of the harmonisation should not necessarily be to ensure equivalence in terms of overall data structures but to **enable a level of integration on two levels:**

- metadata – a form of structured information that describes and explains a given set of data, providing context with details such as the source, type, owner and relationships to other data sets;
- exchange protocol – a method and system that provides for the exchange of information or data between stakeholders of a process by defining a set of common rules of communication that is agreed and applied by all participants.

From this perspective, based on the scope of the future system, the essential elements of data that could be considered for a hybrid record (which can take the shape or either an entity or an asset) could be organised as shown in Figure 20.

Figure 20: Base level of data for a record



Source: Authors' elaboration.

The ensemble of data defining the record, forming the metadata of that record, would serve as the core information package with which all other functional elements (information about transactions and assets) would be associated.

The scope of the metadata would entail describing the package and facilitate the storage of the basic information at the central node. Yet that same metadata would provide support for the first step of the search process (definition of the query). It must be stressed that the use of metadata and of a minimal set of data is also relevant from a legal perspective as stated in the Sections 7 and 8 on legal feasibility and must be treated in accordance with the respective regulations.

From a protocol perspective, the second level of harmonisation, integration should be considered in terms of defining the data format employed for the exchanges between registries and the central or intermediary nodes ([107]). Possible candidates for this could be RDF

([107]) Intermediary nodes are components or devices in a network, that have the role, among others, of facilitating the connection between other components of that network.

(Resource Description Format is the W3C standard for web data) with XML (Extensible Markup Language) or JSON-LD (a lightweight Linked Data format) serialisation.

Keeping the harmonisation requirements to a minimum would in fact enhance the extensibility of the system, providing a flexible and adaptable foundation.

6.4 Defining scenarios for an asset registry

Considering the scope and particularities of the project, an ideal architecture would be:

- highly distributed – data would remain where they are;
- domain agnostic – querying the system would not require knowledge of each asset's features;
- highly secure – data would not be transferred; the processing would happen at the origin and only the necessary results would be communicated to the requestor.

Nonetheless, the variety of assets to be covered and the different issues observed in the case studies (see Annex F) show that for some of the assets, at least for the foreseeable future none of these expectations can be fulfilled due to the status and architecture of the systems supposed to be interconnected with it. The observed situation underpins the need to analyse the theoretical scenarios and extract conclusions that would best fit the technical and legal framework at the time of implementation.

6.4.1 Description, operational feasibility and IT assessment of core scenarios

6.4.1.1 Scenario 1: Creating new registers/databases at national level in addition to the existing ones (Beneficial ownership, bank account) for sources of wealth for which registers do not exist at national level, and their interconnection at European level to form an EU asset registry;

A national registry with its own asset database would have to be implemented by every Member State. This registry would offer the technical foundation for the creation of registries for all the assets that are not yet covered by the respective Member State. The existing category-based registries would establish individual connections with the national registry.

National registries would be able to communicate with them using a gateway that would allow for the integration and secure exchange of data. Competent authorities at the national level would conduct searches in their own repositories and would be able to send queries to other Member State repositories. A portal implemented at the EU level would offer the competent EU-level authorities the ability to search for information in all national registries using the access provided by the gateway.

This scenario would first require the development of national infrastructure that would ensure central access by the competent authorities to all the relevant data at the Member State level. It would also entail the creation of new registries for assets where there currently are none.

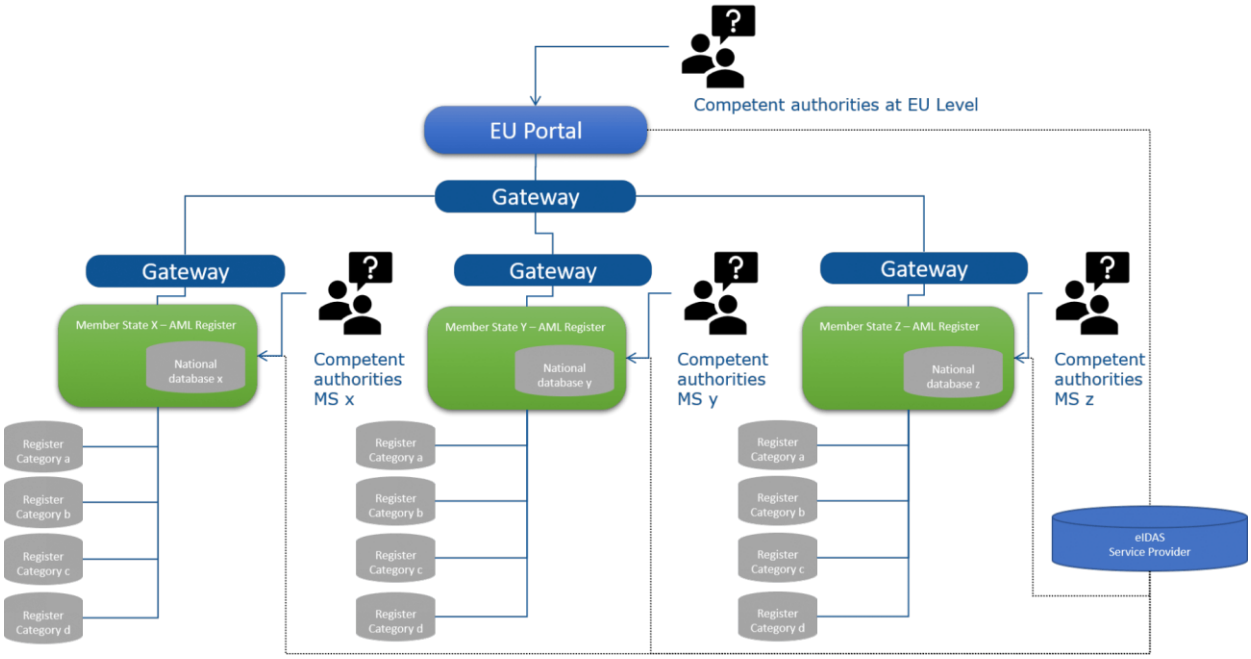
The asset registries that already exists (e.g. properties, land, ships, aircraft and firearms) could be slotted into the proposed architecture without much additional consideration, apart from the operational issues related to these assets highlighted in Section 6.2.1.

This scenario does not effectively leverage the infrastructure of already-interconnected asset registries (bank accounts and LLCs) or potentially EUCARIS, which technically could be interconnected to fully cover all Member States. Instead, each Member State would have to construct a central national register, transmitting data from the registries in question and only then could these registries be interconnected at an EU level, which seems inefficient operationally and would mean higher costs than may be necessary.

Nevertheless, against all these drawbacks it must be stated that this scenario is the preferred option for the projects already implemented at an asset class level. This is due to:

- the lower risks in relation to differences in Member State legislative frameworks and their ability to preserve control over their data, in particular with regard to national and EU GDPR requirements which are assessed in Tasks 5 and 6;
- reduced risks in relation to data quality due to the Member State obligation to provide data that conform with DAC and AML regulations.

Figure 21: Creating a new database at the national level in addition to existing ones



Source: Authors' elaboration.

The scenario offers two implementation options:

- National registries would be developed and maintained independently by each Member State, following common protocols to facilitate the exchanges with the gateway. Each Member State could decide on the complexity of the architecture at the national level and implement the necessary mechanisms to support connectivity for the local/category repositories.
- A common open architecture would be developed at the EU level and deployed by each Member State within its own infrastructure. The Member States would share the cost for the development and maintenance of the platform and only be responsible for the deployment, customisations for the integration of the local/category registries and the overall operations of their instance.

The choice of the two options has an impact on implementation.

Table 32: Assessment of Scenario 1

Factor	Advantages	Disadvantages
Financial	Benefits might appear at the Member State level depending on the current development of their existing registries and the overall development strategy	High cost of implementation and operation
Compliance with the local regulations and legal framework	Allows for full flexibility at the local level, accommodating the existing processes, national customs and specific data models associated with each individual asset class	Requires a high degree of synchronisation among Member States to guarantee the complete coverage of a particular asset class across the EU at the data level. This impacts the overall operational costs for Member States and the Commission. Much more difficult to ensure in terms of governance
Security	Minimises risks associated with a 'high-value target' (central repository) by keeping the data distributed	
Data quality and monitoring	Ensures the essential data qualities that are needed for the scope (consistent, complete and up to date) Monitoring at the Member State level ensures easier and more direct quality assurance of the collected data	Most difficult architecture for facilitating complex queries (e.g. queries involving relations between assets and entities cross-border) The time difference between queries performed on local registries and among those in other Member States might force the user to perform simple targeted queries and assemble the results in a larger analysis outside the system, thus negating some of the expected advantages of the system
Service provided and availability	At the national level it has the potential to provide a much better experience for the competent authorities and potentially to improve public services	High risk of being affected by some registries going offline. A registry not being online means data for that asset class and that Member State will not be available for analysis
Integration and interoperability	Brings the institutional actors in Member States closer to their counterparts, thus creating opportunities for networking, sharing best practices and awareness of one another's experiences and environment	Requires the highest number of interconnectors to be built (impacting on the implementation, maintenance effort and cost) Least potential for further developing the interoperability among Member States; Maintaining synchronisation at the technical level would prove to be difficult since each Member State might have different institutional capabilities for some asset classes (see the Netherlands real estate registry use case in Annex F, compared with the situation in Italy, Germany and France) Least potential to disrupt the status quo in relation to the difficult situations (e.g. asset classes that are getting the least attention in some Member States, registries combining different categories)

Factor	Advantages	Disadvantages
Extensibility and change management	Allows for easy extension and development at the Member State level. In general terms, it does allow for freedom in the choice of technologies and decisions on the development strategy	Any changes at the Member State level could impact the overall functions of the system at the EU level, if not properly coordinated

Source: Authors' elaboration.

6.4.1.2 Scenario 2: Interconnecting only the existing national registers required under EU law (Beneficial ownership, bank account) and real estate registers with an EU asset registry

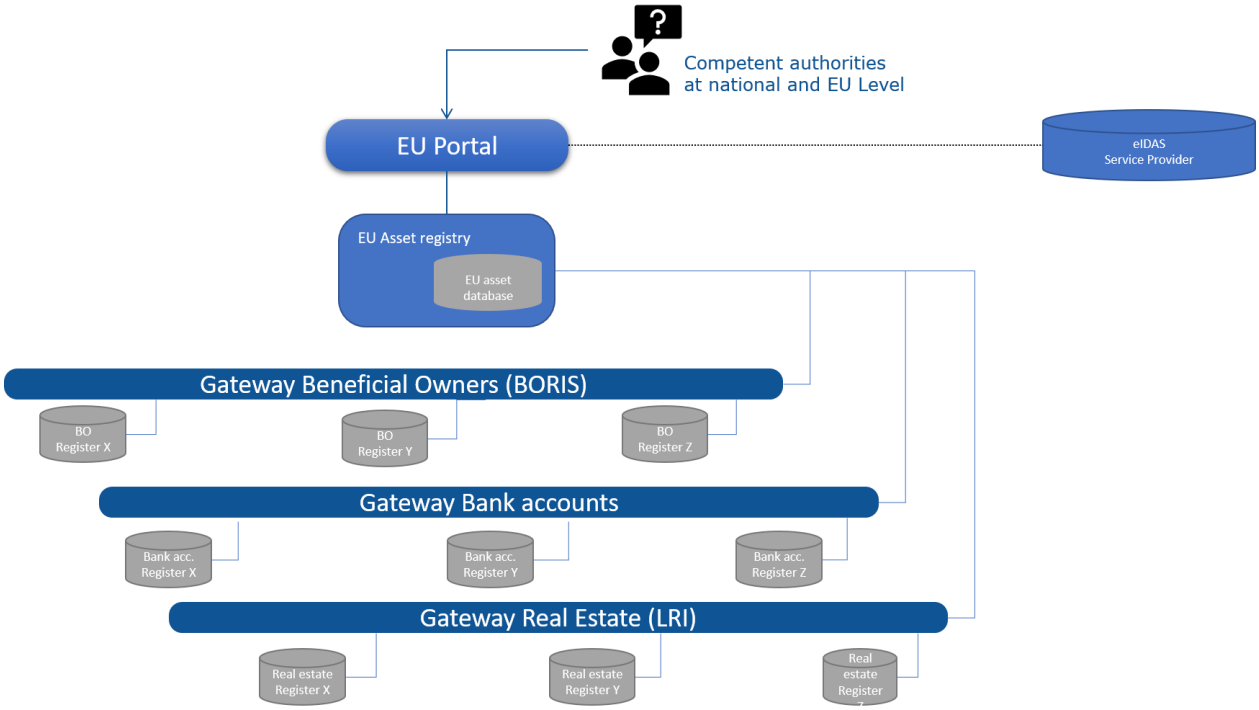
A light approach to interconnection would be to use the architecture of existing modules and facilitate their interconnection using a central registry. All queries by authorised parties at the EU and national levels would take place using a dedicated EU portal, attached to which would be a central registry (see the EU asset registry in Figure 22) where only the data necessary to perform the queries be stored. Crucially, this is different from the 'EU asset registry' under Scenario 3, which entails centralising all the necessary data covering all assets into one registry, not just the building blocks needed to formulate queries.

The system would integrate the most relevant data from components that already exist, namely the beneficial ownership registries, bank account registries and real estate registries.

The interconnection between registries for the same asset class is the most commonly used method for performing integrations between registries at the EU level.

In terms of the operational burden, this is by far the easiest option to implement as some relevant interconnections are already in place or in the process of being established. The operational feasibility of interconnecting real estate registries has already been assessed in Section 6.2.2.2.3 and has been found feasible in principle, with caveats being the heterogeneity of existing national real estate registries in particular.

Figure 22: Interconnecting only the existing national registries



Source: Authors' elaboration.

Table 33: Assessment of Scenario 2

Factor	Advantages	Disadvantages
Financial	The scenario focuses on a minimal set of interconnectors, increasing the chances of success for the chosen asset classes and for the project in general	
Compliance with the local regulations and legal framework	Provides an incremental development on top of the infrastructure and tools already in use for BORIS, BRIS and LRI II, making it the fastest and cheapest option. Adding new features and interconnecting existing networks would be a natural evolution for most users already involved with those systems	Minimises the scope to the fewest asset classes (although important ones in terms of size)
Security	Minimises the potential interest and access for attackers, involving the fewest entry points (registries open and interconnectors)	
Data quality and monitoring	Ensures the essential data qualities that are needed for the scope (consistent, complete and up to date) Monitoring at the Member State level ensures easier and more direct quality assurance of the collected data	
Service provided and availability	The small number of assets involved minimises the potential complexity of	

Factor	Advantages	Disadvantages
	the queries to be developed and implicitly of the query mechanism (fewer parameters to control)	
Integration and interoperability	Benefits directly from best practices developed for the existing interconnectors, in areas like application maintenance and security	
Extensibility and change management	Relatively easy to extend to other asset classes that are using similar interconnector mechanisms and, in some cases, sharing the infrastructure (e.g. EUCARIS, EBOCS and IRI)	Has the least potential development capabilities in terms of adding new features and creating advanced functions, as it would inherit an existing architecture (changing the existing interconnectors would be difficult and hard to justify)

Source: Authors' elaboration.

6.4.1.3 Scenario 3: Establishing a central EU asset registry connecting the existing registers at national level and, for sources of wealth not covered by them, featuring a central EU register/database with national access portals.

A more complex scenario based on the principle of a central EU asset registry could be used to extend the ability of the Member States to capture data for categories that are not already covered.

In such a scenario, all existing registries would have to interact directly with the EU gateways. The essential data provided by each individual registry would be stored in the central asset database with full compliance with GDPR rules. The scenario could accommodate differences in the number and types of registries that each Member State connects to the central registry. Nevertheless, a lack of participation by any one Member State in all the relevant asset classes might undermine the objective of the project.

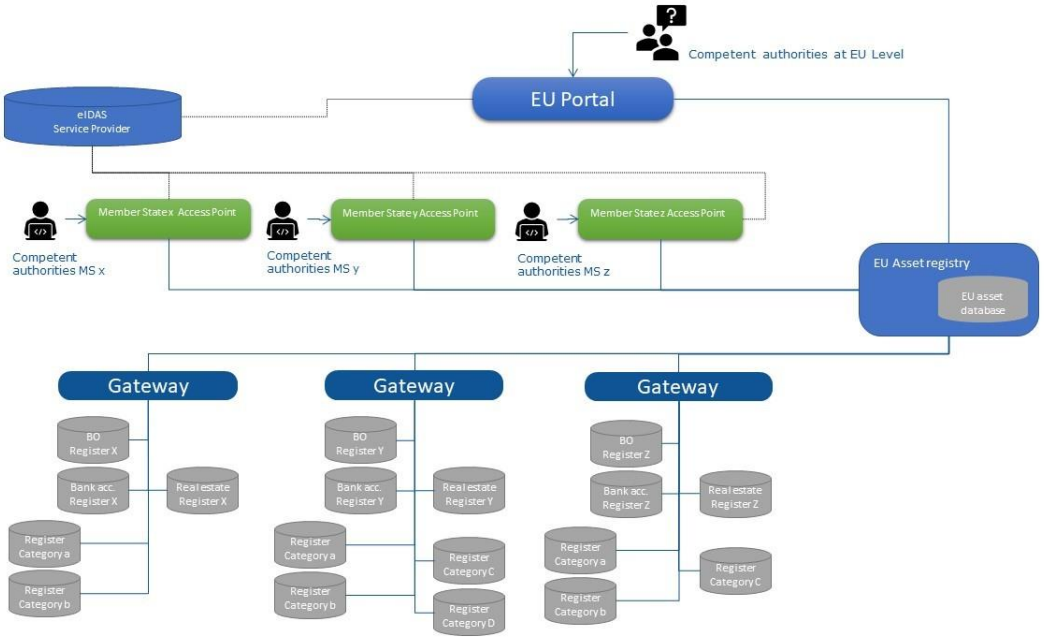
New categories that are not already covered by the Member States with their own registries should ideally be implemented at the level of the EU central registry. Member States should be allowed to implement their own registries following their own constraints but sharing at the same time a common technical infrastructure. This reuse of infrastructure and largely also of the maintenance resources is actually by far the most important advantage of this scenario. Yet this advantage comes as a two-edged sword, as it also brings with it the principle of centralisation of data that is seen as potentially dangerous from the perspective of privacy and security, and subsequently a sensitive aspect for Member States and at least a some of the public.

This scenario offers the necessary freedom to the Member States to implement new registries as they see fit at a later time. The experience of using the architecture and protocols set up at the central level could benefit any local implementation on the long term. Still, in the long-term as well, if there is not conclusive evidence that Member States agree on the terms of this scenario, deploying it comes with risks (functions can easily be suspended due to nonparticipation).

It must be noted that the centralised approach to data storage has had the least traction in EU-wide projects. Exceptions do exist (e.g. data.europa.eu, the official portal for European data;

the European Chemical Agency's central reporting for chemical substances) but their relevance for the subject of the present study is limited as they do not deal with the storage of asset ownership information. It should also be assumed that even in the ideal case of Member States agreeing to transfer their data to a central server as supposed by the scenario, the process would still be affected by factors like the delayed update of data, differences in the frequency of updates, inconsistencies in data or even data quality issues derived from their own source maintenance process. Overall, a major element affecting the feasibility of this scenario is the distance between the collection mechanisms and the final repository of data as well as the complexity of the process (both technical and legal) to move the data from the capture moment to the registry.

Figure 23: Central EU asset registry



Source: Authors' elaboration.

Such a scenario has the potential to facilitate a more rapid implementation of registries for domains that are not covered by some of the Member States. It would also ensure the interoperability at the level of data models between such registries, allowing for much easier exchanges of data.

Another aspect to consider is the fact that the centralisation of the registries could subsequently induce the consolidation of the reporting systems under a unified framework. In this process, data and information that were previously dispersed across multiple registries would be brought together and overseen by a central authority. From an operational perspective, this is a real advantage but would require a governance structure where the European Commission and the Member States agree to share the data controller function.

Table 34: Assessment of Scenario 3

Factor	Advantages	Disadvantages
Financial	The scenario benefits from the usual advantages derived from scale and centralisation. It is by far the most economically sound option from an infrastructure point of view Ensures an easy path for the creation of new registries, by eliminating the costs associated with the implementation	Certain more or less hidden costs (e.g. administrative, pre-processing) might be incurred in relation to operations that take place at the Member State level to support the centralisation
Compliance with the local regulations and legal framework	Easier alignment of the governance mechanism with the AML directives and overall EU regulations	Difficult implementation process due to probable Member State objections on regulations affecting domains of specific interest to them High risk of failure due to subjective factors like lack of awareness or trust Highly susceptible to regulatory hurdles due to the storage of sensitive data at a central level
Security		Increased security risk due to the 'high-value target' factor associated with a centralised location
Data quality and monitoring	More options available to resolve deficiencies in terms of consistency due to concentrated resources; Ability to more easily adapt to developments in terms of data structure and organisation Better integration between data and the functionalities provided would allow for a more efficient use of the existing data	Induces risks in relation to the ability to maintain the essential data qualities (complete and up to date) for the registries that are not using direct interconnection mechanisms to push data to the central repository A certain amount of detail could potentially be lost due to the uniformity of data (e.g. the reporting rules of a Member State with a tradition of operating ships might be different from those of a landlocked country. It would be difficult to impose the same standard on both Member States and as a result a relative uniformity might be applied, impacting quality) Monitoring mostly on the part of the European Commission rather than Member States, with far fewer available resources and much more difficult implementation
Service provided and availability	Centralisation enables the highest capabilities in terms of ensuring data and service availability; Significant speed advantages especially for complex queries due to centrally stored asset repositories enabling synchronised searches across domains and the building of interdependencies between records of different registries	Reluctance of Member States to provide the data and general regulatory hurdles might affect the ability to provide the services for specific, more sensitive areas
Integration and interoperability	Creates the most opportunities to develop true interoperability between Member States by agreeing on common technical and data architecture Better technical integration, facilitating a synchronised advancement of all	The loss of control of the data for the Member States will come with a certain loss of ability to control them in the long term Competence for managing and integrating data and processes at the

Factor	Advantages	Disadvantages
	Member States, particularly from the technology perspective	level of asset classes might be become an attribute found only at the EU central level
Extensibility and change management	Provides an easy path for developing new services around the available data (easier to improve and deploy new data models, with more testing opportunities by creating dedicated test environments closer in terms of data to the production environment, etc.) due to more direct access to a larger part of the data	

Source: Authors' elaboration.

6.4.2 Alternative option

6.4.2.1 Alternative scenario: Interconnection of existing and centralised asset registries (Scenario 2 + Scenario 3)

Considering the advantages and disadvantages of the previously described options and following from the conclusions of the operational assessment (see Section 6.2), this section sets out an alternative option where the advantages of Scenarios 2 and 3 are put together. The positive element of this option comes from the fact that different assets and operational situations at the Member State level could be addressed with different tools (e.g. enforcing the implementation of a more developed ship registry at the Member State level for a Member State that does not have a sufficient market for this asset might be operationally unfeasible).

In such a scenario, the connectivity would depend on the status of each registry at the time of implementation:

- the existing networks (e.g. BRIS, BORIS, EUCARIS and LRI) would be connected to a central EU portal thru dedicated gateways;
- all other existing registries found at the national level would be connected through a national gateway to the main EU asset registry or regularly transfer data dumps to the EU asset registry;
- for the non-existing registries, the Member States would have liberty to decide to either create them on the EU registry infrastructure or develop them locally as needed.

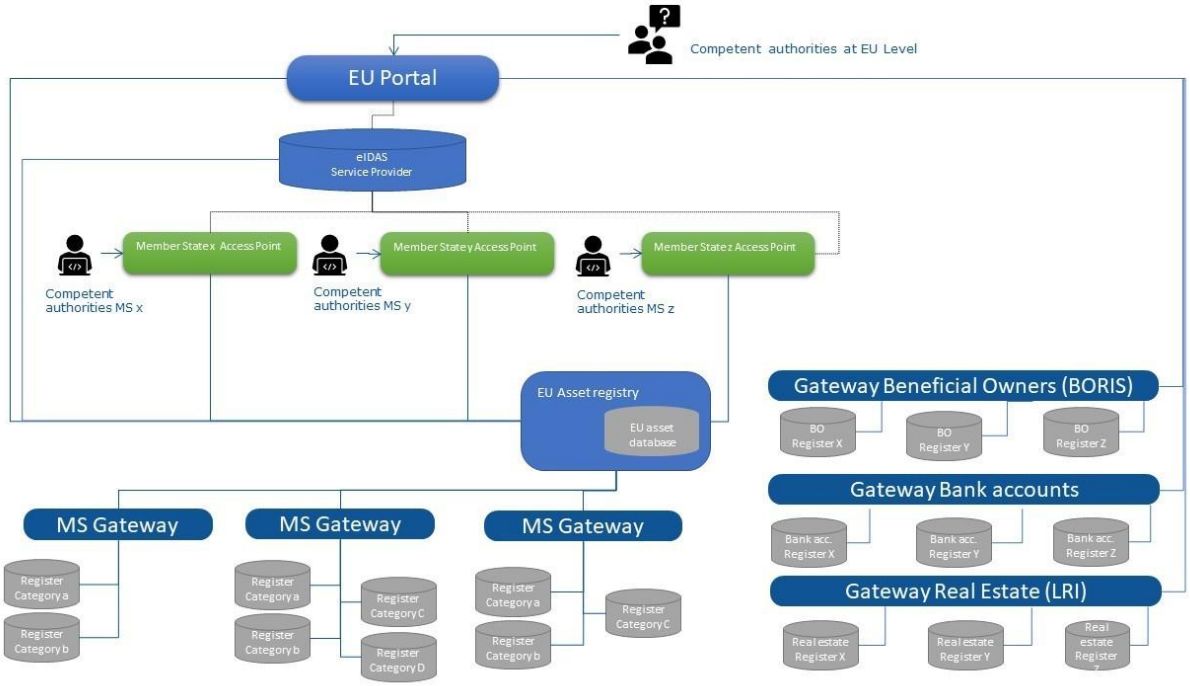
This scenario tries to accommodate the various stages of development of each Member State, registry and asset class at the time of implementation. It also offers the necessary flexibility for the Member States to decide on the best scenario for their registries.

Operationally, the provisions of this scenario to allow the simple integration of existing and to-be-interconnected asset registries (beneficial owners, LLCs, motor vehicles and hypothetically also the existing information on listed securities contained within ESMA's database) would alleviate the burden and costs considerably compared with scenarios where such interconnection first needs to occur. Additionally, by allowing newly created registries to be at the EU level only, no separate registries at the Member State level would need to be created for assets that have a relatively small number of reporting entities or entries as is. This represents another advantage above the remaining options from an operational and cost point of view. However, the remaining existing registries (properties, land where not part of the LRI,

ships, aircraft and firearms) would still require operational resources, as laid out in the operational feasibility assessment, to harmonise the existing data before they are transferred to the central EU asset registry.

While this scenario tries to fill the gaps that respectively affect Scenario 2 and Scenario 3, bringing in even more flexibility (it allows Member States to decide on the preferred option of connecting to or using a central registry), it fails to resolve issues related to data quality. We also must acknowledge that the higher diversity of mechanisms involved would increase the complexity of the project and the risks that come with that from an operational perspective.

Figure 24: Interconnector of existing and centralised asset registries



Source: Authors' elaboration.

6.5 Support functions to serve platform operations

Independent of the scenario chosen, some supporting functions have to be considered relevant for the operational costs. These are continuous activities, their costs and effectiveness, which affect the overall success of the project.

6.5.1 Standardisation and data modelling

Some of the most important benefits the platform would provide are the integration of the data used by means of data models that facilitate the retrieval, correlation and security of the information managed.

Thus, an essential element of the operational activity should be the formal activity of harmonising data by adhering to, or coordinating actions, in areas like:

- open standards associated with relevant domains or technologies,
- reference data vocabularies (ISA Core Vocabularies, Publications Office Authority Tables, ESTAT Classification, etc.) and domain ontologies.

6.5.2 Governance

The main objectives of the governance structure would be to deal with the following aspects:

- Manage change in the development of the various regulations involved (at the Member State and EU level).
- Ensure the application of GDPR regulations in the context of the platform activities.
- Ensure regulatory compliance and the definition of policies in accordance with the legal framework at the EU and Member State levels, including but not limited to security and access rights, notably,
  - the roles and functions of different organisations accessing the system, and
  - the right to access information across the different types of information (including intra- and inter-Member State) at both the organisational and user levels.

6.6 Assessment of costs and benefits

This section assesses the potential costs and benefits ([108]) of three scenarios under consideration:

- **Scenario 1** – creating new registers/databases at national level in addition to the existing ones (Beneficial ownership, bank account) for sources of wealth for which registers do not exist at national level, and their interconnection at European level to form an EU asset registry;
- **Scenario 2** –interconnecting only the existing national asset registries required by EU law (on beneficial ownership, bank accounts) and real estate registries towards achieving an EU asset registry;
- **Scenario 3** – establishing a centralised EU asset registry that connects the existing registries at the national level and, for assets or sources of wealth not covered by them, features a central EU registry/database with national access portals.

The first two scenarios do not envisage the creation of a centralised EU-level database or registry ([109]). These two scenarios would involve the interconnection ([110]) of national registries plus an EU-level platform and access point to retrieve the data from national registries. The third scenario, however, envisages the creation of a central EU-level asset registry or database which would store substantial amounts of information.

For the purpose of this cost-benefit analysis, Scenarios 1, 2 and 3 are evaluated. It is important to note that this analysis builds on the operational feasibility assessment conducted in Section 6.2. Therefore, only assets assessed as operationally feasible are considered due to the

([108]) Incremental costs and benefits relative to the status quo are considered.

([109]) A centralised database is a type of database in which all the data are stored in a single location, such as on a single server or in a single data centre.

([110]) An interconnection system refers to the set of technologies, protocols and standards that enable different systems and databases to communicate and share information with each other. Overall, the goal of an interconnection system is to enable different systems and databases to work together seamlessly, without requiring users to manually transfer data between them. By doing so, interconnection systems can help to increase efficiency, reduce errors, and improve collaboration.

reasons explored in Section 6.2. The exact assets accounted for under each scenario are further explored below.

6.6.1 Assessment of costs

For each of the scenarios, the following costs have been assessed:

- the costs for the EU budget as well as for Member States, and where possible, stakeholders responsible for reporting;
- one-off costs (initial set-up costs), as well as recurring costs (annual).

6.6.1.1 Scenario 1: Creating new registers/databases at national level in addition to the existing ones (Beneficial ownership, bank account) for sources of wealth for which registers do not exist at national level, and their interconnection at European level to form an EU asset registry

Implementation of Scenario 1 would involve:

- creating **new asset registries at the national level** for asset categories that are not already covered by existing registries. The operational assessment in Section 6.2 has found that it is not operationally feasible to create entirely new, comprehensive and complete asset registries for any of the asset categories that currently have no established registries. These categories are therefore not included in the assessment of this scenario: cash in hand, listed securities, unlisted securities, crypto assets, in-game assets, works of art, collectors' items, antiques, jewellery, safe deposit boxes at non-banks and commodities. In terms of the comparative assessment between Scenarios 1 and 3 the operational costs would not differ in any case and would largely be the same irrespective of whether data is submitted to a national central registry under Scenario 1 or an EU registry under Scenario 3 as the same data would need to be collected and reported by intermediaries. It should be noted that the operational costs of establishing new registers, that is for intermediaries to submit data to a register, would exceed the IT cost of a setting up a register. For example, under DAC2, and at international level the OECD Common Reporting Standard (the 'CRS'), costs for financial intermediaries far exceeded that IT costs for Member States for setting up reporting systems of data between Member States. The operating factors for establishing new asset registers are discussed under Section 6.2.2.
- establishing a **centralised (national) AML/CFT/tax evasion registry in every Member State**. Depending on the availability of information in the existing national registers, the category-based registries may need to be harmonised and connected to the central AML/TE registry which would be directly accessible to national FIUs and other national competent authorities, such as tax authorities, and asset recovery authorities. The Study does not provide an estimate of the costs of possible harmonisation of these existing national asset registers, and whether existing information could be used in a standard format to report to the central registry which would not require harmonisation.
- developing an **EU-level infrastructure** that provides FIUs and other competent authorities at the EU level with access to all the national AML registries to which they are entitled. Similar to the approach adopted for BRIS and BORIS, the EU-level infrastructure would consist of two components,
  - a European central platform (ECP) – the engine that would allow the flow of data from national registries, and
  - a European access point (EAP) providing a user interface.

These assumptions are also relevant for Scenarios 2 and 3. For comparative purposes these elements would be the same for all Scenarios.

To assess the feasibility of Scenario 1, it is first important to review the current situation regarding **asset registries at the national level.**

As explored earlier in this section and throughout the feasibility assessment (and as summarised in Table 35), **national registries have been established for nine asset categories** across the Member States.

For these asset categories, under Scenario 1, the key requirement and therefore cost for the relevant authorities within Member States (as well as reporting stakeholders) is harmonisation of the data to enable the interconnection of these registries.

As shown in the table below, **existing data connection efforts are already in motion for six categories.** These are cash at bank, LLCs, properties, land, motor vehicles and safe deposit boxes at banks. This makes these categories, in theory, the easiest to connect under Scenario 1. However, for some of these categories further considerations must be accounted for, such as the completeness and appropriateness of the data for AML/tax evasion monitoring.

For three further categories, *ships, aircraft* and *firearms,* **interconnection has been assessed as feasible.** But gaps in the data would also need to be addressed before interconnection could occur, particularly for ships and aircraft, for which asset values are often missing.

Table 35: Changes required at the national level under Scenario 1

	Current situation	Necessary changes	
Asset	**Are there existing national registries?**	**Feasible for Interconnection**	**Harmonisation efforts needed**
Cash at bank	Yes	Yes (AML framework)	No
Cash in hand	No	No	
Listed securities	No*	No	
Unlisted securities	No	No	
Limited liabilities companies	Yes	Yes (BRIS)	No
Crypto assets	No	No	
In-game assets	No	No	
Land	Yes	Yes (LRI)	Yes
Real estate, other than land	Yes	Yes (LRI)	Yes
Motor vehicles	Yes	Yes (EUCARIS platform)	Yes
Ships	Yes	Yes	Yes
Aircraft	Yes	Yes	Yes
Firearms	Yes	Yes	Yes
Works of art	No*	No	
Collectors' items	No*	No	
Antiques	No*	No	
Jewellery	No*	No	

Safe deposit boxes at banks	Yes	Yes (AML framework)	No
Safe deposit boxes at non-banks	No*	No	
Commodities (precious raw materials)	No*	No	

* Coverage identified in private registries – see the Task 2 mapping exercise.

Source: Authors' elaboration.

6.6.1.1.1 Cost breakdown of implementation

In terms of costs, the implementation of Scenario 1 would involve costs for parties as outlined below.

- The competent authorities of the EU and Member States ([111]) would bear two types of costs,
  - **one-off** costs to set-up the above infrastructure, and
  - **recurring (ongoing) costs** for the maintenance and use of the EU portal, and the central AML registries at the national level.
- Stakeholders would bear incremental costs for reporting to the asset registries.

Considering first the former parties, Table 36 provides a detailed breakdown of the anticipated cost categories and items that authorities at the EU and Member State levels would face.

([111]) National costs would vary among Member States depending *inter alia* on (i) the number of new registries to be created, (ii) the quality and scope of existing registries, (iii) level of centralisation of data, (iv) level of digitalisation and (v) labour costs.

Table 36: Types of costs expected under Scenario 1

Cost level	Cost type	Cost category	Cost item	Initial set-up costs	Ongoing costs
EU level	Cost of setting up and running the EU level infrastructure	IT costs	Work on system design and development of detailed functional and technical specifications for the EU infrastructure (ECP + EAP)	x	
			Security analysis of system requirements	x	
			System development and testing	x	
			Communication with Member States (e.g. technical workshops, setting up a communication platform)	x	
			Project management	x	
			Software hosting		x
			Maintenance, updates and bug-fixes		x
			IT helpdesk		x
		Governance and oversight	Cost of setting up and running an EU-level steering committee	x	x
		Operational/administrative	HR costs: cost of dedicated staff managing the EU-level portal and gateway	x	x
National level	Cost of standardising most relevant data in existing asset registries	IT costs	Changes to existing asset registries to facilitate connection with national gateway	x	
			Changes to existing asset registries to ensure a minimum level of harmonisation/standardisation of data formats and structures across the EU including any data-cleaning efforts	x	
	Costs of setting up and running a national AML/TE registry and gateway	IT costs	Development of detailed technical specifications for a new national AML/TE registry and gateway including API and protocols required for communication with EAP	x	
			Acquisition of requisite hardware and software	x	
			System development and testing		
			Project management	x	
			Software + data hosting		x
			Maintenance, updates and bug-fixes		x

Cost level	Cost type	Cost category	Cost item	Initial set-up costs	Ongoing costs
			IT support (helpdesk)	x	x
		Operational/administrative	HR costs: cost of dedicated staff managing the national AML registry	x	x
			Training of officials (FIUs, LEAs, tax authorities, etc.) in use of the new AML registry	x	x
			Development and updates of user manuals and procedure manuals	x	x
	Cost of introducing legislative changes	Legal cost	Legal advice and support to establish the legal basis for the national AML registry + gateway and new category-based asset registries	x	
			Legal advice and support to ensure compliance with relevant legislation, e.g. the GDPR	x	x

Source: Authors' elaboration.

6.6.1.1.2 Orders of magnitude: EU-level costs

As a starting point, the study looks at the costs incurred by the Commission to develop other relevant interconnection systems. Table 37 provides an overview of the costs incurred by the Commission in the development of BRIS.

Table 37: Cost of implementing BRIS

Cost component	Cost (€)	Period over which cost incurred
Cost of a European Central Platform		
Preliminary work on system design + technical specification	153 500	2012-2013
Further development of technical specification	911 939	2013-2015
Continuation of work on technical specification	762 457	2014
Preliminary work to ensure security of the system	200 000	2015
Security analysis of system requirements	5 000 000	2016-2019
Communication with Member States (technical workshops, communication platform)		
Development of BRIS 2.0 – preliminary work		
Setting up IT environments for software development		
Maintenance and bug fixing		
Software hosting		
Drafting and updating specifications		
Development and testing of BRIS 2.0	1 000 000	
Total cost of setting up + transition to BRIS 2.0	**8 027 896**	**2012-2019**
Maintenance costs	**1 000 000**	**2017-2019**
Cost of a European access point		
Implementation cost	125 082	2015
Maintenance cost	N/a	since 2019
Helpdesk	30 000	

Sources: European Commission (2023a), Report on the technical and financial aspects of the Business Registers Interconnection System (BRIS) Directive (EU) 2012/17, SWD(2023) 79 final and authors' elaboration.

Further information provided by DG JUST and DG DIGIT illuminate costs:

- The initial costs (development of the technical specification and implementation) for launching the first version of BORIS were in the order of EUR 1.6 million (EUR 1 million incurred by DG DIGIT and EUR 600 000 by DG JUST), while the annual running costs for both BORIS and BRIS together are around EUR 300 000.
- The cost of the interconnection systems/registries operated by the EU vary widely, but one could take as a reference EUR 500 000–1 000 000 for the development of an EU-level interconnection system and around EUR 150 000 (annually) for maintenance.

Table 38 provides the recent costs associated with the three most relevant EU-level interconnection systems.

Table 38: Cost for implementing EU-level interconnection systems

IT system and cost type	2021	2022
IRIS (Insolvency registers interconnection system) – costs representing largely the work for maintenance and new Member State onboarding*	€65 000	€81 000
BRIS (Business Registers Interconnection System) – costs mainly relating to development of new functionalities**	€1 200 000	€1 200 000
BORIS EAP (European Access Point for the Beneficial Ownership Registers Interconnection System) – costs largely related to improvements, maintenance and onboarding of new Member States	€150 000	€180 000

* The setting up of the central search pilot system (IRI 1.0) cost approximately EUR 280 000, whereas its adaptation towards the establishment of IRI 2.0 is estimated to amount to approximately EUR 170 000.

** According to data published by the Commission, the costs for the development of the first version of BRIS amounted to approximately. EUR 1 700 000 ([112]).

Source: Interview with DG JUST and DG DIGIT.

Based on the above information, the orders of magnitude for implementing Scenario 1 are estimated as shown in Table 39.

Table 39: Anticipated costs for developing an EU-level asset registry interconnection system

Cost component	Cost (€)
System design & technical specification (ECP + EAS) – one-off	1.5 m to 2.5 m
Annual maintenance costs*	Around 150 000
Helpdesk – annual	30 000 to 35 000

* These would inter alia depend on which hosting option is used: cloud hosting (EUR 15 000) or data centre hosting (EUR 50 000), annually.

Source: Authors' elaboration.

Additionally, the expected time dedicated to the tasks the Commission would need to carry out are in Table 40.

Table 40: Anticipated staff effort at the European Commission level

Task	# Full-time equivalent	DG responsible
Business management of the system and overall responsibility	1.5	FISMA
IT project management: ECP system development and maintenance	1	DIGIT
IT project management: EAP system development and maintenance	0.3	FISMA

Source: Based on BRIS.

([112]) On the interconnection of bank account registries, see Annex 7 in SWD(2021) 190 final – impact assessment on the proposed new anti-money laundering package.

6.6.1.1.3 Orders of magnitude: Member State costs

As previously indicated, implementing Scenario 1 would involve the following costs for Member States:

- upgrading/adapting existing asset registries to ensure a minimum level of data completeness and standardisation;
- setting up a centralised (national) AML/tax evasion registry connecting all category-based asset registries;
- developing an interface linking the centralised AML/tax evasion registry with the EU-level platform (ECP).

6.6.1.1.4 IT costs

It is not possible to precisely determine the cost of implementing Scenario 1 at the national level given the restricted scope of this study, as it would involve the extent to which all existing registries need to be upgraded or adapted.

However, data have been extracted from existing sources to provide an indication of the order of magnitude of the costs that Member States may incur in implementing Scenario 1 (see box below). Importantly, the costs would vary significantly by Member State and depend on factors such as the scale of the gaps to be filled, the current level of centralisation of asset registries and the complexity of centralising asset ownership data where it has not yet been done for this information. Other cost factors include the level of digitalisation of existing registries, labour costs, the volume of data and the complexity/degree of automation of the technical solution that the Member States chooses.

Available estimates of costs for setting up asset registries and interconnections at the national level

A recent study on the interconnection of real estate registries provides **cost estimates for harmonisation and interconnection** at the national level.

These include the costs for infrastructure, system development and integration, system maintenance and operation, and business. For Scenario 2 in Table 41, additional costs associated with harmonising the information include those for design (e.g. identifying missing information) and data collection (e.g. collection & maintenance of missing information) ([113]).

Table 41: Estimated costs of harmonisation and interconnection of real estate registries – Member State level

Scenario		One-off costs	Recurring costs
Scenario 1 – Interconnection refers to the cost estimate to interconnect national real	Range	€81 774 to €272 580	€47 973 to €159 911
	Median costs	€136 290	€79 956

([113]) This report does not explicitly indicate whether the financial burden of each activity falls on the state or the stakeholder.

estate registries/data retrieval systems at the EU level	Total (27 Member States)	€4.08 m	€2.04 m
Scenario 2 – Interconnection and harmonisation refer to the cost estimate to interconnect national real estate registries/data retrieval systems at the EU level and the cost estimate to harmonise the information contained in those real estate registries/data retrieval systems	Range	€95 844 to €319 480	€52 173 to €173 911
	Median costs	€159 740	€186 956
	Total (27 Member States)	€4.81 m	€2.61 m

* Total figures are based on an estimation of each scenario for each Member State considering effort levels involved Scenario 2 additionally includes design costs and data collection costs.

Source: Deloitte Consulting Advisory (2021). Member State-level costs include: (i) infrastructure costs, (ii) system development and integration costs and (iii) business costs.

Cost estimates for BRIS ([114])

Introduction of the system for interconnecting registries required Member States to make the necessary adaptations on their side consisting, in particular, of the development of an interface linking each registry to the platform. To interconnect their business registries to BRIS, Member States developed their IT systems in accordance with the technical specifications published by the Commission.

Initial cost to set up the interconnection

According to the information provided to the Commission by 16 Member State/EEA countries, the budget that business registries have spent to set up the initial interconnection to BRIS varied considerably. Among the 20 countries that replied to this point of the questionnaire, the average cost of setting up such an initial connection to BRIS amounted to EUR 355 718, with the estimated amounts ranging from EUR 43 306 to EUR 1 240 000.

Employees working on the BRIS interconnection

In order to interconnect their business registries to BRIS, Member States had to dedicate personnel to manage the system. Overall, 15 Member States and EEA countries provided the Commission with information on the personnel dedicated to BRIS during the reporting period. Based on the information received, those Member States and EEA countries overall allocated an average of 1.16 full-time equivalents (FTEs) per year for the IT and policy management of BRIS, with average FTEs per country ranging between 0.3 FTEs and 3 FTEs per year.

Insolvency registries interconnection system (IRI)

The setting up of the central search pilot system (IRI 1.0) cost approximately EUR 280 000, whereas its adaptation towards the establishment of IRI 2.0 is estimated to amount to approximately EUR 170 000 ([115]).

Source: Authors' elaboration.

([114]) See the Report on the technical and financial aspects of the Business Registers Interconnection System (BRIS) Directive (EU) 2012/17, European Commission (2023a).

([115]) See the proposal as regards access of competent authorities to centralised bank account registries through the single access point, European Commission (2021c).

On the basis of these wide-ranging figures, this study presents some orders of magnitude of the potential costs of harmonising and interconnecting asset registries at the national level (see Table 42). Note that the table does not include the cost of developing registries already envisaged by legislative measures, as these would not constitute incremental costs for Member States.

The figures presented below should be treated with extreme caution for several reasons. First, there are several unknowns at this stage, most notably the technical and functional specifications of the national registries, the volume of data that would be covered by each asset category, the sources of data, etc. Second, the cost estimates are based on available data relating to existing databases whereas in reality, every database would have unique features (and would thus entail unique challenges and costs). Third, not all Member States would be at the minimum end of the range and likewise, not all would be at the maximum end of the range.

6.6.1.1.5 Costs for stakeholders

In addition to the costs borne by authorities, the harmonisation and interconnection of asset registries under Scenario 1 may lead to additional costs for stakeholders or reporting entities. These costs would be associated with the potential need to provide further information to an asset registry or update reporting functions. However, as no new asset registries would need to be created, these costs are expected to be low, as in most cases, reporting stakeholders would have the same or slightly increased reporting responsibilities.

Table 42: Cost estimates for Scenario 1 – Member State level

Cost item		Average costs	Number of asset categories	Number of Member States	Costs (million)
Connecting current asset registries to the national AML/tax evasion registry (LLCs, bank accounts and safe deposit boxes at banks)	Min	€80 000	2	27	€4.32
	Max	€280 000			€15.12
Harmonising and connecting asset registries to the national AML/tax evasion registry (land and real estate, ships, aircraft, firearms and vehicles)	Min	€95 000	5	27	€12.83
	Max	€320 000			€43.2
Cost of centralised data retrieval system or national AML/tax evasion registry	Min	€1 200 000	N/a	27	€32.40
	Max	€1 500 000			€40.50
Total one-off costs	Min			27	€49.55
	Max				€98.82
Recurring cost (annual)	Min	10 %		27	€4.95
	Max	50 %			€49.41

* The basic cost for establishment of a direct connection to a system varies between EUR 5 000 and EUR 30 000 per system.

** Although motor vehicles already have an interconnected system, analysis has shown significant gaps in the data relating to asset values, and therefore these asset registries are considered to need further harmonisation.

Source: Authors' elaboration.

6.6.1.2 Scenario 2: Interconnecting only the existing national registries required under EU law (beneficial ownership, bank account) and real estate registers with an EU asset registry

Implementation of Scenario 2 would involve the establishment of:

- an EU-level interconnection of bank accounts;
- an EU-level interconnection of national real estate registries (land and other real estate);
- an EU-level interconnection of (modified) national real estate registries – additional data collection/ harmonisation – to allow EU level interconnection.
- a gateway that links EU-level interconnections (for the three asset categories) with an EU asset registry (note that the EU-level interconnection of business ownership data already exists through BORIS);
- an EU-level data retrieval system or a central 'hub' that contains all the information necessary to be able to conduct searches. This is called the 'EU asset registry', but it only serves to conduct searches in the other registries. This is absolutely different to the 'EU asset registry' referred to in Scenarios 3 and 4. There, the idea is that it actually stores all the data for assets, whereas here it only stores centrally the data necessary to be able to conduct searches in the other registries;
- a dedicated EU portal which would allow access to national and EU-level competent authorities with access to the above data.

6.6.1.2.1 Costs at the national level

The main incremental costs at the Member State level would stem from the harmonisation and interconnection of real estate registries and bank account registries at the national level (beneficial ownership databases already have EU-level interconnection via BORIS). The costs for harmonisation and interconnection of real estate registries are presented above in Table 41 (one-off costs estimated to be in the order of EUR 4.81 million and annual costs around EUR 2.61 million).

On harmonisation and interconnection of bank account registries, these costs have previously been considered in the Commission's impact assessment of the AML package ([116]) (but this document does not provide proper cost estimates or any indications of whether costs related to harmonisation and standardisation need to be incurred). In our understanding, connecting an existing national registry to an EU database would primarily entail integration costs. The factors that determine these costs include:

- *Middleware and integration*. If the national database and the EU database use common protocols or have built-in capabilities for external connections, the cost could be relatively low. However, if custom middleware or adaptors are needed, costs could rise.

([116]) See SWD(2021) 190 final – Annex 7 on the interconnection of bank account registries impact assessment on the proposed new anti-money laundering package.

- *Security protocols*. Ensuring data safety during transmission is crucial. It might be necessary to implement security protocols and encryption, and potentially to set up virtual private networks or dedicated lines.
- *Data mapping and transformation*. Even if both databases are 'ready' to connect, their data structures might differ. Some work might be required to map data elements of the national database on the equivalent elements in the EU database.
- *Testing and validation*. Before making any connection live, rigorous testing is essential. This process might result in additional costs or modifications.
- *Documentation and compliance*. Especially when dealing with EU databases, ensuring compliance with the GDPR and other EU data directives might incur additional costs.
- *Maintenance and monitoring*. There would be recurring costs for monitoring the connection, handling errors and updating the integration as necessary.

Given the complexities of databases, especially at the national and EU levels, and based on previous integration projects, the cost to interconnect national registries at the EU level is estimated to lie in the range of EUR 80 000 to EUR 280 000, and EUR 95 000 to EUR 320 000 where harmonisation is also required.

6.6.1.2.2 Costs for stakeholders

It is important to note that under this scenario, no significant new data collection and reporting costs are envisaged. Additional reporting costs would mainly relate to gap filling with respect to land and real estate registries.

6.6.1.2.3 EU-level costs

The incremental costs that are expected to occur at the EU level are presented in Table 43. It is assumed under this scenario that the EU would incur the cost of setting up technical interfaces/gateways for integrating the most relevant data from BORIS and national registries for real estate and bank accounts. In terms of orders of magnitude, the costs are likely to be slightly lower or higher than for Scenario 1 (see Table 39), depending on whether interconnection is established with three EU-level systems (BORIS and eventually, EU-level systems for bank accounts and real estate) or whether interconnection would need be established with BORIS, 27 national registries for bank accounts and 27 national registries for real estate.

Table 43: Types of costs expected under Scenario 2

Cost level	Cost type	Cost category	Cost item	One-off costs	Ongoing costs
EU	Cost of setting up and running the EU asset registry and portal	IT costs	Development of detailed functional and technical specifications for the EU infrastructure (ECP + EAP)	x	
			Security analysis of system requirements	x	
			System development and testing	x	
			Communication with Member States (e.g. technical workshops, setting up a communication platform)	x	

			Project management	x	
			Software hosting		x
			Maintenance, updates and bug-fixes		x
			IT helpdesk	x	x
		Governance and oversight	Setting up and running an EU-level steering committee	x	
		Operational/administrative	HR costs: cost of dedicated staff managing the EU asset registry and portal	x	x

Source: Authors' elaboration.

6.6.1.3 Scenario 3: Establishing a central EU asset registry connecting the existing registers at national level and, for sources of wealth not covered by them, featuring a central EU register/database with national access portals.

Implementation of Scenario 3 would involve the following actions:

- development of an EU level asset registry that
  - includes data collected by existing national registries;
  - captures data for asset categories not already covered. Similar to Scenario 1, this assessment accounts for only those assets for which the creation of an EU asset registry is deemed operationally feasible. Therefore, the following asset categories are not included: cash in hand, listed securities, unlisted securities, crypto assets, in-game assets, works of art, collectors' items, antiques, jewellery, safe deposit boxes at non-banks and commodities;
- development of an EU user interface.

The assets that would be covered by the EU level registry are therefore the same as under Scenario 1.

Table 44 outlines the costs likely to be associated with this scenario. These cost estimates are based on the following assumptions and considerations:

- **Scale and complexity.** Building a centralised database for the entire EU, catering to 27 Member States, would be a massive project, demanding top-tier infrastructure and technical capabilities.
- **Historical data.** The costs listed in the present analysis are based on data gathered from EU reports, existing procurement procedures and interviews with vendors and technical experts correlated with desk analysis and benchmarking techniques. Interviews and desk research performed in Tasks 1, 2 and 4 have gathered information about projects that could be considered relevant for the analysis. While some of the existing studies ([117]) do make reference to platforms and provide a set of budgets associated with the set-up of already operating platforms that are partially similar in scope (e.g. BRIS and BORIS), it is obvious that the respective references

([117]) European Commission (2019b).

do not entirely correspond to the requirements of the platform that is assessed here. Given the complexity and novelty of the project, the costs have had to be evaluated based on extrapolation of existing data and benchmarking with known systems.

- **Integration with national databases.** Given that existing national databases might be using different standards and architecture, integration could be complex and thus costly.
- **Security and compliance**. Given the sensitive nature of the data, top-level security measures, encryption and compliance with EU data protection regulations (like the GDPR) are factored in.
- **User accessibility.** The system needs to be accessible to various authorities across different Member States. Costs related to interface development, translation, authentication and authorisation have been taken into account.

A number of assumptions have been made in order to simplify the estimations, as follows:

- Major registries of the Member States and partner platforms covering specific sectors (e.g. land registries) would cooperate in establishing communication and further alignment with the gateway functions of the platform.
- The timeline envisaged includes 1 year for the procurement (outside the scope of the cost estimation), 2 years for development and testing, and 3 years of operations. It should be noted that a proof of concept could be set up in a much shorter timeframe under the proviso that specific conditions are met (sample data availability, corresponding platform already in place, etc.).
- For the given timeframe, of a maximum of 5 years, no upgrade has been considered for the technology. Only corrective maintenance costs and standard software upgrades have been included in the process.

Table 44: Costs for implementation and maintenance of a centralised EU system

Cost category		Cost item (€)	Estimated range (€)
Infrastructure & set-up	Primary servers	2 000 000	2 200 000
	Backup and disaster recovery servers	1 500 000	1 650 000
Storage	High speed SSDs, backups, archiving	1 500 000	1 650 000
	Networking equipment & bandwidth	1 500 000	1 650 000
Database development	Design, development & testing	3 500 000	3 850 000
	Integration with existing national databases: land/real estate registries	2 561 483	8 268 275
	Integration with bank accounts/safe deposit boxes	2 160 000	7 360 200
	Integration with existing national databases: firearms, aircraft, ships	2 160 000	7 360 200
	Harmonisation/filling data gaps of the motor vehicle registry (50 % of total harmonisation costs)	189 900	633 200
	Security software and services	1 000 000	
	EAP and ECP	1 500 000	2 500 000

	Authorisation & authentication systems for 27 Member States	1 200 000	1 320 000
	Training modules – design & delivery	500 000	750 000
Other costs	Project management	1 500 000	2 000 000
	Legal & compliance	1 000 000	1 500 000
	Contingency	2 000 000	2 200 000
Total one-off costs		27 448 300	50 816 500
Annual recurring costs (10 % of one-off costs)		2 744 800	5 081 650

Notes: Extra costs associated with the harmonisation and standardisation of data are derived from the difference between the cost to interconnect national registries at the EU level, which lie in the range of EUR 80 000 to EUR 280 000 and EUR 95 000 to EUR 320 000 where harmonisation is also required. For Scenario 3, 50 % of the harmonisation costs are expected to apply to Member States while 50 % are borne by the EU. These figures are an estimation and would require more in-depth analysis of the individual registries and division of tasks.

Source: Authors' elaboration.

As shown in the table above, the total cost for implementation and operation of the system on a timeframe of 5 years is estimated at approximatively EUR 26 million to EUR 47 million, with an annual ongoing cost of EUR 2.6 million to EUR 4.7 million.

6.6.1.3.1 Member State costs

In terms of the costs for Member States under Scenario 3, these are projected to be lower than for Scenario 1. This is because much of the harmonisation burden is expected to fall on the EU. In the context of Scenario 3, in line with a study regarding the harmonisation of real estate registries, harmonisation would mean adapting the data to the level deemed necessary ([118]), in this instance for a centralised EU registry.

This point is particularly relevant due to the difference in the quality or uniformity of data within existing registries, for instance reporting rules may be different for ships in a landlocked country than in a country with a coastline. Although the EU would bear a burden of moderating the harmonisation of data at the EU level, Member States would still be responsible for monitoring and ensuring overall data quality for each asset category. While it is difficult to determine the exact breakdown of harmonisation costs, for the purpose of this study the proportion of harmonisation costs for Scenario 3 are determined to be 50 % of the estimated total (between EUR 664 000 and EUR 2.2 million in one-off costs, plus between EUR 20 000 and EUR 661 000 in recurring annual costs). Still, these figures should be treated with caution as the precise breakdown would be dependent on many factors and would vary significantly among Member States.

6.6.1.3.2 Stakeholder costs

It is important to highlight that beyond the IT infrastructure expenses, additional costs associated with increased reporting may also be incurred by reporting entities (such as companies and individuals). While these costs are not expected to be large, given that under

([118]) European Commission (2021c), https://data.europa.eu/doi/10.2874/068191

this scenario the study solely considers those assets that already have established registries, the reporting obligations would only increase when filling gaps in the data on an ongoing basis entailed an obligation on top of current reporting expectations.

An example of the costs associated with reporting to an EU centralised system are those stemming from the EU's Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) Regulation, which requires chemical substances of a certain size to be registered with the European Chemicals Agency. In this instance, the costs arise on a per registration basis, with companies most frequently incurring costs in the range EUR 1 000 to 10 000 for all tonnage bands of chemicals. ([119])

Overall, given the projected small amount of increased reporting for these specific assets, feeding data into one centralised EU registry is assumed to result in little additional cost burden for reporting stakeholders.

6.6.2 Comparative overview of costs

Table 45 provides a comparative overview of the costs of implementing each of the three scenarios. The main takeaways are as follows:

- **Scenario 2 is the least costly** as it involves the interconnection of primarily established registries. It would involve interconnecting existing EU/national asset registries and providing access through one European single access point. As this scenario does not involve any new registries to be developed the incremental burden on reporting stakeholders would also be low, only requiring the cost involved in filling data gaps and harmonisation.
- **Scenario 1 is expected to be the costliest** compared to Scenario 3 as new central asset registers would need to be established at national level for each Member State under Scenario 1 whereas under Scenario 3 one asset register could be established at EU level which would gather data from the specific national asset registers which, resulting in economies of scale**.**

Table 45: Comparative overview of the costs of implementing the three scenarios

([119]) Kastalie et al. (2021).

Criteria	Scenario 1	Scenario 2	Scenario 3
EU level	Relatively low Initial: €1.5 m to €2.5 m Ongoing: up to €200 000 per year	Slightly higher or lower compared with Scenario 1	Initial: €27 m to €50 m Annual ongoing cost of approximatively €2.7 m to €5.8 m
Member State level	Initial: €49.5 m to €98.8 m (across all asset categories with gaps and all Member States) Ongoing: €4.9 m to €41 m	Initial: approximately €10 m Ongoing: €5 m	No substantial costs expected. Member States will share in some of the harmonisation costs. Initial cost of approximately €664 000 to €1.1 m (120)
Costs for stakeholders	Few additional costs envisaged	Few additional costs envisaged	Few additional costs envisaged

Source: Authors' elaboration.

6.6.3 Assessment of benefits

In September 2017, Europol warned that between 0.71 % and 1.28 % of the EU's annual GDP is 'detected as being involved in suspect financial activity'. In 2019 alone, this amounted to a value of between EUR 117 billion and EUR 210 billion of suspicious activities and transactions occurring through the EU's financial system and economy. Only a minor share of these suspect transactions and activities are detected, with about 2 % of assets seized, and only 1 % ultimately confiscated. This allows criminals to invest in expanding their criminal activities and, ultimately, provides them with a greater capacity to infiltrate the legal economy ([121]).

The rationale for setting up an EU-level interconnection system under Scenario 1, or a centralised EU asset registry under Scenario 3, is based on the premise that such an approach will deliver the following benefits:

- **Cost efficiencies for competent authorities** in their fight against illicit financial flows, as a unified repository means quicker cross-referencing and validation. Authorities could trace criminal activities or suspicious transactions faster. For example, if, on average, it takes 5 days for law enforcement agencies to gather data from disparate sources and a centralised database reduces that to just 1 day, there's an 80 % reduction in time spent on data gathering alone.
- **Streamlined processes** would lessen the need for multiple requests to different agencies or cross-border entities, reducing paperwork and manpower.
- **Improvements to cross-border flows of information and cooperation** between national competent authorities would enable them to swiftly identify, track, freeze or seize relevant assets.

([120]) These costs may be lower, as a centralised EU registry would enable data deficiencies and discrepancies to be resolved more easily due to concentrated resources, but this cannot currently be quantified.

([121]) SWD (2021) 190 final – Impact Assessment accompanying the anti-money laundering package.

- **Checks on the correctness and completeness of data in national asset registries** through the process of interconnection would help to improve these aspects in each registry. For example, company data input in real estate/property registries would be checked for coherence with the information input in the beneficial ownership and company registries for the same entity. Discrepancies in the registries could be followed up for risk assessment purposes.
- **The imposition of sanctions would be faster** through immediate access to relevant data, so authorities could act more quickly.
- **A centralised database would allow for the immediate identification and freezing of assets**, ensuring they would not be moved or liquidated.

This is in turn, would contribute to:

- greater tax revenues (recovery of unpaid taxes),
- enhanced asset seizures that could be used for the common good,
- a reduction in the scale of tax evasion/money laundering, as well as potentially other financial crimes due to a deterrence effect and preventative actions.

The time and effort currently required by different competent authorities to identify, track, freeze or seize assets is difficult to assess. Data that would be required to make this assessment are not readily available, as most of them are highly likely to be confidential. These effects are therefore qualitatively assessed in this section. An attempt has been made, however, to provide orders of magnitude of potential benefits in terms of the size of asset bases of the asset categories in Table 13.

To distinguish between the three scenarios, it is first useful to get a sense of asset coverage under the different scenarios (see Table 46). It has been possible to estimate amounts for most financial asset categories, although value amounts are not available for many non-financial asset categories, like vehicles. Instead, the actual quantity of assets have been estimated.

Table 46: Available estimates of asset class sizes

Sub-category	Indicator	Value in bn € or number of assets/transactions	Geography	Data source	Covered in Scenario 1	Covered in Scenario 2	Covered in Scenario 3
Cash in hand	Information not available						
Cash at bank	The total deposits held by monetary financial institutions in the euro area from non-financial companies and households	€15 000	Euro area	ECB (2021)	x	x	x
Listed securities	On-exchange trading volume of equity, bonds and bonds-like instruments in 2020	Approx. €13 300	EEA30	ESMA (2021)			
Unlisted securities	On-exchange trading volume of equity, bonds and bonds-like instruments in 2020	Approx. €13 400	EEA30	ESMA (2021)			
Limited liability companies		According to the European Commission, currently almost 26 000 000 of these enterprises are active across the EU	EU	EC (2019c)	x		x
Crypto assets	The market capitalisation of the largest cryptocurrencies	Approx. €2 500	Euro area	ECB (2022)			
In-game assets	40 % of video game revenues in 2021 (122)	€9 300	EU				
Real estate	• Real estate held in cross-border ownership across the EU in 2018 • Value of real estate market in Europe	• €1 400 • Projected to reach US$166 800 billion in 2024	• EU • Europe (123)	• Ecorys (2021) • Real Estate -	x	x	x

(122) Video games Europe (2019), https://videogameseurope.eu/wp-content/uploads/2019/08/ISFE-Key-Facts-Brochure-FINAL.pdf

(123) Europe consists of: Albania, Armenia, Austria, Azerbaijan, Belarus, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Georgia, Germany, Greece, Hungary, Iceland, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, Moldova, Montenegro, Netherlands, North Macedonia, Norway, Poland, Portugal, Romania, Russia, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Turkey, Ukraine, United Kingdom.

Land				Europe | Statista Market Forecast			
Motor vehicles	The number of EU-registered passenger cars in 2021	253 000 000	EU27	Eurostat (2023)	x		x
Ships	Number of ships and fishing vessels registered in EU Member States in 2021	• 18 293 ships registered in EU Member States (2021) • 75 000 fishing vessels registered in EU Member States	EU	ESMA	x		x
Aircraft	Fleet of planes in Europe in June 2022	6 845	Europe	CH Aviation (2022)	x		x
Firearms	Illicit firearms owned by civilians in the EU in 2017 (56 % of the estimated total firearms)	Around 35 000 000	EU	EC (2022) 'Updated EU rules on import and export of firearms'	x		x
Collectors' items	Information not available						
Works of art	Value of art/antiques sales in the EU market	€7.4	EU	Art Basel & UBS Report (2023)			
Antiques							
Jewellery	Information not available				x		
Commodities (precious raw metals)	Value of imported and exported gold into the EU in 2022	€20 (imported) €10 (exported)	EU	Eurostat (2023)			
Safe deposit boxes at banks	Information not available				x		x
Safe deposit boxes at non-banks	Information not available						

Notes: For the securities, ESMA data give a breakdown of on-exchange vs off-exchange turnover in the annual statistics report, however it is only given separately for bonds and equities. In the above table, these figures have been combined to get total off- and on-exchange figures.

Source: Authors' elaboration.

6.6.4 Ratings-based assessment of benefits

This section provides a ratings-based assessment – using a seven-point Likert scale (see Table 47 and Table 48) – of the anticipated benefits of the different scenarios.

Table 47: Likert scale

+3	+2	+1	0	-1	-2	-3
Large benefit/ advantage compared with 'do nothing'	Moderate benefit/ advantage compared with 'do nothing'	Small benefit/ advantage compared with 'do nothing'	No substantial change from do nothing	Small cost/ disadvantage compared with 'do nothing'	Moderate cost/ disadvantage compared with 'do nothing'	Large cost/ disadvantage compared with 'do nothing'

Notes: All ratings have been assigned relative to the status quo/do nothing option. Thus, the absolute value of 'large' or 'moderate' or 'small' is not really important. All that matters is that within the analysis, the impacts of each option are given appropriate ratings relative to the status quo and each other.

Source: Authors' compilation.

Table 48: Ratings-based assessment

Criteria	Scenario 1	Scenario 2	Scenario 3	Reasoning and supporting evidence
Improved cross-border flows of information and cooperation between national competent authorities	1	2	3	Scenario 3 has the highest potential of the three scenarios to develop effective interoperability between Member States, allowing for fast and efficient cross-border flows. Scenario 1 scores the lowest in this category as the architecture would be least able to facilitate complex queries (e.g. queries involving relations between assets and entities cross-border). Scenario 2 benefits from the existing interconnection infrastructure but covers fewer asset categories than the others, so is less comprehensive.
Detection of tax evasion/money laundering and other illicit financial flows	2	1	3	Databases can facilitate detection of tax evasion, money laundering and other illicit flows by enabling extraction of relevant conclusions through correlating otherwise disparate data. In this respect, Scenarios 1 & 3 represent optimal solutions as they offer the ability to correlate the most data on a larger scale (comprehensive asset and geographical coverage), although Scenario 3 as a centralised database could provide easier correlation of data. On the other hand, as highlighted in Section 3, there are concerns regarding data quality under Scenario 3.

Criteria	Scenario 1	Scenario 2	Scenario 3	Reasoning and supporting evidence
				Under Scenario 2, the benefits would be constrained by the limited types of assets covered. Meanwhile, Scenario 1 would be more focused on performing correlations at the national level.
Faster recovery of assets (assets freezing and seizing)	1	2	3	By enabling quick and direct access to data at the EU level, Scenario 3 could lead to faster investigations and greater success in asset tracking and freezing. Scenarios 1 & 2 would also offer an improvement from the present situation in terms of the speed of retrieval, however, not to the same degree as a centralised system. Scenario 2 scores higher than Scenario 1 here as it would mostly use already established registries that function well and have established best practices, indicating they would work more efficiently.
Reduction in the scale of tax evasion/money laundering and other financial crimes due to a deterrence effect and preventative actions	2	1	3	The more effective an option is with respect to conducting searches, facilitating investigation, etc., the more effective it would be in having a preventative and deterrence effect. Thus, the ratings for this criterion reflect the previous ratings, and considers the scope of coverage in each scenario.

Source: Authors' compilation.

Table 49: Different stakeholder perspectives on the benefits of a central EU asset registry

AML supervisors of financial institutions	FIUs	LEAs	Tax authorities
Quick and easy access to European-wide data Access to better quality data Less time and manpower required to conduct fit & proper due diligence and investigations (on- and off-site inspections) – a more comprehensive supervisory approach Providing a means to confirm/cross-check information reported by obliged entities subject to financial supervision	Rapid and secure access to data Improved accuracy and coverage of data (EU-wide, all asset categories) FIUs would gain efficiency in gathering information, more flexibility and an increase in responsiveness	Quick and direct access to data leading to faster investigations and greater success in asset-tracing and freezing	Improved ability to detect tax evasion – centralisation of information would make it possible for tax authorities to (a) be aware of the existence of information without having to conduct blind searches in several databases (blindly) and (b) connect the dots Uniformity of data would make it easier to use and exchange

Improved transparency and efficiency of investigations	Improved transparency and efficiency of investigations (124) Direct access to data (instead of requesting and receiving/not receiving) the information from another FIU, hence optimisation of workflow in terms of time spent and manpower needed		Faster identification of unfulfilled taxpayers' obligations Faster and more effective tax debt collection leading to an increase in tax revenues

Source: Authors' compilation.

According to the ratings-based assessment, overall Scenario 3 emerges as the most beneficial solution based on expert judgement. Although from a functionality perspective the benefits are likely to be similar to Scenario 1, the scores indicate that there may be added benefits associated with an EU-level asset registry (compared with establishing centralised, national AML/tax evasion registries) due to its more rapid implementation, greater integration and easier standardisation of data.

Scenario 3 was also a popular option among the competent authorities (the second most supported option). Some respondents indicated that this type of registry would make it easier to address current gaps in data (in terms of asset coverage), and that direct access to EU-wide data would make it easier and quicker for authorities to conduct investigations, detect illicit financial flows and take action (asset freezing, tax recovery, etc.). Despite this, there are two significant risks associated with Scenario 3 that must be weighed against these benefits and how likely they are to materialise:

- If new national assets registers were to be established and data then passed directly to an EU asset registry, the EU registry may face a number of issues in terms of data quality and data completeness. Typically, monitoring and enforcement, including sanctions, is administered at national rather than at EU level. Experience with the DAC and the establishment of beneficial ownership registers has indicated that the quality of data exchanged or submitted could be an issue in respect of new asset registers. Such data would need to be checked with national sources, and indeed with other national registers, a feature which may be easier at national level overall. Secondly, cooperative compliance on ensuring that all intermediaries submit data would be easier to monitor at national level than at EU level.
- To date, there have been few successful examples of this type of EU-wide interconnected registry. Authorities typically tend to prefer national registries and interconnection. This trend indicates a potential risk of failure due to lack of awareness and trust by authorities in the functioning of such a registry.

Scenario 2 scores low in many of the benefit categories, mainly due to its more limited coverage of the assets. Nevertheless, surveys and interviews suggest that interconnection

([124]) When investigating/analysing money laundering, FIUs would not have to wait for answers to arrive (if the queries are put to different countries the answers may arrive at different times, complicating investigation). A central EU asset registry would reduce the number of cross-border queries submitted relating to the same case (asking other jurisdictions repeatedly for information about the same entities or transactions from different angles, as new information comes up during investigation).

under Scenario 2 is the most preferred option among stakeholders, and it is also the most cost-effective in our analysis. In addition to the obvious cost advantages, it would be the quickest and least risky to implement as it builds on registries already required by the AMLD. However, as mentioned, the key shortcoming is that Scenario 2 leaves out a number of asset categories.

Finally, though Scenario 1 scores relatively high in terms of benefits, it is the least supported option among stakeholders. Creating a single asset registry that aggregates all national data at the country level could be an efficient means of tackling the limitations of accessing multiple, fragmented databases. Yet, stakeholders expressed some reservations regarding the feasibility of establishing national AML/tax evasion registries in each Member State and interconnecting these at an EU level. Specifically, these reservations concern the additional data harmonisation that would involve, though in principle, these activities would apply to both Scenarios 1 and 3, which also envisage some additional data collection efforts.

Box 2: Summary of operational and IT feasibility assessment, cost and benefit assessment (Task 4)

Operational Feasibility assessment

The operational feasibility of building asset registers, which contain complete, comprehensive and up-to-date information, was assessed in this task. Building new registries is operationally extremely burdensome for assets where no such registries currently exist, and thus, it is not considered feasible in this study. This assessment was made due to the characteristics of these assets, such as a large volume, difficulties with the traceability of ownership, frequent changes in ownership or subjective valuation. The required efforts in data collection and harmonisation needed to achieve the data quality necessary for an asset registry were deemed too high to be realistically feasible.

Conversely, where asset registries administered by public authorities exist, the study analysed the necessary steps needed to coalesce the information into a harmonised EU-wide registry. Existing infrastructure, such as EUCARIS for vehicle-related information or Business Registers Interconnection System (BRIS) for company-related information, was expressly considered in this assessment.

IT Feasibility assessment

The IT feasibility assessment was focused on finding the optimal architecture to fulfil the requirements of allowing the competent authorities in the EU to perform their functions in relation to anti-money laundering and tax evasion investigations on a wide scale. They accomplish this by mostly solving interoperability issues between registers functioning on different technical, operational and legislative frameworks.

In the distributed architecture (Scenario 1), a national registry with its own asset database would have to be implemented by every Member State. The existing asset registries would establish a connection with a central national asset registry. New registries have to be created by every Member State for the asset categories that are not already covered.

Scenario 2 is considered a lighter approach, connecting the already existing registries and facilitating their interconnection using a central registry.

In Scenario 3, all existing registries will have to interact directly with the EU gateways. Data provided by each individual registry will be stored in the central asset database. At the same time, new categories not already covered by the Member States with their own registries will be directly implemented at the level of the EU central registry.

The analysis shows that each option is technically feasible, with the drawbacks being specifically linked with elements that are more related to flexibility, data coverage, complexity of data integration, and security aspects. Based on the analysis of these scenarios, a mixed approach (Scenario 2 and 3) has also been considered, limiting the central storage function at EU level to a semantic architecture of metadata and data that Member States are willing to store at central level, engaging all other data from the place of origin by interconnections. The mixed option would minimise security risks and ease the process of integration from a technical and administrative point of view while also facilitating the construction of a GDPR conformant environment.

Costs and Benefit Assessment (Task 4)

The costs and benefit assessment (incremental costs and benefits relative to the status quo) were considered for the three proposed scenarios.

Data collected through primary interviews with DG DIGIT and DG JUST and surveys run across national competent authorities in the context of AML and the fight against tax evasion and secondary research (documentary and literature review) does not permit a full-fledged quantification of the costs and benefits. Instead, it provides:

- A descriptive overview of the types of costs and benefits that may accrue under the different scenarios;
- Where possible, the orders of magnitude of the likely costs and benefits associated with each of the three scenarios;
- A ratings-based assessment of the relative benefits of each scenario.

The results of our assessment concluded that Scenario 3 emerges as the most beneficial option due to its potentially more comprehensive asset coverage. However, surveys and interviews suggest that interconnection (Scenario 2) is the most preferred option among the surveyed and interviewed stakeholders. It is also the most cost-effective option, but its narrow scope of assets covered limits it. It has obvious cost advantages and would be the quickest and least risky to implement as it builds on existing registries. One way of circumventing this issue is a mix of interconnection (Scenario 2) with a central EU level asset registry (Scenario 3).

The second most supported option for the competent authorities is creating a centralised EU asset registry (Scenario 3). For some respondents, this would help address current gaps in data (in terms of asset coverage and quality of data) and provide direct access to EU comprehensive data, which would make it easier and quicker for authorities to conduct investigations, detect illicit financial flows and take action (asset freezing, tax recoveries etc.). However, there are serious concerns regarding potential data quality issues resulting from EU-led supervision instead of national supervision.

Finally, setting up new national registries (Scenario 1) is the least supported option among stakeholders. While creating a single asset registry that aggregates all national data at country level is widely seen as an efficient measure to tackle the limitations of accessing multiple, fragmented databases, stakeholders expressed major reservations regarding the feasibility of establishing many new registries in each Member State and interconnecting these at the EU level. From a cost perspective, this option would be even more expansive than Scenario 3, which leverages economies of scale due to being only centralised at the EU level.

7. Task 5: Legal feasibility assessment: EU law framework and legal constraints

7.1 EU Legal framework

The current AML legal framework at EU level establishes a right for FIUs and competent authorities to access information on assets and their holders (or certain transactions) in a timely manner at national and cross-border level. For this purpose, EU law sets forth mechanisms for the exchange of information between FIUs and authorities in charge of combating money laundering and predicate offenses such as tax evasion at domestic and cross-border levels. Tasks 5 and 6 seek to assess the legal feasibility of introducing a single point of access to information for FIUs and competent authorities.

Task 5 is structured as follows:

- An outline of the EU AML and Tax Frameworks related to the scope of the Study;
- The interaction of the EU AML and Tax Frameworks with the Charter of Fundamental Rights and GDPR.

7.1.1 State of play: information exchange mechanisms under the AMLD framework

The existing legal framework allows for FIU's to obtain information on asset and their holders domestically and cross-border through the data collected by obliged entities under their reporting and due diligence obligations.

Information exchange: obliged entities' obligation to collect and report information

Obliged entities ([125]) act as the main data collectors and reporting entities under the AMLD framework ([126]). They are bound by record keeping, reporting and due diligence obligations.

Importantly, customer due diligence (CDD) must be performed by obliged entities in certain circumstances including where the value of a transaction exceeds EUR 15 000 or EUR 10 000

([125]) Obliged entities are the trading professionals and financial institutions, directly involved in the asset transactions. The notion of 'obliged entity' includes entities such as credit institutions, financial institutions and a list of natural or legal persons acting in the exercise of their professional activities. Due to their activity, these entities are in a privileged position to collect information on their customers and transactions which present a risk.

([126]) Consolidated text: Directive (EU) 2015/849 of the European Parliament and of the Council of 20 May 2015 on the prevention of the use of the financial system for the purposes of money laundering or terrorist financing, amending Regulation (EU) No 648/2012 of the European Parliament and of the Council, and repealing Directive 2005/60/EC of the European Parliament and of the Council and Commission Directive 2006/70/EC (Text with EEA relevance) Text with EEA relevance

in cash, or regardless of the amount, where there is suspicion of money laundering. Due diligence requires obliged entities to collect information on the identity of the beneficial owner, scrutinise transactions, and where necessary, the source of funds. The CDD obligation covers transactions of assets such as crypto assets, assets held in a safe deposit box at banks, high-value goods (such as works of art), cash in hand and cash at bank, provided the monetary thresholds are reached and obliged entities are involved. However, it is worth noting that a survey conducted in 2016 by the EU FIU Platform found that some FIUs relied on their power to obtain information from obliged entities ([127]).

Cooperation and exchange of information between FIU's and competent authorities

Under Article 32 of the AMLD4, Member States shall ensure that 'their FIUs have access, directly or indirectly, in a timely manner, to the financial, administrative and law enforcement information that they require to fulfil their tasks. FIUs shall be able to respond to requests for information by competent authorities in their respective Member States when such requests for information are motivated by concerns relating to money laundering, associated predicate offences or terrorist financing'. This provision is crucial to ensure firstly that FIU's have access to the relevant information they need for their tasks and, second, that they are therefore able to exchange information with their EU homologues and competent authorities.

Regarding cooperation between FIUs, the AMLD creates an obligation on FIUs to promptly forward reports which concern another Member State to the FIU of that Member State; although a low volume of cross-border reports have been forwarded since the entry into force of this provision in June 2017 ([128]). Further, FIUs must exchange, 'spontaneously or upon request' 'any information that may be relevant for the processing or analysis of information by the FIU related to money laundering or terrorist financing and the natural or legal person involved' as enshrined in Article 53 of the AMLD4. In this context, a request for information issued by an FIU to that of another Member State must be motivated and comprise the relevant facts, background information, the reasons for the request and an explanation of how the information sought will be used. In addition, the AMLD provides that different exchange mechanisms may apply if so, is agreed between the FIUs, in particular as regards exchanges through the FIU.net or its successor.

Once an FIU launches a cross-border request for information to a responding FIU, Article 53(2) requires Member States to ensure that the FIU subject of the request uses the whole 'range of its available powers which it would normally use domestically for receiving and analysing

([127]) EU FIU Platform mapping exercise and gap analysis on FIU's powers and obstacles for obtaining and exchanging information, report adopted by the EU FIU Platform. See Commission Staff Working Document on improving cooperation between EU Financial Intelligence units (SWD(2017) 275 final) – it is also worth noting that reliance on this power by FIU's to obtain information from obliged entities 'does not seem an effective and efficient means to identify assets whose existence and location is not previously known'.

([128]) European Commission (2019a).

information when it replies to a request for information (..) from another FIU'. In addition, the responding FIU must respond to the request in a 'timely manner'.

Crucially the Directive also obliges Member States to ensure their FIUs rely on protected channels of communication between themselves and encourages the use of the FIU/net or its successor. It further provides that Member States shall ensure their FIU's cooperate in the 'application of state-of-the-art technologies' in accordance with their national law. These technologies must allow FIU's to 'match their data with that of other FIUs in an anonymous way by ensuring full protection of personal data with the aim of detecting subjects of the FIU's interests in other Member States and identifying their proceeds and funds'.

More generally, Member States must ensure FIUs cooperate with each other to 'the greatest extent possible, regardless of their organisational status' ([129]). As developed in Section 4.5, there is currently no harmonised organisational status for the FIUs of Member States: 21 FIU's have been established under a police or administrative model; five (CY, DK, EL, HU, NL) are hybrid, and in Luxembourg the FIU pertains to the judicial category ([130]). The differing organisational statuses obstructed the exchange of information as police FIU's could not exchange information with administrative FIU's and vice versa ([131]). Given the wide degree of discretion enjoyed by Member States in the setting up of their FIU's, the latter remain heterogeneous in terms of their rights and powers to access information as seen in Task 2. This creates issues in the context of the cross-border exchange of information because an FIU may have a right to access certain information in its own legal system and send a request to a receiving FIU which may not have the same right or means to access this information under its national law, and will therefore be *de facto* unable to provide the requesting FIU with the relevant information it requests ([132]).

It is important to stress that this study does not consider the different access rights across Member States as an issue to be solved with an EU asset registry. Instead, the objective is merely to ensure that authorities who do have such rights can access the information they need in a cross-border context.

The AMLD seeks to remove other restrictions to the exchange of information between FIU's. For example, Article 50a of the AMLD5 prohibits Member States from placing any unreasonable restrictive conditions on the exchange of information or assistance between competent authorities for the purpose of the AMLD. Competent authorities shall not refuse a request for assistance for the following reasons: the request is also considered to involve tax matters; 'national law requires obliged entities to maintain secrecy or confidentiality, except in

([129]) Article 52 of the AMLD4.

([130]) EU FIUs Platform (2016), 'Mapping exercise of Gap Analysis on FIU's powers and obstacles for obtaining and exchanging information', 5-7, cited in Mouzakiti (2020), p. 354.

([131]) Ibid.

([132]) Ibid.

those cases where the relevant information that is sought is protected by legal privilege or where legal professional secrecy applies'; or that 'nature or status of the requesting counterpart competent authority is different from that of the requested competent authority'. FIU's may only refuse to exchange information in exceptional circumstances where the exchange could infringe upon the fundamental principles of its national law ([133]). In addition, when exchanging information and documents, the transmitting FIU may impose restrictions and conditions for the use of that information which must be complied with by the receiving FIU.

i. The FIU.net network

The FIU.net network was set up informally to enhance and promote cooperation between FIU's. The platform allows FIU's to exchange and cross-check information online anonymously ([134]) through an information system connecting decentralised databases. The FIU.net IT system provides for secure channels of communication between FIU's allowing them to send regular case file requests, forward cross border reports and disseminate reports of other Member States' FIUs ([135]). FIUs provide feedback to private entities on effectiveness of and follow-up to reports of suspected ML/TF ([136]). The system is operated by Europol since 2016. According to the Commission's report on FIU cooperation, at least half of Member States' FIU's rely on the platform as the main tool to communicate with FIUs from other Member States. Nonetheless the platform has been the subject of recurrent technical difficulties due to its need to be upgraded ([137]). Moreover, the FIU platform comprises legal limitations preventing it from producing binding standards to overcome difficulties in the exchange of information ([138]).

In addition, FIU's disseminate the results of their analyses to law enforcement and tax authorities for further investigations and prosecution where there are grounds to suspect money laundering, associated predicate offences or terrorist financing. Moreover, they can order the temporary freezing of transactions.

ii. Existing registers, data retrieval systems, and specific obligations concerning real estate under the AMLD framework and their legal basis

In addition to the general framework for the exchange of information developed above, the AMLD contains specific provisions for certain assets (real estate) and obligations on Member

([133]) See Article 53(3) of the AMLD4.

([134]) Balboni, P, Kroon, U., and Macenaite, M. (2013), 'Data protection and data security by design applied to financial intelligence' in Reimer, H., Pohlmann, N., and Schneider, W. (eds.), *ISSE 2013 Securing Electronic Business Processes* (Springer, 2013) 73.

([135]) European Commission (2019a).

([136]) Impact assessment template (editable version) (europa.eu).

([137]) European Commission (2019a).

([138]) Ibid.

States to hold centralised registries for beneficial ownership of companies and trusts as well as for bank accounts.

BORIS

Article 30 of the AMLD ([139]) titled 'beneficial ownership information' establishes interrelated obligations that relate to the setting up of central registers in Member States.

- Under Article 30(1), 'Member States shall ensure that corporate and other legal entities incorporated within their territory are required to obtain and hold adequate, accurate and current information on their beneficial ownership, including the details of the beneficial interests held'. Thus, these are required to provide information on the beneficial owner and on their legal owner to obliged entities on the context of the exercise of the latter's customer due diligence obligations.
- Moreover, under Article 30(3) this information (mentioned in Article 30(1) and collected by corporate and legal entities) may be collected in accordance with national systems and must be 'held in a central registry in each Member State' such as a commercial register or a public register.
- And, as required under Article 30(4), to ensure the data in the central registers is 'adequate, accurate, and current' Member States must put in place mechanisms whereby obliged entities and competent authorities report any discrepancies between the data in the registry and the information available to them.

The three obligations described above organise the setting up of the national central registries of beneficial owners. These registries allow for the identification of the natural persons who own or control a trust, legal entity, or similar legal arrangements (the beneficial owners).

Concerning the access to the information held in these central registries, Article 30(2) of the AMLD4 provides that FIU's and competent authorities shall have access in a timely manner to this information. More specifically, Article 30(5) stipulates that the information on beneficial ownership is accessible in all cases to (a) competent authorities and FIU's without any restriction; (b) obliged entities within the framework of customer due diligence, (c) and any member of the general public. Importantly, Article 30(6) specifies that FIU's and competent authorities have timely and unrestricted access to all the information in the central registry without alerting the entity concerned. Alongside, the legislation specifies that 'competent authorities' shall include public authorities with designated responsibilities for combating money laundering or terrorist financing, as well as tax authorities, supervisors of obliged entities and authorities that have the function of 'investigating or prosecuting money laundering, associated predicate offences (..), tracing and seizing or freezing and confiscating criminal assets'.

In addition to the obligation on Member States to ensure that beneficial ownership information is collected and stored in central registries, Article 30(10) of the AMLD5 obliges them to make sure the central beneficial ownership registries are interconnected. Their interconnection must be done via the European Central platform set out in Directive 2017/1132([140]) under Article 22

([139]) https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A32015L0849

([140]) Directive (EU) 2017/1132 of the European Parliament and of the Council of 14 June 2017 relating to certain aspects of company law (codification) (Text with EEA relevance.).

which provides that the system of interconnection of registries is composed of Member States' registers, the platform, and the portal serving as the European electronic access point. Importantly, Article 31 of the AMLD5 provides for homologue dispositions concerning trusts and other types of legal arrangements, such as, *inter alia*, fiducie, certain types of Treuhand or fideicomiso, where such arrangements have a structure or functions similar to trusts. Together, Articles 30 and 31 of the AMLD organise the setting up of the national registers containing beneficial ownership information and their interconnection to form the Beneficial ownership Interconnected System (BORIS).

The amendment of the AMLD framework and inclusion of Articles 30 and 31 which provide for the obligation to hold centralised beneficial ownership registers was established using Article 114 TFEU as a legal basis. This is also true regarding the obligation on Member States to ensure their central registers are interconnected via the European Central Platform in accordance with the technical specifications and procedures established by the Commission's implementing acts. The justification for the choice of legal basis is provided in the text of the proposal and lies in the fact that money laundering and terrorist financing, due to the economic losses and disruption on a cross-border level it entails represents a threat to the internal market ([141]).

Bank account registers (BAR)

Article 32a introduced by the AMLD5 creates an obligation on Member States to set up a national centralised automated mechanism such as a central database or data retrieval system (hereinafter: centralised bank account registry) comprising information on the identification of any natural or legal persons holding or controlling payment accounts, bank accounts identified by IBAN, and safe-deposit boxes held by credit institutions within their territory. This information is directly accessible in an immediate and unfiltered manner to national FIUs and to national competent authorities, in order to fulfil their respective obligations. In their turn, under the exchange mechanism indicated in Article 53 AMLD, FIUs must provide other FIUs with relevant information held in the centralised automated mechanisms (either spontaneously or upon request). Moreover, the interconnection of bank account registers is mentioned in Article 32a(5), technically feasible according to the Commission's assessment adopted in July 2019 and a proposal to achieve it has been published.

The obligation for Member States to put in place such systems stem from the amendment to the AMLD framework on the basis of Article 114 TFEU. Article 32a(2) provides that the information held in these registers is directly accessible in an immediate and unfiltered manner to national FIUs and it should be accessible to national competent authorities for fulfilling their obligations under this Directive.

To grant access to the information held in the BAR to competent authorities other than FIUs, Directive 2019/1153 was introduced and based on Article 87(2) TFEU which concerns police cooperation and allows the Union to introduce measures involving Member State's competent authorities (including police, law enforcement authorities) and concern *inter alia* the collection, storage and exchange of information relevant for the prevention, detection and investigation

([141]) Proposal for a Directive of the European Parliament and of the Council on the mechanisms to be put in place by the Member States for the prevention of the use of the financial system for the purposes of money laundering or terrorist financing and repealing Directive (EU) 2015/849.

of criminal offences. Thus, under Article 3 of Directive 2019/1153, Member States are required to designate the competent authorities empowered to access and search its national centralised bank account registry.

To obtain information on persons who hold bank accounts in another Member State, law enforcement authorities may channel requests for such information via FIUs and, when this is not the case, they may exchange the relevant information cross border on the basis of bilateral police cooperation agreements or judicial cooperation instruments. This includes exchanges on the basis of Framework Decision 2006/960 JHA ([142]) (also referred to as the "Swedish Initiative"). Some Member States might require a European Investigation Order (EIOs). Moreover, a deadline of 30 days applies for the recognition or execution of an EIO. This may result in a lengthy process, hampering speedy access to information on persons who hold bank accounts.

Since 19 December 2020, Regulation (EU) 2018/1805 ([143])on the mutual recognition of freezing and confiscation orders applies. This Regulation establishes precise timelines for the recognition and execution of freezing orders. In case immediate freezing is necessary (i.e. because there are legitimate grounds to believe that the property will imminently be removed or destroyed), the executing authority has 48 hours to decide on the recognition of the freezing order. Once the decision on the recognition has been taken, the executing authority has 48 hours to take the concrete measures for the execution of the order. For confiscation orders, there is a maximum timeframe of 45 days to take a decision on their recognition and execution (see in this context Articles 9 and 20 of the Regulation). When issuing certificates for the mutual recognition of freezing and confiscation orders, Member States may indicate the details of the bank account of the affected person. Therefore, knowing where in the EU a suspect holds a bank account is invaluable information for competent authorities to quickly identify to which other Member States, they should request the freezing and confiscation of money stored in those accounts before it is moved somewhere else.

Real estate ownership

Similarly, Article 32b of the AMLD5 provides that Member States 'shall provide FIUs and competent authorities with access to information which allows the identification in a timely manner of any natural or legal persons owning real estate, including through registers or electronic data retrieval systems where such registers or systems are available'. Importantly, Directive 2018/843 and thus the obligation to provide timely information on real estate ownership is based on Article 114 TFEU.

In contrast with the provisions for BO and bank account registers however, EU law solely obliges Member States to ensure FIUs and competent authorities can access information on real estate property ownership in a timely way. Reliance on registers or electronic data retrieval systems to fulfil this obligation is subject to their availability.

([142]) Council Framework Decision 2006/960/JHA of 18 December 2006 on simplifying the exchange of information and intelligence between law enforcement authorities of the Member States of the European Union.

([143]) Regulation (EU) 2018/1805 of the European Parliament and of the Council of 14 November 2018 on the mutual recognition of freezing orders and confiscation orders.

In addition to Article 32b, Article 31(3)(a) of the AMLD5 requires Member States to ensure that beneficial ownership information of express trusts and similar legal arrangements is held in a central beneficial ownership register set up by the Member States where the trustee of the trust is established or resides. Importantly, where the trustee (or person holding an equivalent position in similar legal arrangements) is outside the Union, the AMLD requires that the beneficial ownership information 'shall be held in a central register set up by the Member State where the trustee of the trust or person holding an equivalent position in a similar legal arrangement enters into a business relationship or acquires real estate in the name of the trust or similar legal arrangement'.

7.1.2 State of play: information exchange under the framework for combatting tax evasion

Value Added Tax (VAT)

In what Value Added Tax (VAT) is concerned, the relevant legislative instruments at EU level include Council Regulation (EU) No. 904/2010 ([144]), as well as the accompanying implementing regulations and in particular, Commission Implementing Regulation (EU) No. 79/2012 ([145]).

Under the framework of these instruments, rules have been adopted providing for technical details and mechanisms concerning exchanges of information and its use. Among the key aspects relevant to the current study, the following may be noted:

- The exchange of information between Member States relies on the use of electronic standard forms and may take place either with or without a prior request.
- Under Regulation (EU) 904/2010, exchange of information is conducted via One-Stop Shop, an electronic portal which allows taxable persons to register for VAT in a single Member State (Member State of Identification) and declare and pay VAT due on all intra-EU cross-border business-to-consumer (B2C) e-commerce activities in that Member State, according to the tax rates of the Member State of Consumption.
- Articles 17-24 provide for the electronic storage and exchange of relevant information by Member States which includes, among others, information concerning valid VAT numbers, intra-community transactions or vehicle registrations (access to EUCARIS, connecting information from car national registration authorities across the European Union). For such information, automated access is granted to the competent (tax) authorities of any other Member State.
- Article 33-36 provide for the establishment of Eurofisc - the network of Member States for the swift exchange of information on risky operators from the point of view of cross border VAT fraud (mainly Missing Trader Intracommunity (MTIC) fraud, eCommerce fraud, fraud concerning the customs procedure and fraud involving means of transport). Under the Regulation, Eurofisc may perform joint processing and analysis of data, operations which are carried out through the electronic system - Transaction Network Analysis (TNA).

([144]) Consolidated text: Council Regulation (EU) No 904/2010 of 7 October 2010 on administrative cooperation and combating fraud in the field of value added tax (recast). See EUR-Lex - 02010R0904-20210701 - EN - EUR-Lex (europa.eu)

([145]) Consolidated text: Commission Implementing Regulation (EU) No 79/2012 of 31 January 2012 laying down detailed rules for implementing certain provisions of Council Regulation (EU) No 904/2010 concerning administrative cooperation and combating fraud in the field of value added tax (recast). See EUR-Lex - 02012R0079-20231106 - EN - EUR-Lex (europa.eu)

- In what external information on transactions is concerned, the following points are relevant:
  - Tax authorities have at their disposal information from a customs database (Surveillance) on VAT exempt importations (customs procedure 42);
  - As from 1 January 2024, Payment Services Providers will submit quarterly data on cross-border payments to tax authorities for the control of eCommerce.

Tax Recovery

Assistance and exchange of information between Member States may be necessary in order to ensure the recovery of the tax concerned, in situations where no spontaneous payment of the relevant tax is performed. Such cases may include, for example, situations in which the debtor is resident in another state than the one where tax is due or where the debtor owns assets located in another state (such as bank accounts or immovable property etc.). Exchange of information is important, in these hypotheses, for instance, to (1) locate assets abroad for possible recovery assistance abroad and (2) evaluate the ability to pay in the Member State of establishment, in view of determining to which extent recovery can proceed in the Member State of establishment.

For such situations, EU Directive 2010/24 ([146]) concerning mutual assistance for the recovery of claims relating to taxes, duties and other measures provides mechanisms for exchange of information at EU level regarding the tax debtor's assets. Under Articles 5 and 6 of the Directive, any information which is foreseeably relevant to the requesting authority may be provided, under certain limitations concerning, inter alia, the disclosure of commercial, industrial or professional secrets, or information which would be liable to prejudice the security of, or be contrary to the public policy of the requested Member State.

In practice, the exchange of information for tax recovery purposes is mostly an exchange of information upon request, as regulated by Article 5 of EU Directive 2010/24, and the exchange of information without prior request is limited (as it only applies to refunds of taxes, other than VAT) ([147]).

Direct Taxation

In the field of direct taxation, given that this area is not harmonised across the EU, mechanisms are in place to ensure efficient cooperation between tax authorities for the reduction of tax evasion ([148]). The main legal instrument in the field of administrative cooperation and exchange of information between Member States is set forth in the DAC Directives ('DAC'). For ease of

([146]) https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A32010L0024

([147]) Article 6 of Directive (EU) 2010/24.

([148]) Enhanced administrative cooperation in the field of (direct) taxation (europa.eu).

reference and coherence, we use the term 'DAC' to refer to the consolidated version of Council Directive 2011/16/EU (DAC1) of 15 February 2011 on administrative cooperation in the field of taxation and repealing Directive 77/799/EEC , as successively amended by several directives relevant to the subject matter of this report: Council Directive (EU) 2014/107 (DAC2), Council Directive (EU) 2016/2258 (DAC5), Council Directive (EU) 2018/822 (DAC6) and Council Directive (EU) 2021/514/EU (DAC7) ([149]).

7.1.2.1 Scope and subject matter of DAC

In the field of taxation, DAC lays down the rules and procedures under which the Member States shall cooperate with each other with a view to exchanging information that is foreseeably relevant to the administration and enforcement of the domestic laws concerning certain taxes. The material scope of the directive enshrined in Article 2(1) is broad and includes 'all taxes of any kind levied by, or on behalf of, a Member State or the Member State's territorial or administrative subdivisions, including the local authorities.' Moreover, under the DAC, competent authorities may cooperate and exchange information according to 3 different processes: spontaneous, automatic and on request as developed below.

7.1.2.2 Types of exchanges of information under the DAC Directives

i. **Exchange of information on request**

Article 3(8) defines the notion of exchange of information on request as 'information based on a request made by the requesting Member State to the requested Member State in a specific case'. This type of exchange of information is developed in Article 5 of the DAC and provides that competent authorities may request any information foreseeably relevant and that is possessed or obtained as a result of administrative enquiries, if requested by an authority of another Member State. Importantly, Article 5a defines broadly the notion of 'foreseeable relevance'. Thus, information is considered to have foreseeable relevance where 'at the time the request is made, the requesting authority considers that, in accordance with its national law, there is a reasonable possibility that the requested information will be relevant to the tax affairs of one or several taxpayers, whether identified by name or otherwise, and be justified for the purposes of the investigation'. In addition, the requesting authority must provide 'at least' information regarding the relevant tax purpose and a specification of the information required for the administration or enforcement of national law, consistently with Article 5a(2). In this context, Article 6(1) sets an obligation on the requested authority to arrange for the 'carrying out of any administrative enquiries necessary to obtain the information referred to in Article 5, provided the requesting authority has demonstrated the 'foreseeable relevance' with the adequate information.

This channel for exchanging information comprises limits set forth in Article 17 of DAC, which may be relevant for the purposes of a proportionality assessment. Article 17(1) provides that such a request for information should be relied on in cases where 'the requesting authority has exhausted the usual sources of information which it could have used in the circumstances for obtaining the information requested'. Furthermore, and given the plurality of legal systems in

([149]) DAC consolidated text consulted at: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02011L0016-20230101

the EU, Article 17(2) stipulates that, in the context of the exchange of information on request, there is no obligation on a Member State to carry out enquiries or to communicate information if it would 'be contrary to its legislation to conduct such inquiries or to collect the information requested for its own purposes'.

In addition, Article 17(3) allows for competent authorities of Member States to refuse to provide the information requested 'where the requesting Member State is unable, for legal reasons, to provide similar information'. Other circumstances limiting the possibility to rely on Article 5 for the exchange of information on request include respect for 'industrial, commercial, or professional secret of a commercial process or information whose disclose would be contrary to public policy' - Article 17(4). Thus, the exchange of information on request is subject to national legislation and considerations of public policy and commercial, trade, and professional secrecy. Importantly, the requested authority has an obligation to motivate its decision to refuse to communicate requested information to another authority according to Article 17(5). It follows that a degree of discretion and additional considerations may restrict the scope of the information that can be communicated in the context of requested information under DAC by designated competent authorities of Member States.

The time limits for providing information on request may also be relevant for a legal feasibility evaluation of the individual scenarios. Under Article 7(1) of the DAC, 'the requested authority shall provide the information referred to in Article 5 as quickly as possible and no later than 3 months from the date of receipt of the request'. According to the same provision, in case the requested authority is unable to provide the information within the indicated three-months period, the time limit may be prolonged up to six months from the date of receipt of the request or, in certain special circumstances, other time limits may be agreed upon between the requesting and requested authorities - Article 7(2).

ii. **Mandatory automatic exchange of information**

Under Article 3(9) of DAC1 ([150]), automatic exchange of information is defined, for the purpose of Article 8(1) as 'the systematic communication of predefined information on residents in other Member States to the relevant Member State of residence, without prior request, at pre-established regular intervals'.

In this context, the competent authority of each Member State is required to communicate to the competent authorities of other Member States 'all information that is available at national level concerning residents of that other Member State for the following categories: (a) employment income, (b) directors fees, (c) life insurance products not covered by other Union legal instruments, (d) pensions, (e) ownership of and income from immovable property and (f) royalties. The notion of 'information that is available' refers to 'information in the tax files of the Member State communicating the information, which is retrievable in accordance with the procedures for gathering and processing information in that Member State', as enshrined in Article 3(9) of DAC1. For these categories, Member States have an obligation to communicate to authorities from other Member States, in line with Article 8(1) and in an automatic way, any information available on residents which reside in that other Member State regarding the specific 'categories of income and capital, as they are to be understood under the national

([150]) https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32011L0016

legislation of the Member State which communicates the information'. Under this general rule, the communication of information must take place at least once a year, within six months following the end of the tax year of the Member State during which the information became available.

Thus, pursuant to Article 8(3a), under the reporting and due diligence rules established in Annexes I and II of DAC2 ([151]) and applicable to financial institutions, Member States must communicate by automatic exchange relevant financial account information regarding reportable accounts that includes, among others, 'the name, address, TIN(s) and date and place of birth (in the case of an individual) of each Reportable Person that is an Account Holder of the account and, in the case of any Entity that is an Account Holder and that (...) is identified as having one or more Controlling Persons that is a Reportable Person, the name, address, and TIN(s) of the Entity and the name, address, TIN(s) and date and place of birth of each Reportable Person', as well as information concerning 'the account balance or value (...) as of the end of the relevant calendar year or other reporting period or, if the account was closed during such year or period, the closure of the account'. The information referred to in this paragraph must be communicated annually, within nine months from the end of the calendar year or other reporting period to which the information relates. The information to be communicated under the provisions indicated above is standardized and pre-defined for exchange purposes and should not differ between Member States, as agreed based on the Common Reporting Standard of the OECD.

Reportable cross-border arrangements must also be communicated automatically by Member States based on data collected from intermediaries (in respect of information within their knowledge, possession or control), under the terms provided by Article 8ab DAC6, within one month of the end of the quarter in which the information was filed ([152]).

Furthermore, information reported by platform operators (as defined by the rules in Annex V of DAC7) must be automatically exchanged between Member States pursuant to Article 8ac. This exchange refers to information reported by operators of online platforms through which sellers are connected to users in order to provide, directly or indirectly, activities related to the (a) rental of immovable property, (b) personal services, (c) sale of goods or (d) rental of any mode of transport. Such information must be communicated between Member States 'within two months following the end of the Reportable Period' applicable to the platform operator ([153]).

Under the DAC5 ([154]), in order to combat predicate offenses, tax authorities have access to information collected by obliged entities under the EU AMLD framework. Such information can be used to check, for example, if reporting financial institutions are compliant with customer due diligence rules under DAC2. Under the EU AMLD framework, tax authorities have unrestricted access to information on beneficial ownership information held in national registers.

([151]) https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A32014L0107

([152]) Article 8ab(18) DAC6.

([153]) 153 Article 8ac(3) DAC7.

([154]) https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=celex%3A32016L2258

iii. **Spontaneous exchange of information**

The notion of 'spontaneous exchange' is defined in Article 3(10) of DAC1 as the 'non-systematic communication at any moment and without prior request, of information to another Member State'. This Article provides for an obligation on competent authorities to communicate information where it may be of particular relevance for the purpose of fighting against tax evasion/fraud, where it may be useful to other competent authorities of other Member States.

Under Article 9(1) of DAC1, competent authorities of each Member State have an obligation to provide relevant information to the competent authority of any other Member State in certain specific circumstances. These include situations where (a) there are grounds 'for supposing that there may be a loss of tax in the other Member State'; (b) 'a person liable to tax obtains a reduction in, or an exemption from, tax in one Member State which would give rise to an increase in tax or to liability to tax in the other Member State'; (c) 'business dealings between a person liable to tax in one Member State and a person liable to tax in the other Member State are conducted through one or more countries in such a way that a saving in tax may result in one or the other Member State or in both'; (d) the competent authority of a Member State has grounds for supposing that a saving of tax may result from artificial transfers of profits within groups of enterprises or (e) 'information forwarded to one Member State by the competent authority of the other Member State has enabled information to be obtained which may be relevant in assessing liability to tax in the latter Member State'. Moreover, competent authorities may exchange spontaneously information with other competent authorities of other Member States wherever they deem information may be useful for the purposes of avoiding tax evasion and tax fraud consistently with Article 9(2).

In addition, under Article 10(1) states that information must be forwarded 'as quickly as possible, and no later than one month after it becomes available'. The competent authority receiving the information under the 'spontaneous exchange' by another competent authority is obliged to confirm 'if possible by electronic means' and no longer than seven working days after receipt, that it has received the information.

7.2 Assessment of the legal constraints

For the implementation of any of the three scenarios defined in the previous tasks, the fundamental rights to the respect for private life and the protection of personal data must be considered. According to the GDPR personal data is defined as 'any information relating to an identified or identifiable natural person' ([155]). Therefore, information allowing for the identification of asset owners qualified as 'personal data'. Moreover, the notion of **processing of data** is defined as any operation (whether or not by automated means) performed on

([155]) An identifiable natural person is one who can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity of that natural person.

personal data including the 'collection, recording, organisation, structuring, storage, adaptation or alteration, retrieval, consultation, use, disclosure by transmission, dissemination or otherwise making available, alignment or combination, restriction, erasure or destruction'. Due to the set of operations (described above) performed on the personal data of natural persons, the scenarios entail the restriction on the right to privacy and data protection of persons. For this reason, their implementation raises fundamental rights concerns. The rights to the respect for private life and the protection of personal data are enshrined in Articles 7 and 8 of the CFR ([156]) and EU secondary legislation, including in particular the GDPR ([157]), directly referred to under the DAC and AMLD frameworks. Importantly, the features of a legislative instrument to set up an EU asset registry will have to comply with certain requirements set forth under the GDPR. Thus, the general modalities of the restriction to data protection rights of persons which apply to all three scenarios are developed below.

The action of the EU legislator is bound by the general principles of proportionality and subsidiarity. The legal requirements to satisfy the proportionality and subsidiarity tests, as enshrined in EU primary law and developed by the ECJ.

Charter of Fundamental Rights (CFR)

Regarding private life, Article 7 provides that everyone 'has the right to respect for his or her private and family life, home and communications'. For data protection, Article 8(1) of the CFR stipulates that 'Everyone has the right to the protection of personal data concerning him or her'. Moreover, Article 8(2) specifies that such data must be processed fairly, 'for specified purposes and on the basis of the consent of the person concerned or some other legitimate basis laid down by law. Everyone has the right of access to data which has been collected concerning him or her, and the right to have it rectified.' Lastly, Article 8(3) subjects' compliance with these rules to the control 'by an independent authority'.

Notwithstanding, these rights are not absolute and can be restricted subject to the condition of compliance of the limitation with the conditions listed in Article 52(1) of the CFR. These include that it must be provided for by law; respect the essence of the rights, genuinely meet objectives of general interest recognised by the Union or the need to protect the rights and freedoms of other; and be necessary and proportionate.

This legal framework is the basis of the GDPR which develops in further details the content and possibility to restrict the scope of the right to data protection of persons. It is worth noting that recital (73) of the GDPR states that that these restrictions should be according to the requirements set out in the CFR and in the European Convention for the Protection of Human Rights and Fundamental Freedoms (ECHR) ([158]).

General Data Protection Regulation (GDPR)

Main principles under Article 5 of the GDPR

([156]) https://www.europarl.europa.eu/charter/pdf/text_en.pdf

([157]) Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC.

([158]) European Convention for the Protection of Human Rights and Fundamental Freedoms (ECHR), 4 November 1950, 213 UNTS 221.

The material scope of the GDPR is provided in Article 2(1) and covers 'the processing of personal data wholly or partly by automated means and to the processing other than by automated means of personal data which form part of a filing system or are intended to form part of a filing system'.

Importantly, Article 5 of the GDPR sets forth the principles that relate to the processing of personal data: lawfulness, fairness and transparency, purpose limitation, data minimisation, accuracy, storage limitation, integrity and confidentiality, and accountability.

Lawfulness, fairness and transparency refers to data controllers' obligation to have legitimate grounds for the processing of personal data. To process lawfully signifies to find a lawful ground, and not breach other laws ([159]). The research points to the fact that fairness is connected to awareness, which means that data subjects should be made aware of data processing ([160]). Fairness also means that the data collected should not be misused or mishandled ([161]).

Furthermore, the principle of **purpose limitation** requires that data is to be collected for specified, explicit and legitimate purposes. Under this principle, users should be able to know how and for what purposes their data will be used and decide whether to entrust a data controller with particular data ([162]). In the present study, the purpose for the collection of data will be to support the investigation, detection and prevention of money laundering and tax evasion by FIU's and designated competent authorities.

Regarding **data minimisation**, 'the personal data should be adequate, relevant and limited to what is necessary for the purposes for which they are processed'. In practical terms, this can mean for instance that personal data collected and held by a centralised registry should be strictly necessary for the specifically determined purposes. Thus, the data held on registers has to be limited to such information that was lawfully collected under applicable national law regarding tax evasion, money laundering or terrorist financing. Moreover, it must include solely the minimum set of information needed to support the FIU's and competent authorities' detection and investigations.

On **Security** (integrity and confidentiality) – Recital (39) of the GDPR states that personal data should be processed in a way that guarantees 'appropriate security and confidentiality of the

([159]) Bhaimia, S. (2018), 'The General Data Protection Regulation: the Next Generation of EU Data Protection', *Legal Information Management*, Vol. 8, No 21.

([160]) Costa, L. and Poullet, Y. (2012), 'Privacy and the Regulation of 2012', *Computer Law & Security Review*, Vol. 28, No 254, 256; Article 29 Data Protection Working Party, 'Opinion 8/2014 on the on Recent Developments on the Internet of Things', 16 September 2014.

([161]) Onetrust Blog, 'Understanding The 7 Principles Of The Gdpr', See https://www.onetrust.com/blog/gdpr-principles/

([162]) Article 29 Data Protection Working Party (2014), Opinion 8/2014 on the on Recent Developments on the Internet of Things, 16 September.

personal data, including for preventing unauthorised access to or use of personal data and the equipment used for the processing'.

Observance of these principles is of particular importance in the context of implementing AML policy ([163]). They are further substantiated by the development of the rights of people whose data is being collected and processed, namely, data subjects' rights.

Data subjects' rights and possible restrictions

The GDPR introduced a number of rights for persons whose data is being processed (data subjects). These include for example the right to be informed on the collection and processing of their personal data ([164]) the right to obtain data from the data controller on the purpose for the processing of their personal data, the categories of data concerned and the recipients to whom the data will be disclosed ([165]).

However, these rights are not absolute, and in line with the possibility to restrict data protection rights enshrined in Article 52 of the CFREU and Article 23(1) of the GDPR provide for the possibility for Union or Member States' law to which the data controller or processor is subject, to restrict the scope of data subjects' rights ([166]) by way of a legislative measure. Such a restriction must 'respects the essence of the fundamental rights' and be 'necessary and proportionate in a democratic society to safeguard (…) (d) the 'prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, including the safeguarding against and the prevention of threats to public security'; and 'other important objectives of general public interest of the Union or of a Member State (...) in particular an important economic or financial interest of the Union or of a Member State, including monetary, budgetary and taxation matters'. In the case *Digital Rights Ireland ([167]),* the ECJ recognised that the fight against international and serious crime (such as money laundering and tax evasion) constitutes an 'objective of general interest'.

It follows that the objective of fighting against money laundering and tax evasion in the internal market could justify a restriction on persons right to data protection and privacy provided that the interference with fundamental rights is proportionate ([168]).When implementing Union policy to fight money laundering, particular regard must be given to striking a balance between interference with the fundamental rights to privacy and data protection and the necessary measure to achieve Union goals, as stated by the EDPS in its opinion 5/2020 ([169]).

([163]) EPDS (2020), Opinion 5/2020 on the European Commission's action plan for a comprehensive Union policy on preventing money laundering and terrorism financing.

([164]) Articles 13 and 15 of the GDPR.

([165]) Article 15 of the GDPR.

([166]) These are embedded in articles 12 to 22 of the GDPR.

([167]) Joined Cases C-293/12 and C- 594/12 Digital Rights Ireland Ltd v Minister for Communications, Marine and Natural Resources and Others and Kärntner Landesregierung and Others ECLI:EU:C:2014:238.

([168]) It is worth noting here that, in the Digital Rights Case the ECJ stated that "in view of the important role played by the protection of personal data in the light of the fundamental right to respect for private life and the extent and seriousness of the interference with that right caused by Directive 2006/24, the EU legislature's discretion is reduced, with the result that review of that discretion should be strict", Digital Rights Ireland, paras 45-48.

([169]) EPDS (2020), Opinion 5/2020 on the European Commission's action plan for a comprehensive Union policy on preventing money laundering and terrorism financing.

Interplay between the GDPR and the DAC

Under the DAC, the Commission has joint responsibilities as data processor with Member States. However, Member States retain the role of data controller. The Commission does not have access to data that would identify a taxpayer neither through the data exchanges under DAC1, DAC2 and DAC7, as well as the data contained in the Central Depositories established for DAC3 and DAC6.

Importantly, Article 25(1) of the DAC provides that all exchange of information in its scope shall be subject to the provisions of the GDPR. In addition, it expressly states that Member States shall restrict the scope of the obligations and rights provided for in Article 13 (information that must be provided to the data subject when data is being collected), Article 14(1) and Article 15, to the extent required in order to safeguard the interests referred to in point (e) of Article 23(1) of that Regulation' which include the important objectives of general public interest of the Union and taxation matters.

These rights that may be restricted include the obligation for the controller to communicate the modalities of the exercise of the rights of the data subject; the right to information of the data subject where personal data is collected (including specific obligations on the controller to communicate *inter alia* the name and identity of the controller and contact details of the data protection officer (Article 13)). Further limitation can be exercised regarding the right of access by the data subject (Article 15) to obtain from the controller confirmation as to whether 'his personal data is being processed'. It is worth mentioning that under the existing system for the exchange of information between competent authorities as enshrined in the DAC legal framework, there have not yet been any claims concerning breaches of data subjects' fundamental rights or provisions of the GDPR before the ECJ.

Interplay between the GDPR and the AMLD

Article 41 of the AMLD ([170]) stipulates that the processing of data undertaken in its scope is subject to the GDPR and Directive (EU) 2018/1725 ([171]). In addition, such processing shall be considered to be 'a matter of public interest under the GDPR' therefore allowing for a restriction of data subjects rights to pursue this objective insofar as it is necessary and proportional.

General Principles of EU law

In addition to fundamental rights considerations, any measure would also need to be in compliance with the principles of subsidiarity and proportionality. These principles are described below.

([170]) https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02015L0849-20180709

([171]) Regulation (EU) 2018/1725 of the European Parliament and of the Council of 23 October 2018 on the protection of natural persons with regard to the processing of personal data by the Union institutions, bodies, offices and agencies and on the free movement of such data, and repealing Regulation (EC) No 45/2001 and Decision No 1247/2002/EC (Text with EEA relevance).

Subsidiarity assessment for the creation of a single point of access to information

The principle of subsidiarity is enshrined in Article 5 of the TEU. It requires an assessment of whether the objective of creating a single point of access to information for FIU's and competent authorities (to support them in their duties related to investigations of money laundering and the predicate offense of tax evasion activities and tax recovery measures) cannot be sufficiently achieved by Member States but can rather, by reason of scale or effect of the action, be better achieved at Union level.

The 2019 Commission AML package ([172]) has outlined how criminals have been able to exploit Member State's divergent AML regimes. The fragmentation of national rules leads to 'weak links' in the EU AML framework ([173]). Indeed, the alleged money laundering and tax evasion cases show significant cross-border dimensions that would be impossible to address by Member States acting alone. The absence of an effective AML/tax evasion framework in one Member State or differing domestic laws across Member States may be instrumentalised by criminals and have consequences for other Member States and undermine the proper functioning of the internal market.

To meet the general objective of combatting money laundering and the predicate offense of tax evasion by speeding up access to comprehensive information on asset ownership for FIU's and competent authorities, as well as to ensure the aim of the EU legislative instrument is achieved, each scenario would require action both at national and EU level.

Proportionality assessment

EU institutions are bound in their action by the general principle of proportionality as enshrined in Article 5(4) TFEU. To assess the proportionality of a measure, the ECJ has developed a threefold test ([174]).

- Firstly, the measure must be suitable and appropriate to pursue legitimate objective of the Union.
- Secondly, if fundamental rights are impinged upon, it must be *strictly* necessary to reach the desired end in the sense that no less restrictive alternative could reach the same objective, and when there is a choice between several appropriate measures, recourse must be had to the least onerous one. To assess whether less restrictive alternative measures exist, it is necessary here to examine the mitigating measures in place ([175]). In addition, the mitigating measures show to which cross-border

([172]) Impact assessment template (editable version) (europa.eu)

([173]) Commission Staff Working Document Impact Assessment (2021), Accompanying the Anti-money laundering package, European Commission, Brussels, SWD(2021) 190 final. See EUR-Lex - 52021SC0190 - EN - EUR-Lex (europa.eu)

([174]) See for example R v Minister for Agriculture, Fisheries and Food, Ex p Fedesa (Case C-331/88) [1990] ECR I-4023.

([175]) Therefore, all mitigating measures in place under the current legal framework will be examined under each scenario.

information authorities currently have access to. This is crucial to establish whether it would be necessary to create a single access point at EU level and expand the scope of information accessible to them.

- Thirdly, it must not impose a burden on individuals that is excessive in relation to the objective sought, and, the EU added value of the measure (proportionality *stricto sensu*) ([176]).

To assess the legal feasibility of creating a single point of access to information for FIUs and competent authorities, the main elements that should be considered for each scenario under the necessity and proportionality test are examined in Task 6.

Conclusion

The AMLD and DAC create a framework for the collection of information by obliged entities and the exchange of targeted information cross-border between FIUs and competent authorities. However, certain limitations apply to the exchange of information such as the cumbersome and time-consuming dimension of cross-border data exchange.

The setting up of a single point of access to information on asset ownership will necessarily involve the collecting and processing of personal data and impinge upon individual fundamental rights to data protection and privacy under the CFREU and the GDPR. However, the Charter and the GDPR allow for the restriction of fundamental rights provided that the conditions and requirements developed above are met.

8. Task 6: Legal feasibility assessment – Analysis of the 3 scenarios

In the context of the EU legal framework developed above, the legal feasibility of the 3 scenarios will be assessed in this section. As developed in the previous task, the proportionality and necessity of a legislative proposal must be assessed against the objective that it pursues. Furthermore, particularly the aspect of creating new registries for assets that do not have one is explored as part of the scenarios which prescribe the creation of such registries. In this study, regardless of the selected scenario, the objective of an eventual proposal is the same. It is therefore developed below and will be referred to as 'the objective' in the following sections.

Under each asset category, the following main criteria (as already introduced under Task 5) were assessed: subsidiarity, necessity and proportionality.

Regarding subsidiarity, it is contemplated whether creating a single point of access to information for FIU's and competent authorities can be sufficiently achieved by Member States or, by reason of scale or effect of the action, rather at EU level. As part of this assessment, the status quo regarding existing infrastructure and practices are also weighed under the asset categories.

([176]) Judgments of 25 February 2010, *Müller Fleisch*, C-562/08, EU:C:2010:93, paragraph 43; of 9 March 2010, *ERG and Others*, C-379/08 and C-380/08, EU:C:2010:127, paragraph 86.

On necessity, considerations are made under the asset categories, whether action is necessary, and if so, to what extent steps should be taken. More specifically, the assessment is made based on existing legislation (including mitigating measures) and practices.

For the proportionality assessment, it was examined whether registries related to the asset exists (as already scrutinised under Task 4), and whether reporting obligations are in place to national competent authorities. This has been done under each asset category. Furthermore, the size of the asset base is also weighed (the extent to which non-financial assets are likely to be high-value), as well as the extent to which the asset under analysis is handled by obliged entities. Furthermore, as already described under Task 5, fundamental rights considerations, especially from the perspective of safeguarding privacy rules and data protection rules and principles are added under each asset category, especially if specific issues need to be pointed out.

The objective of creating a single access point of information

The general objective of a unique access point for competent authorities, FIUs and tax authorities to information on the various sources of asset ownership in the EU is to achieve 'speeding up access for competent authorities to financial information and facilitate cross-border cooperation' ([177]) to fight money laundering and the predicate offense of tax evasion.

The interconnection of registries mentioned under the following scenarios is a preferable option for several reasons. Firstly, the specific objective would be to ensure that FIUs, tax authorities and other competent authorities have efficient access, in a timely manner, to necessary and comprehensive information on asset ownership across the Member States to carry out their missions of combatting money laundering and tax evasion. As developed in previous tasks, the extent and nature of such access is always dependent on the type of authority, as well as on the powers, tasks and competencies granted to an authority according to national legislation.

Secondly, interconnection would speed up access for competent authorities to cross-border financial information and facilitate cross-border cooperation. Indeed, immediate and direct access to information is essential to ensure the success of criminal investigations related to money laundering and predicate offenses such as tax evasion.

Thirdly, it would allow for the timely identification of assets in view of the adoption of tax recovery measures and their tracing, freezing and confiscation. It follows that the objective is ultimately to support competent authorities, FIUs and tax authorities in their duties (e.g. imposing sanctions) to fight against money laundering and tax evasion by providing them with access to cross-border information on asset ownership.

([177]) European Commission, Call for tenders FISMA/2021/OP/0001, Tender specifications p. 4.

The ECJ has recognised the fight against money laundering and predicate offenses such as tax evasion as a legitimate objective of the Union capable of 'justifying a barrier to the fundamental freedoms guaranteed by the Treaty' ([178]). This objective is also recognised as a justification to the restriction of data subjects' fundamental rights under Article 32 of the GDPR.

Legal basis under EU law

As established by the ECJ's case law, the choice of a legal basis for a measure must be determined in light of 'the aim and content of the measure' ([179]).

- The existing asset registers and register interconnections at EU level are founded on different legal bases, which could serve as potential paths leading to interconnecting new registries:
- In case of the two registries set up under AMLD (Beneficial ownership registers, bank account registries), the legal basis is Article 114 TFEU. This article allows the EU legislator to adopt rules to achieve the objectives stated in Article 26 TFEU and aims at ensuring the proper functioning of the Internal Market in accordance with the relevant provisions of the Treaties. Article 114 TFEU forms the main legal basis of the AMLD, which acknowledges under Recital (1), that '[f]lows of illicit money can damage the integrity, stability and reputation of the financial sector, and threaten the internal market of the Union as well as international development. Money laundering, terrorism financing and organised crime remain significant problems which should be addressed at Union level. In addition to further developing the criminal law approach at Union level, targeted and proportionate prevention of the use of the financial system for the purposes of money laundering and terrorist financing is indispensable and can produce complementary results'.
- Regarding the Business Registers Interconnection System (BRIS), the legal basis is Article 49 TFEU, ensuring the freedom of establishment of nationals of a Member State in the territory of another Member State.
- Beyond that, Article 115 TFEU could serve as a further option for a legal basis. It could be argued that the approximation of national laws, regulations or administrative provisions of Member States on national registers is feasible, as such national legislation has direct effect on the establishment or functioning of the internal market. For example, Directive (EU) 2011/16/EU ([180]) provides for the cross-border exchange of information on real estate ownership between tax authorities under Article 115 TFEU.

8.1 Scenario 1: The legal feasibility of creating new registers/databases at national level in addition to the existing ones (Beneficial ownership, bank account) for sources of wealth for which registers do not exist at national

([178]) Judgment of 31 May 2018, Zheng, C 190/17, ECLI:EU:C:2018:357, para 38; Judgment of 25 April 2013, Jyske Bank Gibraltar, C 212/11, ECLI:EU:C:2013:270, para 64; Judgment of 30 June 2011, Zeturf, C 212/08, ECLI:EU:C:2011:437, para 45-46.

([179]) See judgment of 11 June 1991, C-300/89, Commission v Council (known as the "Titanium dioxide" case), EU:C:1991:244, p. 10.

([180]) https://eur-lex.europa.eu/legal-content/en/TXT/?uri=celex%3A32011L0016

level, and their interconnection at European level to form an EU asset registry

8.1.1 Description of Scenario 1

The first scenario requires the creation and interconnection of databases at Member State level, to create a single point of access to information for competent authorities.

In particular, this first scenario requires the EU legislator to adopt the following steps:

- To introduce an obligation on Member States to set up national registries and to adopt measures that ensure obtention of (harmonised) information which should be adequate, accurate and current on ownership of assets;
- To provide that the information is held in a centralised asset registry in each Member State.

Once the registries have been set up, the Member States would be required to ensure their central registers are interconnected via the selected platform for interconnection.

The new asset registers would be developed and administered independently by each Member State and embedded in their national legal frameworks. Once every Member State has implemented the new asset registries, they would be interconnected with other registries (existing and new registries) from each Member State.

The operational setting will rely on either a common protocol to facilitate the exchanges with the gateway or a common open architecture developed at EU level and deployed by each Member State on its infrastructure as described in the previous section (see Task 4). Lastly, an EU portal would be set up to provide FIU's and competent authorities from all Member States with access to the interconnected registries via a common gateway.

It follows that, under the first scenario, FIUs, tax authorities, and other competent authorities (excluding AML supervisors of financial institutions ([181])), will be able to access and search the registers of their own Member State and retrieve information from that of other Member States to perform their nationally varying designated duties related to combatting money laundering and tax evasion. As developed in the previous Tasks, relevant information will include at least data allowing for the identification of the owner of the asset.

([181]) Among competent authorities it is important to note that LEAs act in a criminal context in contrast with other supervisory authorities. As developed in the first section, AML supervisors of financial institutions are bound by the obligation of professional secrecy to present confidential information in aggregated form without identifying individual credit and financial institutions. It follows that they do not carry out investigations with regards to private or legal persons and therefore do not need to have access to asset registers to carry out their duties.

The data regarding the ownership of assets would be spread across different national databases. FIU's and competent authorities would be able to rely on the interconnected system to swiftly and effectively access the information and to carry out their duties of detecting, investigating and combating money laundering and tax evasion including freezing and confiscating assets where relevant.

8.1.2 General considerations

Fundamental Rights considerations ([182])

The information collected and aggregated in national databases with the aim of eventually interconnecting them together amounts to data processing under the GDPR and represents an interference with the data subjects fundamental right to privacy and data protection safeguarded under the Charter of fundamental rights of the European Union (CFREU). Hence, the legal framework developed under Task 5 regarding privacy and data protection will apply.

The rights to the respect for private and family life and the right to data protection ([183]) enshrined in Articles 7 and 8 of the CFR would prevent the collection and storage of information in registries for all the assets identified as relevant for AML/CFT and tax evasion purposes under Task 1. In line with the ECJ's case law, collecting a large amount of data on individuals that would allow 'very precise conclusions to be drawn concerning the private lives of the persons whose data had been retained such as the habits of everyday life, permanent or temporary places of residence, daily or other movements, the activities carried out, the social relationships of those persons and the social environments frequented by them' ([184]) violates citizens' rights to privacy and data protection. Thus, any collection of substantial volumes of data concerning a large scope of citizens will be considered by the Court as 'an interference with the fundamental rights of practically the entire European population' ([185]) and would present a high risk of violating citizens fundamental rights.

Therefore, it should be underlined that the potential creation and interconnection of such national registries requires careful considerations from a fundamental right, and particularly a data protection perspective. Future legislative solutions should pay particular attention to lay down rules in accordance with the principle of data minimisation, meaning that only the information *strictly necessary* in the AML/CFT and tax evasion context should be collected and stored. Furthermore, such legislative solutions should specify the framework for the accessibility of data in interconnected registries. In other words, a clear and legitimate interest should be well defined (i.e. in which circumstances can access be requested), as well as the group of public authorities and/or individuals to whom certain pieces of data can be disclosed must also be precisely set out. Related to that, processes in which data is disclosed must be

([182]) It should be noted that the fundamental rights considerations concerning data minimisation, storage of personal data, and data security applicable to scenario 1, in particular in terms of interconnection of national level registries (once additional registries have been set up) also apply to scenario 2 which envisages the interconnection of existing registries under EU law. This is particularly true regarding reliance on metadata to ensure observance of the principle of data minimisation by limiting the set of information accessible through a search to what is *strictly* necessary.

([183]) The right to data protection is also enshrined in Article 16(1) TFEU.

([184]) Joined Cases C-293/12 and C- 594/12 *Digital Rights Ireland Ltd v Minister for Communications, Marine and Natural Resources and Others and Kärntner Landesregierung and Others* ECLI:EU:C:2014:238. para 56.

([185]) Ibid.

regulated as well (e.g., the possibility of a foreign public authority to directly access the national registry without any constraints, or only after, for example, notifying the public authority maintaining the national registry in question).

Data controllership

Under Article 4(7) of the GDPR, the data controller means 'the natural or legal person, public authority, agency or other body which, alone or jointly with others, determines the purposes and means of the processing of personal data; where the purposes and means of such processing are determined by Union or Member State law, the controller or the specific criteria for its nomination may be provided for by Union or Member State law'.

Thus, the data controller will have certain duties and bear the legal responsibility for the processing of the data in a lawful way. *Inter alia*, the controller is in charge of determining the purpose and means of the processing of personal data. It is also 'responsible for the lawfulness of the processing, for the protection of the data, and respecting the rights of the data subjects' ([186]). Lastly, data subjects who wish to exercise their rights must submit their requests to the data controller ([187]).

Under Scenario 1, in principle, data controllership would be located at national level since the personal data is stored in national registries and the determination of the purposes and means by which the data is processed occurs at national level. This conclusion is true provided that the central routing component does not store personal data (as is the case for BRIS for example where the EU component stores core profile data for the purpose of searches) ([188]). If the decentralised national level databases function in the same way as the interconnected registries for insolvency ([189]), the information that is displayed after a search on the central platform is merely 'transient data' which is not stored in the EU portal.

It follows that under Scenario 1, data controllership will be spread across Member States including among the designated authorities administering the registries who, in principle, will be the decision makers in terms of means and purpose for the data processing. Each Member State would have to ensure that a data controller is designated, in line with the GDPR requirements, to determine the means and purpose of the data processing for each registry. This means that some discretion would be left to Member States as regards to the entity/type of authority designated for this purpose. For example, drawing from the existing BORIS registers, data controllers of national registries can take the form of Registry Agencies ([190]), national data protection agencies ([191]) or Ministries ([192]), depending on the Member State.

([186]) For more information on the role of the data controller see: https://edps.europa.eu/data-protection/data-protection/glossary/d_en

([187]) Ibid.

([188]) Ibid.

([189]) Available at Insolvency Registers (europa.eu).

([190]) As is the case in Bulgaria, for example. For more information on authorities designated as data controllers for the national Beneficial Ownership registers see: BORIS_DP_info_EU_EU_en.pdf

([191]) Ibid. See for example the CNIL in France.

([192]) Ibid. See for example the Ministry of Justice in Lithuania.

As such, data controllers will have to safeguard compliance with the principles deployed in Article 5 of the GDPR including lawfulness fairness, transparency, purpose limitation, data minimisation, accuracy and security. Arguably, this may be seen as an advantage of Scenario 1 compared with Scenario 3 (where the Commission would endorse the role of data controller). This is explained by the fact that Member States could be seen as better placed to ensure the enforcement and monitoring of GDPR requirements compared with the Commission.

8.1.3 Legal feasibility for creating new asset registries for assets in Scenario 1

8.1.3.1 Cash in hand

General proportionality test of creating asset registries for cash in hand

Appropriateness and suitability of creating asset registries for cash in hand

As depicted in

Table 14 and Table 15 under Task 2, no register for cash in hand exists in any of the Member States.

To provide adequate support to FIU's and competent authorities in the context of their duties, cash in hand registries would have to comprise data that is accurate, complete and updated. As shown in Task 4, this is *de facto* not operationally feasible, partly due to the high rate of change in ownership (high frequency of cash transactions), and the non-traceability and anonymity which characterise cash in hand transactions.

Necessity of creating asset registries for cash in hand

Importantly, mitigating measures are in place at EU level to tackle the inherent risks of cash in hand when it comes to tax evasion and money laundering. These include the due diligence obligation of obliged entities where payments are either made or received in the form of cash equal or above the amount of EUR 10 000. Moreover, an EU-wide ceiling of EUR 10 000 is intended to be set for cash payments to render money laundering for criminals more difficult ([193]). Member States will be able to maintain or enact a lower ceiling discretionarily in their legal systems.

In addition to the AMLD, the 'Cash Controls Regulation' ([194]) requires any traveller entering or leaving the EU territory and with an amount of cash equal or above EUR 10 000 to lodge a cash declaration to the relevant competent authority of the Member State through which they are entering the Union. Under Article 3(2) the information which must be provided in the declaration is listed and includes, inter alia, 'the carrier, including full name, contact details, including address, date and place of birth, nationality and identification document number'; and 'the owner of the cash, including the full name, contact details, including address, date and place of birth, nationality and identification document number, where the owner is a natural person, or the full name, contact details, including address, registration number and, where available, VAT identification number, where the owner is a legal person'. Thus, under the Cash Regulation, FIU's have access to information related to transactions registered under the Cash Regulation. They may also transmit this data to other competent authorities for the investigation of predicate offenses such as tax evasion under the EU AMLD framework. As explained in Section 6.2.2.1.1, this measure is not enough to ensure a complete, full and accurate dataset of cash, even solely considering cash crossing borders.

Proportionality (stricto sensu) of creating asset registries for cash in hand

According to the SNRA report ([195]), almost 57 000 cash cases attributed to criminal activities were detected in 2021 representing a total amount of EUR 37 billion. Further, customs controls detected almost 9 000 cases where cash was either incorrectly or not declared all together, representing around EUR 254 million. The true number of misuse of cash in criminal activities is likely to be significantly larger, but likely undetected.

([193]) For more information see: Anti-money laundering: Council agrees its position on a strengthened rulebook - Consilium (europa.eu)

([194]) Regulation (EU) 2018/1672 of the European Parliament and of the Council of 23 October 2018 on controls on cash entering or leaving the Union and repealing Regulation (EC) No 1889/2005.

([195]) European Commission (2022b).

In order to establish complete and accurate cash registers, not only relying on the CCR data, creating reporting obligations on individuals and businesses to report information related to cash in hand transactions would be manifestly disproportionate and impeded by the legal framework on data protection as it would be impossible to comply with the GDPR principles enshrined in Article 5 (data completeness, data accuracy, data currency). It would also involve the processing of immense amounts of data in light of the volume of cash transactions and the rapidity in the change of ownership of this asset. Thus, the existing mitigating measures are less onerous and allow for the collection of information on high value cash transactions between FIU's, tax authorities and other competent authorities in a cross-border manner.

Conclusion

The Cash Controls Regulation and AMLD frameworks are mitigating measures that serve the purpose of allowing FIU's to exchange cross-border information related to cash holders relevant for their duties in specific situations. In addition, Task 4 has shown that it is not operationally feasible to create national registries for this asset due to the technical and operational barriers impeding the collection of information that would be complete, accurate and frequently updated. It appears that the setting up of registries for cash in hand is not appropriate and manifestly disproportionate to meet the objective.

8.1.3.2 Listed securities

General proportionality test of creating asset registries for listed securities

Appropriateness and suitability of creating asset registries for listed securities

A listed security is a financial instrument which is traded through an exchange (i.e. registered with a CSD). As developed under Task 2, although databases covering listed securities exist in all Member States (see *inter alia* tables 14-17), they are all managed entirely by private entities (CSD's act as the primary registry for the holders of securities - see Task 2).

According to ESMA ([196]), EUR 13 400 trillion EUR in portfolios securities is held cross-border reflecting a high level of cross-border ownership of this asset. As further developed below, the current legal framework for securities ensures that information is collected by obliged entities under their obligation to conduct due diligence with their customers.

The information collected by CSD' and financial intermediaries does not systematically include the identity of the beneficial owner of the asset. This is partly due to the differing national definitions of shareholder ([197]) which in some cases does not allow for the identification of the end investor (or beneficial owner) due to the complex chains of intermediaries and reliance on omnibus accounts. Therefore, as reflected in the operational feasibility assessment under Task 4, it is currently not possible for all Member States to obtain complete and accurate data on ownership of securities that would provide competent authorities, FIUs and tax authorities with relevant information for their missions.

([196]) ESMA (2021).

([197]) Company law remains an area of shared competence between the EU and Member States.

In addition, the high volatility of the price of securities and rapid change in ownership means that collecting and storing information that is accurate in national registries would entail frequent updates and significant resources as outlined in Task 4.

The main challenges in terms of appropriateness of setting up national registries for listed securities would lie in the difficulty to obtain relevant and pertinent information for the purpose of detecting money laundering activities and predicate offenses of tax evasion and allowing for tax recovery measures ([198]) (as further developed below this is due to the existing legal frameworks at national and EU level). In addition, as shown in Task 4 (section 6.2.2.1.2), the frequent updates that would be required due to the regular and fast paced fluctuations (and difficulty of valuation) of security prices also entail significant challenges.

Necessity of creating asset registries for listed securities

Reporting obligations related to listed securities for tax evasion and money laundering purposes are in place under the AMLD and the DAC. Moreover, other pieces of EU legislation create reporting obligations and increase transparency in this sector for the purpose of safeguarding the integrity of the single capital markets union.

Mitigating measures exist at EU level under the AMLD, the CSD Regulation, Market in financial Instruments ('MiFiD' I and II) ([199]) and the DAC which provides automatic exchange of information on an annual basis between the Member State where the financial account is held and the Member States where the financial account holder is resident. Under the current EU AML framework, the AMLD under its article 2(1)(3)(i), (ii), and (iii) includes natural or legal persons acting in the exercise of their professional activities including managing of clients' money, securities or other assets and opening/management of securities accounts. Hence, entities conducting management of securities and opening account for such assets are obliged entities under the Directive and will therefore be subject to the customer due diligence obligations set out in Chapter II. These due diligence obligations include *inter alia*: customer identity verification on the basis of documents, data or information obtained from a reliable and independent source; identification of the beneficial owner; assessing the nature and intended purpose of the business relationship and conducting ongoing monitoring of the business relationship including security of transactions (…) and ensuring the customer information documents/source of the funds is kept up to date ([200]).

DAC2 ensures the cross-border exchange of data related to financial asset values, gross proceeds and passive financial incomes such as dividends and interest income of equity securities held by a custodian (in fact in the EU all listed securities are defined as held by a

([198]) Particularly on the beneficial ownership of listed securities.

([199]) Consolidated text: Directive 2014/65/EU of the European Parliament and of the Council of 15 May 2014 on markets in financial instruments and amending Directive 2002/92/EC and Directive 2011/61/EU (recast) (Text with EEA relevance).

([200]) See Article 13 DAC.

custodian given that they are registered in a national CSD). This data is exchanged cross-border between the competent authorities for tax purposes if the investor has a financial account in another Member State.

The Shareholder rights Directives (I and II) ensure the possibility for listed companies to identify their shareholders. However, due to the different definitions of shareholders in Member State's legal systems, the identification of beneficial ownership is not yet possible across all Member States ([201]). Thus, the creation of national databases before the adoption of an EU definition means the data would be heterogeneous and, in some cases, lack relevance for the work of FIUs and competent authorities (for example if it does not allow for the identification of the beneficial owner but merely identifies the Central Security Depository as the security owner in some cases).

Alongside, the Central Security Depository Regulation ([202]) ('CSDR') and MiFiR both contribute to the financial and securities sector transparency. Among the reporting obligations, the CSDR settlement internalises must report to their competent authorities on a quarterly basis the aggregated value and volume of all securities transactions settled outside the securities settlement systems State; competent authorities must transfer this data to ESMA and inform of any potential risk resulting from that settlement activity. Due to the aggregated nature of this data, it will not be useful to create national registries which will have to draw on specific information on securities and their holders rather than aggregated data. The CSDR provides for dematerialisation and record keeping obligations, but this does not necessarily allow for the identification of the beneficial owner since often only large omnibus clients as asset owners are identified ([203]). MiFiR introduced transparency requirements for equities, bonds, structures finance products, etc., thereby increasing the transparency of the market.

A recent COMM Proposal proposes to create reporting obligations on (EU and non-EU) financial intermediaries concerning information on listed securities income flow to be exchanged with the securities issuers in Member States for withholding tax relief purposes ([204]). Within the scope of this reporting obligations are equity securities, and Member States may choose to also include debt securities. This reporting obligation will provide tax authorities with additional cross-border information on payment of dividends or interest to relevant tax administration, allowing the latter to identify more effectively the legal and beneficial owner of the listed security.

Existing database for financial instruments

Article 26 of Regulation 600/2014 (MiFIR ([205])) creates an obligation on trading venues to supply transaction data to competent authorities, to which ESMA has access to combat financial abuse purposes. This data allows for systematic and harmonised data collection on

([201]) For example, in Austria, Belgium, Bulgaria, Croatia, Estonia and the Netherlands, the national definition of shareholder allows for the identification of the beneficial owner.

([202]) Regulation (EU) No 909/2014 of the European Parliament and of the Council of 23 July 2014 on improving securities settlement in the European Union and on central securities depositories and amending Directives 98/26/EC and 2014/65/EU and Regulation (EU) No 236/2012 Text with EEA relevance.

([203]) This is further developed under Task 1.

([204]) COM(2023) 324 of 19.06.2023 Council Directive on Faster and Safer Relief of Excess Withholding Taxes.

([205]) Regulation (EU) No 600/2014 of the European Parliament and of the Council of 15 May 2014 on markets in financial instruments and amending Regulation (EU) No 648/2012.

securities holdings and transfers as developed in Task 4. Nevertheless, the purpose of this data is not to fight money laundering and tax evasion, or to allow for tax recovery measures to be taken. Hence as developed above, the information to which ESMA has access does not systematically include all information relevant for the work of FIUs and competent authorities such accurate data on the beneficial owner of the security or its value. It is important to note that this applies to all securities except derivatives, for which accurate identification of owners is possible across all Member States in a systematic manner. More specifically, as previously mentioned under Task 4, existing omnibus accounts allow that only the owners can be identified, however, the beneficial owners of the account cannot be identified.

Proportionality (stricto sensu) of creating asset registries for listed securities

As touched upon it earlier under Task 4 (see point 6.2.2.1.2), data collected through transmission of transaction information (for example to national regulators or ESMA) has reporting obligations (and infrastructure) in place (paving the way to collect relevant and up-to-date data). However, shortcomings can be detected, because the current reporting obligations fall short on identifying beneficial owners, as well as in terms of purpose, as the reported data is to ensure the functioning of the capital market and not to combat tax evasion and money laundering. An additional practical challenge is due to the large amount and frequency of securities exchanges, in many cases taking place cross-border (according to ESMA; approximately 13.400 trillion euros in portfolio securities), which testifies the complexity of the market.

Nevertheless, if the legal standing of the current reported data were to be adjusted, this existing database could be used to acquire data related to listed securities, so creating a new register could be avoided. It could be argued, that creating a completely new registry on securities ownership data would be burdensome for both issuers (as large amount of data would have to be reported) and competent authorities (as a centralised infrastructure would have to be created and data reporting be enforced). It should be added, that this reporting obligation could raise issues related to legal certainty, as issuers (or financial intermediaries) could be uncertain to what pieces of data to report on, hence they could either underreport (thus risking sanctions from an authority) or overreport (thus risking an overreach in terms of reporting on data that are not required, consequently violating data protection rules) in a fast-evolving market.

Conclusion

As developed above, the listed securities market is characterised by a degree of transparency due to the existing legislation (CSDR, MiFid II and MiFiR, SRD I and II) and registers already exist in many Member States, though they are held by highly regulated private entities ([206]). Moreover, cross-border exchange of information between tax authorities related to financial incomes such as dividends and interest income of equity securities held by a custodian is provided under the DAC. This is also the case under the AMLD whereby obliged entities (such as custodian banks) collect information on the opening and managing of accounts for their customers which is accessible for FIU's and must be transmitted to another Member State's FIU if suspicions related to money laundering activities are detected. It follows that, mitigating

([206]) Importantly these registers are currently all administered by private entities. The possibility to transfer such data for the purpose for aggregating it in a publicly managed asset registry may be a possibility but heavy legal implications would be at play – *inter alia* the purpose would have to be compatible with that of combating AML and tax evasion.

measures exist and FIUs and competent authorities have access to information on this asset via these channels of information exchange.

8.1.3.3 Unlisted securities

General proportionality test of creating asset registries for unlisted securities

Appropriateness and suitability of creating asset registries for unlisted securities

The operational feasibility assessment conducted under Task 4 has shown that currently, for these assets, there is a lack of complete, accurate and current data (section 6.2.2.1.3). The operational difficulties to collect complete, current and accurate information on unlisted securities may render the setting up of national registers highly challenging (and perhaps inappropriate) thereby explaining the reasons why there are currently no such registers for these assets as established in Task 2.

This may be due in part to the fact that this asset is less regulated than listed securities and is the subject of fewer reporting obligations, as developed in previous tasks. If ensuring that the data is current and up to date is *de facto* complicated, this undermines the appropriateness of setting up national registers. Information which is not up to date would not meet the objective as it would not provide the authorities with usable data for their missions.

At the same time, it should be mentioned that for the two largest sub-groups of unlisted securities (as mentioned under Task 4), namely closely owned legal persons and widely held unlisted securities, the potential for mitigating measures varies:

- On the one hand, for closely owned legal persons, it is likely that the beneficial ownership registry would already indicate who the owners are, as the beneficial ownership threshold for identification purposes in case of legal persons is above 25%, thus a significant amount of information is already available through BORIS or BRIS.
- On the other hand, widely held unlisted securities resemble difficulties related to listed securities, because ownership is fragmented (as such the direct and indirect ownership will be below the 25% threshold for beneficial ownership identification purposes). In addition, widely held unlisted securities are less regulated under the EU financial regulation framework and they are also not covered by AEOI under DAC2 if they are considered as non-custodial. A provision under DAC8 will require automatic exchange of information between Member States on non-custodial dividends. However, as with the existing non-financial categories under the DAC, the automatic exchange of information will depend on the availability of information at national level in the sending Member State.

Necessity of creating asset registries for unlisted securities

Under the AMLD, some obliged entities may be involved in the management of funds including unlisted securities and therefore collect information via their due diligence obligations on holders of unlisted securities. However, this would in no way ensure the collection of information on holders of all unlisted securities.

For tax purposes, the DAC provides for the automatic exchange of information on income on custodial dividends. However, most unlisted securities are not custodial and will therefore not be the subject of the obligation to automatically exchange information. Remedying this, an

amendment to DAC8 will ensure that information on non-custodial securities is also exchanged automatically between tax authorities ([207]). However, this exchange will only take place if the information is already available to the tax authority to be reported.

In line with the findings of Task 4 (section 6.2.2.1.3) and due to the nonexistence of current and harmonised data on holders of unlisted securities, the collection of such data would create a consequent burden on issuers and financial intermediaries. The burden is even more significant considering the high volume of unlisted securities issues, their rapid change in ownership and large number of potential reporters.

Proportionality (stricto sensu) of creating asset registries for unlisted securities

In terms of unlisted securities, the difficulties are reminiscent to the points mentioned for listed securities, although complexities stemming from the cross-border element is less relevant for unlisted securities ([208]). According to Oxera, the equity markets of 23 Member States revealed a foreign ownership share for listed and unlisted securities of more than 30% in 2018, which indicates that unlisted securities show a growing asset base in the EU ([209]).

Currently, there is a lack of regulatory framework in the EU. Consequently, harmonised data on holder of unlisted securities do not exist (although for closely owned legal persons, BORIS and BRIS could provide some valuable information). In case of widely held unlisted securities, finding relevant data and properly reporting on it would create a notable financial and reporting burden on issuers (and financial intermediaries) and competent authorities alike. Furthermore, as already described under Task 4, the market of unlisted securities is complex, as the number of issuers of unlisted securities is wide-ranging. This could raise questions of legal certainty for issuers, as they might fail to report necessary data (thus risking sanctions from the authority) or might opt for overreporting pieces of data (which could lead to violating data protection rules).

Conclusion

Available data on unlisted securities is scarce, especially regarding widely held unlisted securities. As things stand, relevant information through BORIS could be collected concerning closely owned legal persons (as also underlined by the ESMA report in Task 4), offering an opportunity for data collection on some unlisted securities. However, widely held unlisted securities have fragmented ownership and low level of regulation under the EU financial regulatory framework. This means that collecting information on these types of unlisted securities would significantly increase the burden stemming from reporting obligations for issuers and for competent authorities as well.

([207]) Excluding dividends exempt from the Parent Subsidiary Directive: Council Directive 2011/96/EU of 30 November 2011 on the common system of taxation applicable in the case of parent companies and subsidiaries of different Member States (europa.eu)

([208]) https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=SWD%3A2014%3A0127%3AFIN

([209]) Oxera (2020).

8.1.3.4 High value goods (Works of Art, Antiques, Collectors' items, Jewellery) and firearms

General proportionality test of creating asset registries for high-value goods

Appropriateness and suitability of creating asset registries for high-value goods

Task 2 mapping has indicated that, while all Member States have in place registers for firearms administered by public entities, works of art, collectors' items, antiques and jewellery have no comparable national registers, but rather information gathered through transaction data.

The appropriateness and suitability of setting up national registers for high-value assets must be assessed considering the level and coverage of the existing data, its accessibility, as well as the operational challenges outlined in Task 4.

The implementation of national registers for works of art, antiques, collectors' items and jewellery is problematic due to factors such as the private character of existing data, lack of structured reporting obligations, unclear criteria and unreliability of data. Additionally, the case Study for Malta (Annex F) shows that drawing information from existing private registers to substantiate national centralised registers would be extremely challenging. Examples of these challenges include a lack of incentive for private entities to share their data, outdated IT systems, and the absence of a legislative framework to ensure structured reporting obligations.

Moreover, as shown under Task 4, there is currently no competent authorised body to record the information and no adequate legal framework to ensure the reporting of structured data ([210]). Operationally it is not currently possible for Member States to obtain data that is complete, accurate and current, which is a pre-requisite to ensure that the data centralised in registers is useful to authorities and *in extenso* meets the sought objective.

Additionally, as the relevant sections in Task 4 and the SNRA have indicated, most of the financial crime involving these high-value assets occurs with illicit trades of stolen goods using private parties instead of professional traders or auctioneers. These goods and their transactions would therefore not appear in any database constructed based on legitimate trades of high-value goods, rendering these databases of limited use.

Necessity of creating asset registries for high-value goods

The AMLD obliges art dealers and auction houses to conduct customer due diligence on high value transactions of an amount equal or above EUR 10 000. Article 21(31)(c)(i) and (j) of AMLD includes as obliged entities 'persons trading or acting as intermediaries in the trade of work of art, including when it is carried out by art galleries and auction houses, where the value of the transaction or a series of linked transactions amounts to EUR 10 000 or more' and 'persons storing, trading, or acting as intermediaries in the trade of works of art when is it carried out by free ports, where the value of the transaction or a series of linked transactions

([210]) Regulation (EU) 2019/880 creates a licensing framework for the import in the internal market of certain cultural goods. However, it is limited to the import of certain cultural goods and therefore does not exhaustively cover all high value goods – including those already in circulation in the internal market.

amounts to EUR 10 000 or more'. Data must be collected by these intermediaries and service providers (as 'obliged entities') for transactions reaching the EUR 10 000 thresholds under due diligence obligations.

In what high-value assets are concerned, the AMLD framework only refers to works of art traded by persons, galleries, auction houses or other intermediaries, if their value is of EUR 10,000 or more, but does not deal separately with other potentially high-value items such as antiques or jewellery. This could be explained, on the one hand, by the general approach of the AMLD to set up effective AML/CFT mechanisms without creating too burdensome reporting obligations for obliged entities, bearing in mind 'their size, characteristics and the nature of the business', as well as the need to balance the associated compliance costs – see for instance Recitals 2 and 59 of the AMLD. Additionally, given the lack of national registers and the private character of data, as well as the overall unreliability of data concerning especially items such as jewellery and antiques which are not traded as works of art, the imposition of AML reporting and due diligence obligations could be considered highly disproportionate for these categories of goods.

Further, the legal framework which applies to cash at hand transactions developed above also applies to high value goods thereby preventing the dissimulation of buyers' identity through cash payments. As developed in the previous chapter, and in line with the general provisions in Article 33 AMLD, all suspicious transaction must be reported by obliged entities to FIUs and regardless of the amount, on the basis of the risk-based approach.

In addition, Regulation (EU) 2019/880 on the introduction and import of cultural goods creates a framework to address the risks of money laundering and stemming from the illicit trade in pillaged cultural goods ([211]). Under the first Recital of the Regulation, common rules on trade with third countries are necessary 'as to ensure the effective protection against illicit trade in cultural goods and against their loss or destruction, the preservation of humanity's cultural heritage and the prevention of terrorist financing and money laundering through the sale of pillaged cultural goods to buyers in the Union'.

The Regulation requires Member States to issue import licences for the import of cultural goods regardless of their value (such as archaeological objects or parts of monuments at least 250 years old). For goods of a value above EUR 18 000 and older than 200 years, an importer statement must be submitted by the importer to customs. In line with Article 5(2)(a) and (b) of Regulation (EU) 2019/880, the statement must comprise *inter alia* 'a declaration signed by the holder of the goods stating that the cultural goods have been exported from the country where they were created or discovered' and 'a standardised document describing the cultural goods in question in sufficient detail for them to be identified by the authorities and to perform risk analysis and targeted controls'. Moreover, under Article 8(1) of the same Regulation, the 'storage and the exchange of information between the authorities of the Member States, in particular regarding import licences and importer statements, shall be carried out by means of a centralised electronic system' set up and maintained by the Commission. Although the collection of data to substantiate national registers could draw on this existing database, it must

([211]) Recitals (1) and (11) of Regulation (EU) 2019/880 of the European Parliament and of the Council of 17 April 2019 on the introduction and the import of cultural goods; OJ L 151, 7.6.2019, p. 1–14.

be noted that only certain specific categories of high value goods (cultural goods with certain characteristics entering the internal market) are covered.

Currently there are no specific mechanisms for automatic information exchange for high-value goods under the DAC. Tax authorities in Member States can avail of the requests for information provisions under the DAC but they would need to know in the first place where the asset is located and could not rely on blanket requests.

Additional sources of information exist that could facilitate the tracing of high-value assets, such as import licensing documentation, VAT declarations and insurance information. In light of this, the current state of play suggests that the existing mitigating measures covering high value goods (or their potential improvement) may amount to less restrictive means to achieve the objective of ensuring FIU's and competent authorities to access and exchange the information necessary for their duties.

Proportionality (stricto sensu) of creating asset registries for high-value goods

Moreover, given the highly heterogenous landscape of registers across the EU for this category of assets, as well as the absence of clear and uniform reporting criteria, creating an obligation on Member States to collect such information would amount to a significant administrative and financial burden. In addition, the collection and storage of data on all art ownership and high-value assets would present a high risk of violation of citizens fundamental rights to data protection and private life as enshrined in the GDPR and the CFR.

Conclusion

As outlined under Task 4, it is not somewhat feasible to ensure the collection of complete, accurate and up to date information on high value good and their holders. The available data, however, would only cover legitimate trades involving legitimate parties, not private trades with stolen goods. In addition, the lack of any designated competent authority or adequate legislative framework to ensure the collection of harmonised data means that the administrative burden on Member States would be significant (see Task 4). The current state of play suggests that enhancing the existing mitigating measures outlined above may amount to less restrictive means to achieve the objective of ensuring FIUs and competent authorities can access and exchange the information necessary for the performance of their duties.

Furthermore, in order to counteract the illegal trade of stolen works of art or other valuables, more effort could be put into developing databases of stolen goods. This would allow for traders to accurately identify whenever an illicit party wants to traffic a stolen good in their premises and report such trade if necessary.

8.1.3.5 Commodities (precious raw materials)

General proportionality test of creating asset registries for commodities

Under Task 1, commodities were defined as 'precious raw metals' comprising gold bars, uncut diamonds, and other such fungible assets of high value. The findings suggest that this asset is of high relevance to the fight against money laundering and tax evasion. Indeed, the literature review conducted in Task 1 shows that the material characteristics of commodities (i.e. low storage costs due to small volume, easily convertible in cash or other commodities) make it a privileged asset for the purpose of evading tax and laundering money.

Appropriateness and suitability of creating asset registries for commodities

The mapping of existing registers conducted in Task 2 shows that in all Member States data on these assets are exclusively held at company level, providing only private access to information. Currently, there is divergence in legislation surrounding commodities in Member States. Yet, but the Cash Controls Regulation lays down certain reporting obligations to the largest groups of diamonds traded, as well as related to gold bars, nuggets and pure gold coins. As a result, no publicly administered national registers for commodities were identified in the Member States. Moreover, commodities are not covered by the automatic exchange of information rules under DAC. It should be noted that the AMLD6 package foresees adding entities dealing with diamonds and gold to have reporting obligations.

In this context, setting up additional national registers in the Member States and interconnecting them for the purpose of fighting against money laundering and tax evasion in respect of this category of assets could appear as a suitable and adequate measure. Considering the high risk that commodities pose, as well as their characteristics which make them suitable to be easily transported across borders (and thus difficult to control), the potential solution proposed under Scenario 3 could also be considered from the perspective of adequacy and suitability.

Necessity of creating asset registries for commodities

Regulation 2018/1672 ([212]), or the 'Cash Regulation', provides in Article 2(1)(a)(iii) that commodities used as a 'highly-liquid stores of value' are considered as 'cash' under its scope. Moreover, Article 2(1)(e) defines the notion of 'commodity used as a highly-liquid store of value' meaning a good which is either coins with a gold content of at least 90% and (b) bullion such as bars, nuggets or clumps with a gold content of at least 99,5% ([213]). Thus, under the Regulation, other items such as diamonds are not considered commodities in the sense of the Regulation and, *in extenso,* 'cash'.

The framework developed in the section on cash in hand will therefore be relevant. Given that gold coins and bars clumps with a gold content of at least 99,5% gold will amount to 'cash', they will be covered by the obligations stemming from the Cash Regulation and developed above, in the 'cash in hand' section. These include lodging a declaration with the competent authorities including details on the carrier, the owner and the cash itself when entering and leaving the Union with a cash value above EUR 10 000. Under Article 9 of the Cash Regulation, information obtained by the competent authorities is recorded, transmitted to the FIU of the respective Member State and is subsequently subject to exchange with the relevant FIUs of the other Member States in accordance with Article 53(1) AMLD. Furthermore, pursuant to

([212]) Consolidated text: Regulation (EU) 2018/1672 of the European Parliament and of the Council of 23 October 2018 on controls on cash entering or leaving the Union and repealing Regulation (EC) No 1889/2005.

([213]) Annex 1 of Regulation (EU) 2018/1672 of the European Parliament and of the Council of 23 October 2018 on controls on cash entering or leaving the Union and repealing Regulation (EC) No 1889/2005

Article 10 of the same regulation, 'the competent authority of each Member State shall, by electronic means, transmit the following information to the competent authorities of all the other Member States: (a) *ex officio* declarations composed under Article 5(3); (b) information obtained under Article 6; (c) declarations obtained under Article 3 or 4, where there are indications that the cash is related to criminal activity and (d) anonymised risk information and risk analysis results'. Although the Cash Regulation appears to cover only gold, the AMLD provides for obligations regarding transactions in cash above EUR 10 000 and any occasional transaction above EUR 15 000, as described in Task 5. Moreover, regardless of the amount involved, all obliged entities must report suspicious transactions, in line with the risk-assessment approach. It follows that gold and other raw precious metals, provided they either present a suspicious character or reach the indicated monetary threshold, will be the subject of due diligence and reporting obligations under the current AMLD framework. In addition, according to the Commission's proposed regulation on AML/CFT, 'cash' will be defined to include, among others, 'commodities used as highly-liquid stores of value' ([214]) thus falling under the newly proposed rules on limits to cash transactions (over EUR 10 000), in addition to reporting and due diligence obligations concerning suspicious transactions.

Currently there are no specific automatic exchange of information mechanisms for commodities (precious raw materials) under the DAC. Tax authorities in Member States can avail of the requests for information provisions under the DAC but they would need to know in the first place where the asset is located and could not rely on blanket requests.

Proportionality (stricto sensu) of creating asset registries for commodities

Whilst the setting up of additional national databases could provide FIUs and competent authorities with quick and efficient access to information (for authorities having the legal right to access to such information) given the high relevance of commodities for money laundering and tax evasion, it will also impinge upon the fundamental rights of individuals. Indeed, the collection and processing of information regarding holders and commodities will be a limitation of the rights to privacy and data protection. Consistently with the principle of data minimisation, the data collected will have to be limited to what is strictly necessary to pursue the declared objective of the AMLD, namely the 'protection of the financial system by means of prevention, detection and investigation of money laundering and terrorist financing', ([215]) (for instance, the name and details of the asset holder and specific characteristics of the asset, which type of commodity, what is its storage liquid value). Another important consideration is the scope of the data subjects covered by the new registries. Given that a high number of persons may be concerned as holders of commodities, this will increase the seriousness of the limitation.

However, given the value of these types of assets and their high potential for use in money laundering/ terrorist financing activities, the interference with data subjects' rights may be considered justified and strictly necessary to achieve the legitimate aims pursued. In this sense, it is also to be noted that under the proposal for a new AML/CFT regulation, the explanatory notes expressly reference 'dealers in precious metals and stones, who, given the

([214]) See Article 2(30) of the proposed AML/CFT regulation, available at https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52021PC0420.

([215]) Recital 64 of the AMLD. See also Recitals 1 and 2 – AMLD text available at https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32015L0849.

exposure to money laundering and terrorist financing risks of the sector, should continue to apply AML/CFT requirements'.

Consequently, for the purposes of the balancing exercise required under the strict proportionality assessment, the high value of the goods (over EUR 10 000) and their potential misuse for money laundering and/or terrorist financing activities should be weighed against the principles of privacy, data protection and data minimisation.

Conclusion

Member States chose different regulatory solutions regarding commodities, leading to a fragmented legislative landscape in the EU. At the same time, some types of commodities fall under the scope of the Cash Controls Regulation (for example certain types of diamonds), which lays down specific reporting obligations on assets under its scope. Creating a database could ensure a fast and efficient access to relevant information on ownership of commodities. Particular attention should be paid to limit data collection to what is strictly necessary under the objective of the AMLD, to safeguard privacy and to uphold data protection rules.

8.1.3.6 Safe-deposit boxes at non-banks

Safe-deposit boxes are provided by two types of service providers. On the one hand, banks (e.g. financial institutions) and on the other non-banks, private companies.

General proportionality test of creating asset registries for safe deposit boxes at non-banks

Appropriateness and suitability of creating asset registries for safe deposit boxes at non-banks

At least one provider of safe-deposit boxes at non-banks (registering transaction data) was encountered in each Member State in the course of the mapping of existing registers under Task 2. Due to the lack of reporting obligations on the content of safe deposit boxes, creating registries that do not include this information would not be appropriate as it would not be completely useful for the competent authorities. The enactment of legislation allowing for the collection of information on the content of safe deposit boxes at non-banks would therefore be a pre-requisite to the setting up of national registers.

Necessity of creating asset registries for safe deposit boxes at non-banks

Safe-deposit boxes at banks are covered by the AMLD and are included in a national database in accordance with Article 32a (which will be explored in the analysis for Scenario 2). Conversely, providers of safe deposit boxes at non-banks are not included as 'obliged entities' under the AMLD framework, although Member states may, under the risk-based approach, extend the scope of the AMLD obligations 'in whole or in part to professions and to categories of undertakings, other than the obliged entities referred to in Article 2(1), which engage in activities which are particularly likely to be used for the purposes of money laundering or terrorist financing' (Article 4 AMLD) and may retain or adopt stricter provisions than those of the AMLD, within the limits of Union law (Article 5 AMLD). For instance, as developed in Task 3, in the Netherlands, providers of non-bank safe deposit boxes are considered obliged entities and therefore supervised by the Dutch Central Bank which carries out due diligence obligations and customer records. Furthermore, safe deposit boxes at non-banks are not covered by the automatic exchange of information mechanisms under the DAC.

Given that safe deposit boxes at non-banks may carry a similar potential for use in money laundering and/or terrorist financing activities as safe-deposit boxes at banks, the setting up of national registers might be considered as a necessary measure, contributing to the attainment of the objectives concerning the prevention, detection and investigation of money laundering and terrorist financing as well as those regarding the fight against tax evasion and recovery of taxes. However, if such measures are to be modelled by the rules concerning safe deposit boxes at banks, data retained in respect of these assets would generally include the name of the lessee, complemented by additional identification data/a unique identification number and the duration of the lease period (see Article 32a(3) of the AMLD). It should therefore be considered to what extent this set of data would be necessary to tackle the risks of ML/TF and/or tax evasion, or to what extent such data should be further complemented by information on the content of the safe deposit box – a component that is further analysed under the strict proportionality test.

In a recent study to evaluate free zones in the EU, FIU and tax authorities were asked whether they considered that free zones had a higher risk of money laundering and tax evasion than other sectors in their jurisdiction ([216]). Some free zones, like the Luxembourg free zone, offer storage facilities for high-value goods. Competent authorities confirmed that they did not consider free zones at a high risk of money laundering and tax evasion compared to other sectors. Further, they confirmed that their free zones were subject to their national AML framework.

Currently there are no specific automatic exchange of information mechanisms for safe deposits at non-banks under the DAC. Tax authorities in Member State can avail of the requests for information provisions under the DAC but would need to know in the first place where the asset is located and could not rely on blanket requests.

Proportionality (stricto sensu) of creating asset registries for safe deposit boxes at non-banks

The operational feasibility assessment (Task 4 - section 6.2.2.1.6) shows that the burden on providers of safe deposit boxes would be consistent. This is further sustained by the fact that privacy is often an important element of the business model of providers of safe deposit boxes. Therefore, small companies providing these services could endure a significant financial loss as a result of the setting up of registers allowing for the identification of customers and including information on the content of the boxes and relevant owner (if distinct from the person relying on the service).

In addition, the creation of a reporting infrastructure and obligations would also amount to an administrative burden on Member States and providers of safe deposit boxes.

For the purposes of the balancing exercise required under the strict proportionality analysis, the aspects to be considered include, on the one hand, the relative importance of safe-deposit boxes at non-banks in ML/TF and/or tax evasion activities and, on the other hand, (i) the potentially limited use of data concerning owners/lessees of safe deposit boxes in attaining the legitimate objectives sought, (ii) the intrusive character of potential measures requiring the disclosure of the contents of the safe deposit boxes in relation to privacy and data protection

([216]) European Commission (2023b).

obligations as well as (iii) their impact on the business model of providers of safe deposit boxes and the compliance burden which would be placed on small businesses offering such services.

Conclusion

There is a risk of arbitrage when banks are obliged entities under the EU AMLD framework while non-banks which may provide the same service are not. Nevertheless, the lack of data on high-value assets held by non-banks limits any assessment of the impact for money laundering and predicate offences in light of the burden imposed on storage intermediaries and citizen being required to collect and report data.

8.1.3.7 Crypto Coins and NFTs

General proportionality test of creating asset registries for Crypto coins and NFTs

As indicated in the literature review (see section 3.2.3.3), reliance on this asset has increased with a total market capitalisation of EUR 2.5 trillion in 2021 globally ([217]) and over 9 000 virtual assets available on today's market ([218]). In addition, new virtual assets are emerging at a fast pace ([219]). Although little data is available on the size of the illegal activity market in proportion to the overall cryptocurrency market ([220]), these assets should nonetheless be the subject of particular attention when avoiding money laundering activities and reducing tax evasion.

Appropriateness and suitability of creating asset registries for Crypto Coins and NFTs

As previously shown, there are currently no registries for cryptocurrencies and NFTs at national level. In addition, the Task 4 acknowledges the purely technical feasibility of setting up decentralised registries containing information on wallet holders, although there is currently no competent authority or authorised body to collect and record the information. It shows that ensuring that the data is updated poses further challenges which could raise doubts on the appropriateness of setting up national registries as a means to exchange information cross-border. However, it is important to emphasize that establishing a registry is currently not seen as feasible due to the outstanding amount of harmonisation required across Member States, as elaborated in Section 6.2.2.1.7.

([217]) https://www.ecb.europa.eu/pub/financial-stability/fsr/special/html/ecb.fsrart202205_02~1cc6b111b4.en.html

([218]) Tax transparency rules for crypto-asset transactions (DAC8) (europa.eu)

([219]) Study on developments with regard to virtual assets users and the possibility to set-up and maintain a central database registering users' identities and wallet addresses accessible to financial intelligence units (FIUs)" - Study on developments with regard to virtual assets users and the possibility to set-up and maintain a central database registering users' identities and wallet addresses accessible to financial intelligence units (FIUs)" - Publications Office of the EU (europa.eu)

([220]) Ibid.

Although custodian wallet providers and providers engaged in exchange services between virtual currencies and fiat currencies fall in the scope of the AMLD as obliged entities and therefore collect information under their due diligence obligations, this does not allow for the collection of information on assets and their holders as transactions remain largely unregulated. However, the upcoming amendment of the proposal to recast the regulation on information accompanying transfers of funds and certain crypto-assets ('TFR' ([221])) and the Proposal for a Regulation on Markets in Crypto-assets ('MiCA' ([222])) will greatly increase transparency in the virtual assets market (this is further explored below when examining the mitigating measures in place). Until this legislation is implemented, Member States would not be able to draw on existing data collection obligations to substantiate potential registers. In addition, in Member States which have not enacted a travel rule for crypto assets, it is not yet possible to identify the originator and the beneficiary of a transaction between two Virtual Asset Service Providers ('VASPs'). Currently, crypto asset can be transferred anonymously and quickly. This renders the collection of information on beneficiary holders of assets impossible – although the TFR Regulation explored below will change this.

In the long-term, once the recent EU legislation (MicA and TFR) is implemented in all Member States, the setting up of additional registers could eventually be envisaged. However, until the EU legal framework allows firstly, for the systematic identification of holders of crypto assets and, secondly, requires the collection of harmonised data on transactions of virtual assets, setting up national registries will be extremely challenging in the short term. Further, the setting up of a centralised database at EU level, as mentioned in Article 65 AMLD may be more appropriate as it would de facto overcome the challenge of ensuring the interoperability of national centralised registries. In light of these considerations, the setting of national registries may not be deemed as an appropriate tool to tackle the risks of this asset and meet the objective.

Necessity of creating asset registries for Crypto coins and NFTs

The AMLD5 has extended the material scope of the AMLD legal framework by introducing harmonised definitions of 'virtual asset' and 'custodian wallet provider' ([223]). The notion of 'virtual currency' is defined as 'a digital representation of value that is not issued or guaranteed by a central bank or a public authority, is not necessarily attached to a legally established currency and does not possess a legal status of currency or money but is accepted by natural or legal persons as a means of exchange and which can be transferred, stored and traded electronically'. Regarding 'Custodian wallet providers', they are entities providing services to safeguard private cryptographic keys on behalf of its customers, to hold, and transfer virtual currencies'. Moreover, providers engaged in exchange services between virtual currencies and

([221]) Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on information accompanying transfers of funds and certain crypto-assets (recast) COM/2021/422 final.

([222]) Proposal for a REGULATION OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL on Markets in Crypto-assets, and amending Directive (EU) 2019/1937.

([223]) It should also be noted that the Commission's upcoming AMLD proposal will include all crypto asset service providers (and not only wallet providers and exchange services) in the scope of 'obliged entities' and ban the possibility for professionals to provide anonymous wallets of crypto assets.

fiat currencies are included as 'obliged entities' in the scope of the AMLD and must therefore comply with the relevant due diligence obligations stemming from this legislation. This includes, as developed in the previous chapter, *inter alia,* collecting information on their customers when they enter into a transaction or business relationship, and submitting reports to national FIUs for suspicious virtual assets transactions with the collected information under due diligence obligations. Alongside, and to tackle issues related to anonymity, Article 47(1) of the AMLD regarding 'supervision' creates an obligation upon Member States to ensure that providers of exchange services between virtual currencies and fiat currencies and custodian wallet providers are registered.

Regarding VAT taxation, the ECJ has ruled in the case C- 264/14 Hedqvist that the cryptocurrency Bitcoin fell under the scope of the exemption of VAT enshrined in Article 135(1)(e) of the VAT Directive as it constituted a 'currency'. Therefore, no information is collected via VAT requirements.

In addition, under the DAC Directives, tax authorities are challenged as they are not always able to track the capital gains stemming from trading virtual assets ([224]). Given the high risks concerning virtual assets and related transactions, the absence of mandatory information exchange obligations or declaration obligations between FIUs and EU tax authorities may pose a challenge to the adequacy of the current legal framework ([225]). Thus, the taxation of income or gains from crypto assets remains in the remits of domestic tax law of each Member State. To address this, the recent amendment to the DAC ('DAC8') focusses on the exchange of information about 'gains and profit made from crypto transactions' by EU users ([226]). Information will be exchanged as from 1 January 2026 by the competent authority of the Member State where the crypto asset is held with the competent authority of the Member State where the investor is resident ([227]). CASPs will have to report income earned by their users for taxation purposes.

The MiCA And the TFR

The Markets in crypto assets Regulation ('MICA') recently adopted replaces the national legal governance of issuing, trading and custody of virtual assets and complement the existing AML framework. The MICA Regulation introduces requirements related to transparency, authorisation, supervision of transactions and disclosure ([228]), and creates a single harmonised regulatory licensing framework for crypto asset issuers and crypto asset service providers ('CASP'). For example, Article 16(3)(b) will require issuers of crypto assets to apply for an

([224]) See: Tax transparency rules for crypto-asset transactions (DAC8) (europa.eu)

([225]) It is worth noting that the DAC 8 proposal would cover virtual assets and enable EU tax authorities to receive and exchange information about virtual assets by including due diligence obligations and reporting obligations for service providers in this area.

([226]) Tax transparency rules for crypto-asset transactions (DAC8) (europa.eu)

([227]) Council Directive (EU) 2023/2226 of 17 October 2023 amending Directive 2011/16/EU on administrative cooperation in the field of taxation.

([228]) Crypto-assets: green light to new rules for tracing transfers in the EU | News | European Parliament (europa.eu)

authorisation and provide proof of the absence of a criminal record in respect of convictions related to anti-money laundering legislation. Further, CASPs will be obliged to conduct due diligence and report suspicious transactions. It will also allow national competent authorities and FIU's to exchange information on suspected money laundering activities involving crypto assets.

In addition, the TFR Regulation will introduce the 'travel rule' in all Member States. The travel rule requires that information on the source of an asset and its beneficiary (data allowing for the personal identification of the parties to the transaction) is transmitted from the CASP of the sender of the funds to the CASP of the beneficiary. Therefore, CASPs will be required to collect and store information on crypto assets transactions ([229]). Importantly, NFTs are currently excluded from the scope of the TFR.

The implementation of the MiCA and the amended TFR amount to considerable mitigating measures to the risks of cryptocurrencies and allow for the identification of asset holders and the collection of information on transactions.

Proportionality (stricto sensu) of creating asset registries for Crypto coins and NFTs

The recent legislative developments amending the TFR and introducing the MiCA will create the basis for the collection of information on crypto assets in Member States. However, at this early stage where the collection of information on asset holders is highly challenging and not always possible (and no such registry exists), requiring Member States to set up registries would create a significant financial and administrative burden on them as depicted under the Cost-benefit analysis (CBA) carried out in Task 4 which will likely be deemed disproportionate.

In addition, the EU added value of setting up national registers for this asset would be limited in light of the existing mitigating measures described above which will create a solid framework for the reporting of information related to cryptocurrency transactions and allow for the traceability of crypto asset holders.

Conclusion

The current lack of EU Regulation for NFTs renders the collection of data on asset holders difficult to envisage for now. On the other hand, cryptocurrencies will be the subject of major legislative developments that will allow for the identification of parties involved in transactions and the collection and storage of information by CASPs on transactions. Once this legislation is implemented, the setting up of national registers could potentially be envisaged – but this would occur at a much later stage.

Alternatively, the new legislative developments could significantly lessen the need for a registry altogether, if implemented correctly and working as envisioned. DAC8 as a mitigating measure introduces automatic exchange of information on crypto assets between Member States to counter tax abuse. In light of the operational challenges outlined in Task 4 (e.g. difficulty to

([229]) It is worth mentioning that under the new TFR, private unhosted wallets that are not regulated by CASPs are excluded from the rules. This means that private wallets (unhosted wallets) and person-to-person transfers without a provider will not be subject to the travel rule. However, crypto exchanges above EUR 1000 will need to be reported to national authorities (including where unhosted wallets owned by private individuals are involved).

ensure the updates of the information), setting up national registries appears to be disproportionate and unnecessary.

8.1.3.8 In-game assets

General proportionality test of creating asset registries for in-game assets

In-game assets are embedded in video games and include, for example, character customisation items, items that modify the in-game statistics of a player, complete game accounts or loot boxes. Players of video games or third parties can trade these items for fiat currencies within the videogame or through a third-party platform. Nonetheless and in contrast with cryptocurrencies, there is no official mechanism to directly convert in-game assets into fiat currencies. This is done in practice via third-party offerings. Their relevance in the context of money laundering and tax evasion was assessed and confirmed in the previous sections of this report.

Appropriateness and suitability of creating asset registries for in-game assets

In the course of the mapping of national databases conducted under Task 2, no existing registry covering in-game assets was identified. This may be due to the novelty of the area, meaning that regulators at Member State level have not yet enacted legislation to create a framework for the trading of in-game assets.

Indeed, there appears to be, to an extent, a regulatory lacuna regarding the sale and re-sale modalities of in-game assets notwithstanding efforts by certain platforms such as Blizzard and eBay to decrease their illicit trading ([230]). It follows that currently the trading of these digital tools does not comprise a robust reliable and secure system to protect the integrity of their exchange in Member States ([231]). Moreover, the research carried out under Task 1 has shown that, although in-game assets are relevant, it is difficult to trace and identify the beneficial owners in practice.

Due to the difficulty of correct identification and verification, the creation of databases would require, *ex-ante*, a legislative framework enabling, *inter alia,* the identification of the correct owners ([232]). Secondly, and even when the legal framework allows for the identification of the owner, Member States would have to obtain this information for the purpose of holding it in a centralised registry via reporting obligations placed on in-game asset traders.

As underlined in Task 4, there is a lack of any competent authority and existing infrastructure to capture accurate and complete data on in-game assets both at EU and Member State level. In light of these challenges and the current regulatory landscape, the appropriateness and suitability of setting up national registers for in-game assets is questionable.

Necessity of creating asset registries for in-game assets

([230]) See research conducted under Task 1.

([231]) Ibid.

([232]) Through the introduction of EU harmonised notions in the field of in-game assets, for instance.

There are current no specific provisions under the DAC for automatic exchange of information on in-gaming assets. Tax authorities can rely on information on request but would need to know in the first place where an asset is held and could not rely on blanket enquiries.

The risks of in-game assets could be tackled by less onerous measures on individuals, rather than creating registries in each Member State. For example, providing a legal right to FIU's and tax authorities to access this information and ensuring its adequate implementation in practice by creating reporting obligations.

Proportionality (stricto sensu) of creating asset registries for in-game assets

In the absence of existing registers and an EU legal framework for data collection on beneficial holders of in-game assets, creating an obligation on Member States to collect such information would amount to a significant administrative and financial burden as developed in the CBA under Task 4.

It should also be borne in mind that the sale of in-game assets in 2021 has generated a revenue amounting to 40% of the overall EUR 23.3 billion value of the European video game market. Thus, this asset base is worth EUR 9.3 billion which constitutes a relatively small value, though no data is available on the cross-border aspect (in comparison with real estate, the cross-border value of property reaches EUR 1.4 trillion). The value is relatively small compared with other assets (e.g. real estate property) and no data is available on the cross-border aspect. The EU added value may therefore be limited especially in light of the complexity and challenges involved.

Conclusion

Due to their relatively low value (as compared with other assets such as real estate) and the high burden that the collection of data for this asset would create on Member States, companies and data subjects, the creation of in-game asset registries in Member States may be disproportionate.

Alternative less restrictive means to solve this issue can be envisaged, such as obligations to conduct more stringent user verification on the side of the asset sellers or measures curbing re-sale of such assets on third-party websites.

8.1.4 Leveraging existing national registries for assets in Scenario 1

8.1.4.1 Motor-vehicles, ships and aircraft

General proportionality test

Appropriateness and suitability of leveraging existing registries

Under Task 1, these vehicles are classified as having a high relevance in the context of money laundering and tax evasion. The existence of networks of illicit dealership in this area ([233]) for the purpose of tax evasion and money laundering further testifies the relevance of this asset.

Registries covering these assets were identified in all Member States (see Task 2).

GDPR

In line with the principle of 'data limitation' enshrined in Article 5(b) of the GDPR, any existing data that could leverage existing registries must have been initially processed for a purpose that is compatible with the secondary processing, namely, the prevention of money laundering and tax evasion. In this respect, as seen in Task 5, all information collected by obliged entities (on all types of assets) under the AMLD and DAC will have been processed for a purpose that is compatible with the objective of creating a single access point of information under this Study. Therefore, it will be possible for Member States to rely on this collected data to leverage their existing registers:

- **Vehicles**: regarding the information in the EUCARIS system, as well as the national registries which constitute it, the compatibility of purposes could be found as it relates to the prevention of criminal offences (including traffic offenses). Therefore, it may be possible to reuse this data or draw on these existing datasets to substantiate additional national registries. This is however not the case for certain existing registries mapped under Task 2 for aircraft for example (see case study Malta – Annex F).

Necessity of leveraging existing registries

In terms of assessing the necessity to ensure access to authorities to information on vehicles, under the findings of Task 2, over half of the surveyed FIUs have indicated that they are legally entitled to access data relating to vehicles for investigations, and the majority have current access to information on vehicles assets and their holders through a registry. Regarding tax authorities, most of them indicated being legally entitled to access this data for investigation and litigation. On the current access, more than half of the surveyed tax authorities relied on registers. Given the high percentage of accessibility of FIUs and tax authorities in this area, creating additional registries does not seem necessary, as authorities already have access to the information.

Further, regarding the necessity of ensuring the possibility to reuse the data, the existence of additional cooperation mechanisms concerning vehicles, such as EUCARIS, should also be assessed. As indicated in Task 3, EUCARIS is a data exchange mechanism (not a database) developed by governmental authorities that enables countries to share vehicle, driving licence

([233]) See https://www.eurojust.europa.eu/news/eurojust-supports-major-crackdown-vat-fraud-car-sales

and other transport related data ([234]). This exchange relies on the connection between vehicle and driving licence authorities in Europe and serves to detect and address car theft and registration fraud.

Relying on the EUCARIS system, national registration authorities and government institutions can obtain data concerning vehicles and their holders from another country. Under the current EUCARIS architecture, 3 differing procedures exist. The first concerns the exchange of vehicle and driving licence data (15 Member States are part of this EUCARIS Treaty); the second vehicle owner/holder and insurance data (currently, 6 Member States have implemented this but, in line with the EU Council Decision 2008/JHA/615 ([235]) on cross-border crime, soon all Member States will be involved) and data on traffic offenders. Relying on EU Council Decisions 2008/615/JHA and 2008/616/JHA ([236]), police and law enforcement authorities (LEAs) can obtain data for the purpose of criminal prosecution and counter terrorism purposes**.** Therefore, the reuse of the data exchanged via the EUCARIS system for AML and tax evasion purposes will be nevertheless limited as it is primarily collected for the purposes of combatting road-related crimes. However, police and LEAs can exchange information on vehicles insofar as it serves the purpose of criminal prosecution. It should also be noted that, as seen in task 4, information on the value of assets is not included in the existing national registries although it is key in the context of AML/Tax evasion. Thus, (and as seen in task 4), although existing information could be leveraged on, data on the asset value would still need to be collected.

The cooperation between national registration authorities and police across the EU under EUCARIS does not rely on the creation of central registers, but rather on a system of data exchange and the connection of national authorities. Moreover, no data exchange systems exist that are as systematically and broadly available for ships and aircraft.

Proportionality (stricto sensu) of leveraging existing registries

From a strict proportionality perspective, the balancing of rights and interests at play under the current scenario for vehicles should consider several relevant aspects.

The existence of national registers for the assets in this category in all Member States implies that no additional registries would need to be created. Additionally, the interconnection of national databases could prove more efficient in ensuring swift access to the relevant data on a cross-border level, given the intrinsic limitations of current existing frameworks under the AMLD and DAC, as well as the different legal purpose and mechanisms used under EUCARIS. Moreover, since the assets included in this category are of high relevance in the fight against money laundering and tax evasion, the interference with individual's fundamental rights could be justified by the need to pursue these legitimate objectives.

([234]) EUCARIS – EUCARIS is the EUropean CAR and driving license Information System.

([235]) Council Decision 2008/615/JHA of 23 June 2008 on the stepping up of cross-border cooperation, particularly in combating terrorism and cross-border crime.

([236]) Council Decision 2008/616/JHA of 23 June 2008 on the implementation of Decision 2008/615/JHA on the stepping up of cross-border cooperation, particularly in combating terrorism and cross-border crime.

From this point of view, the EUCARIS data exchange mechanism as well as the existing cooperation framework under DAC and AMLD may be considered less intrusive measures than the creation and interconnection of national databases.

These targeted systems amount to a less serious interference on data subjects' rights. At the same time, as only high value vehicles are relevant for the purpose of the current study, national databases and their interconnection should focus only on vehicles which meet a certain value threshold to be a proportionate measure. As in the case of certain types of assets under this category only a very small percentage of goods are likely to meet the 'high value' threshold (most likely larger ships and aircraft). Issues in defining such a threshold and attaching monetary value to an asset can be addressed by deriving the value of an asset from its characteristics (model, year, existing market prices).

Conclusion

The existing framework may be considered less intrusive to data subjects' rights, although in the case of some of the mechanisms in place, the purpose is not directly related to money laundering and tax evasion.

8.1.4.2 Cash at bank/Safe deposit boxes at banks

As developed in Task 4, EU law already requires Member States to hold bank account registries (and data on lessees of safe deposit boxes at banks) and has envisioned their interconnection under the AML/tax evasion framework ([237]). Therefore, these existing registries could be slotted into the proposed architecture for Scenario 1 as developed under the IT feasibility assessment in Section 6.4.1.1 in Task 4.

The main legal considerations at play for the interconnection of bank account registries with other asset registries are explored under Scenario 2 ('The legal feasibility for interconnecting only the existing national registries required by EU law (Beneficial ownership, bank account) and real estate registries with an EU asset registry').

8.1.4.3 Limited Liability company

Similarly, beneficial ownership registers are required under EU law. As developed in Task 5, Directive (EU) 2017/1132 in its article 22(1) relating to certain aspects of company law ([238]) establishes a system of interconnection of registers composed of Member States' registers. Therefore, the legal feasibility of creating and interconnecting these registers is assumed and will not be assessed. As for cash at bank, these existing registries would be slotted into the proposed architecture for an EU asset registry as developed under Task 4.

([237]) Therefore the legal feasibility of creating and interconnecting the registries will not be conducted in this section.

([238]) Consolidated text: Directive (EU) 2017/1132 of the European Parliament and of the Council of 14 June 2017 relating to certain aspects of company law (codification) (Text with EEA relevance)Text with EEA relevance EUR-Lex - 02017L1132-20220812 - EN - EUR-Lex (europa.eu).

8.1.4.4 Real Estate

Article 32b of the AMLD obliges Member States to provide FIUs and competent authorities with access to information allowing for the identification, 'in a timely manner, of any natural or legal persons owning real estate, including through registers or electronic data retrieval systems where such registers or systems are available'. Against this background, real estate registries are found in all Member States although their purpose is not consistently related to AML/tax evasion.

Regarding tax evasion, Article 8(1) of the DAC provides that the competent authority of each Member State shall, 'by automatic exchange, communicate to the competent authority of any other Member State all information that is available concerning residents of that other Member State, on the following specific categories of income and capital as they are to be understood under the national legislation' that includes (e) ownership of and income from immovable property ('DAC1 IP AEOI'). It follows that the current framework provides for the automatic exchange of 'available information' such as the ownership of and income from immovable property. However, it must be noted that the information exchanged depends on the accessibility and availability of the data, at national level, to the competent authority sending the information (as described in Task 5).

The legal feasibility assessment of interconnecting these registries is conducted under Scenario 2 ([239]). The main legal challenges identified in Scenario 2 include differences in existing registries in terms of the scope of information they contain, their purpose, organisational mode, the accessibility to entities and search functionalities they offer. A further challenge lies in the degree of fragmentation between Member States real estate law.

8.1.5 Conclusion – Scenario 1

The legal feasibility assessment of Scenario 1 reveals the following challenges:

- Restriction on individuals' fundamental rights;
- Increased administrative burden imposed on the Member States;
- Disadvantages in terms of further developing interoperability between Member State registries which is an essential component to guarantee the efficiency of the interconnected system (see Task 4);
- Higher costs compared to Scenario 3.

At the same time, various benefits have been identified in relation to the creation of the new registries and their interconnection such as:

- Certain flexibility on behalf of Member States which would build their own registers, since the new asset registers would be developed and administered independently by each Member State and embedded in their national legal frameworks;
- Ensuring data quality and data security;
- Enhancing cooperation between Member States institutional actors;

([239]) Therefore, the legal feasibility of creating and interconnecting the registries will not be conducted in this section.

- Interconnected registries would allow real time access to information on assets held by tax authorities in the EU. This is an improvement as compared to existing EU frameworks under the DAC and the Tax Recovery Directive where information is exchanged on an annual basis and where the cooperation relies on the competent authority of the Member State where the asset is located to reply to information on request from the Member State where the taxpayer is resident.

The conducted analysis shows that the disadvantages outweigh the advantages. For instance, the costs to implement Scenario 1 would be high as each Member State would have to set up new registries which involves operational costs and the costs connected to the amendment of national legislation to ensure reporting infrastructures (see the CBA under Task 4 for more precise information on the estimation of costs). The maintenance (recurring) costs are also significant due to the complex IT architecture, as well as the challenges regarding ensuring and securing access to a high number of actors with different legal access rights and organisational nature (see Task 2 for more information).

Yet, the GDPR obligations in establishing interconnected national registers under Scenario 1 would be less onerous in terms of responsibilities and the related liabilities as compared to Scenario 3.

Overall, the analysis asset per asset conducted above shows that setting up national registries (in view of their eventual interconnection) would place a significant burden on Member States and entities in charge of collecting the information, thereby arguably amounting to a disproportionate measure. This is the case where there is no legal framework in place to ensure the collection of information in a harmonised way or to allow for the identification of beneficial ownership of assets (e.g. cash in hand, crypto assets, in-game assets and securities). Naturally, for the group of assets where registers already exist, this is less of an issue. For those assets for which registers have been created but for other purposes than fighting money laundering or tax evasion, the question of the possibility to reuse the data exchanged via the EUCARIS system for AML and tax purposes needs to be addressed.

8.2 Scenario 2: The legal feasibility for interconnecting only the existing national registers required under EU law (Beneficial ownership, bank account) and real estate registers with an EU asset registry

8.2.1 Description of Scenario 2

The second scenario involves the interconnection of existing national registers required under EU law. In line with the AMLD framework, these include registers on beneficial ownership and bank accounts. As developed in Task 5, although there is no EU law obligation on Member States to have in place real estate registries, the AMLD requires the exchange in a timely fashion of information on real estate ownership, including via national registers where available. Under this scenario, as developed in Task 4, a central component of the registry will

contain data making the building of queries possible ([240]) on a real-time or close to a real time basis. Unlike Scenario 1, Member States will not have to create and implement additional registries. Rather, registers access point would build upon existing national databases under EU law and their interconnection ([241]).

To perform the legal feasibility in relation to this scenario, first, a legal basis stemming from EU law is identified. General considerations which apply to the interconnection of all asset registers are then set forth. The legal feasibility assessment in particular regarding the proportionality and necessity tests is developed for the existing national registers namely real estate and bank-account/safe deposit boxes at bank.

8.2.2 General considerations

Data controllership

Scenario 2 involves the interconnection of existing registries under EU law. Thus, queries are conducted through the EU portal by FIUs and competent authorities who will then access the EU asset database. As for Scenario 1, the information is stored -and the registries are administered- at national level. It follows that similar considerations regarding data controllership as those set out under Scenario 1 apply to Scenario 2.

Insofar as no information is stored in the EU portal (meaning that any data on the EU portal it is merely transient data), the purpose and means of the data processing are determined by an entity at national level and the data controllers of national registries are those designated under Member State law. To provide some examples, regarding the BORIS national registries, Member States have designated the following data controllers ([242]):

- the Austrian Federal Ministry of Finance (Austria)
- the Registry Agency (Bulgaria)
- the Financial Agency (Croatia)
- National Tax and Customs Authority (Hungary)
- Central Register of Beneficial Ownership of Companies and Industrial and Provident Societies (Ireland)
- Ministry of Justice (in Spain and Lithuania)

([240]) LLC registers will not be considered for the purposes of this study since the interconnection of business registries is foreseen by BRIS. See https://e-justice.europa.eu/content_business_registers_at_european_level-105-en.do

([241]) It is important to note that regarding real estate, all Member States currently have registers in place although they differ largely in terms of their organization structure, their purpose, the information they contain, and the territorial scope they cover (see Task 2 – mapping of existing registries). This aspect is further explored in the section below regarding real estate registries.

([242]) For more information see: BORIS_DP_info_EU_EU_en.pdf.

The GDPR considerations under Scenario 2 are similar to Scenario 1. However, given that Scenario 2 involves only real estate, beneficial ownership and bank account registers, GDPR considerations would be more limited with respect to scope. In terms of comparison of the three scenarios, if the Commission was designated as a data processor under Scenario 2, then it would incur less obligations and responsibilities than under Scenario 3 where personal data on asset ownership would be contained in a database to be set up by the EU.

Central registries and harmonisation of the data

Given that under this scenario national registers will be embedded in domestic legislation, different national legal frameworks will apply. Therefore, the modalities of administration of the national registers may also differ (e.g. deadline for updating the registry, type, scope and format of the data included, category of assets effectively covered, etc. ([243])([244])).

To be easily searchable and useful for the purposes of investigating money laundering and tax evasion, the harmonisation - to an extent – of the scope and type of data to be included regarding holder and assets may be desirable where relevant (as is currently the case for bank account registers but not for real estate registers). For example, obligations for Member States to include identification data (such as a unique identification number, or name accompanied with valid identification documentation of a natural or legal person) on the beneficial ownership of an asset will ensure this information is available across all Member States' databases. This would render the interconnected system more appropriate and effective to meet its objective. Task 3 of this study indicates that only a limited number of Member States have interconnected their national real estate registers with their national beneficial ownership registers. This finding is also confirmed with information on income and ownership data exchanged between Member States under the DAC where beneficial ownership information is not available.

Moreover, regarding the existing registries, requirements under EU law to hold centralised databases (as is the case for bank account registers) rather than multiple databases for a category of asset will increase the operational feasibility and benefits of later interconnection.

Overall conclusions on the general considerations

The challenges in interconnecting registers already encountered relate to the fact that national registers differ, both in terms of the scope and type of information covered, as well as their format. Hence, the economic and technical viability of the interconnection and the lack of homogeneity that make up the different assets of that registry must be considered.

([243]) National case studies in for example CZ, LU, CY point to the privacy concerns.

([244]) For example, interview with FIU Croatia.

8.2.3 Assets considered under Scenario 2

8.2.3.1 Beneficial Ownership registries ('BORIS')

As developed in task 5, Article 22(1) of Directive 2017/1132 relating to certain aspects of company law ([245]) sets up (2) a system of interconnection of registers composed of Member States' registers, the European central platform and the portal serving as the European electronic access point. Moreover, the Commission Implementing regulation ([246]) establishing the technical specifications and procedures required for the system of interconnection of central registers has already entered into force ('BORIS' project). Therefore, the legal feasibility of the interconnection of the existing registries for beneficial ownership under EU law is assumed and will not be conducted.

8.2.3.2 Cash at bank/Safe deposit boxes at banks

Literature findings emphasized in Task 1 showed that the use of bank accounts to hide wealth offshore has lost attractiveness in recent years due to stricter AML legislation and other national measures. Yet, bank account registers still remain of high relevance for money laundering and tax evasion, enforcement measures and asset recovery.

Current legal framework

Importantly, Article 10(1) of the AMLD creates an obligation on Member States to prohibit their financial institutions from 'keeping anonymous accounts, anonymous passbooks or anonymous safe deposit boxes'. Thus, EU law prohibits anonymity of bank account registers and safe deposit boxes at bank, and these holders will therefore have to be known to the financial institutions.

EU obligation to have in place a centralised bank account registry and scope of the information contained therein

Under article 32a of the AMLD, Member States have an obligation to set up a national centralised automated mechanism such as a central database or data retrieval system (hereinafter: centralised bank account registry) comprising information on the identification of any natural or legal persons holding or controlling payment accounts, bank accounts and safe deposit boxes held by credit institutions within their territory.

The minimum information which must be included in these national systems State/databases is set forth under Article 32a(3) and includes data on both the customer-account holder/any person purporting to act on behalf of the customer and the beneficial owner of the customer-account holder (name, complemented by further data enabling identification). Moreover, the database must also include the IBAN number and date of the account opening and closing;

([245]) Consolidated text: Directive (EU) 2017/1132 of the European Parliament and of the Council of 14 June 2017 relating to certain aspects of company law (codification) (Text with EEA relevance)Text with EEA relevance EUR-Lex - 02017L1132-20220812 - EN - EUR-Lex (europa.eu).

([246]) COMMISSION IMPLEMENTING REGULATION (EU) 2021/369 of 1 March 2021 establishing the technical specifications and procedures required for the system of interconnection of central registers referred to in Directive (EU) 2015/849 of the European Parliament and of the Council https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32021R0369&rid=6

and where a safe deposit box is involved the name of the lessee, complemented by additional identification data/a unique identification number and the duration of the lease period. Importantly, Member States have the discretion to require other information to be included where it is deemed essential for FIUs and competent authorities to fulfil their missions.

Access of FIUs and competent authorities to the databases

On the ability to search and find this data, Article 32a(2) specifies that the information held in these centralised databases must be directly accessible 'in an immediate and unfiltered manner' to national FIUs and competent authorities for fulfilling their obligations under the AMLD. Consistently with Article 4 of the Directive (EU) 2019/1153 laying down the rules facilitating the use of financial and other information for the prevention, detection, investigation or prosecution of certain criminal offences ([247]), FIUs and competent authorities can access and search information on bank accounts for 'purposes of preventing, detecting, investigating or prosecuting a serious criminal offence or supporting a criminal investigation concerning a serious criminal offence, including the identification, tracing and freezing of the assets related to such investigation'.

Hence access to the information is not publicly available and limited to specific designated competent authorities and FIUs and for the specific purpose of preventing, detecting and investigating or prosecuting 'serious criminal offences' or supporting a criminal investigation.

Further, regarding the modalities of the access, Directive (EU) 2019/1153 provides that 'Member States must ensure that the relevant authorities have the power to access and search, directly and immediately, bank account information where necessary for the performance of their tasks'. Accessibility of this data to FIUs from any other Member States is also compulsory under Article 32a(2) of the AMLD.

Additional fundamental rights and data protection safeguards

Considering the sensitive nature of financial information, safeguards to protect the fundamental rights and in particular the right to data protection of EU citizens are embedded in the existing legal framework. In line with the CFR and the GDPR, the current Directive (EU) 2019/1153 stipulates in Article 5(2) that the staff involved in the access and search of the national bank account registers must maintain high standards of confidentiality and data protection and have the appropriate integrity and skills.

Furthermore, access and logs to the centralise bank account registers are recorded and verified by the relevant data protection officers. Relying on these logs, the latter shall check the admissibility of a request and the lawfulness of data processing to safeguard data security.

In addition, under Article 6(4) of the Directive 2019/1153 Member States are required to ensure that authorities operating the centralised bank account registers take appropriate measures,

([247]) Directive (EU) 2019/1153 of the European Parliament and of the Council of 20 June 2019 laying down rules facilitating the use of financial and other information for the prevention, detection, investigation or prosecution of certain criminal offences, and repealing Council Decision 2000/642/JHA.

such as specialised training programmes, to ensure that staff members are aware of the relevant data protection rules.

Foreseen interconnection of central bank registers

Article 32a(5) of the AMLD required the Commission to submit a report to the European Parliament and to the Council assessing the conditions and the technical specifications and procedures for ensuring a safe and efficient interconnection of the centralised automated mechanisms and where appropriate, a legislative proposal in that regard. The report on the interconnection of national centralised automated mechanisms (central registries or central electronic data retrieval systems) of the Member States on bank accounts ([248]) was adopted in July 2019 and set out a number of elements to be considered for a possible interconnection of bank account registries and data retrieval systems. In July 2021, the Commission proposed ([249]) to interconnect the centralised automated mechanisms via the bank account registers (BAR) single access point which will be developed and operated by the Commission. The proposal provides that FIUs are to be granted immediate and unfiltered access to the information on payment and bank accounts, and safe deposit boxes in other Member States available through the single access point interconnecting the centralised automated mechanisms. Another proposal ([250]) by the Commission requires that the competent national authorities designated pursuant to Article 3(1) of Directive 2019/1153 are granted the power to access and search, directly and immediately, bank account information in other Member States available through the bank account registers (BAR) single access point when necessary for the performance of their tasks for the purposes of preventing, detecting, investigating or prosecuting a serious criminal offence or supporting a criminal investigation concerning a serious criminal offence, including the identification, tracing and freezing of assets related to such an investigation.

Information comprised in the interconnected databases

The interconnection of the databases will necessarily comprise personal data concerning legal and natural persons thereby impacting persons fundamental rights to privacy and data protection.

Article 32a of the AMLD is limited to the **information strictly necessary to identify a bank account holder / beneficial owner, the relevant bank, and details on the account** (date of opening and closing) as developed above.

([248]) European Commission (2019b).

(249) Proposal for a Directive of the European Parliament and of the Council on the mechanisms to be put in place by the Member States for the prevention of the use of the financial system for the purposes of money laundering or terrorist financing and repealing Directive (EU) 2015/849, COM(2021) 423 final.

(250) Proposal for a Directive of the European Parliament and of the Council amending Directive (EU) 2019/1153 of the European Parliament and of the Council, as regards access of competent authorities to centralised bank account registries through the single access point, COM(2021) 429 final.

Sensitive data such as the balance on the accounts or specific transactions are not included in the interconnected databases.

However, where deemed necessary for the purpose of an investigation, authorities may request further information to other Member State's authorities (such as details on a transaction) relying on police and judicial cooperation mechanisms.

It follows that the interconnection of bank account registers across the EU would need to be restricted to the processing of strictly necessary and limited set of information in order to comply with the principle of data minimisation established in the GDPR. These safeguards should also include mechanisms to ensure the data accessed is limited to FIUs and competent authorities according to their respective competences, and data access and data processing is done in line with data protection rules. Moreover, mirroring the current EU legal framework, robust safeguards and data protection protocols to ensure the staff involved has received adequate trainings and knowledge of the data protection rules will be necessary to protect the fundamental rights to privacy and data protection of individuals.

8.2.3.3 Real estate registries

Under Task 1, real estate was categorised as having a high relevance for both money laundering and tax evasion. Indeed, the real estate sector is highly exposed to risks of money laundering and tax evasion as EUR 1.4 trillion is held in cross-border ownership ([251]).

Current legal framework

General AMLD and DAC obligations

Real estate agents are covered by the scope of the AMLD and considered obliged entities under Article 2(1)(3)(d). Moreover, they are also covered when acting as intermediaries in the 'letting of immovable property' where the transactions for which the monthly rent is equal or superior to EUR 10 000. Therefore, as developed in Task 5, real estate agents, as 'obliged entities', will have to comply with the relevant due diligence obligations enshrined in the AMLD including the reporting of suspicious transactions on the basis of the risk-assessment approach. Due to this obligation, real estate agents could provide information on the ownership of the asset (as the transaction could be linked to the owner of the asset).

Regarding specifically tax evasion, the DAC Directives ('DAC1') under Article 8(1) provides that the competent authority of each Member State shall, 'by automatic exchange, communicate to the competent authority of any other Member State all information that is available concerning residents of that other Member State, on the following specific categories of income and capital as they are to be understood under the national legislation' including (e) ownership of and income from immovable property. It follows that the current framework provides for the automatic exchange of 'available information' such as the ownership of and income from immovable property. However, it must be noted that the information exchanged

([251]) European Commission (2019c).

depends on the accessibility and availability of the data, at national level, to the competent authority sending the information (as developed in Task 5).

EU law obligation for Member States to provide FIUs with access to information on real estate ownership including through registers or electronic data retrieval systems where available

Under the AMLD Article 32b, Member States have an obligation to provide FIUs and competent authorities with access to information allowing for the identification, 'in a timely manner, of any natural or legal persons owning real estate, including through registers or electronic data retrieval systems where such registers or systems are available'.

Unlike Article 32a concerning bank account registers, Article 32b does not oblige Member States to have in place real estate registers, nor does it specify the minimum set of information that must be accessible to FIUs. However, all Member States must ensure FIUs and competent authorities have access to information which allows for, at least, the identification of natural or legal persons holding real estate assets. Given that EU law does not harmonise the scope of the information that must be included - except in the sense that the identification of the owner must be possible for FIUs and competent authorities - information contained in registries is heterogenous and subject to national legislation. Indeed, there are discrepancies in the information contained in different real estate registries ([252]).

Moreover, and in contrast with Article 32a on bank account registers, the current wording of Article 32b does not impose the creation of a single centralised registry per Member State. Rather, it creates an obligation to ensure 'access to information (..) including through registers or electronic data retrieval systems where such registers or systems are available'. Thus, under the current national regulatory landscapes Member States may have more than one real estate registry covering immovable property on their territories, as seen in France, Germany and Italy for instance ([253]). In the framework of the AMLD6 proposal, the requirement for Member States to provide FIUs and other competent authorities access to national real estate registers is maintained and the minimum extent of such access is defined. Under the AMLD6 proposal, Member States have the obligation to notify those competent authorities to the Commission, which can have access to such registers (including on the extent of their access).

Fragmented national level frameworks for land registers

At national level, it is therefore allowed for Member States to have more than one real estate register and no harmonisation of the scope and nature of the information contained therein currently exists at EU level. For example, Austria, Czechia, and Finland ([254]) have multiple

([252]) As seen with the ELRA project: European e-Justice Portal - Land registers - EU level (europa.eu)

([253]) European Commission (2022a).

([254]) The presented table used European Commission, DG Justice, Report on Land Registers Interconnection feasibility and implementation analysis, produced by iLICONN Consortium, 2014, p.134-143 as a basis and updated the information provided

registers relating to real estate assets on their territories with no consolidation at the level of a central register ([255]). The modalities governing access to competent authorities and the entities in charge of ensuring the provision of information also differ according to national-level frameworks ([256]). Importantly, there is currently no harmonised framework in the field of property law. For instance, across different legal systems, the instruments transferring ownership of real estate property vary and may include 'partition', 'instruments creating or transferring a right 'in rem' or different types of leases. This is also true regarding the organisations models across Member States which include, for instance, regional registers, cadastres and address registry ([257]). In addition, some Member States have set up separate databases for tax identification of the real estate property as in France and Portugal ([258]). Thus, the plurality of administrative structures, practices, and legal regimes across the Member States complicates the exchange of cross-border information between designated competent authorities ([259]).

Hence, the EU law system and the current registers differ in terms of the scope of information they contain, their purpose, organisational mode, the accessibility to entities and search functionalities they offer. As developed above, the legal framework relating to real estate law is largely fragmented. Intervention at EU level to introduce harmonised definitions in the field of property law thereby safeguarding consistency across the data contained in the registers could be considered subject to compliance with areas of Union's competence.

Foreseen harmonisation and interconnection of the existing real estate national registers

Notwithstanding these national level discrepancies, Article 32b(2) stipulates that the Commission shall submit a report 'assessing the necessity and proportionality of harmonising the information included in the registers and assessing the need for the interconnection of those registers' and where appropriate accompany it with a legislative proposal. This report has been issued and its implications in the present Study are further explored in below section.

Interestingly initiatives to increase cross-border registration and transparency across Member State's land registers exist. For example, as previously stated in Task 4, the LRI (Land registers Interconnection project) aims to provide a single access point for persons and professionals to 'query and retrieve relevant information via a single, adaptive, multi-lingual interface, in compliance with the national legal and technical capabilities' ([260]). Moreover, the European Land Registry Association (ELRA) launched the IMOLA-project to produce a model for 'standardised land registry output' collaborating with various associations in this area ([261]). Many stakeholders worked together to create an 'interoperability framework' for these registers and such projects provide valuable insights regarding the modalities of a legal instrument to interconnect these existing national databases although the scope of the information and

by the respondents of the survey. If an update to the information was provided, it is indicated in the table (column "Update provided"). If no answer or partial information has been provided, the information might be outdated and/or not accurate.

([255]) Land Registers Interconnection Gabriel Sima European Commission, DG Justice and Consumers (europa.eu) p 6

([256]) For more information see: https://www.elra.eu/imola/

([257])Pdf available at: European e-Justice Portal - Land registers in EU countries (europa.eu)

([258]) European Commission (2022a).

([259]) https://www.elra.eu/imola/

([260]) Land Registers Interconnection Gabriel Sima European Commission, DG Justice and Consumers (europa.eu)

([261]) https://www.elra.eu/imola/

accessibility will have to be adjusted for the objective of combatting money laundering and tax evasion.

Necessity and proportionality of interconnecting real estate registers

The interconnection of national real estate registers and their inclusion in a single point of access infrastructure at EU level could facilitate cross-border access to information contained in Member States registers. However, because the purpose of the existing registries is, in most cases, not to combat money laundering and avoid tax evasion (as seen above) and, in line with the GDPR, the initial processing purpose (registering property rights, planning) may not be compatible with the secondary processing purpose (as required by the principle of data limitation). This questions the appropriateness of interconnecting these registers under Scenario 2.

Importantly, AML authorities are overall satisfied with the speed of cross-border information exchange on real estate between FIUs and competent authorities and with the information contained in real estate registers since, in general, it allows for the identification of proprietorship and information on real estate ([262]). However, and as outlined by the Commission's report, complex ownership structure involving trust and companies set up in low-transparency jurisdictions and complex ownership schemes ('shell companies') is frequent in the real estate sector. This means that FIUs and competent authorities cannot access, through existing registers, to all the information relevant for their work. For example, identification of the beneficial ownership is not always possible. This is also true regarding the price of the property which is crucial information from the angle of AML and tax avoidance detection/recovery purposes in the real estate sector ([263]).

In terms of DAC1, while most Member States exchange information on IP ownership () under the DAC, a limited number of Member States do not exchange such information or only exchange information on income derived from IP. Accordingly, in terms of necessity, an assessment would need to be made to determine key aspects of the functioning and effectiveness of AEOI under DAC1 including completeness of exchanges between Member States and the availability of information.

Therefore, and as outlined in the Commission's report, the information that is currently available in the Member State's real estate registers does not include seller/buyer identification data, property price, date of sale or ownership change (record of ownership transfers) which is what is typically needed by FIUs and competent authorities. This means that an obligation to collect additional data that is relevant (and currently missing from the national registers) would be a prerequisite to any legislative measure seeking to harmonise and/or interconnect national real estate registers. Moreover, and given the widely heterogenous set of data currently contained in registers, the information would also have to be harmonised to facilitate the identification of property ownership both at domestic and cross-border level, and therefore to be useful to authorities for the purpose of the legitimate objective.

([262]) Report European Commission (2021), p. 87, 91 (content of real estate registers), and 101-102 (cross-border exchange of information).

([263]) Ibid.

Necessity of interconnecting real estate registers and mitigating measures in place

It must be noted that the existing real estate registers can be used complementarily with the interconnected beneficial ownership registers (BORIS) accessible to FIUs and competent authorities. This mitigates the risks of reliance on 'shell companies' or complex ownership structures involving companies and trusts. In addition, as developed above, the DAC and AMLD provide for the cross-border exchange of information on real estate ownership, thereby amounting to rather robust mitigating measures. Especially, as high-risk real estate is held through complex chains, an interconnection between BORIS and real estate registers would be favourable.

8.2.4 Conclusion – Scenario 2

Scenario 2 which involves interconnecting existing registries presents the following challenges:

- **Impact on persons fundamental rights to privacy and data protection**

The interconnection of the databases will undoubtedly include personal data concerning legal and natural persons and thus will impact persons fundamental rights to privacy and data protection. Thus, interconnection will have to consider GDPR principles. As an example, the interconnection of bank account registers across the EU would need to be restricted to the processing of strictly necessary and limited set of information.

- **Fragmentation of the existing legal framework**

When it comes to real estate registries, there are currently differences in terms of the scope of information they contain, their purpose, and organisational mode, which questions the appropriateness of measure. It has to be taken into account that the intervention at EU level to introduce harmonised definitions in the field of property law thereby safeguarding consistency across the data contained in the registers could be considered respecting Union's competences.

- **The economic and technical viability of the interconnection and the lack of homogeneity**

There are, however, certain benefits when it comes to the interconnection of the existing registries. Since it draws on the existing registries and interconnection projects (BORIS, BAR), the administrative and financial burden on Member States is reduced. Considering the findings in Task 4, it would be the least costly option and the fastest to implement among the proposed scenarios.

Although Task 4 shows that it would be the most operationally feasible option, the challenges associated with the fundamental rights to privacy and data protection as well as the operational burden on Member States to ensure the harmonisation of information must be taken into account. In particular, aspects of necessity, especially with regard to the effectiveness and enforceability with current mitigating measures, for example exchanges of information on IP ownership and income under DAC1, would need to be further assessed.

8.3 Scenario 3: The legal feasibility of establishing a central EU asset registry connecting the existing registers at national level and, for sources of wealth not covered by them,

featuring a central EU register/database with national access portals

8.3.1 Description of Scenario 3

In terms of scope, all the assets assessed under Scenario 1 would also be included in Scenario 3. The assessment below on Scenario 3 focusses on the significant differences between Scenarios 3 and Scenarios 1 and 2, particularly regarding GDPR and Fundamental Rights considerations.

As developed in Task 4, the third scenario requires the creation of an EU central asset registry. Member States would be allowed to implement their registries following their own constraints, whilst sharing a common infrastructure. Under this scenario, the data comprised in each individual registry at national level would be stored in the central EU level database. Information regarding additional categories of assets would be captured and implemented directly in the EU database subsequently, allowing each Member State to implement their own registers in light of specific constraints, while sharing a common technical infrastructure.

Thus, unlike Scenarios 1 and 2, the data would be centralised and aggregated rather than spread out across national level registers. This would ensure, according to the findings of Task 4, technical advantages related to data quality, interoperability and integration based on the use of a common (EU central) architecture and protocols – as compared to Scenarios 1 and 2 – although the centralisation of data could also raise both technical and legal issues concerning the level of participation, privacy, security and loss of control at Member State level. Furthermore, as observed in Task 4, in general the centralised approach for data storage has had the least traction in EU wide projects.

Under current EU law AML/CFT and tax evasion rules, as resulting from the findings of Task 5, no EU-level registers exist, except for some very limited instruments such as the One-Stop Shop scheme provided by Regulation (EU) 904/2010 in the field of VAT. Rather, existing mechanisms for collecting and using relevant information under the AMLD and DAC frameworks, apart from DAC6 for reporting cross-border tax arrangements in an EU Central Depository, rely on exchanges between Member States (automated or upon request), within the limits and for the categories of assets dealt with under the specific instruments. Even highly interconnected EU-level instruments such as EUCARIS clarify that these are not ‘a database nor a central repository but an exchange mechanism’ connecting the relevant Member States’ authorities.

Consequently, for the hypothesis presented under Scenario 3 and its legal feasibility, several key considerations relating to its impact upon fundamental rights (and in particular, privacy and data protection) must firstly be analysed. Subsequently, aspects regarding potential quality, monitoring and enforcement issues will be considered.

Scope of the information and interference with fundamental rights

Under Scenario 3, the scope of the data processed is large and relates to many different types of information with varying levels of sensitivity (financial, ownership, identification, etc.). As developed under the other scenarios, and in line with the principle of data minimisation, the scope of the information should be restricted to what is strictly necessary to pursue the objective sought. This will usually include data on the identity of the legal and beneficial asset holders (e.g. name, unique identification number or subject to documentation provided under

national law). Moreover, data regarding the asset will also be pertinent. Depending on the nature of the asset, this input will vary (IBAN for cash at bank; registration number for vehicles; geographical location for real estate; etc.).

In Scenario 3, all this data would be centralised and stored in one database for the purpose of consolidating the EU asset register. Unlike scenarios 1 and 2, the data would not be spread across a plurality of databases in Member States, but rather aggregated at EU level. It follows that the aggregation of this data and high-level asset coverage highly increases the seriousness of the infringement upon fundamental rights. In this context, many aspects of data subjects' lives may be captured. The scope of information would be extremely wide and encompass numerous assets and asset holders through its transversal nature.

In line with the *Digital Rights Ireland* observations developed in Task 5, the access and searchability of this information may enable the drawing of specific conclusions on persons' habits, daily lives and social relationships. It follows that it is likely to amount to a manifestly disproportionate interference with the fundamental rights to privacy and data protection of persons. Nonetheless, organisational and technical security measures could be considered such as pseudonymization to implement data protection principles and mitigate some of the risks.

Moreover, research and interviews conducted with stakeholders at national level have depicted the possibility and dramatic consequences of a cyberattack if data were centralised ([264]). Even if adequate risk mitigating security safeguards are set up, the possibility for a leakage exists and must be considered when assessing the appropriateness and proportionality of such a legal instrument.

Scope of the data subjects

Importantly and as in the case of *Digital Rights Ireland* regarding the Data Retention Directive, the establishment of an EU asset register under Scenario 3 is likely to involve the collection and processing of information regarding a very high number of EU persons. Especially given that the asset coverage is high (many assets holders will be concerned) in order to provide FIUs and competent authorities with the relevant information for their missions, this will further increase the seriousness of the interference on data subjects rights.

Access to the EU asset registry

In line with the ECJ case law developed and the principles of security and confidentiality under the GDPR (Task 5), data comprised in the EU register should evidently not be accessible to the public and competent authorities' access should be carefully secured by the data controller (EU institutions) and data protection officers.

Similarly to the other scenarios, the modalities of accessibility of the EU registry would have to be restricted to FIUs and competent authorities in the framework of their duties related to detecting, investigating and preventing money laundering and tax evasion. This is even more important under Scenario 3 because of the high level of data aggregation and storage it

([264]) Interview with the French ACPR (Authority for Prudential Control and Resolution).

requires. Furthermore, to ensure observance of the principle of proportionality and the protection of personal data, the staff members of entities that would have access to the EU asset registry would be determined at Member State level and EU law should ensure they are adequately trained in terms of data protection requirements.

Thus, robust safeguards for fundamental rights would have to accompany the modalities of access to the EU register and comply with the principle of data security. These could include, *inter alia*, a system relying on recorded logs whereby data protection officers shall ensure the lawfulness of the processing of data and the admissibility of a request. The channels of access to the centralised database should be secured and include adequate organisational and IT security measures such as strong authentication requirements. In addition, the staff involved in the administering of the system should have proven integrity and be highly trained and knowledgeable about the data protection rules applicable.

Data controllership

GDPR considerations as outlined in Scenario 1 are also applicable for Scenario 3. Under Scenario 3, data provided by each national register, as well as data on assets not already covered at national level, would be stored in an EU central asset database.

For comparison of all three scenarios, an assessment should also be made of potential responsibilities and liabilities for the Commission in establishing registers in terms of its role as data processor and/or data controller. The role of the Commission as data processor and/or data controller is still to be determined for a number of related systems like BORIS and BARIS, while in the case of DAC and the FIU network, Member States retain the role of data controller and the Commission acts as a data processor.

Irrespective of whether the Commission would be assigned joint data controller with Member States or act as data processor, establishing an EU registry under Scenario 3 would clearly incur more responsibilities and potential liabilities for the Commission than under Scenarios 1 and 2. While only Member States competent authorities would use personal data and not the Commission, maintaining a database would incur more accountability in terms of security and data storage compared to the case where only a web portal was set up to access the 27 national asset registers (as is the case for Scenarios 1 and 2).

Additionally, given the very high volume of data centralised at EU level, a key aspect to be considered in relation to Scenario 3 is whether the Commission is, in light of the principles of subsidiarity and proportionality, better placed to manage the data under CFREU and GDPR rules as compared to the Member States, not only from the perspective of the collection and storage of data, but also from the perspective of the exercise of the rights of data subjects and their proximity to the data controller.

Importantly, the data controller will have an obligation to ensure compliance with the GDPR and the data protection standards applicable. Consistently with Article 26(1) of the GDPR, the EU level data controller will be in charge of implementing ‘appropriate technical and organisational measures’ and implement appropriate data protection policies to ensure compliance with the GDPR.

Moreover, Article 65 of the GDPR provides that ‘persons who have suffered material or non-material damage as a result of an infringement of this Regulation shall have the right to receive compensation from the Union institution or body for the damage suffered, subject to the

conditions provided by the Treaties' creating a remedy for data subjects who have seen their rights infringed by the data controller.

Other proportionality considerations specific to Scenario 3

In what collection and reporting aspects are concerned, the findings under Task 4 have shown that from an operational perspective, Scenario 3 presents the advantages deriving from scale and centralisation, in the sense that it could ensure a better alignment with the requirements of the current legal frameworks, and could achieve a much higher level of interoperability and integration of collected data. However, from a legal perspective, this scenario would entail a higher degree of interference with the competences of the Member States and would impose a heavier administrative burden on local authorities. Further, the costs of establishing an EU registry for Member States and the Commission would be smaller under Scenario 3 compared to Scenario 1 where the Member States would need to individually establish their asset registries at national level.

8.3.2 Conclusion – Scenario 3

Under Scenario 3, the data would be centralised and aggregated rather than spread out across national level registers. The analysis performed under Scenario 3 presents the following challenges:

- Aggregation of data and high level of asset coverage increases the seriousness of the infringement upon fundamental rights to a great extent. The access and searchability of information contained in a database may enable the drawing of specific conclusions on persons' habits, daily lives and social relationships. Consequently, it is likely to amount to a manifestly disproportionate interference with the fundamental rights to privacy and data protection of persons.
- Access to EU central registry would have to be restricted to FIUs and competent authorities in the framework of their duties related to detecting, investigating and preventing money laundering and tax evasion considering the principles of security and confidentiality of GDPR. The restriction regarding accessibility applies to all scenarios, but is even more important under Scenario 3 due to the high level of data aggregation and storage required.
- Research and interviews conducted with stakeholders at national level have revealed the possibility of a cyberattack in case the data is centralised.

An important point to be considered is in light of principles of subsidiarity and proportionality if the EU institutions are better placed to manage data based on CFREU and GDPR compared to the Member States.

- Since no similar database was created in the past, there are risks related to the successful implementation of this scenario, as well as certain hidden costs which may incur (see also the conclusions of the operational feasibility assessment – Task 4).
- There are risks in relation to the ability to maintain the essential data qualities, see also Section 6.4.1.3. If new registers were to be established at Member State level, the EU would need to have mechanisms to ensure effective monitoring and enforcement for data quality and data completeness purposes that would be submitted directly by intermediaries to the EU asset registry. So far in the fields of taxation and money laundering, these measures have been a national competence rather than an EU competence.

The measures proposed under Scenario 3 could be considered disproportionate in light of the excessive restrictions of individuals' fundamental rights. The centralisation and storage of a wide scope of data regarding a high number of persons, even if it pursues a legitimate objective enshrined in EU law, will amount to a serious, extensive and long-lasting interference with persons' fundamental rights. On the other hand, organisational and technical security measures could be considered such as pseudonymization to implement data protection principles and mitigate some of the risks. Further, from an IT perspective, the third scenario offers the benefits of a high-level interoperability and extensibility.

Overall, it is uncertain if the benefits (e.g. timely identification of assets in several jurisdictions, direct access to EU wide data, possibility to correlate data at a larger scale) resulting from the creation of an EU central registry could outweigh the disadvantages of the proposed measure, especially those related to the interference with the fundamental rights.

8.4 Alternative scenario: The legal feasibility of interconnecting existing and centralised asset registries

8.4.1 Description of alternative scenario

Under Task 4, a fourth scenario drawing from Scenarios 2 and 3 was developed ("alternative scenario"). Such scenario would interconnect the existing networks to a central EU Portal. Alongside, all existing national registries (BRIS, BORIS, LRI, EUCARIS, EBOCS) would be connected via a national gateway to the main EU asset register or would transfer data to the EU register regularly. For non-existing registries, Member States would have the discretion to either set up registries at national level or directly create them on the EU register infrastructure (see Task 4). It has to be taken into account that the diversity of mechanisms involved under this scenario will impact the complexity of the project and may create additional risks from an operational point of view (see Section 6.4.2.1). For non-existing registries, in case Member States decide to setup the registries on the national level, the findings presented in Scenario 1 have to be considered, and operational costs and costs connected to the amendment of national legislation have to be taken into account.

A legislative proposal to implement this scenario would entail the following:

- Setting up the EU asset registry (including the managing EU institutions or agencies; the designation of the data controller, the modalities of access to the register; and the technical specification *inter alia* on the storage, maintenance, update modalities).
- Creating an obligation on Member States to ensure their existing registries under EU law are interconnected via the relevant EU platform.
- Creating an obligation on Member States to connect their national registries through a national gateway to the main EU asset registry.
- Creating an obligation on Member States to ensure competent authorities including tax authorities and FIUs have access to the interconnected system.

When implementing this scenario, the restrictions it poses on the fundamental rights have to be carefully considered (see previous analysis of three scenarios), such as concern related to data security as described under Scenario 3.

In principle, while applying this scenario, it is necessary to ensure compliance with CFREU, the GDPR and the data protection standards.

8.4.2 Conclusion – alternative scenario

This alternative scenario presents the following challenges:

- The scenario still raises concerns related to data quality.
- As stated under the operational feasibility assessment, some existing registries (properties, land where not part of the LRI, ships, aircraft, firearms) still need operational resources in order to harmonise the existing data before it is transferred to the central EU asset registry.
- Concerns related to the interference with fundamental rights have to be taken into account.

Regarding the last point, the concerns raised under Scenario 3 (particularly regarding compliance with the principle of data security as enshrined in Article 5 GDPR) would be partly mitigated by the semi-centralised structure of the single point of access to information. In addition, data controllership would be spread between the EU level and Member States, rendering the system complex, but not impossible to implement in compliance with the GDPR and Articles 7 and 8 of the CFREU.

Furthermore, the following benefits could be pointed out:

- It allows for some flexibility for the Member States when it comes to the best scenario for their registries.
- The operational burden and costs will be decreased considering that the provisions of this scenario will allow simple integration of already existing and to be interconnected asset registries. From an operational and cost perspective, it is also advantageous for registries (particularly for assets that have a relatively small number of reporting entities or entries) which are to be created at EU level, since no separate registries on Member State level would need to be created.

Box 3: Summary of legal feasibility assessment (Tasks 5 and 6)

Task 5 includes the analysis of the general framework on information exchange mechanisms under the AMLD and tax evasion frameworks. It also describes some specific provisions, including obligations of the Member States with regard to already existing registries, such as registries for beneficial ownership of companies and trusts, bank accounts and real estate registries. Additionally, relevant legislative provisions on information exchange under the framework for combatting tax evasion are presented concerning value-added tax, tax recovery and direct taxation.

Furthermore, Task 5 analyses the main legal constraints that apply to all three scenarios. It shows that for implementing all three scenarios, it is important to consider the provisions of the GDPR, the Charter of Fundamental Rights of the European Union and the general principles of EU law, including proportionality, necessity and subsidiarity. Establishing new registries with the purpose of their interconnection, interconnection of already existing registries or the establishment of central EU registry will undoubtedly involve processing personal data and impinge upon individual fundamental rights to data protection and privacy under the CFREU and the GDPR. However, in some instances, restrictions on a person's right to data protection and privacy can be justified provided there is an essential objective of general public interest, such as the objective of fighting against money laundering and tax evasion in the internal market, and the interference is proportionate.

Task 6 of the legal feasibility assessment presents a detailed analysis of the three scenarios and briefly discusses a fourth alternative scenario identified under Task 4. Under each asset category, the criteria of subsidiarity, necessity and proportionality are assessed.

This task shows that, as also reflected in Task 5, when considering implementing any of the proposed scenarios, the interference with fundamental rights to privacy and data protection has to be given special attention. This is especially relevant for Scenario 3, when the aggregation of data and the high level of asset coverage increase to a great extent the seriousness of the infringement of fundamental rights. In principle, organisational and technical security measures could be introduced to implement data protection principles to mitigate some of the risks.

The analysis shows that another point relevant to all scenarios is the rules on accessibility. A clear and legitimate interest should be well defined as well as those to whom certain pieces of information can be disclosed. Other legal issues have to be taken into account, such as fragmentation of the existing legal framework, as mentioned under Scenario 2.

The conclusion for each scenario includes an overview of the benefits as well as challenges considering legal, IT, operational aspects, which help to draw a clear picture of the success of each scenario.

9. Task 7: Technical study – Options for the design and functionalities of an EU asset registry

This section contains a qualitative analysis of the implementation of the EU asset registry, as part of the feasibility assessment.

Findings from the surveys (as described in Section 4 on mapping registries) have illustrated that the majority of FIUs, LEAs and tax authorities have basic access to most asset information needed for their actions. Nonetheless, they face several limitations when it comes to accessing information and communicating with other competent authorities at the national or EU level. It has also been shown that the level of access attained includes details of relevant asset holders. From a practical perspective, this observation has a major impact on drafting the optimal scenario for the implementation of the platform. The development of the platform must not necessarily be seen as a method to provide basic access (such access already exists), but more so as one to improve the abilities of the FIUs and relevant institutions to:

- perform quick analysis of relevant data;
- retrieve information in a timely manner.

Both actions are needed, not only to identify actual cases but also to make decisions on the recovery of assets, efficiently block evasion and freeze assets. These objectives are nevertheless mentioned in the legal framework as justified needs in themselves.

In this respect, the platform must not be viewed as just a repository of data but more as an integrative mechanism to serve the needs of the relevant Member State institutions in terms of information exchange and collaboration. This is important from the perspective of ensuring the required functions on a longer lifecycle of the system.

The results of both, technical evaluation of the three scenarios (IT feasibility) and the cost benefit analysis, have shown that pure implementations of data centralisation and interconnection of systems are not preferable due to factors that extend from difficulty in attaining data harmonisation to insufficient coverage of the different asset groups.

Analysis of the legal framework for cooperation between FIUs (see Tasks 5 and 6 on the legal feasibility assessment) has confirmed that the exchange of data is not only supported but also encouraged up to the level of ‘spontaneously or upon request’. What is more, the framework prohibits Member States from placing unreasonably restrictive conditions on the exchange or assistance between authorities involved in AML activities. The evaluation of the legal feasibility of the three scenarios in the previous section has also confirmed that each of them can be implemented under the principles of use of a ‘minimum amount of data’ and of ‘legitimate interest’. Such conditions, correlated with existing platforms, which perform more limited, domain-specific, but still similar functions (see Table 31 in Section 6.3 on IT feasibility), demonstrate that implementation based on interconnections between existing registries is both functionally and legally possible.

Yet, the number of domains that are underserved by dedicated registries, the difficulties in retrieving necessary information as outlined in Task 2 on mapping registries and the cost-benefit analysis have all demonstrated the need to find a solution for completing the picture. This is particularly relevant for domains that fall into the virtual realm (e.g. crypto and in-game assets). These are domains that by definition deal with data that are both already outside standard border controls and exposed to breaches. As shown in the legal feasibility assessment, the fourth scenario that mixes the interconnection and centralisation options offers the highest level of coverage but raises concerns in relation to fundamental rights and risks associated with access to data. In this respect, the careful technical implementation of such a scenario could be feasible if designed to follow the principle of minimum amount of data. This would involve only centralising data that are relevant for the needs of the interested institutions (e.g. description of transactions and relations between entities), while preserving personal information on Member State systems (either with an extended use of dedicated

identifiers and/or by employing the use of blockchains). This would avoid the direct risks associated with a breach of the central system.

To address these issues, this section focuses on the options for designing a system that would allow for interconnection to facilitate the exchange of data, as well as the limited centralisation of information where and when this is feasible.

9.1 Methodology

To ensure efficiency, technical options have been studied from the perspective of system lifecycle. Under this approach, defining options involves identifying the limits, constraints, and areas of possible development that the system would encounter over a given lifespan.

Based on the scope of the project, as well as the data gathered and presented in previous sections, several assumptions can be made:

- the system will function in a highly distributed environment;
- the system will interact with a variety of data sources;
- the systems providing source data do not follow specific models in terms of information structure, and even less so in terms of architecture or technologies engaged;
- the system will be active over a period that will span a number of technical evolutions, affecting both its own components and the sources of information in a non-synchronous way (each individual domain and particular registry will evolve in its own way, which will not necessarily follow the same steps and choices made for the components of the system).

Given these conditions, attaining the necessary adaptability will require the architecture to be both resilient and adaptable to possible changes that might occur at any lower node (in an individual registry in any of the Member States). Such adaptability could be achieved through adopting a neutral stance in terms of the technical approach to implementation (defining just the protocols, and not the exact technologies to be used for example, or always offering at least two options for a function that is involved in interactions).

The activities performed in Task 7 have followed this principle.

9.2 Description of the main parameters of the IT infrastructure

9.2.1 Technical considerations

Establishing the parameters of the IT infrastructure requires an understanding of the elements that define the operations of the system. These elements should be seen not necessarily from a technical point of view (storage needs, processing needs and interconnections) but more from a functional perspective. This offers a long-term view and a better focus on the underlying needs of the involved parties. In that sense, the key aspects of interest for the envisaged system are:

- data security
- data heterogeneity (in terms of source, structure, script, language)
- common identifiers and methods of correlation between records
- governance of the system and data access.

Each of these aspects must be considered within the context of the construction of the European platform as well as existing technical platforms. These are foundational elements that can serve both as support and limitations for the system.

9.2.2 Security

Due to the wide array of information the platform would interact with or store, namely highly sensitive and personal data about private and legal entities, it is easy to assume that it has the potential to become a high-value target for different types of attacks.

An obvious method to avoid such a risk is to use a dedicated network like the Trans European Services for Telematics between Administrations (TESTA) or the Common Communications Network and Common Systems Interface (CCN/CSI), instead of generic internet-based access. Such a closed network is by definition constructed to limit access to just the systems that are entitled to do so. While this approach could be a candidate for evaluation, there are reasons to support an internet-based approach, for example:

- An internet-based approach would be easier for smaller-scale registries. The closed architecture of the CCN/CSI would limit the ability of certain registries to connect and interact with the platform, due to the complexity and limitations imposed by proprietary technology.
- Due to the open architecture, the overall cost, complexity, and more importantly the maintenance needs would be lower.
- The scope and general features of the platform would have the potential to attract a wide range of stakeholders who would be more interested in testing and engaging with the system through a generic internet-based network. A small registry administered by a private entity would be more inclined to invest in extending its system using open technologies due to the inherently easier access to sufficient resources and reduced costs.

This generic internet-based approach opens up a series of issues related to the security of data. In this respect, during the data lifecycle, the most critical moment in terms of security is the exchange of data between system nodes (e.g. the transfer of data from a national registry to any other similar system from a different Member State, or to a superior element of the system such as one of the gateways). To attain the necessary level of security for data exchanges, the platform should consider the use of encryption technology. An obvious option would be the use of the eDelivery building block described in Task 4.

A secondary issue, though not the least important, is the security of stored data. To increase protection, it is advisable to apply a multi-level approach combining hard mechanisms (protection associated with a service offered by a system that is set up and operated by EU-LISA, DIGIT or a similar agency) with softer approaches (e.g. attention to the use of anonymised metadata instead of full records, separate processing and storage of metadata). Under such a scenario, the EU-LISA infrastructure would offer the advantages of a solid IT platform that is designed to offer protection of the data hosted on it, while the exchange of anonymised metadata would minimise the risks induced by possible interception during the exchanges of packets between individual registries and the central system that initiates the interrogations.

9.2.2.1 Heterogeneity of the data

The ability of a given system to process data is highly dependent on its capacity to interpret that data. In a system that directly integrates data from multiple registries sourced from all Member States, the ability to integrate data and use them regardless of the script and language is essential. In this respect, the scope of the envisaged system should cover data that are diversified in terms of structure (data schemes specific to the asset and Member State, for example ship registration in Ireland in comparison with other Member States), script and language.

Applying standards like ISO 20020 to facilitate a level of standardisation (as indicated for example by Commission Implementing Regulation 2019/363 specifying technical standards with regard to the format and frequency of reports on the details of securities financing transactions to trade repositories) can offer an integrative approach for other domains. However, it must be acknowledged that no standardisation effort would be able to cover the needs associated with every asset, in every Member State, due to differences in the structure of the information maintained by different registries, as signalled in the previous tasks.

In support of this, it is noted that structured data are considered by two regulations:

- Commission Implementing Regulation 2022/2120, laying down implementing technical elements regarding data standards and formats, templates and procedures for reporting information on projects funded through crowdfunding platforms,
- Commission Implementing Regulation 2019/363 mentioned above.

These two cases exemplify how the specific needs of each asset affect the ability of the authority to impose a data scheme that is applicable to all types of assets. This limits common areas to fields such as:

- name of entity (legal or physical person)
- identifiers of the entities involved (for the subject of the query, source registry, etc.)
- countries associated with the record/transaction
- code of the sector
- numerical amounts (value of object, value of transaction)
- currency applied to the amounts
- type of record/operation
- dates associated with the record.

As limiting as they seem, however, these fields might be essential for tax evasion and AML usage scenarios. Correlating information about each individual transaction allows institutions to keep track of the sums involved, helping them identify anomalies that require additional investigations. Contrary to the principle of concentrating large amounts of data about each individual entity based on information coming from outside the system, this approach focuses on organically identifying individual transactions made by and between entities, facilitating the identification of trends or suspicious individual transactions and taking into account different data protection legislations across member states. The result would be a system that stores the minimal amount of data. This has the potential to both increase data security and ensure compliance with GDPR rules. The metadata fields that are critical for implementation should be called minimum mandatory information and should be defined in accordance with the decisions of the governance body put in place.

Already implemented solutions, applied across the EU in different domains (including BRIS), have demonstrated that the use of controlled vocabularies ([265]), linked data ([266]) and semantic data models ([267]) can open a range of opportunities in terms of ensuring interoperability. Together with the Core Vocabularies already used by BRIS, one could add a model like FOAF for identifying relations between persons. It is advisable nevertheless to consider developing a dedicated money laundering and tax evasion ontology. This would help integrate the various elements and allow for the efficient use of data.

9.2.2.2 Common identifiers and methods of correlation between records

The employment of unique keys is an essential aspect in the design of IT systems working with large amounts of data. This aspect becomes even more important when the system manages records that are received from disparate sources or associated with different data schemes. The assumption is that, in the tax evasion and AML context, use of a common identifier will facilitate the matching of records associated with the same entity (natural or legal persons). This matching will help associate each asset or transaction with the entity, building the profile of that entity, as required by FIUs and other interested authorities.

While the general assumption of the importance of common identifiers is valid, as such an identifier would simplify the matching of direct transactions undertaken by individuals, the identifier would lose its relevance for indirect transactions carried out with the intent of hiding transfers of values. The analysis conducted in Task 1 shows specific scenarios (e.g. commodities) where such intents are obvious. Considering that such cases are more susceptible to investigation by users of the AML network, it can be observed that the existence of a tax identification number (TIN) and/or Tax Pass identifier would not have an impact on their work. This supports the conclusion that the use of a common identifier like TIN and/or Tax Pass would be useful but not particularly critical for the scope of the envisaged system as it does not provide serious advantages in fighting cases of intended tax evasion or money laundering.

However, it is important to bear in mind that identifiers play an important role in system operations. Following a technical analysis of the data architecture, it can be observed that using identifiers already set at the Member State level, combined with unique keys associated with the respective identification system, could identify the entity with an adequate level of detail.

As signalled in Task 4, a wide range of codes implemented at the Member State and sector level are already in use. These include:

- Personal identification code
- National ID

([265]) The controlled vocabularies are lists of terms, more or less hierarchically organised and maintained under a strict policy, that serve as a way to exchange, tag and ultimately map records. Examples of controlled vocabularies are lists of countries, institutions, regulations, etc.

([266]) Linked data are essentially structured data, which are interconnected with other data to create a web of information. Based on their properties and these interconnections, such data can be explored or queried using semantic queries.

([267]) A semantic data model is an implementation of a method to describe the structure of a certain type of data including the meaning of its different parts and the relationships between them and with the outside data.

- TIN
- National registration number
- VAT code
- European Unique Identifier (EUID)
- Legal Entity Identifier (LEI).

Acknowledging that any one entity can be identified by any of these codes, we should also acknowledge that entities can use, either intentionally or not, different identifiers depending on the context (e.g. an individual might cross borders using their national ID, carry out a transaction in a different jurisdiction under a VAT code, and mention a TIN in a contract). A flexible approach to dealing with this situation could be to devise a mapping mechanism that establishes the correlation between different identifiers, so that an entity could be recognised independently of the identifier used. In such a scenario, any record from any registry could be assigned to the proper entity, even if the original intent was to hide that relation by using a less common or outdated identifier. This would help the tracking functions of the system to establish relations between records in more complex situations. Such a mechanism has the potential to:

- ensure faster implementation (using technical means to attain a functional scope);
- minimise the economic pressure/impact on the existing registries (using data that already exist in the registry);
- facilitate application in scenarios where imposing a new identifier would prove impractical (accessing data stored in old systems or archived/historical data).

The mechanism could practically serve as a connector between the different identifiers and in addition offer a level of personal data protection that would further support the implementation of the system.

Irrespective of this, it should be noted that the addition of a TIN and/or Tax Pass identifier could and should most probably take place not as a necessity of the system but as a mechanism to clarify transactions for reasons outside the present project scope.

9.2.2.3 Governance: Access to data and responsibilities for maintenance

As shown in the legal feasibility assessment (see Task 5), the framework within which the system would have to function would be enforced by both legislation on money laundering and authorities balancing enforcement with other elements of society such as human rights. Due to the implications, neither of the respective legal elements would be very direct. Depending on the context, authorities might be allowed by money laundering regulations, in conjunction with a legitimate interest, to interact with data that might otherwise fall under the GDPR. A lot of decisions must therefore be made following principles that are based on the responsibility of the actors involved. This could only be done by enforcing policies and having strong governance.

The governance architecture controlling the functioning of the system would have to adapt to the type of interconnection selected, the types of data that the system interacts with, how those interactions and interconnections would take place and the impact they might have on each relevant legal aspect, from EU law to Member State law. To be able to do that, an active governance structure must be put in place. This structure should be able to take decisions and define the exact policies applicable under a legal framework that by definition is not exact and can be dynamic.

Competences for decision-making on operational issues, and on policies of access and engagement, should be attributed starting from the elements that define the interaction between national registries and the central system. A more distributed system would hand over most of the governance to the corresponding Member State authorities, while a more centralised option would result in more direct responsibility over governance.

As a factor of simplification, it should be considered that governance could and would integrate specificities associated with each particular domain (e.g. ESMA data governance and the future MiCA Regulation, already referred to in Task 6). These umbrella elements at a domain level would practically help the system in terms of describing the exact criteria to be respected at a central level, taking priority over Member State particularities.

Ideally, given the complexity of the architecture and the wide pool of participants, a multilevel governance framework should be set up. The framework should include a steering committee (for operational aspects) and a general assembly. The general assembly would set the policies associated with access to data and interconnection matters.

9.2.3 External architectural references

To facilitate rapid advancement, references have been taken from two systems that are relevant for scope, namely:

- the Business Registers Interconnection System (BRIS)
- the Beneficial Ownership Registers Interconnection System (BORIS).

In the case of BRIS, the system has set up electronic communication between national business registers to provide cross-border public access to information on companies and their branches. At the same time, BORIS interconnects the national central registers of beneficial ownership, allowing interested parties to collect beneficial ownership data.

Both platforms have been constructed as decentralised systems that interconnect national registers of businesses and beneficial owners with the European e-Justice portal through the European central platform. The systems share similar components and, in this respect, prove that from an architectural standpoint, the approach is feasible.

The central element of BRIS and BORIS is the European central platform, established by Article 22(1) of Directive (EU) No 2017/1132. Further supported by the Commission Implementing Regulation 2020/2244 on the technical specifications and procedures for the system of interconnection of registers (establishing BRIS) and Commission Implementing Regulation 2021/369 on the technical specifications and procedures required for the system of interconnection of a central register, (associated with BORIS), the legal framework clearly specifies the mechanism that can facilitate an effective exchange of information at the necessary scale.

The analysis shows that the two systems establish a formal approach for domains where national central registers are already in place or about to be established. While the regulations clarify the mechanisms for two sectors that are already strongly centralised and well regulated, it must be acknowledged that not all verticals are in the same situation.

9.2.4 Main parameters

The parameters of the system can be defined as shown in Table 50.

Table 50: System parameters

Methods of communication	• The system should use service-based methods of electronic communication, such as web services, for the purpose of interconnecting registries • The communication between the portal and the platform, and between a registry and the platform, should be one to-one communication • The communication from the platform to the registries may be one-to-one or one-to-many
Communication protocols	• Secure internet protocols, such as Hypertext Transfer Protocol Secure (HTTPS), should be used for communication between the portal, the platform, the registries and the optional access points • Standard communication protocols, such Representational State Transfer (REST) or Simple Object Access Protocol (SOAP), should be used for the transmission of structured data and metadata
Security standards should cover	• Measures to ensure confidentiality of information, including by using secure channels (HTTPS) • Measures to ensure the integrity of data while being exchanged • Measures to ensure the non-repudiation of origin of the sender of information • Measures to ensure logging of security events in line with recognised international recommendations for IT security standards • Measures to ensure the authentication and authorisation of any registered users • Measures to verify the identity of systems connected to the portal, the platform or the individual registries • Measures to protect against automated searches and copying of registers, such as limiting the results returned by each registry to a given limit (e.g. using a CAPTCHA functionality)
Structure of the standard message format	• The exchange of information between registries, the platform and portal should be based on standard data structuring methods and should be expressed in a standard message serialisation format such as XML, JSON-LD or similar • The exchange of information should also include messages necessary for the operation of the system, such as for acknowledgement of receipt, logging and reporting
Data structure	• Any storage of data would follow strict policies in agreement with EU and Member State regulations • There would be a clear separation between anonymised metadata and actually identifiable metadata • Data across the platform would follow Linked Data principles

Source: Authors' elaboration.

9.2.5 Key architectural principles

As shown in Task 4, a number of systems already function in this space (e.g. BRIS, BORIS, ECRIS and IRI). The similarities between them demonstrate that the functional principles are relatively finite. The main difference in the operation of such large-scale platforms is not necessarily in the user interface/front-end functionalities (there are few differences that could be implemented there) as in the underlying architecture of the system and data that support the respective front end. The difference can be observed more specifically in the reaction time such systems provide depending on their level of centralisation.

As defined by the requirements, the scope of the platform would be to facilitate immediate and direct access to the information held in centralised registries. The speed of searches on this platform would be crucial for the success of criminal investigations and timely identification, tracing and freezing of related assets. It should also be noted that 'speed of searches' here does not refer to the actual/exact reaction time of the search interface. Rather, this refers to the time taken by the platform to perform the correlations and to provide a report for a given query. A response time of up to 1 hour can be considered optimal.

Considering the current status of many of the registries targeted across the Member States, as exemplified in Task 4, even that rate of speed would not be reached without a certain level of centralisation. Differences between Member State legislation, and even some technical limitations, also need to be considered. To accommodate limitations, a middle ground could be found in the use of a minimal mandatory set of information.

Reducing the type of information retained as much as possible to metadata and using linked data principles extensively have the potential to further ensure interoperability through the use of common models and specific vocabularies. This same approach also opens the architecture to the possibility of centrally storing some of the data, more specifically metadata, while keeping the actual data at the source. This would enhance the system's ability to perform its main function (search) in a reasonable amount of time, avoiding the complexity and delays induced by a fully decentralised system.

For specific registries that are not yet or just partially implemented at the Member State level (niche sectors or those too small to justify a sufficiently developed architecture), the partially centralised approach could bring the support of EU authorities closer to the respective sectors and enhance their ability to report. Such sectors would also benefit from system features through the integration of each registry in the larger European environment. To be viable, this option would need to be built on strong data management architecture. To minimise risk and maximise efficiency, the focus of data centralisation would fall on:

- using minimal data that are essential for the initiation of a query (entities metadata);
- taking a concerted approach on metadata regarding transnational transactions (independent of the domain);
- including virtual assets (e.g. crypto assets).

Any additional data stored centrally would fall under the regulation of the specific domain (legal framework at the EU level as defined in Task 5), complemented where needed by policies set up by a dedicated governance structure.

9.3 Description of the technical architecture

The solution envisaged relies on a **central repository module** serving as the backbone for the construction of all other features. Storage would take place in dedicated vaults that would separate metadata from any data that are present in the system.

The module would take advantage of the Linked Data principle and ensure a series of functions. These include mapping the minimal mandatory set of information of all entities

defined, and their corresponding instances found in the individual registries (maintained either in their specific Member State environment on the central server or at the national level).

The search function, provided by the **user interface module,** would function on the metadata vault and would ensure:

- rapid retrieval of a minimal set of mandatory information on entities,
- initiation of requests for provision of extended information from the subsequent vaults (at either the local or Member State level),
- the building of connections between entities.

A dedicated **national workspace module** would serve as a facility for the construction of dedicated registries for any situation where the Member State itself was either not able or had no justification to build one. The workspace module would offer direct interfaces to virtual data capsules. These would have dedicated functions to construct independent registries following their own data model, built as a complement to the minimal set of mandatory information.

A **monitoring and analytics module** would serve platform administrators at the central and local individual registry level, providing them with the necessary reporting features.

Depending on the connection scenario available at each registry level, periodic transfers would take place using web services, the eDelivery channel or the **data acquisition module**. Follow-up exchanges would take place between the platform and any partner platform or registry (either locally in the national workspace or at the Member State level) that has the technical capability and regulatory permission to top up the platform's metadata with additional information, using the **gateway module**.

9.4 Description of the interface

Evaluations of existing registries in previous tasks (see Section 4.3 and Section 6.1) have shown that fewer than half of them offer some form of access for direct interrogation. As the case studies evaluated in Task 4 have shown (see the annexes), the small number of such directly accessible registries can be easily explained by the specificities of the sector such as:

- the small volume of records maintained in the registry
- the small volume of requests for access
- the lack of financial and/or regulatory incentives for the registry owner
- supporting technologies/age of the system (outdated systems)
- the complexity of the parameters involved.

Starting from the captured data, it can easily be observed that implementing a large-scale interface on top of existing services would be difficult and even more so to practically justify (both financially and technically).

To compensate for this, the interface, in whatever form, would have to be supported by a mechanism that could supplement the lack of directly accessible services with data from the central data repository. Such data could be obtained in two ways:

- through the direct and periodic transfer of data from existing registries to the central platform
- through the development of a central EU mechanism that would provide ad hoc registry functionalities to enrolled Member States.

Implementing either of these mechanisms would offer the advantage of direct access to a central data repository that could be interrogated directly by both humans and machines. Considering the existing building blocks identified in Task 4, and more specifically the foundations set up by two major initiatives (BRIS and BORIS), it is envisaged that the platform would most easily be implemented in two phases. The first stage would involve advanced API access, with a web-based direct access interface for user queries in the second phase.

Outside the main search function described below in Section 9.5, the interface would include a dedicated module to provide Member State sector administrators or governing agencies with the ability to construct their own workspace. This would facilitate collection, analysis and reporting functions. Such a module would enable the collection of data on sectors that would otherwise be hard to gather or outside the regulatory framework. This workspace should be offered to Member States under a specific service-level agreement that would define the exact framework of operations for both sides. The module, as with any other components with direct user interaction, would be accessible using the standard EU login component.

9.5 Description of search possibilities

Information captured and synthetised in previous tasks has already shown that the majority of catalogues offer search functions that are not necessarily confined to a limited set of search fields but instead use a variety of parameters:

- Numbers/identifiers linked to individuals and legal entities
  - national registration number
  - TIN
  - VAT code
  - international identifiers (e.g. EUID and LEI).
- Personal names
  - name of the owner or person involved in the transaction
  - artist names (for works of art)
  - breeder name, kennel (for animals)
  - artisan/cabinetmaker name (for high-end furniture).
- Selected indicators, of which many are sector-specific and/or country-specific
  - brand name, serial numbers (for vehicles like vessels or aircrafts)
  - cadastral number, street number, city (for real estate)
  - dates (for collectors' items)
  - shipyard name and contact details (for yachts and other vessels)
  - age, height, weight, breed (for animals)
  - identification number (for vehicles)
  - weight, purity (for precious metals)
  - token identifier (for NFTs).

In conclusion, imposing constraints on the search process by limiting the interface to specific, predefined fields could be considered unproductive. This would limit the records retrieved to the particular assets that contain the respective parameters in their definition. To evaluate the probable needs of a user interacting with the system, two possible scenarios of use have been considered:

- search of a given individual – to identify the individual's associated assets;
- search by select indicators specific to a given sector – to identify assets that could be associated with an individual.

In the first scenario, the search mechanism could be simplified by means of direct mapping between the different registration/identification codes or name fields used by the national registries. This mapping would be the result of alignments between individual and company registration data models at the Member State level. Such alignments between parameters that are standardised have already been developed for projects described in Task 4.

For the second scenario, the search interface should instead be able to cover any of the parameters existing in the database, or combinations of them (cumulative manner). Taking these conditions into account, the interface mechanism would have to allow for incremental development of the search criteria, starting with generic parameters (such as national registration numbers) and progressing to more specific ones.

A search interface providing the necessary combination of features requires complex interactions with the user if not properly designed. To facilitate engagement with both expert and less advanced users, the system should introduce two options.

(1) The first option is a **wizard-type interface** where the criteria would be defined in stages as shown below.



This interface would guide the user to incrementally define the search scenario as follows:

1. **Select scope**
   Here the user would identify the specific sector the search would apply to (financial transactions, real estate, automobiles, etc.).
2. **Select metadata**
   Here the user would indicate the fields chosen to conduct the search on, from a given set that is specific to the sector selected (e.g. country, city, cadastral number, parcel number, name of the owner for real estate properties). To accommodate the specificity of each sector and to adjust to the particularities of the Member States, certain fields could be marked as required by default (e.g. country).
3. **Define conditions**
   Here, for each of the selected fields, the user would enter the values available.

Following these steps, the user would initiate the search and be presented with the results.

(2) The second option is a **query-based interface** where advanced users would be allowed to define the search conditions using:
- specific filters (like the JIRA ticket filters)
- a standard dedicated language (like SPARQL, Gremlin).

9.6 Calculation of costs of establishment and maintenance

Interviews and desk research conducted for mapping the relevant list of assets, gap analysis, IT feasibility and cost-benefit analysis have gathered information about projects that could be

considered relevant for the analysis. While some existing studies ([268]) make reference to platforms and provide a set of budgets associated with the setup of existing platforms that are partially similar in scope, it is clear that the references do not entirely correspond to the requirements of the platform assessed here. Given the complexity and novelty of the project, the costs have had to be evaluated based on the extrapolation of existing data and benchmarking with known systems (Table 51).

Table 51: Cost factors to consider in implementing and maintaining the system

Steps and processes	One-off costs 5 years (in EUR million)	Ongoing costs 1 year (in EUR million)
Development of the detailed technical specification for the solution Assumption: It is assumed that the specification will include detailed descriptions of the technical interfaces, data structures and testing procedures	0.180	
Development, licences, annual hosting and maintenance of the central repository module Assumption: It is assumed that a commercial instance of a platform with the necessary characteristics will be used. The development of the repository module will include all aspects associated with the definition and maintenance of the data model/ontology, integration of standard vocabularies (e.g. taxonomy of services, list of registries).	2.100	0.850
Data acquisition module Assumption: The cost includes the development, maintenance and support of the module that will allow the population of the repository with bulk data received from national registries. It should facilitate operations of data load and quality control on the data received. The annual costs would also cover the necessary adaptations to deal with registry structures that were not originally envisaged.	0.250	0.142
National workspace module for ad hoc registries Assumption: The module will provide the organisation, management and acquisition functions necessary for deploying and operating ad hoc registries on the central platform. This is a service for Member State registries that are either not operating or require upgrades to be able to integrate with the central platform, and where costs are not justified at the Member State level. The annual costs also cover the necessary adaptations to deal with minor functionality updates and helpdesk operations to support Member State operations.	0.450	0.125
General gateway and integration module Assumption: The module ensures necessary data exchange and messaging capabilities to support integration with existing systems (BRIS, BORIS, IRI) and create the standard interface services for national registries that have the necessary capabilities to interconnect.	0.520	0.148
User interface module Assumption: The module will provide the generic web-based interface of the system including all functionalities associated with the construction of queries for basic and advanced searches. The cost includes the purchase of a dedicated search engine licence and customisation to ensure the operations of the system under the parameters defined in the detailed specification.	0.375	0.254

([268]) European Commission (2019b), https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=CELEX %3A52019DC0372

Monitoring and analytics module Assumption: The module will provide metrics and reports at the central and Member State levels as well as helpdesk support for the operation of the module.	0.145	0.058
Platform management and operations costs Assumption: The cost will cover technical maintenance and support (e.g. system administration, application management), general project and change management (e.g. ticket management for maintenance, conference system), governance activities (e.g. procurement, coordination with Member States, organisation of coordination meetings at the technical and administrative levels). At least two coordination meetings, one governance meeting and two technical meetings would take place yearly.	0.346	0.428

Source: Authors' elaboration.

As shown in the table above, the total cost for the implementation and operation of the system on a timeframe of 5 years would be approximatively EUR 14.4 million. The actual implementation of the modules and all associated functions would cost under EUR 4.5 million, with an annual ongoing cost of approximatively EUR 2 million.

9.6.1 General assumptions

A number of assumptions have been made in order to simplify the estimations, as follows:

- The cost estimates are for a project implemented at the central level in accordance with a mixed scenario (combining options 1 and 3), as defined in Task 4.
- Major registries of the Member States and partner platforms covering specific sectors (financial transactions, land registries, etc.) would cooperate in establishing the communication and further alignment with the gateway functions of the platform.
- Minor registries would coordinate with the platform providing periodic uploads of the minimal mandatory set of information for their specific sectors.
- The timeline envisaged includes 1 year for the procurement (outside the scope of the cost estimation), 2 years for development and testing, and 3 years of operations. It should be noted that a proof of concept could be set up on a much shorter timeframe under the proviso that specific conditions are met (sample data availability, the corresponding platform is already in place, etc.).
- For the given timeframe of a maximum of 5 years, no technological upgrade has been considered. Only corrective maintenance costs and standard software upgrades have been included in the process.

The costs listed in the present analysis are based on data gathered from EU reports, existing procurement procedures and interviews with vendors and technical experts, correlated with desk analysis and benchmarking techniques.

9.6.2 Volumetric information

While the scope of the project does mention that the functionalities provided by the system should cater to the needs of AML authorities, FIUs and tax authorities, it is not possible to target an exact cost/performance, as the need for the services to be provided has not been exercised on a similar existing system. Even considering a given number of interested parties (based on the number of relevant authorities of each Member State), the possibility of correctly calculating the cost of operations is greatly affected by the inability to estimate the number of queries initiated by each of the parties involved. Thus, depending on the variation of these numbers, both the estimated operating cost and the time for responses might differ.

9.6.2.1 Inquiries

The evaluation started with an estimation of the volumes involved and considered several existing sources at the EU ([269]), international and Member State levels ([270]). It can safely be assumed that the number of cases processed annually by Eurojust (600 cross-border money laundering cases in 2021) does present a picture of the current situation, if the scope of participating parties were to be enlarged. Simply making such a service available could dramatically increase the number of requests for information ([271]). Consequently, it could be expected that over a longer period (more than 5 years), the estimates provided are likely to be outdated. Based on a reasonable assumption of data trends and in view of the increase in the number of stakeholders (scaling to the number of agencies at the EU and Member State levels), an average of 50 000 inquiries per year should be considered for the purpose of this exercise.

9.6.2.2 Storage space and records

In contrast, the volume of data to be stored depends on the type and number of entities deemed to be within the scope of the platform and can be more effectively estimated. As shown in the inventory developed during Task 1, sources of wealth can be linked with both natural and legal persons. To be able to extract valuable information, the system would then need to be able to correlate data about both natural and legal persons. Starting from the estimated number of active economic entities across the EU ([272]) and considering the annual increase and a minimal number of properties associated with each one, it is estimated that a volume 160 million resources, 5 billion properties and a total of 90Tb would be in use by the central system. A different approach would be to base the calculation on the EU banking population (>EUR 400 million). This, however, has the potential to tweak the analysis in the direction of financial transactions, while the actual focus of the platform is on more generic registries.

Assuming that the scope of the system would scale over a period of 5 years and also that additional properties would be added beyond the minimum mandatory information, a reserve of 50Tb should be also considered.

9.7 Defining the implementation strategy

At this preliminary assessment stage, evaluating detailed implementation options such as derivate technological options (e.g. graph databases versus triple stores) could be considered outside the scope of the study. Nevertheless, alternative options for implementation do exist. Such options could have an impact on many areas of strategy, starting with the duration of implementation up to the performance offered by the system.

It is therefore recommended to approach the subject by setting up a proof of concept (PoC) focused on a specific, minimal set of services (small-scale bulk load, communication and

([269]) Eurojust Report on Money Laundering, Eurojust, European Union Agency for Criminal Justice Cooperation (europa.eu).
([270]) Germany Mutual Evaluation Report.
([271]) Transparency and Exchange of Information for Tax Purposes.
([272]) Business demography statistics – Statistics Explained (europa.eu).

search functions). The PoC will help to calibrate expectations and identify the main requirements from a direct user close to operational settings.

9.7.1 Issues and possible solutions that have to be considered in the implementation strategy

Building and maintaining a central database that integrates information which varies in terms of systems providing the source data, sectors of applicability, the language used, script of content and country of origin, is a process that comes with a number of issues that have to be considered.

From a technical perspective, the process of setting up the solution could be affected by the following major complications:

- **Connectivity – in relation to accessing source data**
  - Access to data would not solely be dependent on the ability to connect to source registries but also on the capacity to work with data dumps provided by national registries and with locally registered data developed within the national workspace environment.
- **Content diversity – in terms of language and script**
  - Diversity in terms of language and scripts would be dealt with through the use of controlled multilingual vocabularies.
- **Data heterogeneity – in terms of data structure**
  - A base level of uniformity would be obtained through the use of standardised minimal mandatory sets of information. This information would serve to establish the connection between entities or map registry records with entities. Any additional information would serve a purpose but would not be part of the functional aspects of the platform.
- **Regulatory limitations – at the EU or Member State level**
  - The platform would specifically use anonymised minimal mandatory sets of information.

Beyond these, the solution would come with a set of more specific technical issues, as follows:

- ensuring the necessary uptime of the system;
- maintaining the alignment between data structures of the different registries;
- accommodating new sectors added to the catalogue;
- managing the addition of new registries;
- accommodating the increasing needs of authorities using the system.

Managing these issues would be part of the risk management process that would be implemented in the overall governance process of the platform.

9.7.2 Analysis of the implementation strategy

Building a functional ecosystem takes time. The time factor becomes even more critical with the increase in the number of participants in that ecosystem. By performing reverse

engineering on the business process associated with the function of a specific platform, a strategy can be devised to implement that platform in a given amount of time.

A generic strategy to implement the platform includes the following four phases:

- Phase 1 – Establish the techno-organisational foundation of the platform;
- Phase 2 – Engage the main stakeholders with a demonstrable value offering;
- Phase 3 – Build exposure by enlarging the pool of users;
- Phase 4 – Scale the platform functions and services to the planned level (and/or respond to users' needs).

Considering the critical functions of the platform, the strategies below could be devised.

9.7.2.1 A 5-year implementation strategy

Based on the analysis, a 5-year strategy could be considered the base reference construct. In this scenario, the platform business plan would incrementally follow a standard process, including the six main steps in Table 52.

Table 52: 5-year implementation strategy

1	Procurement	Formal procurement stage
2	Coordination	Coordination with the Member States, individual stakeholders and relevant registries on policies, general governance and data model
3	Development	Implementation of the main repository
4	Limited operationalisation	Deployment in production for a reduced number of sectors (3-5)
5	Extension	Incremental implementation of additional functions
6	Production release	Full operationalisation with the complete set of functions for all sectors of interest

Source: Authors' elaboration.

The implementation strategy in such a scenario would be built on a combination of strict project management and rapid application development.

9.7.2.2 A 3-year implementation strategy

Shortening the estimated implementation timeframe to a 3-year period could be done by using a similarly structured approach but reducing the scope of each phase correspondingly. The strategy would take the shape of a proof-of-concept implementation, with the reduction affecting each phase as shown in Table 53.

Box 4: Summary of options for the design and functionalities of an EU asset registry (Task 7)

In terms of user interface, the proposed solution defined a mechanism that uses the semantic relations between records to facilitate a sequential building of queries based on the information available at central level and known by the user at the time of request. This mechanism will use the vague semantic relations between records to enable the construction of queries based on incomplete values (e.g. list legal entities that own a ship or an airplane located in a different member state than the place of registration). The mechanism will be doubled by a written query-based system that allows advanced users to incrementally develop queries, to include multiple sources as they get connected and to work with sources that include properties that are outside the range of the central system (e.g., the main query at EU level can include subqueries run on properties that are specific to a domain that only one country uses).

Considering the findings in terms of architecture, plus a number of assumptions and a given list of modules listed in the technical description, a calculation of costs has been developed. The calculation was constructed to accommodate a scenario of usage described in terms of storage volume and estimated number of queries per year (average of 50 thousand). All these elements coalesce into a calculation that gave a total estimated budget of approximately EUR 14.4 million distributed over a period of 5 years.

Finally, the possible strategies were evaluated, pointing to the importance of setting up a Proof of Concept to validate the assumptions. It was found that implementing the platform in a period of 3 years is principally possible, but it will induce a number of limitations that are not necessarily technically related. An analysis of technical, generic organisational and project management principles showed that in this respect, we can expect a full implementation of the platform to take around five years. At the same time, the operational phase could begin much faster.

Table 53: 3-year implementation strategy

1	Procurement	The scope of procurement action would be reduced to the acquisition of the necessary commercial components
2	Coordination	Coordination would take place solely with a limited number of main stakeholders (e.g. EU actors with direct interest in AML activities)
3	Development	Implementation of the main repository
4	Limited operationalisation	Deployment in a demonstrative environment for a single sector
5	Extension	Incremental implementation of additional sectors (up to five) to facilitate a more effective exploitation of the platform
6	Production release	Operationalisation in a production environment with a limited set of functions for a given set sectors presenting the most interest

Source: Authors' elaboration.

While this secondary scenario does come with clear limitations in terms of both domain and functional coverage, it also offers certain advantages in terms of a more direct correlation with the realities of each stakeholder and domain. It can therefore be considered a valid approach.

Conclusions

The **objective** of this study was to assess the feasibility of providing a unique access point for competent authorities involved in the fight against money laundering and tax evasion to relevant asset registers in the EU Member States and to highlight and weigh three scenarios under which this could be achieved. The study first defined the relevant assets for money laundering and tax evasion, then mapped existing registries and identified gaps. It assessed the feasibility of three potential scenarios from an operational, legal and IT perspective and provided a cost-benefit analysis.

The study highlighted the immense difficulties in creating new asset registries for which none previously existed. Key challenges included ensuring high-quality and complete data along with major operational challenges in constructing registries denoting full and comprehensive ownership. Additional possibilities for gaining information on ownership and value of these assets were also highlighted, mostly relying on the collection and standardisation of transaction data. The heavy burden this would place on potential reporting entities such as traders was acknowledged as well as the difficult task for authorities to verify accurate reporting.

Similarly, challenges to harmonise existing registries were highlighted, notably the fragmented landscape of real estate and land registries across Member States, as well as issues encountered in similar interconnection exercises, such as BRIS, BORIS or DAC.

Under Scenario 1, the main advantages were found to lie mainly in the high asset coverage. At the same time, it would be costly and resource-consuming to implement it in practice.

In comparison with Scenario 1, Scenario 2 might be swifter and less costly to implement as no new registries would need to be created. In addition, it would imply the processing of a narrower scope of personal data, given that only certain categories of assets would be covered (real estate, companies and trusts, bank account registries and safe deposit boxes at bank). In line with the GDPR and the ECJ's case law, this may reduce the seriousness of the interference on fundamental rights. On the other hand, the reduced asset coverage would provide competent authorities with less extensive information on asset holders.

Scenario 3 was found to most seriously restrict citizens' fundamental rights to data protection and privacy. Nevertheless, it comprises a significant number of technical advantages and a wide coverage of assets, rendering it appropriate and necessary to reach the objective. Implementation might be challenging, however, as the EU would be in charge of ensuring monitoring contrary to most interconnection efforts so far. The legal feasibility study attested that this option is the most likely to not meet the required criteria due to the aforementioned restrictions.

The study also proposed an alternative scenario, blending Scenarios 2 and 3, which could be the object of further study.

In terms of **user interface**, the study proposed a mechanism that makes use of the semantic relationship between records. This facilitates a sequential building of queries based on the information available at a central level and known by the user at the time of request. This mechanism would use the vague semantic relationships between records to enable the construction of queries based on incomplete values (e.g. to list legal entities that own a ship or an aircraft located in a different Member State than the place of registration). The mechanism would be doubled by a written query-based system that allows advanced users to incrementally develop queries, to include multiple sources as they get connected and to work with sources

that include properties that are outside the range of the central system (e.g. the main query at the EU level can include sub-queries run on properties that are specific to a domain that only one country uses) (Task 7).

In terms of **needed architecture**, a total estimated budget of approximately EUR 14.4 million was calculated to be distributed over a period of 5 years. In terms of the timeline, implementing the platform in a period of three years would be possible in principle, but it would induce a number of limitations that are not necessarily technically related. Technical, generic organisational and project management principles showed that in this respect, we can expect that a full implementation of the platform would take around five years, while the operational phase could begin much faster (Task 7).

Overall, Scenario 2 was found to be the most realistic and straightforward to implement. This is due to the comparatively few operational obstacles, mostly relying on already available and partly ready interconnected registries. There are few legal obstacles, as the data remains with the Member States and is not centrally stored. It is important to note that even these tasks involve a considerable degree of harmonisation and coordination and should by no means be interpreted as easily achievable. The major drawback of this scenario is the low coverage of assets, potentially limiting its usefulness. Scenarios 1 and 3 do exhibit a broader coverage of assets but are found to be either operationally (Scenario 1) or legally (Scenario 3) questionable.

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ANNEX A: RESEARCH METHODOLOGY

This study has prioritised countries at risk of money laundering and terrorist financing (source countries), and taken account of countries at risk of facilitating money laundering and terrorist financing (international financial centres). The rationale for this selection is to tackle the cross-border aspects of financial crime, and to provide better networks and understanding of the international aspects of this issue. The preliminary country selection was based on a mapping exercise that identified where access to and the linking of asset registries would be most beneficial (Table 54).

Table 54: Country selection matrix

Member State characteristics					Quantitative selection criteria						Preliminary selection			Final selection based on surveys		
Member State	Size	EU region	Accession	Legal system	Basel AML	CPI	KYC	FSI	Average score	Enterprises having ERP	Selection for interviews	Selected for IT case studies	Selected for legal case studies	Tasks 1-3	Tasks 4-7	Tasks 5-6
AT	Small	West	Old	Civil law	56.2	77	80.03	44	64.308	43	No	No	No			
BE	Small	West	Old	Civil law	60.2	75	75.9	55	66.525	53	No	No	No			
BG	Small	East	New	Civil law	68.8	43	73.57	51	59.093	23	No	No	No			
CY	Small	South	New	Mixed	51.9	58	69.45	39	54.588	33	Yes	No	Yes		Yes	Yes
CZ	Small	East	New	Civil law	57.2	56	76.49	45	58.673	38	Yes	No	Yes	Yes		Yes
DE	Large	West	Old	Civil law	55.8	80	77.8	48	65.4	29	Yes	Yes	No	Yes		Yes
DK	Small	North	Old	Civil law	65.4	87	85.61	55	73.253	50	No	No	No			
EE	Small	North	New	Civil law	76.4	74	84.48	57	72.97	26	No	No	No			
ES	Large	South	Old	Civil law	63.4	62	75.24	56	64.16	43	Yes	No	No			Yes
FI	Small	North	Old	Civil law	70.3	86	86.75	48	72.763	43	No	No	No			
FR	Large	West	Old	Civil law	60.8	69	77.57	50	64.343	48	Yes	No	No	Yes		Yes
GR	Small	South	Old	Civil law	62.7	48	75.38	49	58.77	38	Yes	Yes	No			Yes
HR	Small	South	New	Civil law	60.5	47	79.92	45	58.105	26	Yes	Yes	No			Yes

Member State characteristics					Quantitative selection criteria						Preliminary selection			Final selection based on surveys		
Member State	Size	EU region	Accession	Legal system	Basel AML	CPI	KYC	FSI	Average score	Enterprises having ERP	Selection for interviews	Selected for IT case studies	Selected for legal case studies	Tasks 1-3	Tasks 4-7	Tasks 5-6
HU	Small	East	New	Civil law	50.1	44	74.24	46	53.585	14	Yes	No	Yes			Yes
IE	Small	North	Old	Common law	55.4	74	75.63	52	64.258	28	No	No	No		Yes	
IT	Large	South	Old	Civil law	53.9	53	72.8	50	57.425	35	Yes	No	No		Yes	Yes
LT	Small	North	New	Civil law	64.9	60	83.22	50	64.53	50	No	No	No			
LU	Small	West	Old	Civil law	52.6	80	74.78	45	63.095	41	Yes	Yes	Yes			Yes
LV	Small	North	New	Civil law	53.8	56	75.86	41	56.665	32	Yes	Yes	Yes	Yes		Yes
MT	Small	South	New	Mixed	45.2	54	82.14	38	54.835	32	Yes	No	Yes		Yes	Yes
NL	Large	West	Old	Civil law	54.4	82	71.92	33	60.33	48	Yes	Yes	Yes		Yes	Yes
PL	Large	East	New	Civil law	56.4	58	74.74	44	58.285	29	Yes	No	No			
PT	Small	South	Old	Civil law	63.4	62	74.38	46	61.445	43	No	No	No			
RO	Small	East	New	Civil law	52.1	44	71.73	37	51.208	23	Yes	Yes	Yes	Yes	Yes	Yes
SE	Small	North	Old	Civil law	68.1	85	87.46	54	73.64	37	No	No	No			
SI	Small	South	New	Civil law	66.5	60	83.81	62	68.078	33	No	No	No			
SK	Small	East	New	Civil law	60.5	50	73.35	49	58.213	31	No	No	No			

Source: Authors' compilation.

All Member States were screened. Countries with the lowest average scores were then prioritised for a feasibility analysis on creating an EU asset registry, in the context of anti-money laundering (AML) and preventing tax evasion. Our proposed country selection also considered the representativeness of samples based on the size of the Member State in question, geography, duration of EU membership and the legal system. Some 15 Member States with the highest number of combined risk elements were identified (scoring below 61.5 points). The only exception is Luxembourg, which was proposed for selection based on it being a major financial investment centre. The aim was to ensure the feasibility of creating an EU asset registry to effectively combat money laundering and tax evasion. For this reason, the focus was on countries with the lowest average score in all indices.

Table 55 provides the preliminary list for selecting countries for interviews and case studies. The final country selection was validated and adjusted based on the quality and response rate received through the survey questionnaires circulated among the registries, financial intelligence units (FIUs), law enforcement authorities (LEAs), AML supervisors of financial institutions and tax authorities. As a result, we conducted interviews in 14 countries (CY, CZ, DE, ES, FR, GR, HR, HU, IE, IT, LV, MT, NL and RO).

Table 55: Survey Assessment Matrix

Task	Sub-task	Activity	Methodology	Reference text from ToR	# Questions	Questions
Task 1: Inventory of different assets/sources of wealth	1.1: Outline of different assets and sources of wealth	Assets mapping	Literature review			
Task 1: Inventory of different assets/sources of wealth	1.1: Outline of different assets and sources of wealth	Assets mapping	Desk research			
Task 1: Inventory of different assets/sources of wealth	1.1: Outline of different assets and sources of wealth	Assets mapping	Interviews with NGOs	*Inventory must indicate how relevant direct access to particular sources of information is for the purposes of AML and fighting tax evasion*		
Task 1: Inventory of different assets/sources of wealth	1.1: Outline of different assets and sources of wealth	Assets mapping	Interviews with FIUs/LEAs/AML supervisors of financial institutions	*Inventory must indicate how relevant direct access to particular sources of information is for the purposes of AML and fighting tax evasion*	3.2, 3.3, 3.4	3.2 Which additional assets/sources of wealth currently not included in the list in Question 1 on the previous sheet would you consider relevant in the context of AML/tax evasion and should be added to the list? 3.3 For which of the assets you consider relevant in the context of AML/tax evasion and for which currently no asset register exists in your country would you like to have access to ownership information? 3.4 Which assets/sources of wealth would you consider removing from the list?
Task 1: Inventory of different assets/sources of wealth	1.1: Outline of different assets and sources of wealth	Assets mapping	Interviews with tax authorities	*Inventory must indicate how relevant direct access to particular sources of information is for the purposes of AML and fighting tax evasion*	3.2-6	3.2 Please indicate how important your own national asset registers are in combating tax evasion or for tax claim recovery purposes? 3.3 Under Directive 2011/16/EU for Administrative Cooperation in direct taxation (the DAC), Regulation 904/2010/EU on administrative cooperation and combating fraud in the field of value added tax (VAT) and Directive 2010/24/EU concerning mutual assistance for the recovery of claims relating to taxes, duties and other measures, a Member State's tax authority can

Task	Sub-task	Activity	Methodology	Reference text from ToR	# Questions	Questions
						request the assistance of the tax authorities of other Member States. Do you consider that access to the national asset registers of other Member States, for example through the interconnection of national asset registers, would significantly improve the fight against tax evasion and assistance for tax claim recovery purposes? 3.4 Assets can also be held through legal persons, such as private limited liability companies, and legal arrangements such as trusts. In terms of availability of beneficial ownership information, i.e. the natural person owns the asset indirectly through the legal person/arrangement, do you consider that this has been an obstacle in combating tax evasion and in conducting tax claim recovery, in both national and cross-border contexts? 3.5 Which additional assets/sources of wealth currently not included in the list in Question 1 on the previous sheet would you consider relevant in the context of AML/tax evasion and should be added to the list? 3.6 Which assets/sources of wealth would you consider removing from the list?
Task 1: Inventory of different assets/sources of wealth	1.1: Outline of different assets and sources of wealth	Definitions of assets	Literature review			
Task 1: Inventory of different assets/sources of wealth	1.2. Indication whether assets are held directly or indirectly	Definition of ownership	Survey among registry administrators	*A distinction would have to be made as to whether assets may be held directly or indirectly*	2.17	2.17 Is the identity and type of legal owner of the asset indicated in the register?
Task 1: Inventory of different assets/sources of wealth	1.2. Indication whether assets are held directly or indirectly	Definition of ownership	Desk research			
Task 1: Inventory of different assets/sources of wealth	1.3. Analysis of relevance	Analysis of relevance	Interviews with FIUs/LEAs/AML supervisors of financial institutions/tax authorities	*Inventory must indicate how relevant direct access to particular sources of information is for the purposes of AML and fighting tax evasion*	2.1	2.1 According to the current understanding within your organisation, please rate on a scale of 0 to 4 the importance (e.g. the size of the asset base) of each asset in the investigation and enforcement activities of your organisation. According to the current understanding within your organisation, please rate on a scale of 0 to 4 the risk (i.e. the inherent vulnerable risk factors of

Task	Sub-task	Activity	Methodology	Reference text from ToR	# Questions	Questions
						an asset, e.g. easy to conceal/non-regulated market) of each asset being used for tax evasion
Task 1: Inventory of different assets/sources of wealth	1.3. Analysis of relevance	Analysis of relevance	Literature review			
Task 2: Mapping of existing registers	2.1: Description of scope and format of data stored	Design of scorecards	Own expertise			
Task 2: Mapping of existing registers	2.1: Description of scope and format of data stored	Mapping of registries	Desk research			
Task 2: Mapping of existing registers	2.1: Description of scope and format of data stored	Design of scorecards	Desk research			
Task 2: Mapping of existing registers	2.1: Description of scope and format of data stored	Assessment	Analysis of legislation			
Task 2: Mapping of existing registers	2.1: Description of scope and format of data stored	Assessment	Desk research			
Task 2: Mapping of existing registers	2.1: Description of scope and format of data stored	Assessment of scope and format	Survey among registry administrators	*A description of data stored (exact scope and format of information recorded in the register, including whether the registers include information on the value of the asset held, VAT number or other tax code, identification data of legal representatives). A description of how information can be retrieved, e.g. whether specific data must already be known in order to launch research in a database (which would imply that no research is possible unless the required data is known at the start of the research)*	2.1, 2.10-12, 3.25-30	2.1 Scope of the registry 2.10 Do you use a data catalogue to catalogue all the data on assets? 2.11 If you use a data catalogue, what kind of data cataloguing do you use? 2.12 If you do not use any data cataloguing, what would the costs be of implementing/migrating to it? 3.25 In which format is the data in the registry stored? 3.26 What are the categories of information available in the registry? 3.27 What is the type of data available in the registry? 3.28 What is the size of the registry in megabytes (MB)? [Optional] 3.29 What is the number of records in the registry (known or estimated)? [Optional] 3.30 Considering the current operation conditions, what is the extra load your asset data systems can accept (queries per hour)? [Optional]
Task 2: Mapping of existing registers	2.2: Description of digitalisation and aggregation level	Assessment of searchability	Survey among registry administrators	*A description as to whether the registers are digitalised allowing for search functions*	2.8, 2.9	2.8 Are your data on assets stored for the long term in such a way that both humans and machines can easily access them, i.e. are they

Task	Sub-task	Activity	Methodology	Reference text from ToR	# Questions	Questions
				and web services, and whether digitalisation of registers is contemplated at national level		retrievable by their identifier using a standardised communication protocol? 2.9 Which indicators can be used to search the register?
Task 2: Mapping of existing registers	2.2: Description of digitalisation and aggregation level	Assessment of text harmonisation	Survey among registry administrators	*A description as to whether the registers are digitalised allowing for search functions and web services, and whether digitalisation of registers is contemplated at national level*	2.13, 3.31-36	2.15 Are the registry entries encoded using the standard UTF encoding? 3.31 Is all the information for your territory included in the registry? 3.32 Is the data unique within the registry (i.e. no double entries within the registry)? 3.33 Is the data in your database structured such that they can be easily used by both humans and machines? 3.34 Do you have a data quality system implemented to check the quality of the registry continuously or periodically? 3.35 If you have implemented a data quality system, what kind of system have you built to check each entry's quality? 3.36 How frequently is your registry updated?
Task 2: Mapping of existing registers	2.2: Description of digitalisation and aggregation level	Assessment of metadata	Survey among registry administrators	*A description as to whether the registers are digitalised allowing for search functions and web services, and whether digitalisation of registers is contemplated at national level*	2.7	2.7 Are your data on assets easily searchable both by humans and machines (i.e. the data are accompanied by machine-readable metadata)?
Task 2: Mapping of existing registers	2.3: Assessment of availability and accessibility of national BO registers	Mapping of registries	Desk research			
Task 2: Mapping of existing registers	2.3: Assessment of availability and accessibility of national BO registers	Assessment of accessibility	Survey among registry administrators	*A description as to whether information on the beneficial owner (BO) is available or can be easily obtained by consulting the national register of BOs with the information available in the register originally consulted*	2.18-21	2.18 Is information on BOs of the asset included in the register? 2.19 Is information on previous legal owners of assets recorded in the registry? 2.20 Is information on previous BOs of assets recorded in the registry? 2.21 If yes, which indicators are recorded for previous legal and BOs of assets recorded in the registry?

Task	Sub-task	Activity	Methodology	Reference text from ToR	# Questions	Questions
Task 2: Mapping of existing registers	2.3: Assessment of availability and accessibility of national BO registers	Assessment of accessibility	Desk research			
Task 2: Mapping of existing registers	2.4: Accessibility assessment – type of competent authority	Assessment of accessibility	Survey among registry administrators	*Access to the registers, with a particular focus on FIUs, LEAs and tax authorities. The mapping must define access by type of competent authority involved in the fight against money laundering and tax evasion and indicate whether the access is direct or indirect*	3.9-13	3.9 Does your domestic FIU have access to your register? 3.10 Does your domestic tax authority have access to your register? 3.11 Does your domestic LEA have access to your register? 3.12 Are there any differences in access conditions to registries for tax authorities, FIUs and AML national competent authorities? 3.13 Are there any differences in access conditions to registry for national or non-national authorities (e.g. authorities from other Member States)?
Task 2: Mapping of existing registers	2.4: Accessibility assessment – type of competent authority	Assessment of accessibility	Interviews with FIUs/LEAs/AML supervisors of financial institutions/tax authorities	*Access to the registers, with a particular focus on FIUs, LEAs and tax authorities. The mapping must define access by type of competent authority involved in the fight against money laundering and tax evasion and indicate whether the access is direct or indirect*	2.1	2.1 Indicate detailed assets considered (see definitions in the 'Asset definitions' sheet – multiple options possible) Whether your organisation has the legal right to access the information on the asset and the owner
Task 2: Mapping of existing registers	2.5: Accessibility assessment – format of information	Assessment of accessibility	Survey among registry administrators	*Access to the registers, with a particular focus on FIUs, LEAs and tax authorities. The mapping must define access by type of competent authority involved in the fight against money laundering and tax evasion and indicate whether the access is direct or indirect*	3.14-24	3.14 What are the options available to access/retrieve the information contained in the registry? 3.15 What method do you use to manage access to the data in the registry? 3.16 What are the minimum legal requirements to access the data in the registry for the general public? 3.17 What is currently required by the registry to access the data in the registry for tax authorities? 3.18 What is currently required by the registry to access the data in the registry for FIUs? 3.19 What is currently required by the registry to access the data in the registry for AML-competent authorities?

Task	Sub-task	Activity	Methodology	Reference text from ToR	# Questions	Questions
						3.20 Is the access to the registry free of charge for FIUs, tax authorities or AML-competent authorities? 3.21 For what unit is the payment? 3.22 How does the payment take place? 3.23 Is your authority connected to the TESTA network? 3.24 If your authority is connected to the TESTA network, what was the cost of connecting it?
Task 2: Mapping of existing registers	2.5: Accessibility assessment – format of information	Assessment of accessibility	Analysis of legislation			
Task 2: Mapping of existing registers	2.5: Accessibility assessment – format of information	Assessment of accessibility	Desk research			
Task 3: Mapping of other potential asset registers	3.1: Analysis of how to best retrieve financial or non-financial information	Mapping of sources	Desk research			
Task 3: Mapping of other potential asset registers	3.1: Analysis of how to best retrieve financial or non-financial information	Mapping of sources	Interview with ESMA			
Task 3: Mapping of other potential asset registers	3.1: Analysis of how to best retrieve financial or non-financial information	Mapping of sources	Interviews with professional associations			
Task 3: Mapping of other potential asset registers	3.2: Gap analysis	Mapping of gaps	Compilation based on previous findings			
Task 3: Mapping of other potential asset registers	3.2: Gap analysis	Conducting analysis	Case studies			
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.1: Operational and IT feasibility assessment	Key characteristics of the database	Survey among registry administrators			Whole questionnaire
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.1: Operational and IT feasibility assessment	Information analysis	Survey among registry administrators	*IT standards and models use*	2.2-6	2.2 Is the registry connected to other registries (e.g. part of other national registries, standalone registry)? 2.3 If this registry is connected to other registries, please mention which registries it is connected to. 2.4 If this registry is not connected to other registries, do you manually search other registries?

Task	Sub-task	Activity	Methodology	Reference text from ToR	# Questions	Questions
						2.5 If yes, which registries do you manually search? 2.6 Do you provide any documentation or help in order to support the use of the registry?
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.2: Define scenarios for the asset registry	Define scenarios	Compilation based on previous findings			
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.3: Cost-benefit analysis	Definition of costs	Survey among registry administrators	*These estimates should take into account all aspects of the establishment of such a central database (logistical, technical, licensing, human resources, etc.), as well as the cost of a standard maintenance and evolution mode (special secure equipment, support, training ...) on a yearly basis*	3.43-45	3.43 If a centralised EU asset registry were put in place, would you expect any significant recurring costs to arise for your organisation? 3.44 If a centralised EU asset registry were adopted, would you expect any significant one-off costs to arise for your organisation? 3.45 If a centralised EU asset registry were adopted, would you change your pricing model to access the registry?
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.3: Cost-benefit analysis	Definition of costs	Interviews with FIUs/LEAs/AML supervisors of financial institutions	*These estimates should take into account all aspects of the establishment of such a central database (logistical, technical, licensing, human resources, etc.), as well as the cost of a standard maintenance and evolution mode (special secure equipment, support, training ...) on a yearly basis*	4.8, 4.10	4.8 Which foreseeable benefits, other than those mentioned before, can you expect for your organisation if a centralised EU asset registry were adopted? 4.10 If a centralised EU asset registry were adopted, do you expect any substantial costs to arise for your organisation (e.g. connecting your systems to the new registry)?
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.3: Cost-benefit analysis	Definition of costs	Interviews with tax authorities	*These estimates should take into account all aspects of the establishment of such a central database (logistical, technical, licensing, human resources, etc.), as well as the cost of a standard maintenance and evolution mode (special secure equipment, support, training ...) on a yearly basis*	4.11	4.11 If a centralised EU asset registry were adopted, do you expect any substantial costs to arise for your organisation (e.g. connecting your systems to the new registry)?
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.3: Cost-benefit analysis	Definition of costs	Desk research			

Task	Sub-task	Activity	Methodology	Reference text from ToR	# Questions	Questions
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.3: Cost-benefit analysis	Definition of costs	Interviews with business representatives			
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.3: Cost-benefit analysis	Definition of costs	Interviews with Member States (case studies)			
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.3: Cost-benefit analysis	Definition of benefits	Survey among registry administrators	*Benefits associated with both creating a new database (collection of data) and interconnecting database for the purpose of the fight against money laundering and tax evasion*	3.46-48	3.46 What foreseeable significant one-off benefits can you expect for your organisation if a centralised EU asset registry were adopted? 3.47 What foreseeable significant recurring benefits can you expect for your organisation if a centralised EU asset registry were adopted? 3.48 Have you been involved in any similar integration exercise before?
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.3: Cost-benefit analysis	Definition of benefits	Interviews with FIUs/LEAs/AML supervisors of financial institutions	*Benefits associated with both creating a new database (collection of data) and interconnecting database for the purpose of the fight against money laundering and tax evasion*	4.9	4.9 What foreseeable benefits could be expected from a system of exchange of data between Member States?
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.3: Cost-benefit analysis	Definition of benefits	Interviews with tax authorities	*Benefit associated with both creating new database (collection of data) and interconnecting database for the purpose of the fight against money laundering and tax evasion*	4.10	4.10 What foreseeable benefits, other than those mentioned before, can you expect for your organisation if a centralised EU asset registry were adopted?
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.3: Cost-benefit analysis	Definition of benefits	Desk research			
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.3: Cost-benefit analysis	Definition of benefits	Interviews with business representatives			
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.3: Cost-benefit analysis	Definition of benefits	Interviews with Member States (case studies)			
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.4: Analysis of the administrative costs	Definition of costs	Desk research			

Task	Sub-task	Activity	Methodology	Reference text from ToR	# Questions	Questions
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.4: Analysis of the administrative costs	Definition of costs	Interviews with business representatives			
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.4: Analysis of the administrative costs	Definition of costs	Interviews with Member States (case studies)			
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.4: Analysis of the administrative costs	Definition of benefits	Desk research			
Task 4: Operational/IT feasibility assessment and cost/benefit analysis	4.4: Analysis of the administrative costs	Definition of benefits	Interviews with business representatives			
Task 4: Operational/IT feasibility assessment and cost-benefit analysis	4.4: Analysis of the administrative costs	Definition of benefits	Interviews with Member States (case studies)			
Task 5: Legal feasibility assessment of establishing additional databases/registries	5.1: Identification of legal obstacles	Data collection	Interviews with data protection authorities			
Task 5: Legal feasibility assessment of establishing additional databases/registries	5.1: Identification of legal obstacles	Data collection	Interviews with FIUs/LEAs/AML supervisors of financial institutions/tax authorities	*Legal obstacles relating to the provisions of the EU Treaties, the objectives of the GDPR, the EU Charter of Fundamental Rights, the policies pursued by the European Data Protection Supervisor and stemming from the case-law of the Court of Justice of the European Union*	4.1	4.1 What are the difficulties you currently encounter when you want to check different relevant asset registries, both in your Member State and cross-border?
Task 5: Legal feasibility assessment of establishing additional databases/registries	5.1: Identification of legal obstacles	Data collection	Interviews with Member States (case studies)			
Task 5: Legal feasibility assessment of establishing additional databases/registries	5.1: Identification of legal obstacles	Data collection	Literature review			
Task 5: Legal feasibility assessment of establishing additional databases/registries	5.2: Necessity	Compliance with the EU Treaties	Legal analysis			

Task	Sub-task	Activity	Methodology	Reference text from ToR	# Questions	Questions
Task 5: Legal feasibility assessment of establishing additional databases/registries	5.2: Necessity	Compliance with national law	Legal analysis			
Task 5: Legal feasibility assessment of establishing additional databases/registries	5.2: Necessity	Assessment	Desk research			
Task 5: Legal feasibility assessment of establishing additional databases/registries	5.3: Subsidiarity assessment	Assessment	Desk research			
Task 6: Legal feasibility assessment of interconnecting databases/registries	6.1: Proportionality assessment	Data collection	Interviews with FIUs/LEAs/AML supervisors of financial institutions	*The proportionality of interconnecting databases would need to be assessed against at least the following objectives: faster and more effective access to information, detection of tax evasion and cross-border money laundering, and freezing and confiscation of assets*	4.2-5	4.2 Do you believe these difficulties could be overcome with the setting up of new national registries? 4.3 Do you believe these difficulties could be overcome by setting up interconnection at European level of some national registries, instead of a global centralised EU asset registry? 4.4 Do you believe these difficulties could be overcome by setting up a potential centralised EU asset registry? 4.5 In your view, would a mere exchange of data between Member States, instead of a centralised EU asset registry, be a viable and possibly better alternative?
Task 6: Legal feasibility assessment of interconnecting databases/registries	6.1: Proportionality assessment	Data collection	Interviews with tax authorities	*The proportionality of interconnecting databases would need to be assessed against at least the following objectives: faster and more effective access to information, detection of tax evasion and cross-border money laundering, and freezing and confiscation of assets*	4.2-4, 4.7	4.2 Do you believe these difficulties could be overcome with the setting up of new national registries? 4.3 Do you believe these difficulties could be overcome by setting up interconnection at European level of some national registries, instead of a global centralised EU asset registry? 4.4 Do you believe these difficulties could be overcome by setting up a potential centralised EU asset registry? 4.7 In your view, would a mere exchange of data between Member States, instead of a centralised EU asset registry, be a viable and possibly better alternative?
Task 6: Legal feasibility assessment of	6.1: Proportionality assessment	Data collection	Desk research			

Task	Sub-task	Activity	Methodology	Reference text from ToR	# Questions	Questions
interconnecting databases/registries						
Task 6: Legal feasibility assessment of interconnecting databases/registries	6.1: Proportionality assessment	Assessment	Legal analysis			
Task 6: Legal feasibility assessment of interconnecting databases/registries	6.2: Scenario assessment	Necessity test	Interviews with FIUs/LEAs/AML supervisors of financial institutions	*Under this task, the contractor will perform a legal assessment (i.e. identification of legal obstacles and necessity and subsidiarity assessment)*	4.6-7	4.6 What type of information should be made accessible through the interconnection platform for the centralised EU asset registry? 4.7 Could blockchain technology for the collection, communication, and storage of information help in avoiding the current time gaps that do not allow audit activities and the efficient processing and evaluation of information?
Task 6: Legal feasibility assessment of interconnecting databases/registries	6.2: Scenario assessment	Necessity test	Interviews with tax authorities	*Under this task, the contractor will perform a legal assessment (i.e. identification of legal obstacles and necessity and subsidiarity assessment)*	4.5-6, 4.8-9	4.5 According to your organisation, what would be the main legal obstacles faced by a centralised EU asset registry during its creation and operation? 4.6 What would be the main reasons for establishing a centralised EU asset registry? 4.8 What type of information should be made accessible through the interconnection platform for the centralised EU asset registry? 4.9 Could blockchain technology for the collection, communication, and storage of information help in avoiding the current time gaps that do not allow audit activities and the efficient processing and evaluation of information?
Task 6: Legal feasibility assessment of interconnecting databases/registries	6.2: Scenario assessment	Proportionality test	Legal analysis			
Task 6: Legal feasibility assessment of interconnecting databases/registries	6.2: Scenario assessment	Proportionality test	Desk research			
Task 6: Legal feasibility assessment of interconnecting databases/registries	6.3: Interconnection assessment	Legal obstacles	Legal analysis			
Task 6: Legal feasibility assessment of	6.3: Interconnection assessment	Legal obstacles	Desk research			

Task	Sub-task	Activity	Methodology	Reference text from ToR	# Questions	Questions
interconnecting databases/registries						
Task 7: Technical options for the design and functionalities of an EU asset registry	7.1: identification of the main parameters of the plausible IT infrastructure	Data collection	Results from Task 4			
Task 7: Technical options for the design and functionalities of an EU asset registry	7.1: identification of the main parameters of the plausible IT infrastructure	Data collection	Survey among registry administrators	*For each technical option identified for the design of an EU asset registry, the contractor must precisely identify the main parameters of the IT infrastructure (conditions for access by users, search functionality, governance structure, data controllership)*	3.1-4	3.1 What types of entities supply data for the registry? 3.2 From how many entities do you obtain data? 3.3 What is the legal basis for data collection? 3.4 What are the main obligations imposed to ensure data is processed properly?
Task 7: Technical options for the design and functionalities of an EU asset registry	7.1: identification of the main parameters of the plausible IT infrastructure	Data collection	Interviews with Member States (case studies)			
Task 7: Technical options for the design and functionalities of an EU asset registry	7.2: Description of technical specifications	Data collection	Survey among registry administrators	*Technical specification, including network used and data security, routing component (purely distributed system or distributed system with central routing component), data exchange protocol*	3.37-42	3.37 Do you protect the data entries in the registry by using any encryption of the storage support? 3.38 If you have implemented solutions for data protection at storage, what solutions for the asset data? 3.39 What is the ongoing cost of the implemented solutions for data protection at storage? 3.40 Do you protect the data in the registry by using any encryption for the storage support? 3.41 If you have implemented means for ensuring the traceability of the data usage, what do they cover? 3.42 Is the origin of each entry in the registry documented?
Task 7: Technical options for the design and functionalities of an EU asset registry	7.4: Description of Search possibilities	Data collection	Survey among registry administrators	*Describing the functionalities of a search function that would enable efficient access to information by users for the purpose of the fight against money laundering and tax evasion*		*Derived from search questions*

Task	Sub-task	Activity	Methodology	Reference text from ToR	# Questions	Questions
Task 7: Technical options for the design and functionalities of an EU asset registry	7.5: Calculation of costs of establishment and maintenance	Data collection	Survey among registry administrators	*For each technical option identified for the design of an EU asset registry, the contractor must precisely identify costs of establishment and maintenance*		*Derived from cost questions*
Task 7: Technical options for the design and functionalities of an EU asset registry	7.5: Calculation of costs of establishment and maintenance	Data collection	Results from CBA			
Task 7: Technical options for the design and functionalities of an EU asset registry	7.6: Defining implementation strategy	Data collection	Results from the previous sub-tasks			
Task 7: Technical options for the design and functionalities of an EU asset registry	7.6: Defining implementation strategy	Data collection	Results from Task 5 and 6			

Source: Authors' compilation.

ANNEX B: DEFINITIONS OF IDENTIFIED ASSETS AND SOURCES OF WEALTH

Table 56: Identified assets and sources of wealth

Category	Asset group	Asset	Definition	Examples	Existing definition in EU law	Obliged entities involved with asset under AMLD5	Automatic exchange of information under the DAC for tax purposes
Cash at bank and in hand	Cash in hand	Cash in hand	Bearer-negotiable instruments including monetary instruments such as travellers cheques, negotiable instruments (notably cheques, promissory notes and money orders) that are either in bearer form, endorsed without restriction, made out to a fictitious payee, or otherwise in such form that the title thereto passes upon delivery, and incomplete instruments (including cheques, promissory notes and money orders) signed, but with the payee's name omitted, as defined in Article 2 of EU Regulation 1889/2005	Coins, notes, cheques, promissory notes, money orders	YES	YES	NO
	Cash at bank	Cash at bank	Payment accounts and bank accounts identified by IBAN as defined by EU Regulation 260/2012 held or controlled by natural or legal persons	Payment account, savings account, investment account	YES	YES	YES
Financial assets	Securities	Listed securities	Financial instruments either with a trading obligation or traded on an exchange, for which current ownership and change of ownership is typically recorded by a CSD (central securities deposit) or by a trade repository. Every financial instrument can be intermediated by a CSD, such as transferable securities, sovereign debt, money market instruments or CFDs as laid out in Article 1(2) of EU Regulation 909/2014, which refers to the list of financial instruments in Section C of Annex I in EU Directive 2014/65 (derivatives are excluded from this category). In addition, derivatives subject to trading obligations as defined in Article 34 of EU Regulation 600/2014 are included in this category	Listed shares, sovereign bonds, ETFs, UCITS, emission allowances, CFDs, derivatives subject to trading obligations	YES	YES	YES
		Unlisted securities	Financial instruments without trading obligations and traded OTC off-exchange, not in the scope of EU regulations, which would mandate reporting to CSDs or trade repositories and consequently not to the ESMA	Unlisted shares, unlisted AIFs, unlisted derivatives, etc.	NO	NO	YES*

	Crowdfunding-platform intermediated assets	Crowdfunding-platform intermediated assets	Contracts involving activities destined to raise funds by means of equity, debt or rewards through the use of crowdfunding platforms. Crowdfunding consists of any of the following activities: (i) the facilitation of granting of loans; (ii) the placing without a firm commitment basis of transferable securities and admitted instruments for crowdfunding purposes issued by project owners or a special purpose vehicle, and the reception and transmission of client orders, in relation to those transferable securities and admitted instruments for crowdfunding purposes, as defined in Article 2 of EU Regulation 2020/1503	Equity crowdfunding, debt crowdfunding, rewards crowdfunding	YES	NO	NO
	Life insurance policies/pension products	Life insurance policies	Contract that ensures, that a policy holder's beneficiaries are paid a sum by the insurer upon a life event. Life insurance can have premiums, be paid out on different life events, and have varying cancellation policies, all of which are laid out in Article 2(3) of Directive 2009/138/EC	Insurance on survival to a stipulated age, investment-based insurance products, mutual life insurance	YES	YES	YES
		Pension products	Pension, superannuation or similar scheme that provides retirement benefits to employees, where contributions are made by way of deduction from wages, and the scheme rules do not permit the assignment of a member's interest under the scheme, as defined in Annex II (2c) of Directive 2015/849/EU, excluding insurance policies for pension schemes as well as life insurance policies	Public pension fund, private pension fund, individual pension fund	YES	YES	YES
	Asset insurance policies	Asset insurance policies	Contract that ensures that a policy holder is paid a sum by the insurer upon loss or damage to the insured asset	Fire insurance, theft insurance	NO	NO	NO
	Limited liability companies (LLCs)	LLCs	Companies with share capital and with legal personality possessing separate assets which alone serve to cover its debts. Legal forms of companies considered LLCs in Europe are contained in Annex 2 of the Accounting Directive (Directive 2013/34/EU). Direct, as well as beneficial owners of LLCs, are recorded and kept in a centralised registry via legal obligation (Directive 2015/849/EU)	GmbH, SpA, SA	YES	YES	YES
	Gambling currency	Gambling currency	Physical or digital units provided by gambling providers to gamblers in exchange for currency. These units can be used exclusively to partake in services that involves wagering a stake with monetary value in games of chance at the provider's establishment (physical or at distance), and can also be exchanged into currency by the gambler	Casino chips, online representation of casino chips	NO	YES	NO

	Prepaid cards and vouchers	Prepaid cards and vouchers	Payment methods with a monetary value that can be used to make purchases as an alternative to cash or other payment means, connected to a bank account or wallet. Prepaid cards are defined as non-nominal cards that store or provide access to a monetary value or funds which can be used for payment transactions, for acquiring goods or services or for redemption of currency, as defined in EU Regulation 2018/1672. They are not linked to a bank account. Vouchers are defined as instruments where there is an obligation to accept it as consideration or part consideration for the supply of goods or services, and where the goods or services to be supplied (and/or the identities of their potential suppliers) are either indicated on the instrument itself or in related documentation, including the terms and conditions of the use of such instruments, as defined in Article 1 of EU Council Directive 2016/1065	Pre-paid phone cards, gift cards, payroll cards, travel vouchers, restaurant vouchers	YES	YES	NO
Intangible assets	Crypto assets	Crypto coins	Digital or virtual currencies, which are digital representations of value that are not issued or guaranteed by a central bank or a public authority, as defined in Article 1 of EU Directive 2018/843, and whose transactions are verified and records maintained by a decentralised system using cryptography. Crypto coins are not necessarily attached to a legally established currency, and do not possess the legal status of currency or money. But they are accepted by natural or legal persons as a means of exchange, which can be transferred, sorted and traded electronically	Bitcoin, Ethereum, Litecoin, Cardano, Polkadot, Stellar	YES	YES	NO
		NFTs	Unit of data stored on a digital ledger that keeps records of the purchase and prevents forgery. NFTs are unique digital assets that can be sold and bought like tangible property, and can be seen as virtual proof of ownership	Tokenised art, tokenised music, tokenised domain name, tokenised event tickets	NO	NO	NO
	In-game assets	In-game assets	Digital items or credits that represent a monetary value in video games, which can be traded for cash between participants within the video game or through third-party platforms. This category excludes crypto assets	Customisation items for characters, items that modify the in-game statistics of a player, loot boxes	NO	NO	NO

	Intellectual property	Patents	Legal title that can be granted for any inventions which are susceptible of industrial application, which are new, and which involve an inventive step, as defined in Article 52 of European Patent Convention (1973). An invention shall be considered to involve an inventive step if, having regard to the state of the art, it is not obvious to a person skilled in the art. The legal title gives holders the right, for a limited period, to prevent others from making, using or selling their invention without their permission in the countries for which the patent has been granted. This covers the patents granted by the competent national intellectual property authorities in EU countries or by European patents granted centrally by the European Patent Office. Patents can be licensed or assigned only by the owner of the patent, either in an exclusive or non-exclusive manner. In some countries, exclusive patent licences must be registered in a national registry. This category includes industrial designs, as defined in Article 1 of Directive 98/71/EC, which means the appearance of the whole or a part of a product resulting from the features of, in particular, the lines, contours, colours, shape, texture and/or materials of the product itself and/or its ornamentation. Only the exterior design of a product, visible for the consumer, can be protected	Patent for an apparatus for transmitting vocal or other sounds telegraphically (telephone), electromagnetic inductive suspension and stabilisation system for a ground vehicle (Maglev train)	YES	NO	NO
		Trademarks	Signs, in particular words, including personal names, or designs, letters, numerals, colours, the shape of goods or of the packaging of goods, or sounds, that are capable of: (a) distinguishing the goods or services of one undertaking from those of other undertakings; and (b) being represented in the Register of European Union trademarks, in a manner which enables the competent authorities and the public to determine the clear and precise subject matter of the protection afforded to its proprietor. The definition of trademarks is defined in Article 4 of Regulation 2017/1001	Coca cola brand logo, Twitter logo, Apple logo, Nike logo	YES	NO	NO
		Copyrighted music	Economic exploitation rights of music copyrights	Authors of musical works, music composers, music arrangers, music publishers and authors of written parts of the work, i.e. lyrics; music is defined as an arrangement of sounds having melody, rhythm, and usually harmony	YES	NO	NO

		Copyrighted literary works	Economic exploitation rights of literary copyrights	Literary works include books, brochures, speeches, computer programmes, lectures, addresses, reports, sermons	NO	NO	NO
		Copyrighted audio-visual arts	Economic exploitation rights of audio-visual arts	Audiovisual sketches, documentaries, advertisements, photographs	NO	NO	NO
		Copyrighted graphic arts	Economic exploitation rights of graphic arts (excluding trademarks)	Works of drawing, painting, architecture, engraving, lithography	NO	NO	NO
	Concessions	Concessions	A contract for pecuniary interest concluded in writing by means of which one or more contracting authorities or contracting entities entrust the execution of works and other services, as defined in Article 5(1) of EU Directive 2014/23	Franchise, mining rights	YES	NO	NO
Tangible assets	Real estate	Properties	Buildings that are constructed on a piece of land which are categorised, constructed, recorded and built, as defined in national legislation	Apartment, house, garage	NO	YES	YES
		Land	Delimited, measured and owned part of the earth's surface, recorded and delimited by national legislation	Agricultural land, industrial land	NO	YES	YES
	Vehicles	Motor vehicles	Motor vehicles with a maximum authorised mass not exceeding 3 500 kg and designed and constructed for the carriage of no more than eight passengers in addition to the driver, as defined in Article 4(4) of Directive 2006/126/EC describing the driving licence category B. Trailers, buses, heavy cargo vehicles as defined in Article 3(17) of Directive 2018/858 EU, agricultural machinery as defined in Article 4 of Directive 2006/126/EC. Motorcycles as defined in EU Regulation 168/2013	Cars, agricultural machinery, motorcycles	YES	NO	NO
		Ships	Vessel of any type whatsoever operating or having operated in the marine environment, which includes submersibles, floating craft, floating platforms, self-elevating platforms, floating storage units and floating production storage and offloading units, as well as a vessel stripped of equipment or being towed as defined in Article 3(1) of Directive 1257/2013/EU	Rowboats, steamships, cruise ships	YES	NO	NO
		Aircraft	Machine that can derive support in the atmosphere from the reactions of the air other than the reactions of the air against the earth’s surface, as defined in Annex I, subpart A to Regulation 1178/2011/EU	Jets, aeroplanes, propeller machines	YES	NO	NO

	Other high-value goods	Firearms	Portable barrelled weapon that is designed to expel or may be converted to expel a shot, bullet or projectile by the action of a combustible propellant, unless it is excluded from this definition for one of the reasons listed in Part III of Annex I to Directive 2021/555/EU. Alternatively, an object shall be considered to be capable of being converted to expel a shot, bullet or projectile by the action of a combustible propellant if it has the appearance of a firearm; and as a result of its construction or the material from which it is made, it can be so converted. Both cases are laid out in Article 1(1) of EU Directive 2021/555	Pistols, rifles	YES	NO	NO
		Works of art	Objects with high monetary and aesthetic values created in low numbers directly by artists as a means of expression. In the broader sense this may include fine art, such as paintings or sculptures, or entire installations. More precisely, works of art are considered to be objects belonging to the following TARIC codes: 9701 (pictures, paintings, collages), 9702 (engravings, prints and lithographs), 9703 (original sculptures and statuary), 5805/6304 (tapestries and wall textiles), as well as individual pieces of ceramics executed entirely and signed by the artist, enamels on copper and photographs taken by the artist, printed by or under the supervision of the artist, signed and numbered and limited to 30 copies, all sizes and mounts included, as laid out in Annex IX, Part A to Council Directive 2006/112	Paintings, sculptures	YES	YES	NO
		Collectors' items	Objects that are of value to collectors, either due to their rarity or due to other desirable qualities. TARIC code 9704 specifies that postage stamps shall be considered collectible. TARIC code 9705 also denotes any items that can be of various interest to collectors (zoological, botanical, mineralogical, anatomical, historical, archaeological, palaeontological, ethnographic or numismatic) as collectors' items. All items summarised under these two codes are considered collectors' items, as laid out in Annex IX, Part B to Council Directive 2006/112	Trading cards, stamps, coins	YES	NO	NO
		Antiques	Goods, other than works of art or collectors' items, that are more than 100 years old and fall under TARIC code 9706, as laid out in Annex IX, part C to Directive 2006/112/EC	Furniture, ornaments, vases (all more than 100 years old)	YES	NO	NO
		Jewellery	In the notes laying out objects belonging to TARIC code 7113 covering jewellery, which is defined as any small object of personal adornment (gem-set or not) and articles of personal use of a kind normally carried in the pocket, in the handbag or on the person. These articles may be combined or set, for example, with natural or cultured pearls, precious or semiprecious stones, synthetic or reconstructed precious or semiprecious stones, tortoise shell, mother-of-pearl, ivory, natural or reconstituted amber, jet or coral	Rings, bracelets, necklaces, brooches, earrings, watch chains, fobs, pendants, tie pins, cufflinks, dress studs, religious or other medals, insignia	YES	NO	NO

		Worked specimens	Specimens (meaning animals or plants, whether dead to alive belonging to any species listed in Annexes A-D of Council Regulation 338/97) that were significantly altered from their natural raw state for jewellery, adornment, art, utility, or musical instruments, to the satisfaction of the management authority of the Member State concerned, acquired in such conditions, as laid out in Article 2(w) of Council Regulation 338/97	Ivory decorations	YES	NO	NO
	Animals	Livestock	Livestock including farm animals raised in an agricultural environment, kept or traded as a source of income	Cattle, pigs, sheep, goats, rabbits, poultry and hens, bees from an apiary	NO	NO	NO
		Pigeons (racing)	Pigeons that are kept or traded for racing, and which carry life rings that identify the specific pigeon	Racing pigeons	NO	NO	NO
		Equine animals	Purebred breeding animal of the species Equus caballus and Equus asinus, entered or eligible for entry in the main section of a breeding book established by a breed society or breeding body recognised in accordance with Articles 4 or 34 of Regulation (EU) 2016/1012. Alternatively, a kept animal of the species Equus caballus registered with an international association or organisation, either directly or through its national federation or branches, which manages horses for competition or racing (registered horse). Both options are laid out in in Article 3(25) of the Commission Delegated Regulation 2020/688	Horses, sport horses	YES	NO	NO
		Exotic animals (exotic pets)	Species that are non-native to and not normally domesticated in the EU	Invertebrates, fish, amphibians, reptiles, birds, mammals (including non-human primates)	NO	NO	NO
Commodities	Commodities (precious raw materials)	Commodities (precious raw materials)	Any goods of a fungible nature that are capable of being delivered, including metals and their ores and alloys, agricultural products, and energy such as electricity, as defined in Article 2(6) of Regulation 2017/565/EU	Gold, silver, platinum, palladium, copper, crops, electricity	YES	NO	NO
Other assets	Safe deposit boxes	Safe deposit boxes at banks	Individually secured containers used to contain valuable items, held at credit institutions, as defined in Article 32a(1) of EU Directive 2015/849	Safe, vault (owned by a bank	YES	YES	NO
		Safe deposit boxes at non-banks	Individually secured containers used to contain valuable items, not held at credit institutions	Safe, vault (not owned by a bank)	NO	NO	NO

*Only applicable to information on income on custodial securities.

Source: Authors' compilation.

ANNEX C: ASSETS TO BE REMOVED FROM THE LIST BASED ON SURVEY RESPONSES (SHARE OF COMPETENT AUTHORITIES by AUTHORITY CATEGORY)

Table 57: Assets to be removed based on survey responses

Assets	FIUs	Tax authorities	LEAs	AML supervisors of financial institutions	Total*
Cash at bank	0 %	0 %	0 %	0 %	0 %
Cash in hand	6 %	24 %	0 %	12 %	12 %
Unlisted securities	6 %	0 %	0 %	6 %	3 %
Crowdfunding-platform intermediated assets	6 %	6 %	0 %	6 %	5 %
Life insurance policies	6 %	12 %	0 %	18 %	10 %
Pension products	11 %	6 %	0 %	24 %	12 %
Limited liability companies	6 %	0 %	0 %	0 %	2 %
Crypto coins	6 %	0 %	0 %	0 %	2 %
NFTs	6 %	0 %	0 %	0 %	2 %
In-game assets	6 %	6 %	33 %	6 %	9 %
Land	0 %	0 %	0 %	0 %	0 %
Properties	0 %	0 %	0 %	0 %	0 %
Aircraft	0 %	0 %	0 %	0 %	0 %
Motor vehicles	0 %	0 %	0 %	0 %	0 %
Ships	0 %	0 %	0 %	0 %	0 %
Antiques	6 %	0 %	0 %	6 %	3 %
Collectors' items	11 %	0 %	0 %	6 %	5 %
Firearms	6 %	0 %	17 %	6 %	5 %
Jewellery	6 %	6 %	0 %	0 %	3 %
Worked specimens	17 %	6 %	0 %	12 %	10 %
Works of art	6 %	0 %	0 %	0 %	2 %
Commodities (precious raw materials)	17 %	6 %	0 %	6 %	9 %
Safe deposit boxes at banks	0 %	0 %	0 %	6 %	2 %
Safe deposit boxes at non-banks	11 %	6 %	0 %	12 %	9 %

Notes: Total number of responses per type of competent authority: FIUs (18), tax authority (17), LEAs (6) and AML supervisors of financial institutions (17).

* Total share of responses over total number of competent authorities.

Source: Authors' compilation.

ANNEX D: LIST OF ASSOCIATIONS INTERVIEWED

Name of association/authority	Asset groups covered	Date of interview
ESMA	Listed securities	20.02.2023
EFAMA	Unlisted securities	23.02.2023
CINOA	High-value goods, commodities	23.01.2023
IADAA	High-value goods	11.01.2023

ANNEX E: LIST OF AUTHORITIES INTERVIEWED

Name of authority	Asset groups covered	Date of interview
Romanian naval authority	Ships	15.02.2023
Latvian road traffic safety authority	Cars, trailers, heavy cargo motor vehicles, agricultural machinery, motorcycles, registered vehicles of historical interest	21.02.2023
German federal ministry for digital and transport	Cars, trailers, heavy cargo motor vehicles, agricultural machinery, motorcycles, registered vehicles of historical interest	20.02.2023
Czechia ministry of transport, car registry	Cars, trailers, heavy cargo motor vehicles, agricultural machinery, motorcycles, registered vehicles of historical interest, ships	3.03.2023
French general directorate for civil aviation	Aircraft	02.02.2023

ANNEX F: CASE STUDIES FOR IT FEASIBILITY

Table 58: Aircraft, vehicles (Malta)

Tags	Malta, aircraft, vehicles, tangible assets, national authority
Date	29 November 2022
Member State represented	Malta (MT)
Notes on the case study	The case study questionnaire was partially filled in during a videoed, structured interview with the director of the air vehicle registry. The full questionnaire – Task 2 and the IT questionnaire were then sent to the responder (the interviewee) for additional details, which could not be covered during the interview.
What authority do you represent?	Transport Malta is the competent registry for land, sea and air vehicles.
What data assets are available at the authority you represent?	Air vehicle registry (aircraft type, maker, technical parameters, registration number, owner and operator identity (name, address))
For each data asset, can you mention the current use of the data asset in anti-money laundering (AML)?	Presently, the data asset is not used for AML purposes. The aircraft registration requires proof of source of funds (for the owner) issued by a competent authority, such as a bank.
For each data asset, can you mention other potential uses of the data asset to fight money laundering?	Although the registry contains both the owner's and the operator's identity, the authority cannot identify the final beneficiary owner if an operator or a leasing company owns the aircraft. Typically, airline operators like Ryanair and WizzAir lease aircraft from leasing companies. Thus, the beneficial owner remains hidden from the authority. In the responder's opinion, this type of asset (aircraft) might not be suitable for money laundering. The high maintenance costs of aircraft, the registration, insurance, labour and fuel expenses surpass the initial cost of investment even in the short run.
What is the role of your authority in anti-money laundering (AML)?	Transport Malta has no specific role in AML besides cooperation with FIUs, LEAs and tax authorities.
What is your specific role in fighting money laundering?	See the previous answer. Transport Malta does not have a specific role in AML.
What is the role of the authority regarding the data assets?	Transport Malta is the competent national authority for land, sea and aircraft registration. It collects data concerning vehicles in line with its role as the national registry and acts as the data controller.
Data available in a digital format	Yes
Data maintained in a structured form	Yes
Register accessible to third parties	Yes
Register accessible for machine-to-machine interaction	No
Open/extensible data model	No (the public presentation is a PDF file containing the records; it can be assumed that any existing system is very limited in capability).
Open/extensible architecture	N/A
Challenge	No dedicated system is in place.
Relevance of the case	The case study exemplifies the situation of an authority and domain with a small number of records.

Comments: The interviewee agreed to fill in the questionnaire used in Task 2 and decided to delegate to the person responsible for IT the task of answering the detailed IT questionnaire in the form of a structured interview (the approved IT questions were sent in advance via email).

Throughout the interview, it was apparent that Transport Malta (TM) has all the data/information needed to fulfil its mission. TM is not connected to other national, supranational or international registries directly. TM responds to infrequent inquiries from LEAs and other authorities and extracts information from its registry on a case-by-case basis using a manual search of its information system hosted in a central location. Captain Pace mentioned that it would help TM if they could query the registry of their peer-competent national authorities in other Member States.

Researcher's note: According to the Maltese Air Regulations Policy, section 1.1, '[t]he owner is responsible for the continuous airworthiness of an aircraft. When the aircraft is leased, the responsibilities of the owner are transferred to the lessee if the lessee is stipulated on the Certificate of Registration of the aircraft.'

Table 59: Crypto coins (Cyprus)

Tags	Cyprus, cryptocurrencies, national authorities
Date	20 December 2022
Member State represented	Cyprus (CY)
What authority is the subject of the study?	Cyprus Securities and Exchange Commission (CySEC)
Notes on the case study	The case study questionnaire was filled in following desk research activities. *Sources:* CySEC, https://www.cysec.gov.cy Chambers and Partners, https://chambers.com ESMA, https://www.esma.europa.eu/press-news/esma-news/esma-finds-shortcomings-in-supervision-cross-border-investment-activities-and
What data assets are available at the authority?	• CASP (Crypto Assets Services Providers) Register • EEA CASP Register
What is the role of the authority regarding the data assets?	Following the registration of the CASPs, CySEC: • maintains a register of locally registered CASPs and EEA CASPs that have notified the institution of their operations in Cyprus; • maintains a register of CASP beneficiaries and for persons holding a management position. CySEC publishes the CASP Register on its website, which is accessible to the public, providing the following information: • Name, trade name, legal form and legal entity identifier of the CASP • Physical address of the CASP • Services and activities exercised by the CASP • Domain/website of the CASP.
What is the role of the authority in fighting money laundering?	CySEC supervises Cypriot cryptocurrency businesses, under the Prevention and Suppression of Money Laundering and Terrorist Financing Law of 2007 (the AML/CFT law) which also determines activities of CASPs. The CySEC Directive (on the Register of Providers of Services Regarding Crypto Assets) defines the rules that regulate the creation, maintenance, operations and changes of the CASP Register. The CASP Register contains information about all crypto asset companies that provide services in or from Cyprus. New crypto asset businesses must register with CySEC before commencing their operations in or from Cyprus.

	CySEC imposes a regulatory system based on a number of specific licences that define the types of operations a registered provider is allowed to perform. It also stipulates that in order to apply for a crypto licence, an organisation needs to have an entity incorporated in Cyprus and at least one fit and proper shareholder. A crypto provider is required to have a fully operational office located in Cyprus where resident staff members need to be hired on a full-time basis. A minimum number of personnel must be hired, based on the type of licence the crypto provider is applying for. A CASP is required to: • design internal policies for customer identification and for the tracking of sources of funds; • create workflows and systems for secure and effective management of data; • monitor clients' crypto asset transactions and wallet addresses and report suspicious activities; • perform know your client and other client due diligence checks; • identify the source of client funds; • identify and report suspicious transactions; • undertake a comprehensive risk assessment in relation to clients' activities and take proportionate measures per client, activity and crypto asset in question; • comply with the AML/CFT law and specific rules and regulations.
What is the specific role of the authority in fighting money laundering?	CySEC enforces the application of the AML/CFT law in accordance with its defined mandate. During registration, CySEC requires the CASPs to: • provide a formal and adequate AML manual; • appoint an in-house AML compliance officer; • submit an internal operation manual, describing internal control mechanisms, effective risk-assessment procedures and safeguard arrangements for data storage and information processing systems.
Data available in a digital format	No
Data maintained in a structured form	No
Register accessible to third parties	No
Register accessible for machine-to-machine interaction	No
Open/extensible data model	No
Open/extensible architecture	No
Challenges	• No dedicated system is in place. • The registrar does not collect information about the actual transactions carried out by the CASPs. • There is no framework to incentivise and support the CASPs to report transactions operated.
Relevance of the case	The case study demonstrates the limits of a theoretically strong legal framework, in a highly dynamic technological environment. While a register is maintained (the CASP Register), it does not meet the needs of a proper monitoring activity as it does not focus on the relevant type of information (the transactions that should be reported by the CASPs).

Comments: The peer-reviewed report and the resulting recommendations made by ESMA to the Cyprus Securities and Exchange Commission show that contrary to the impression given by the formal approach taken by CySEC, the actual activities of the financial services providers registered in Cyprus, and specifically the cross-border activities, are not satisfactory. The fact that ESMA requested an increase of resources dedicated to supervision (authorisation, ongoing supervision and enforcement) supports the deduction that the application of AML regulations is similarly affected by a lack of resources.

Researcher's note: The type of CASP register published by CySEC (a list on a static webpage) provides an example of the variety of technical implementation issues to resolve. CySEC's low-tech example demonstrates that a high-level solution, using an API-based harvesting mechanism, is not aligned with the realities and the technical level of the registers found at the Member State level.

Table 60: Land, real estate (Netherlands)

Tags	Netherlands, land register, real estate, ships, vehicles, tangible assets, national authority
Date	28 December 2022
Member State represented	Netherlands (NL)
What authority is the subject of the study?	Netherlands' Cadastre, Land Registry and Mapping Agency – Kadaster
Notes on the case study	The case study questionnaire was filled in following desk research activities. *Sources*: https://www.kadaster.nl https://www.elra.eu
What data assets are available at the authority?	Kadaster's register includes: • spatial data on property • the rights involved • information on energy labels of houses • official real estate estimations, for taxation purposes, which are updated regularly (once a year; in a few cases, twice a year; fewer not so regularly) • underground cables and pipelines • ships • aircraft • telecom networks. Kadaster maintains a great number of key registers of the Dutch government (Rowland, 2022), including: • the Base Register for Addresses and Buildings • the Base Register for Topography • the Base Register for Large-scale Topography. All of these datasets are available as open data. In addition, Kadaster actively publishes and maintains these and other geospatial assets as linked (open) data. The information managed by Kadaster is available predominantly through online web services (e.g. https://www.kadasterdata.nl).
What is the role of the authority regarding the data assets?	• Kadaster collects and registers administrative and spatial data on property and the rights involved, including information on energy labels of houses and underground cables and pipelines. The register also covers ships, aircraft and telecom networks. • Kadaster maintains the Key Registers Cadastre and Topography. • Kadaster is responsible for national mapping and maintenance of the national reference coordinate system.

For each data asset, can you mention other potential uses of the data asset to fight money laundering?	Kadaster offers the possibility to calculate/estimate a (possible) value of a property by using a price index method, based on the registered purchase prices throughout the years, dividing properties by housing type, period and region.
What is the role of the authority in anti-money laundering (AML)?	Kadaster has no legal obligations on reporting possible money laundering-affected transactions.
What is the specific role in fighting money laundering?	• Kadaster has appointed a counsellor for AML. • Kadaster is in the process of researching the possibility of reporting money laundering-affected transactions.
Data available in a digital format	Yes
Data maintained in a structured form	Yes
Register accessible to third parties	Yes
Register accessible for machine-to-machine interaction	Yes
Open/extensible data model	Yes
Open/extensible architecture	Yes
Challenges	• Reaching agreement with similar registers in terms of the data model • Adapting the system to the communication protocol required
Relevance of the case	Kadaster demonstrates the advantages of employing new technologies that are apt for application to the specificity of a sector.

Comments: Kadaster is a good example of an authority that uses new technologies (e.g. street view mapping and semantic technologies) in a very practical and efficient manner. Kadaster is an active player in the linked open data environment, having already released several datasets in the public domain.

Table 61: Movable assets (Romania)

Tags	Romania, deeds, mortgages, financial assets, securities, national authority
Date	11 January 2023
Member State represented	Romania (RO)
What authority is the subject of the study?	Ministry of Justice – Supervisory Authority of the National Registry of Movable Property
Notes on the case study	The case study questionnaire was filled in following desk research activities *Sources*: https://ccir.ro https://mj.rnpm.ro
What data assets are available at the authority?	The National Registry of Movable Property (RNPM) is a database at the national level, the sole recording system for movable property, prioritising mortgage securities, and is classified according to persons and property. The RNPM records operations regarding mortgages and securities transactions associated with mortgages, other rights provided by law, legal and publicity operations required by law, for legal acts such as: • guarantee contracts, claims, rental contracts, consignment, buy and sale, agricultural duties • seizure insurance

	• records in connection with public finance operations • records in connection with various goods – vehicles – agriculture, other than motor vehicles – receivables – bank accounts – shares, securities and other financial instruments – insurance policies – universality (stocks, goodwill or other similar universality) – equipment, facilities and other assets designed to serve the operation of an undertaking – forests to be cut – minerals to be extracted – general crops and crops to be harvested – livestock, etc. By being registered in the electronic archive, guarantees and other deeds on goods fulfil advertising conditions and enjoy its effects. Records can be searched using the following type of information: • identification number • creditor's identification data • debtor's identification data • identification elements of the movable asset for each type of asset (motor vehicle, movable asset attached to an immovable asset, assets of other types) • identification data of the owner of immovable property, to which the movable property is attached • identification data of a given third party. For exact retrieval of information, the best fields to use are: • fiscal code – for the legal entity • personal identification code – for the natural person • chassis series – for motor vehicles • contract number, insurance policy number, manufacturing series of an equipment, etc. – for other goods.
What is the role of the authority regarding the data assets?	• Authorises operators who conduct operations in the registry • Supervises operational activity in the registry carried out by the operators and their agents • Administers and maintains the registry's computer security system • Controls the activity of the operators and their agents, as well as the operation of the registry, in order to ensure compliance with the legal provisions in force • Examines any complaint received regarding non-compliance and orders the measures appropriate for remedying the situation, according to the provisions of the governing law and of the regulation • Supervises the activity of the operators and their agents in compliance with the competition rules • Suspends or revokes the right of an operator or an agent to register in the registry, under the terms of the law • Cooperates with the body of operators, as well as, directly, with operators and agents, for the purpose of improving the technical qualities and performance • Ensures access to reading information recorded in the registry, only through the registry software application on the official websites • Takes necessary measures to restore the functionality of the registry, in the face of factors that could disrupt the registry's operation and to substantial difficulties in fulfilling the purpose for which it was created.

What is the role of the authority in anti-money laundering (AML)?	The authority has no direct roles in relation to AML.
What is the specific role of the authority in fighting money laundering?	It has no specific role in fighting money laundering.
For each data asset, are there other potential uses of the data asset to fight money laundering?	Access to the registry might provide information on particular assets – information that could be used in activities related with AML.
Data available in a digital format	Yes
Data maintained in a structured form	Yes
Register accessible to third parties	Yes
Register accessible for machine-to-machine interaction	Yes
Open/extensible data model	No
Open/extensible architecture	Yes
Challenges	• Reaching agreement with similar registries in terms of the data model • Extending the data model • Adapting the system to the communication protocol required
Relevance of the case	The Romanian case demonstrates the situation of a domain with a legal framework that incentivises actors (in this case 'operators') to directly engage with the register.

Table 62: Commodities (Italy)

Tags	Italy, precious metals, gold, tangible assets, private register
Date	5 January 2023
Member State represented	Italy (IT)
Notes on the case study	The case study questionnaire was filled in following desk research activities *Sources:* https://www.italpreziosi.it https://www.bancaditalia.it
What authority do you represent?	Italpreziosi S.p.A. Italpreziosi it is not an authority but a refinery operating in the production and trading of precious metals, ingots and investment coins, based in Arezzo.
What data assets are available at the authority you represent?	All transactions are recorded in a company register. The data required for registration are: • identity document (identity card, passport) • tax code/VAT number • telephone number • email address. Attached to the transaction are the following documents: • CGV(A) – general conditions of sale (purchase), the contract that binds the parties;

	• order confirmation – a document that summarises products, amounts, addresses and IBAN; • privacy policy – information pursuant to Article 13 of Regulation (EU) No 679/2016 (GDPR); • transport document – a document justifying the transfer of goods between two subjects.
For each data asset, can you mention the current use of the data asset to fight money laundering?	The registration of each transaction follows legal requirements, which state that any sale pursuant to Article 1(2) of Law 17.01.2000, No 7 must be declared to the financial information unit of the Bank of Italy. This legislation is specifically intended to fight money laundering.
What is the role of your authority in anti-money laundering (AML)?	As a private enterprise acting in the business of transactions and storage of gold for the purpose of investment, Italpreziosi S.p.A. is required to record and report all transactions in accordance with Italian legislation.
What is your specific role in fighting money laundering?	To record and report transactions of precious metals to the Bank of Italy
What is the role of your authority regarding the data assets?	To record and maintain a database of transactions
Data available in a digital format	Yes
Data maintained in a structured form	Yes
Register accessible to third parties	No
Register accessible for machine-to-machine interaction	N/A
Open/extensible data model	N/A
Open/extensible architecture	N/A
Challenge	• Clarifying if the transactions also have to be reported to European AML authorities or solely the Bank of Italy
Relevance of the case	The case shows the impact of strong enforcement (by law) on the activity performed by actors in a particular domain (formal obligation to register the transactions).

Comments: Italian legislation also has specific requirements for cross-border transactions: 'Anyone who arranges or carries out the transfer of gold to or from abroad, or the trade of gold in the national territory or other gold transactions, even free of charge, has the obligation to declare the transaction to the Italian exchange office if its value is equal to or greater than 20 million lire. The professional operators referred to in paragraph 3 are also required to declare, whether they operate on their own account, or on behalf of third parties.'

Table 63: Ships (Ireland)

Tags	Ireland, ships, vehicles, tangible assets, national authority
Date	10 January 2023
Member State represented	Ireland (IE)
What authority is the subject of the study?	Mercantile Marine Office (MMO)
Notes on the case study	The case study questionnaire was filled in following desk research activities *Sources*:

	https://www.gov.ie https://www.imdo.ie https://www.seafarers.ie https://www.sailing.ie
What data assets are available at the authority?	No data are actually stored by the Mercantile Maritime Office. The General Register of Shipping is a register maintained by the Revenue Commissioners for the purpose of recording details of Irish ships, shares therein and the ownership thereof.
What is the role of the authority regarding the data assets?	• The Department of Transport is the national authority with responsibility for the registration of Irish ships. The function is defined by the Mercantile Marine Act of 1955, as amended by the Merchant Shipping (Miscellaneous Provisions) Act of 1998, and Sea-Fisheries and Maritime Jurisdiction Act of 2006. The legislation lays down the legal provisions for ship registration, ownership and registry of Irish ships, for the mortgage, sale, transfer and measurement of tonnage of such ships and other connected matters. • The actual registration of ships takes place through one of the approved registrars of ships (also Port of Registry). A Port of Registry is a port designated by the Mercantile Marine Act, at which a ship is registered. There are currently 13 Ports of Registry in Ireland, namely: Arklow, Cork, Drogheda, Dublin, Dundalk, Galway, Limerick, Skibbereen, Sligo, Tralee, Waterford, Westport and Wexford. There is a Registrar of Ships appointed for each Port of Registry. • The registration of the vessel is subject to meeting a number of prescribed standards and conditions, specifically ruled by the Mercantile Marine Act of 1955, as amended. • Exceptions may occur, however. As a general guideline, in order to register a commercial ship in Ireland the owner or majority owner must be a national of an EU Member State or a corporate body established under and subject to the law of an EU Member State and having its principal place of business in an EU Member State.
For each data asset, are there other potential uses of the data asset to fight money laundering?	No specific regulations have been found in relation to the use of the registry information for anti-money laundering.
What is the role of the authority in anti-money laundering (AML)?	N/A
What is the specific role in fighting money laundering?	N/A
Data available in a digital format	No information available
Data maintained in a structured form	No information available
Register accessible to third parties	No
Register accessible for machine-to-machine interaction	N/A
Open/extensible data model	N/A
Open/extensible architecture	N/A
Challenges	• No dedicated system is in place. • The legal framework is linked to local traditions
Relevance of the case	The case study exemplifies the situation of a domain having strong local traditions. While the registration process is carried out, the exemptions in the legal framework do not justify a more dynamic and technologically apt approach from the AML perspective.

Comments: The following categories of ships are exempt from the obligation to register, but their owners may opt to do so:

- ships under 15 net register tons burden (15GT or 15m LOA if a fishing vessel), provided they are used only on the rivers, canals, lakes or coasts of Ireland, Great Britain, the Channel Islands or the Isle of Man, or within the territorial waters off their coasts
- ships acquired before the passing of the Mercantile Marine Act in 1955
- ships where the minister has consented to their registration under the laws of another country
- ships owned by Irish citizens not ordinarily resident within the state
- sea-fishing boats of less than 15 metres in length overall and required to be registered in the Register of Fishing Boats.

Queries relating to registration information and other forms and documents issued by the Revenue Commissioners should be directed to a chosen registrar of ships.

The information in the study refers mainly to commercial ships given that, in the case of recreational crafts, no strict regulation has been put in place. Recreational ship owners might decide to register following the same procedures as for a commercial ship but they are not required to do so.

For a number of years, a sailing association provided an Irish Sailing Small Craft Registry Certificate to boat owners who needed documentation when visiting foreign ports. This was not a government-endorsed document, and the Ireland Department of Transport requested the association to cease operating their registry.

Table 64: Works of art, antiques (Malta)

Tags	Malta, works of art, antiques, tangible assets, private register
Date	2 December 2022
Member State represented	Malta (MT)
Notes on the case study	The case study questionnaire was partially filled in during a structured video interview with the owner of Belgravia Auction Gallery – Malta. The interview is a follow-up to the questionnaire, as the responder Mr Joseph Sammut provided his consent in his role as the auction gallery owner. The complete questionnaire (Task 2), was filled in as part of the corresponding task.
What authority do you represent?	Belgravia Auction is not an authority but one of Malta's oldest and most renowned auction houses.
What data assets are available at the authority you represent?	The data we store serves the interest of our business. Since 2006, we have yet to have reporting responsibility towards the Revenue Office.
For each data asset, can you mention the current use of the data asset to fight money laundering?	Most of the sales of our auctions came out of one's heritage. We specialise in silver items, more precisely, Maltese silver. These items are passed along family lines and eventually might end up in auctions. As mentioned, we have a private database that records all purchases and sales of our auction house. We keep a record of the items, the purchase and sale value, and the name and address of the seller and buyer. We do not record, however, tax identification numbers as we are no longer required to do so.
For each data asset, can you mention other potential uses of the data asset to fight money laundering?	The nature of our business is trust. Our customers (sellers) trust us, and we have known our customers for generations. These silver items represent little value. Typically, there is a few hundred or a few thousand euro-worth of silver cutlery, saucers, etc., that the family has passed down through generations. None of these

	assets represent accumulated wealth. Secondly, the price of silver is relatively low. One would require a truckload of silver to accumulate substantial wealth.
What is the role of your authority in anti-money laundering (AML)?	We don't have such a role.
What is your specific role in fighting money laundering?	There are certain perks related to living on a tiny island. We conduct KYC for every item; it is easy to do as we know the person selling or buying the item. In this business, we know each other.
What is the role of your authority regarding the data assets?	Decades ago, I paid a software developer to create our system. I have used it [the system] ever since.
Data available in a digital format	Yes
Data maintained in a structured form	Yes
Register accessible to third parties	No
Register accessible for machine-to-machine interaction	No
Open/extensible data model	No
Open/extensible architecture	No
Challenges	• Outdated system • No legal framework • No incentive to participate • Financial limitations.
Relevance of the case	The case study exemplifies the situation of a domain traditionally not regulated. It also demonstrates the case of a domain where the need for a registry is well understood by the actors

Comments: At this point of the interview, it was clear that a small auction house like Belgravia, although one of the most prominent players in the country, is not equipped to share data they collected solely to support their business needs. Therefore, Mr Sammut was asked for his opinion on how much it would take to expose their data in a way that might serve the needs of FUIs and LEAs. His response was that Belgravia would need to rewrite the entire software to make it suitable for such purposes. A decade ago, it cost a few thousand euro, so in today's money this would be between EUR 20 000 to EUR 100 000. This is obviously not an amount he would be likely to invest on his own. For the follow-up question, what fee would he consider a reasonable charge for accessing his records? He estimated EUR 100 for each record.

ANNEX G: CASE STUDIES ON ASSET CATEGORY NETWORKS

Table 65: Camden Asset Recovery Inter-Agency Network

Tags	AML networks, asset recovery networks, Europol, CARIN
Date	12 December 2022
Member State represented	N/A
Notes on the case study	The case study questionnaire was filled in following an electronic exchange with the contact, then desk research activities to complement necessary details.
What authority do you represent?	CARIN is an informal network of contacts and a cooperative group concerned with all aspects of recovering the proceeds of crime. **Network structure and functioning** The permanent CARIN Secretariat is hosted by Europol and based at Europol headquarters in The Hague. CARIN members meet regularly at an annual general meeting. Access to the CARIN network and its website is restricted only to contacts (members, observers and associates) of the network. The organisation is governed by a steering group of nine members and has a rotating presidency. **Steering group** CARIN has a steering group consisting of up to nine members, from which one member is elected president. The steering group period runs from 1 June each year and concludes on 31 May. Membership of the steering group rotates periodically with up to three members offering to stand down each year to enable other members to join the steering group. If the number of candidates exceeds the number of vacancies a vote of all members is taken. The steering group oversees the administration of the network, receives applications for membership and observer status and decides if a state, jurisdiction, or third party meets the criteria for membership. Europol and Eurojust have permanent observer status in the steering group. The steering group may establish working groups to examine and report on legal and practical issues. The steering group also assists in the preparation of the annual conference agenda and identifies areas for consideration at plenary meetings.
What is the role of the authority in anti-money laundering (AML)?	CARIN is a network of practitioners from 62 jurisdictions and 13 international organisations. Funded as an outcome of an asset recovery seminar held in Dublin in 2004, CARIN encompasses asset recovery practitioners – both police and judicial experts – within and outside the EU.
What is the authority's specific role in fighting money laundering?	First, it is important to emphasise the informal nature of CARIN and the distinction between formal requests for mutual legal assistance and informal assistance. Second, the asset recovery inter-agency networks (ARINs) help asset recovery specialists to access the appropriate networks, and to cooperate on crucial efforts. Although ARINs comprise practitioners nominated by their corresponding member jurisdictions, they function through the voluntary actions of professionals and their best efforts alone. A typical response time for a request can be anything between a few hours (1 hour) to a few months (3 months) depending on the availability and the circumstances of a particular practitioner. Still, CARIN is an effective tool in the hands of asset recovery professionals. Each participating practitioner of the CARIN network is connected directly to a corresponding practitioner within Asset Recovery Offices (AROs) and LEAs on the one hand, and a member of the judiciary/prosecutor on the other. All the practitioners are accessible via the secretariat. The CARIN secretariat is in contact with other informal networks of the same or similar purposes. These are the ARIN-CARIB, ARINSA, ARIN-EA, ARINWA, ARINAP, ARINWCA the AROs.
	ARIN-CARIB The Asset Recovery Inter-Agency Network for the Caribbean (ARIN-CARIB) is a network of law enforcement and judicial practitioners in the field of asset tracing,

freezing, seizure and confiscation. Each Member State is represented by a law enforcement officer and a judicial expert (prosecutor, investigating judge, etc.) depending on the legal system.

ARIN-CARIB is an informal network of law enforcement and judicial practitioners specialising in the recovery of the proceeds of crime and the prosecution of related criminal offences including money laundering and the trafficking of drugs, firearms, and people.

Founded in 2017, the Caribbean model complements existing networks in Europe, Latin America, Asia, and Africa. Over 30 countries in the wider Caribbean region are now members of the network, which provides a platform for expert practitioners working in this area of key importance, helping them to target organised crime with a particular focus on enhancing cross-border and inter-agency cooperation to deprive criminals of the proceeds of crime. Each member country is represented by a law enforcement officer and a prosecutor or judicial expert, who act as national contact points for sharing information with their counterparts.

The network provides a focus for the four legal traditions found in the Caribbean region: those of the English, Dutch, French, and Spanish-speaking jurisdictions and territories. The Asset Recovery Unit of the Regional Security System (a multinational body comprising seven jurisdictions of the Eastern Caribbean) supplies the permanent secretariat to the network. The presidency of the network rotates annually.

Source: https://arin-carib.org/

ARINSA

ARINSA is a multi-agency informal network of participating countries: Botswana, Lesotho, Mauritius, Namibia, South Africa, Swaziland, Tanzania, Zambia, and Zimbabwe based on the CARIN model. Its mission is to share information, model legislation and country laws in asset forfeiture, confiscation, and money laundering. Other services include newsletters, publications, events, and links to other networks. ARINSA's objectives are to:

- focus on the proceeds and instrumentalities of all crimes, within the scope of international obligations;
- establish itself as a centre of excellence on all aspects of tackling the instrumentalities and proceeds of crime;
- promote the exchange of information and good practice;
- establish a network of contact points;
- facilitate and promote, the establishment, where possible, of national centres of excellence in all aspects of tackling the proceeds of crime;
- make recommendations to other bodies such as the Eastern and Southern African Anti-Money Laundering Group and SADC, relating to all aspects of tackling the proceeds of crime;
- act as an advisory group to appropriate authorities;
- facilitate, where possible, training in all aspects of tackling the proceeds of crime;
- recognise the importance of cooperation with the private sector in achieving its aim.

Source: https://www.arinsa.org

ARIN-EA

ARIN-EA stands for Asset Recovery Inter-Agency Network for Eastern Africa. ARIN-EA is an informal network for the informal exchange of information on individuals, assets, and companies at the regional and international levels, to facilitate the effective tracing and recovery of the proceeds of crime and to deprive criminals of their illicit gains. ARIN-EA membership is drawn from eight Eastern African countries: Burundi, Djibouti, Ethiopia, Kenya, Rwanda, South Sudan, Tanzania, and Uganda. Each member country has nominated two focal points to the network, one from an anti-corruption agency and the second from the prosecutorial/judicial/law enforcement authority (as may be relevant to each jurisdiction) – following the principles and modus operandi of the CARIN network.

ARIN-EA recognises the need to promote cooperation at the regional level as well as international levels, to effectively trace and recover stolen assets within and beyond the territorial boundaries of Eastern Africa, in collaboration with relevant partners.

Source: https://eaaaca.com/about-arinea

ARINWA

The Asset Recovery Inter-Agency Network for West Africa (ARINWA) is an informal network for information exchange on asset recovery, confiscation, and money laundering in West Africa. Each member is represented by three focal points or national contact points from the administrations in charge of justice, security and finance.

ARINWA, through an inter-ministerial approach, aims to enhance the effectiveness of its members in their efforts to deprive criminals of their illicit profits. ARINWA consists of lusophone, francophone and English-speaking West African countries: Benin, Burkina Faso, Ivory Coast, Guinea, Mali, Mauritania, Niger, Senegal, Togo, Cape Verde, Guinea-Bissau, Sao Tome and Principe, Gambia, Ghana, Liberia and Nigeria.

Source: https://en.arinwa.net

ARIN-AP

The Asset Recovery Interagency Network – Asia Pacific (ARIN-AP) is an informal network of experts and practitioners in the field of asset tracing, freezing and confiscation, which intends to serve as a cooperative group in all aspects of tackling the proceeds of crime in the Asia-Pacific region. The members are Australia, Brunei Darussalam, Cambodia, Chinese Taipei, Cook Islands, India, Indonesia, Japan, Kazakhstan, South Korea, Malaysia, Maldives, Mongolia, Myanmar, New Zealand, Nepal, Kyrgyzstan, Pakistan, Palau, Papua New Guinea, Philippines, Singapore, Sri Lanka, Thailand, Timor-Leste, Tonga, Vietnam, and Tuvalu.

ARIN-AP was officially launched at an inaugural meeting held in Seoul between 19–20 November 2013. The meeting brought together 28 law enforcement or prosecutorial agencies from 21 jurisdictions in Asia-Pacific and 6 international organisations. As of 3 March 2021, there were 28 member jurisdictions of ARIN-AP and 9 observers. ARIN-AP is modelled on CARIN. The secretariat of ARIN-AP is in the Supreme Prosecutors' Office of the Republic of Korea. ARIN-AP is to exchange information on individuals, through promoting informal direct communication among contact points for efficient asset recovery prior to or during formal mutual legal assistance. ARIN-AP has established itself as a centre of information and expertise and it promotes the exchange of information and best practices. ARIN-AP works for the mutual benefit of countries in the region as well as for inter-regional cooperation with other regional asset recovery interagency networks.

Sources: Wikipedia and http://www.arin-ap.org

ARIN-WCA

The Asset Recovery Inter-Agency Network West and Central Asia (ARIN-WCA) is an informal network of contacts of law enforcement and prosecution practitioners and a cooperative group in all aspects of tackling the proceeds of crime. It aims at increasing the effectiveness of network members' efforts, on a multi-agency basis, in depriving criminals of their illicit profits. With the working languages of English, Russian and Persian, the network comprises of the following members: Afghanistan, Iran, Kazakhstan, Kyrgyzstan, Pakistan, Tajikistan, Turkmenistan, and Uzbekistan. The latter provides a permanent secretariat located on the premises of the General Prosecutor's Office of the Republic of Uzbekistan. The secretariat aims to:

- provide professional experience, history, and continuity for the network to function effectively;
- provide administrative support to ARIN-WAC members, observers, and associates;
- support the presidency and the steering group and manage the ARIN-WCA website (under construction since 2021)
- maintain the ARIN-WCA contact list;
- draft documents related to the network's activities and devise action plans to implement ARIN-WCA outcomes and recommendations;
- promote ARIN-WCA and the concept in meetings, conferences, and other events;
- facilitate initial contact between members, observers, associates and other ARINs.

	Sources: Author's notes, desk research, UNODC, https://eaaaca.com/collaboration/ARIN %20WCA **ARO(s)** The EU Asset Recovery Offices (AROs) are national central contact points that facilitate the EU-wide tracing of assets derived from crime. Council Decision 2007/845/JHA ('the Decision') obliges Member States to set up or designate such offices. It requests AROs to exchange information under the conditions laid down in Framework Decision 2006/960/JHA and in compliance with the applicable data protection provisions. The Decision is also intended to support CARIN, the global network aimed at enhancing mutual knowledge about methods and techniques for cross-border identification, freezing, seizure and confiscation of illicitly acquired assets. Member States are required to ensure that AROs cooperate with each other and help to facilitate the tracing and identification of proceeds of crime and other crime-related assets by exchanging information and best practices, both upon request and spontaneously. The exchange of information and best practices between AROs should take place regardless of their status (administrative, law enforcement or judicial authority). The purpose of AROs is to remove the financial gain from crime. *Sources:* Wikipedia; the ARO website was not accessible at the time of writing **GAFILAT Asset Recovery Network (RRAG)** GAFILAT (Financial Action Task Force of Latin America) is one of the regional groups of the Financial Action Task Force (FATF) and is made up of Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Cuba, Dominican Republic, Ecuador, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Paraguay, Peru and Uruguay. Within GAFILAT, in accordance with the mandate imposed by members and in pursuit of its objectives to promote the continuous improvement of policies to combat money laundering, terrorist financing, and other related crimes, the effort to recover assets generated by illicit activities is critical. The tools that allow their tracking, seizure and confiscation are essential to reduce the economic potential of criminal organisations and discourage criminal activities. GAFILAT has obtained the status of associate member of the FATF and adheres to the 40 recommendations issued by this organisation. These good practices are the most recognised international standard worldwide in the prevention and combat of money laundering and terrorist financing. Currently, contact points of all GAFILAT member countries are designated, representing police authorities, prosecutors or public prosecutors, financial intelligence units or other relevant central authorities in the matter. GAFILAT observer countries have also designated contact points, which are Spain, France and Italy; and Andorra has also been approved for inclusion in the network. Through the initiative of the United Nations Office on Drugs and Crime (UNODC), and with the support of the Inter-American Drug Abuse Control Commission of the Organisation of American States (CICAD/OAS), Interpol and the executive secretariat of GAFILAT, it was proposed to generate a regional platform to develop a network of contacts, with the objective of facilitating the identification, location, and recovery of assets, products or instruments of illicit activities. That is how, based on the CARIN model, the GAFILAT Asset Recovery Network (RRAG) was formed. Each GAFILAT member country has designated two contact points, generally the prosecutor's office and the police, although in some cases other offices relevant to the functions of the network, such as asset recovery and management offices or FIUs, are also involved. Since October 2010, the RRAG has exchanged information through an electronic platform that ensures the protection and security of requests and responses. This platform is located and maintained in the Financial Intelligence Unit of Costa Rica.
Data available in a digital format	N/A (no data are managed directly by the authority)
Data maintained in a structured form	N/A
Register accessible to external third parties	N/A
Register accessible for machine-to-machine interaction	N/A

Open/extensible data model	N/A
Open/extensible architecture	N/A
Challenge	No dedicated system is in place.
Relevance of the case	The case study demonstrates the practical unifying advantage of a loosely distributed architecture (also informal in this particular case)

Comments: CARIN and asset recovery practitioners might have no direct access to data themselves; thus, they might not be the best examples in the interest of an AML database, but they are well connected and organised and can provide valuable intelligence on what databases might exist in a certain jurisdiction. The practice of asset recovery requires a high level of understanding of what information can be used from what source, through what channel to trace, seize and recover stolen assets.

Researcher's note: CARIN practitioners can be contacted through the secretariat, with a concise invitation (respecting the informal nature of the network) to join up and create a matrix of data points existing in different jurisdictions in the EU (who has what data). This approach could potentially be highly beneficial in terms of quality control and efficiency.

Table 66: European Land Registry Association

Tags	Land registry networks, cadastre, land registry, real estate
Date	5 January 2023
Member State represented	N/A
Notes on the case study	The case study questionnaire was filled in following desk research activities. *Source:* ELRA https://www.elra.eu
What authority is the subject of the study?	• The European Land Registry Association (ELRA) includes 31 organisations representing the land registries of 25 European countries. • The European Land Registry Network (ELRN), an initiative of the ELRA, includes 26 associations from 21 European countries. • ELRN is a cooperation instrument for the implementation of the Regulation on Succession and other European initiatives. It also gives assistance to other European legal networks made up of judges, bailiffs, lawyers, notaries and other legal practitioners involved in everyday land registry matters.
What data assets are available at the authority?	The association does not store any cadastral/land data assets per se, but provides a framework for collaboration between the members and other actors. The members of the association, Member State registrars, are not the sole possible sources of relevant data for the cadastre/land, given that such data might be controlled by other actors such as the following: • business registers • insolvency registers • registers or information held by municipalities or other local authorities • utility providers (water, drainage, sewage, electricity, etc.) • tax authorities • landlords, commonhold associations, management companies, communities of owners (especially in the case of apartments) • deeds not registered in the land register. ELRN provides a set of contact points that can serve to establish a connection between interested actors. The list is available via the ELRA website. From the website, information and requests for assistance from users and officials of one country will be forwarded to the intended recipient of another.

What is the role of the authority in fighting money laundering?	ELRA does not act directly in the area of AML but facilitates the development of frameworks to support cooperation and interoperability. Such actions have an impact on AML.
What is the specific role of the authority in fighting money laundering?	The objectives of ELRA in the scope of AML are the following: • harmonisation of land registry information, • support for the definition of a structured set of data shared at the level of Member States to provide information that facilitates the identification of natural or legal persons owning real estate, • facilitation of efficient management of data and use of guidelines, to trigger warnings in the case of suspicious transactions. ELRA is currently developing two projects, backed by the European Commission, that could serve the purpose of AML: • IMOLA aims to produce a model for standardised land registry output, to increase the accessibility and transparency of land registry information and to facilitate the registration of cross-border documents. • CROBECO concerns cross-border registration in foreign land registers. It introduces an alternative process for foreign buyers of real estate.
Data available in a digital format	N/A (no data are managed directly by the authority)
Data maintained in a structured form	N/A
Register accessible to external third parties	N/A
Register accessible for machine-to-machine interaction	N/A
Open/extensible data model	N/A
Open/extensible architecture	N/A
Challenge	No dedicated system is in place.
Relevance of the case	The case study gives a practical example of a relevant asset category where efforts to find common ground between Member State systems has not resulted in a working model of integration at the system and data architecture levels.

Comments: Land registration systems reflect the legal traditions of the territory in which they operate and therefore vary from place to place. As a result, land registers in each participating country record different information. Some records are more informative than others, but even the most comprehensive registers may not record every legal fact that might serve AML purposes.

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